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Jae-Hyun Yang

Generalized Heisenberg Groups and the Schrödinger–Weil Representation

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Jae-Hyun Yang

Generalized Heisenberg Groups and the Schrödinger–Weil Representation

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Preface

A certain real nilpotent Lie group plays an important role in the study of the foundations of quantum mechanics (cf. [48] and [50]) and the study of theta functions (see [3, 5, 6, 18, 32, 34, 38, 41, 49, 52, 56, 60, 63]).

For any positive integers m and n, we consider the Heisenberg group

$$H_{\mathbb{R}}^{(n,m)} \stackrel{\text{def}}{=} \left\{ (\lambda, \mu, \kappa) \mid \lambda, \mu \in \mathbb{R}^{(m,n)},\ \kappa \in \mathbb{R}^{(m,m)},\ \kappa + \mu\, {}^t\lambda \text{ symmetric} \right\}$$

endowed with the following multiplication law:

$$(\lambda, \mu, \kappa) \circ (\lambda', \mu', \kappa') = (\lambda + \lambda', \mu + \mu', \kappa + \kappa' + \lambda\, {}^t\mu' - \mu\, {}^t\lambda').$$

We let

$$H_{\mathbb{Z}}^{(n,m)} = \left\{ (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)} \mid \lambda, \mu \in \mathbb{Z}^{(m,n)},\ \kappa \in \mathbb{Z}^{(m,m)} \right\}$$

be the integral subgroup of $H_{\mathbb{R}}^{(n,m)}$ that is discrete. The Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ is embedded in the symplectic group $Sp(2(m+n), \mathbb{R})$ via the mapping

$$H_{\mathbb{R}}^{(n,m)} \ni (\lambda, \mu, \kappa) \longmapsto \begin{pmatrix} I_n & 0 & 0 & {}^t\mu \\ \lambda & I_m & \mu & \kappa \\ 0 & 0 & I_n & -{}^t\lambda \\ 0 & 0 & 0 & I_m \end{pmatrix} \in Sp(2(m+n), \mathbb{R}).$$

This Heisenberg group is a 2-step connected real nilpotent Lie group and is important in the study of smooth compactifications of the Siegel modular variety. In fact, $H_{\mathbb{R}}^{(n,m)}$ is obtained as a unipotent radical of a parabolic subgroup of a rational boundary component (cf. [2, p. 134], [9, pp. 122–123], [35, p. 21]). $H_{\mathbb{R}}^{(n,m)}$ also plays an important role in the study of partial Fourier expansions of Siegel modular forms initiated by Shimura (cf. [41, pp. 63–66]). For the case $m = 1$, studies on

this Heisenberg group were done by many mathematicians, e.g., P. Cartier [5], J. Igusa [18], D. Mumford [32, 34] and many analysts (cf. [3]) explicitly. For the case $m > 1$, the multiplication law is a little different from that of the Heisenberg group which is usually known and needs much more complicated computation than the case $m = 1$. Note that our $H_{\mathbb{R}}^{(n,m)}$ is isomorphic to the perhaps more familiar Heisenberg group

$$\mathscr{H}_{\mathbb{R}}^{(n,m)} \stackrel{\text{def}}{=} \left\{ \langle p, q, u \rangle \mid p, q \in \mathbb{R}^{(m,n)},\ u \in \mathrm{Sym}(m, \mathbb{R}) \right\}$$

equipped with the multiplication

$$\langle p, q, u \rangle \star \langle p', q', u' \rangle \stackrel{\text{def}}{=} \left\langle p + p', q + q', u + u' + \frac{1}{2} \mathbb{D}((p, q), (p', q')) \right\rangle,$$

where $p, q, p', q' \in \mathbb{R}^{(m,n)}$ and $u, u' \in \mathrm{Sym}(m, \mathbb{R})$. Here $\mathrm{Sym}(m, \mathbb{R})$ denotes the space of all $m \times m$ symmetric real matrices and

$$\mathbb{D} : \mathbb{V} \times \mathbb{V} \longrightarrow \mathrm{Sym}(m, \mathbb{R}), \qquad \mathbb{V} \stackrel{\text{def}}{=} \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}$$

is the alternating bilinear form on $\mathbb{V} \times \mathbb{V}$ defined by

$$\mathbb{D}((p, q), (p', q')) \stackrel{\text{def}}{=} p\,{}^t q' + q'\,{}^t p - p'\,{}^t q - q\,{}^t p', \quad p, q, p', q' \in \mathbb{R}^{(m,n)}.$$

Indeed, the group isomorphism

$$\Phi : (H_{\mathbb{R}}^{(n,m)}, \circ) \longrightarrow (\mathscr{H}_{\mathbb{R}}^{(n,m)}, \star)$$

is given by

$$\Phi((\lambda, \mu, \kappa)) \stackrel{\text{def}}{=} \left\langle \lambda, \mu, \kappa + \frac{1}{2}(\mu\,{}^t\lambda - \lambda\,{}^t\mu) \right\rangle, \quad (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}.$$

But we have found our model $H_{\mathbb{R}}^{(n,m)}$ to be more convenient in many situations.

The aim of this book is to investigate the generalized Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ in more details and understand its relations to theta functions, the Weil representation, the Schrödinger–Weil representation and theta sums of higher index. Using the properties of the Schrödinger–Weil representation, we obtain the transformation formulas of theta sums of higher index. The Schrödinger–Weil representation and theta sums play an important role in the theory of a pair correlation function. In the previous papers [52, 56, 60], the author decomposed the L^2-space $L^2\left(H_{\mathbb{Z}}^{(n,m)} \backslash H_{\mathbb{R}}^{(n,m)}\right)$ with respect to the right regular representation of $H_{\mathbb{R}}^{(n,m)}$ explicitly and related the study of $H_{\mathbb{R}}^{(n,m)}$ to that of theta functions. We need to

investigate $H_{\mathbb{R}}^{(n,m)}$ in detail for the study of Jacobi forms (cf. [54, 55, 57–59, 61, 66]), the study of harmonic analysis on the Jacobi group and degeneration of abelian varieties (cf. [9]).

This book is organized as follows. The book consists of two chapters. In Sect. 1.1, we introduce the four Heisenberg groups $(H_{\mathbb{R}}^{(n,m)}, \circ)$, $(\mathscr{H}_{\mathbb{R}}^{(n,m)}, \star)$, $(\omega(H_{\mathbb{R}}^{(n,m)}), \cdot)$ and $(\mathbb{H}_{\mathbb{R}}^{(n,m)}, \diamond)$ which are isomorphic as groups to each other. We find the Lie algebra of $H_{\mathbb{R}}^{(n,m)}$ and obtain the commutation relations for $H_{\mathbb{R}}^{(n,m)}$. In Sect. 1.2, we realize the Schrödinger representation of $H_{\mathbb{R}}^{(n,m)}$ as the representation of $H_{\mathbb{R}}^{(n,m)}$ induced by the unitary character of a certain subgroup of $H_{\mathbb{R}}^{(n,m)}$. In Sect. 1.3, we prove that for any positive definite symmetric, half-integral matrix $\mathscr{M}$ of degree m, the lattice representation $\pi_{\mathscr{M}}$ of $H_{\mathbb{R}}^{(n,m)}$ is unitarily equivalent to the $(\det 2\mathscr{M})^n$-multiples of the Schrödinger representation $U^{\mathcal{S},\mathscr{M}}$. In Sect. 1.4, we consider the Fock representation $\left(U^{F,\mathscr{M}}, \mathscr{H}_{F,\mathscr{M}}\right)$ of $H_{\mathbb{R}}^{(n,m)}$. We prove that for a positive definite symmetric half-integral matrix $\mathscr{M}$ of degree m, the Fock representation $U^{F,\mathscr{M}}$ is unitarily equivalent to the Schrödinger representation $U^{S,\mathscr{M}}$. We also find an orthonormal basis for the representation space $\mathscr{H}_{F,\mathscr{M}}$. This section is mainly based on the papers [38, 39, 62]. In Sect. 1.5, considering the Schrödinger representation $\left(U^{S,I_m}, L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)\right)$, we study the Hermite operators and the Hermite functions. We prove that Hermite functions defined in this section form an orthonormal basis for $L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$ and eigenfunctions for Hermite operators, the Fourier transform and the Fourier cotransform. We mention that Hermitian functions are used to construct non-holomorphic modular forms of half-integral weight (cf. [56]). In Sect. 1.6, we investigate the irreducible components of the Hilbert space $L^2\left(H_{\mathbb{Z}}^{(n,m)} \backslash H_{\mathbb{R}}^{(n,m)}\right)$. We describe the connection among these irreducible components, the Schrödinger representations, the Fock representations and the lattice representations explicitly. We also provide the orthonormal bases for these representation spaces, respectively. A decomposition of $L^2(\Gamma\backslash G)$ for a general nilpotent real Lie group G and a discrete subgroup Γ of G was investigated by C. C. Moore (cf. [31]). In Sect. 1.7, we overview Kirillov's beautiful work on the orbit method for a connected simply connected real nilpotent Lie group done early in the 1960s and find the coadjoint orbits of $H_{\mathbb{R}}^{(n,m)}$. Furthermore, we describe the connection between the coadjoint orbits and the irreducible unitary representations of $H_{\mathbb{R}}^{(n,m)}$ following the work of A. Kirillov (cf. [21–23]). In Sect. 1.8, we give an explicit description of theta functions due to J. Igusa (cf. [18] or [32]) and prove that the theta functions are realized as the smooth functions on the Heisenberg group $(\mathbb{H}_{\mathbb{R}}^{(n,m)}, \diamond)$ satisfying some conditions (see Theorem 1.8.17). In Sect. 1.9, we show that a theta function is characterized as a global section of a holomorphic line bundle over a complex abelian variety.

In Sect. 2.1, we briefly review the symplectic group and its action on the Siegel upper half plane. We present some basic properties of the geometry on the Siegel upper half plane which are used in the subsequent sections. In Sect. 2.2, we construct the universal covering group of the symplectic group using the theory

of the Maslov index. This section is based on the book of Lion and Vergne (cf. [26]). In Sect. 2.3, we study the Weil representation associated with a positive definite symmetric real matrix of degree m. We describe the explicit actions for the Weil representation. We review the results on the Weil representation which were obtained by Kashiwara and Vergne [20]. In Sect. 2.4, we construct the covariant maps for the Weil representation. The covariant maps are applied in Sect. 2.6. In Sect. 2.5, we review various types of theta series associated with quadratic forms and their transformation formulas. In Sect. 2.6, we investigate the relation between the Weil representation and the theta series. We construct modular forms using the covariant maps for the Weil representation. In Sect. 2.7, we review the Jacobi group and its action on the Siegel–Jacobi space which is needed in the subsequent sections. We describe the Lie algebra of the Jacobi group and give an Iwasawa decomposition of the Jacobi group. We briefly provide new basic geometric results on the Siegel–Jacobi space without proofs. In Sect. 2.8, we construct the canonical factor of automorphy for the Jacobi group on the Siegel–Jacobi space in a geometrical method. This automorphic factor is used to define the notion of Jacobi forms. In Sect. 2.9, we review the automorphic factor $J_{1/2}$ of weight $1/2$ for the metaplectic group $Sp(2n, \mathbb{R})_*$ on $\mathbb{H}_n$ constructed by Takase [46], Jacobi forms of integral weight and Siegel modular forms of half integral weight. In Sect. 2.10, we define the Schrödinger–Weil representation $\omega_{\mathscr{M}}$ associated with a positive definite symmetric matrix $\mathscr{M}$ and provide the basic actions of $\omega_{\mathscr{M}}$ on the representation space $L^2(\mathbb{R}^{(m,n)})$ explicitly. Furthermore, we construct the automorphic factor $J^*_{\mathscr{M}}$ of the two-fold covering group of the Jacobi group G^J using the automorphic factor $J_{1/2}$ and then find covariant maps for the Schrödinger–Weil representation $\omega_{\mathscr{M}}$. As an application of the Schrödinger–Weil representation, we construct Jacobi forms of half integral weight and index $\mathscr{M}$. The Schrödinger–Weil representation is used in the study of theta sums of higher index in Sect. 2.12. In Sect. 2.11, we discuss the theta series with harmonic coefficients. Pluriharmonic polynomials play an important role in the study of the Weil representation. We prove that the theta series with pluriharmonic polynomials as coefficients is a modular form for a suitable congruence subgroup of the Siegel modular group. This section is mainly based on the book [34], which deals with the case $m = 1$. In Sect. 2.12, we define the theta sum $\Theta_f^{[\mathscr{M}]}(\tau, \phi\,; \lambda, \mu, \kappa)$ of *higher index* and use the properties of the Schrödinger–Weil representation $\omega_{\mathscr{M}}$ to obtain some properties of the theta sum. The theta sum $\Theta_f^{[\mathscr{M}]}(\tau, \phi\,; \lambda, \mu, \kappa)$ is a generalization of the theta sum defined by J. Marklof [30]. Theta sums are applied in the study of a pair correlation function, which measures the statistics of gaps between pairs of energy levels of quantum systems.

Unfortunately, at this moment, it is very difficult and complicated to develop the spectral theory of the Laplace operator on $\Gamma_{n,m}\backslash\mathbb{H}_{n,m}$ and also to decompose the right regular representation of the Jacobi group G^J on $L^2(\Gamma^J\backslash G^J)$ into irreducible ones.

We want to mention that a Heisenberg group was given attention by some differential geometers, e.g., M. L. Gromov, in the sense of a parabolic geometry. A Heisenberg group is regarded as a principal fibre bundle over a Euclidean space

with a vector space or a circle as fibres and may be also regarded as the boundary of a complex ball. The geometry of this group is quite different from that of a Euclidean space. We refer to the interesting paper [6] of P. Deligne.

Professor Dr. Friedrich Hirzebruch passed away on May 27, 2012. He was very kind to me during my stay at MPIM, Bonn. I would like to honour him by dedicating this book to him. Finally, I hope that this book will be very useful in the study of the theory of harmonic analysis on the Siegel–Jacobi space.

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Incheon, Republic of Korea Jae-Hyun Yang

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Notations Frequently Used

$\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ integers, rational, real, complex numbers
$\mathbb{C}^\times$ multiplicative group of nonzero complex numbers
$\mathbb{C}_1^\times = \{z \in \mathbb{C} \mid |z| = 1\}$
$\mathbb{Z}^+$ (resp. $\mathbb{Z}_+$) positive (resp. nonnegative) integers
$\mathrm{Re}\, Z$, $\mathrm{Im}\, Z$ real and imaginary parts of Z
$\det(A)$ determinant of a square matrix A
$\sigma(A)$ trace of a square matrix A
$\mathrm{diag}(a_1, \ldots, a_n)$ diagonal matrix
I_n the identity matrix of degree n
$\emptyset$ empty set
$\sharp S$ or $|S|$ number of elements in S
$\dim V$ dimension of vector space or manifold
V^* dual space of vector space V
${}^tA$ transpose of matrix A
A^* conjugate transpose of matrix A
$\overline{z}$ complex conjugate of z
$GL(n, \mathbb{F})$ invertible $n \times n$ matrices with entries in a field $\mathbb{F}$
$\mathbb{F}^{(m,n)}$ $m \times n$ matrices with entries in a field $\mathbb{F}$
$GL(V)$ invertible linear maps of a vector space V onto V
$\mathrm{End}(V)$ linear maps on a vector space V into V
$\mathfrak{g}$ (resp. $\mathfrak{k}$) Lie algebra of a Lie group G (resp. K)
Ad_G adjoint representation of a Lie group G
Ad_G^* coadjoint representation of a Lie group G
$\mathrm{ad}_\mathfrak{g}$ adjoint representation of a Lie algebra $\mathfrak{g}$
$\mathrm{ad}_\mathfrak{g}^*$ coadjoint representation of a Lie algebra $\mathfrak{g}$
$\mathbb{B}$ Killing form
$\bigoplus V_i$ direct sum of vector spaces V_i
$\mathrm{Sym}(n, \mathbb{F})$ $n \times n$ symmetric matrices with entries in a field $\mathbb{F}$

$S(m, \mathbb{R}) = \mathrm{Sym}(m, \mathbb{R})$ additive group of all $m \times m$ real symmetric matrices
C^∞ infinitely differentiable
$A[B] = {}^tBAB$ Siegel's notation
$A\{B\} = B^*AB = {}^t\overline{B}AB$ Siegel's notation
δ_{ij} Kronecker delta (1 if $i = j$, 0 otherwise)
$\mathrm{Span}(S)$ linear span of subset S in a vector space
S_n permutation group of order n
$SL(n, \mathbb{F}) = \{g \in GL(n, \mathbb{F}) \mid \det(g) = 1\}$ special linear group
$O(n, \mathbb{F}) = \{g \in GL(n, \mathbb{F}) \mid {}^tgg = I_n\}$ orthogonal group
$O(n) = O(n, \mathbb{R})$ real orthogonal group
$J_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$
$Sp(2n, \mathbb{R}) = \{g \in GL(2n, \mathbb{R}) \mid {}^tgJ_ng = J_n\}$ symplectic group
$\Gamma_n = Sp(2n, \mathbb{Z})$ Siegel modular group
$K = Sp(2n, \mathbb{R}) \cap O(2n)$
$U(n) = \{A \in GL(n, \mathbb{C}) \mid A^*A = I_n\}$ unitary group
$H_{\mathbb{R}}^{(n,m)} = \{(\lambda, \mu, \kappa) \mid \lambda, \mu \in \mathbb{R}^{(m,n)},\ \kappa \in \mathbb{R}^{(m,m)},\ \kappa + \mu\,{}^t\lambda \text{ symmetric}\}$
generalized Heisenberg group
$H_{\mathbb{Z}}^{(n,m)} = \{(\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)} \mid \lambda, \mu, \kappa \text{ integral}\}$ integral Heisenberg group
$\mathfrak{Z}$ the center of $H_{\mathbb{R}}^{(n,m)}$
$\mathfrak{Z}_0$ the center of $H_{\mathbb{Z}}^{(n,m)}$
$G^J = Sp(2n, \mathbb{R}) \ltimes H_{\mathbb{R}}^{(n,m)}$ Jacobi group
$\Gamma^J = Sp(2n, \mathbb{Z}) \ltimes H_{\mathbb{Z}}^{(n,m)}$ Jacobi modular group
$\mathbb{H}_n = \{\Omega \in \mathbb{C}^{(n,n)} \mid \Omega = {}^t\Omega,\ \mathrm{Im}\,\Omega > 0\}$ Siegel upper half plane
$\mathbb{H}_{n,m} = \mathbb{H}_n \times \mathbb{C}^{(m,n)}$ Siegel–Jacobi space
$\mathbb{D}_n = \{W = {}^tW \in \mathbb{C}^{(n,n)} \mid I_n - \overline{W}W > 0\}$ generalized unit disk
$\Psi : \mathbb{D}_n \longrightarrow \mathbb{H}_n$ Cayley transform
ds^2 invariant Siegel's metric on $\mathbb{H}_n$
Δ Laplace operator of $(\mathbb{H}_n, ds^2)$
$dv_n(\Omega)$ $Sp(2n, \mathbb{R})$-invariant volume element on $\mathbb{H}_n$
$\tau(L_1, L_2, L_3)$ Maslov index
$L(H, \chi_H)$ holomorphic line bundle over an abelian variety
$\widehat{A} := \mathrm{Pic}^0(A)$ dual abelian variety of an abelian variety A
G^0 identity connected component of a group G
$\mathscr{M}$ $m \times m$ half-integral semi-positive symmetric matrix
$C(X)$ space of all continuous functions on a manifold X

$C_c^\infty(X)$ space of all compactly supported and infinitely differentiable functions on a manifold X

$\mathcal{S}(X)$ Schwartz space of infinitely differentiable functions on a manifold X that are rapidly decreasing at infinity

$$\mathbb{Z}_{\geq 0}^{(m,n)} = \left\{ J = (J_{ka}) \in \mathbb{Z}^{(m,n)} \mid J_{ka} \geq 0 \text{ for all } k, a \right\},$$

$$|J| = \sum_{k,a} J_{ka} \quad \text{for } J = (J_{ka}) \in \mathbb{Z}_{\geq 0}^{(m,n)},$$

$$J \pm \epsilon_{ka} = (J_{11}, \ldots, J_{ka} \pm 1, \ldots, J_{mn}),$$

$$J! = J_{11}! \cdots J_{ka}! \cdots J_{mn}!.$$

For $\xi = (\xi_{ka}) \in \mathbb{R}^{(m,n)}$ or $\mathbb{C}^{(m,n)}$ and $J = (J_{ka}) \in \mathbb{Z}_{\geq 0}^{(m,n)}$, we write

$$\xi^J = \xi_{11}^{J_{11}} \xi_{12}^{J_{12}} \cdots \xi_{ka}^{J_{ka}} \cdots \xi_{mn}^{J_{mn}}.$$

Two unitary representations $(\pi_1, \mathscr{H}_1)$ and $(\pi_2, \mathscr{H}_2)$ of a topological group G are said to be *equivalent* if there is an intertwining operator $\Phi : \mathscr{H}_1 \longrightarrow \mathscr{H}_2$ such that $\Phi \circ \pi_1(g) = \pi_2(g) \circ \Phi$ for all $g \in G$.

$\widehat{G}$ the *unitary dual* of G, that is, the (topological) space of all unitary equivalence classes of irreducible unitary representations of G

Chapter 1
Generalized Heisenberg Groups



1.1 Definitions of the Generalized Heisenberg Groups

For any two positive integers m and n, we consider the following Heisenberg group,

$$H_{\mathbb{R}}^{(n,m)} \stackrel{\text{def}}{=} \{ (\lambda, \mu, \kappa) \,\big|\, \lambda, \mu \in \mathbb{R}^{(m,n)},\ \kappa \in \mathbb{R}^{(m,m)},\ \kappa + \mu\, {}^t\lambda \text{ symmetric}\}, \tag{1.1.1}$$

endowed with the following multiplication law:

$$(\lambda, \mu, \kappa) \circ (\lambda_0, \mu_0, \kappa_0) = (\lambda + \lambda_0, \mu + \mu_0, \kappa + \kappa_0 + \lambda\, {}^t\mu_0 - \mu\, {}^t\lambda_0). \tag{1.1.2}$$

We observe that $H_{\mathbb{R}}^{(n,m)}$ is a connected 2-step nilpotent real Lie group. For the case $m > 1$, the multiplication law is a little different from that of the usual Heisenberg group. The inverse of an element $(\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$ is given by

$$(\lambda, \mu, \kappa)^{-1} = (-\lambda, -\mu, -\kappa + \lambda\, {}^t\mu - \mu\, {}^t\lambda).$$

The Lie algebra $\mathfrak{h}_{\mathbb{R}}^{(n,m)}$ of $H_{\mathbb{R}}^{(n,m)}$ is given by

$$\mathfrak{h}_{\mathbb{R}}^{(n,m)} = \{(P, Q, R) \mid P, Q \in \mathbb{R}^{(m,n)},\ R \in \mathrm{Sym}(m, \mathbb{R})\}$$

with the bracket

$$[(P_1, Q_1, R_1), (P_2, Q_2, R_2)] = (0, 0, P_1\, {}^tQ_2 + Q_2\, {}^tP_1 - P_2\, {}^tQ_1 - Q_1\, {}^tP_2),$$

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where $(P_i, Q_i, R_i) \in \mathfrak{h}_{\mathbb{R}}^{(n,m)}$, $i = 1, 2$. The adjoint representation $\mathrm{Ad} : H_{\mathbb{R}}^{(n,m)} \longrightarrow \mathrm{Aut}(\mathfrak{h}_{\mathbb{R}}^{(n,m)})$ of $H_{\mathbb{R}}^{(n,m)}$ is given by

$$\mathrm{Ad}(g)(P, Q, R) = (P, Q, R + \lambda\,{}^t Q + Q\,{}^t\lambda - P\,{}^t\mu - \mu\,{}^t P),$$

where $g = (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$ and $(P, Q, R) \in \mathfrak{h}_{\mathbb{R}}^{(n,m)}$.

We put $\mathbb{V} = \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}$. Let

$$\mathbb{D} : \mathbb{V} \times \mathbb{V} \longrightarrow \mathrm{Sym}(m, \mathbb{R}) \tag{1.1.3}$$

be the alternating bilinear form on $\mathbb{V} \times \mathbb{V}$ defined by

$$\mathbb{D}((p, q), (p', q')) \stackrel{\mathrm{def}}{=} p\,{}^t q' + q'\,{}^t p - p'\,{}^t q - q\,{}^t p', \quad p, q, p', q' \in \mathbb{R}^{(m,n)}.$$

Let

$$\mathscr{H}_{\mathbb{R}}^{(n,m)} \stackrel{\mathrm{def}}{=} \left\{ \langle p, q, u\rangle \mid p, q \in \mathbb{R}^{(m,n)},\ u \in \mathrm{Sym}(m, \mathbb{R}) \right\} \tag{1.1.4}$$

be the usual generalized Heisenberg group equipped with the multiplication

$$\langle p, q, u\rangle \star \langle p', q', u'\rangle \stackrel{\mathrm{def}}{=} \left\langle p + p', q + q', u + u' + \frac{1}{2}\,\mathbb{D}((p, q), (p', q')) \right\rangle, \tag{1.1.5}$$

where $p, q, p', q' \in \mathbb{R}^{(m,n)}$ and $u, u' \in \mathrm{Sym}(m, \mathbb{R})$. The inverse of an element $\langle p, q, u\rangle \in \mathscr{H}_{\mathbb{R}}^{(n,m)}$ is given by

$$\langle p, q, u\rangle^{-1} = \langle -p, -q, -u\rangle.$$

We define the mapping

$$\Phi : (H_{\mathbb{R}}^{(n,m)}, \circ) \longrightarrow (\mathscr{H}_{\mathbb{R}}^{(n,m)}, \star)$$

by

$$\Phi((\lambda, \mu, \kappa)) \stackrel{\mathrm{def}}{=} \left\langle \lambda, \mu, \kappa + \frac{1}{2}(\mu\,{}^t\lambda - \lambda\,{}^t\mu) \right\rangle, \quad (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}. \tag{1.1.6}$$

The inverse Ψ of Φ is given by

$$\Psi(\langle \lambda, \mu, \kappa\rangle) = (\lambda, \mu, \kappa + \frac{1}{2}(\lambda\,{}^t\mu + \mu\,{}^t\lambda)), \quad \langle \lambda, \mu, \kappa\rangle \in \mathscr{H}_{\mathbb{R}}^{(n,m)}.$$

Since

$$\Phi((\lambda, \mu, \kappa) \circ (\lambda', \mu', \kappa')) = \Phi((\lambda, \mu, \kappa)) \star \Phi((\lambda', \mu', \kappa')),$$

where $(\lambda, \mu, \kappa),\ (\lambda', \mu', \kappa') \in H_{\mathbb{R}}^{(n,m)}$, Φ is a group isomorphism. Thus we identify $H_{\mathbb{R}}^{(n,m)}$ with $\mathscr{H}_{\mathbb{R}}^{(n,m)}$.

The mapping $\omega : H_{\mathbb{R}}^{(n,m)} \longrightarrow Sp(2(m+n), \mathbb{R})$ defined by

$$\omega((\lambda, \mu, \kappa)) \stackrel{\text{def}}{=} \begin{pmatrix} I_n & 0 & 0 & {}^t\mu \\ \lambda & I_m & \mu & \kappa \\ 0 & 0 & I_n & -{}^t\lambda \\ 0 & 0 & 0 & I_m \end{pmatrix}, \quad (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)} \tag{1.1.7}$$

is an embedding of $H_{\mathbb{R}}^{(n,m)}$ into $Sp(2(m+n), \mathbb{R})$. The image $\omega\left(H_{\mathbb{R}}^{(n,m)}\right)$ of ω is a connected 2-step real nilpotent Lie group with the matrix multiplication and is important in the study of smooth compactifications of the Siegel modular variety. In fact, $\omega\left(H_{\mathbb{R}}^{(n,m)}\right)$ is obtained as a unipotent radical of a parabolic subgroup of a rational boundary component (cf. [2, p. 134], [9, pp. 122–123] or [35, p. 21]). The group $\omega\left(H_{\mathbb{R}}^{(n,m)}\right)$ is used in the study of partial Fourier expansions of Siegel modular forms (cf. [41, pp. 63–66]). Since ω is a group homomorphism, we identify $H_{\mathbb{R}}^{(n,m)}$ with $\omega\left(H_{\mathbb{R}}^{(n,m)}\right)$.

We consider another group

$$\mathbb{H}_{\mathbb{R}}^{(n,m)} \stackrel{\text{def}}{=} \left\{[\lambda, \mu, \kappa] \mid \lambda, \mu \in \mathbb{R}^{(m,n)},\ \kappa \in \mathrm{Sym}(m, \mathbb{R})\right\} \tag{1.1.8}$$

equipped with the following multiplication:

$$[\lambda, \mu, \kappa] \diamond [\lambda_0, \mu_0, \kappa_0] = [\lambda + \lambda_0, \mu + \mu_0, \kappa + \kappa_0 + \lambda\, {}^t\mu_0 + \mu_0\, {}^t\lambda]. \tag{1.1.9}$$

The inverse of $[\lambda, \mu, \kappa] \in \mathbb{H}_{\mathbb{R}}^{(n,m)}$ is given by

$$[\lambda, \mu, \kappa]^{-1} = [-\lambda, -\mu, -\kappa + \lambda\, {}^t\mu + \mu\, {}^t\lambda].$$

It is easily seen that the mapping $\Psi : H_{\mathbb{R}}^{(n,m)} \longrightarrow \mathbb{H}_{\mathbb{R}}^{(n,m)}$ defined by

$$\Psi((\lambda, \mu, \kappa)) = [\lambda, \mu, \kappa + \mu\, {}^t\lambda], \quad (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)} \tag{1.1.10}$$

is a group isomorphism. Therefore we identify $H_{\mathbb{R}}^{(n,m)}$ with $\mathbb{H}_{\mathbb{R}}(n, m)$.

The following matrices,

$$X_{st}^0 := \begin{pmatrix} 0\,0\,0 & 0 \\ 0\,0\,0 & \frac{1}{2}(E_{st}+E_{ts}) \\ 0\,0\,0 & 0 \\ 0\,0\,0 & 0 \end{pmatrix}, \quad 1 \le s \le t \le m,$$

$$X_{ka} := \begin{pmatrix} 0 & 0\,0 & 0 \\ E_{ka} & 0\,0 & 0 \\ 0 & 0\,0 & -{}^tE_{ka} \\ 0 & 0\,0 & 0 \end{pmatrix}, \quad 1 \le k \le m,\ 1 \le a \le n,$$

$$\widehat{X}_{lb} := \begin{pmatrix} 0\,0 & 0 & {}^tE_{lb} \\ 0\,0 & E_{lb} & 0 \\ 0\,0 & 0 & 0 \\ 0\,0 & 0 & 0 \end{pmatrix}, \quad 1 \le l \le m,\ 1 \le b \le n,$$

form a basis of the nilpotent Lie algebra $\widetilde{\mathfrak{H}}_{\mathbb{R}}^{(n,m)}$ of the real Heisenberg group $\omega\left(H_{\mathbb{R}}^{(n,m)}\right)$. Here E_{st} denotes the $m \times m$ matrix with entry 1 where the s-th row and the t-th column meet, all other entries 0, and E_{ka} (resp. E_{lb}) denotes the $m \times n$ matrix with entry 1 where the k-th (resp. the l-th) row and the a-th (resp. the b-th) column meet, all other entries 0. By an easy calculation, we see that the following vector fields,

$$D_{st}^0 := \frac{\partial}{\partial \kappa_{st}}, \quad 1 \le s \le t \le m,$$

$$D_{ka} := \frac{\partial}{\partial \lambda_{ka}} - \left(\sum_{p=1}^{k} \mu_{pa} \frac{\partial}{\partial \kappa_{pk}} + \sum_{p=k+1}^{m} \mu_{pa} \frac{\partial}{\partial \kappa_{kp}} \right),$$

$$\widehat{D}_{lb} := \frac{\partial}{\partial \mu_{lb}} + \left(\sum_{p=1}^{l} \lambda_{pb} \frac{\partial}{\partial \kappa_{pl}} + \sum_{p=l+1}^{m} \lambda_{pb} \frac{\partial}{\partial \kappa_{lp}} \right),$$

form a basis for the Lie algebra of left-invariant vector fields on the nilpotent Lie group $H_{\mathbb{R}}^{(n,m)}$, where $1 \le k, l \le m$ and $1 \le a, b \le n$.

Lemma 1.1.1 *We have the following Heisenberg commutation relations:*

$$[\, D_{kl}^0, D_{st}^0 \,] = [\, D_{kl}^0, D_{sa} \,] = [\, D_{kl}^0, \widehat{D}_{sa} \,] = 0,$$
$$[\, D_{ka}, D_{lb} \,] = [\, \widehat{D}_{ka}, \widehat{D}_{lb} \,] = 0,$$
$$[\, D_{ka}, \widehat{D}_{lb} \,] = 2\,\delta_{ab}\, D_{kl}^0,$$

where $1 \le k, l, s, t \le m,\ 1 \le a, b \le n$ *and* δ_{ab} *denotes the Kronecker delta symbol.*

Proof The proof follows from a straightforward calculation. □

We put

$$Z^0_{kl} := -\sqrt{-1}\, D^0_{kl}, \quad 1 \le k \le l \le m,$$
$$Y^+_{ka} := \frac{1}{2}\,(D_{ka} + \sqrt{-1}\, \widehat{D}_{ka}), \quad 1 \le k \le m,\ 1 \le a \le n,$$
$$Y^-_{lb} := \frac{1}{2}\,(D_{lb} - \sqrt{-1}\, \widehat{D}_{lb}), \quad 1 \le l \le m,\ 1 \le b \le n.$$

Then it is easy to see that the vector fields Z^0_{kl}, Y^+_{ka}, Y^-_{lb} form a basis of the complexification of the real Lie algebra $\widetilde{\mathfrak{H}}^{(n,m)}_{\mathbb{R}}$.

Lemma 1.1.2 *We have the following commutation relations:*

$$[Z^0_{kl}, Z^0_{st}] = [Z^0_{kl}, Y^+_{sa}] = [Z^0_{kl}, Y^-_{sa}] = 0,$$
$$[Y^+_{ka}, Y^+_{lb}] = [Y^-_{ka}, Y^-_{lb}] = 0,$$
$$[Y^+_{ka}, Y^-_{lb}] = \delta_{ab}\, Z^0_{kl},$$

where $1 \le k, l, s, t \le m$ *and* $1 \le a, b \le n$.

Proof It follows immediately from Lemma 1.1.1. □

We let $E^\bullet_{kl} := E_{kl} + E_{lk}$ for $1 \le k \le l \le m$. We put

$$R_{kl}(r) := \exp\left(2r X^0_{kl}\right) = (0, 0, r E^\bullet_{kl}), \quad r \in \mathbb{R},$$
$$P_{sa}(x) := \exp\left(x X_{sa}\right) = (x E_{sa}, 0, 0), \quad x \in \mathbb{R},$$
$$Q_{tb}(y) := \exp\left(y \widehat{X}_{tb}\right) = (0, y E_{tb}, 0), \quad y \in \mathbb{R},$$

where $1 \le k \le l \le m$, $1 \le s, t \le m$ and $1 \le a, b \le n$. Then these one-parameter subgroups generate the Heisenberg group $H^{(n,m)}_{\mathbb{R}}$. They satisfy the Weyl commutation relations :

$$P_{sa}(x) \circ Q_{sa}(y) = Q_{sa}(y) \circ P_{sa}(x) \circ R_{ss}(xy) \quad (\text{all others commute}),$$

where $1 \le s \le m$ and $1 \le a \le n$.

J. von Neumann [48] and M. Stone [45] proved the following uniqueness theorem simultaneously and independently.

Theorem 1.1.3 *Let π_1 and π_2 be two irreducible unitary representations of the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ such that*

$$\pi_1((0,0,\kappa)) = \pi_2((0,0,\kappa)) \qquad \text{for all } \kappa = {}^t\kappa \in \mathbb{R}^{(m,m)}.$$

Assume that both the central characters of π_1 and π_2 are nontrivial. Then π_1 is unitarily equivalent to π_2.

We omit the proof of the above theorem. We refer to [26] for the proof of Theorem 1.1.3 in the case $m = 1$ and also to [5] and [34] for more details.

We return to the Heisenberg group $(\mathbb{H}_{\mathbb{R}}^{(n,m)}, \diamond)$ (see (1.1.8)). We set

$$\mathbb{A} = \left\{ [0,\mu,\kappa] \in \mathbb{H}_{\mathbb{R}}^{(n,m)} \,\middle|\, \mu \in \mathbb{R}^{(m,n)},\ \kappa \in \mathrm{Sym}(m,\mathbb{R}) \right\}. \tag{1.1.11}$$

Then $\mathbb{A} \cong \mathbb{R}^{(m,n)} \times \mathrm{Sym}\,(m,\mathbb{R})$ is a commutative normal subgroup of $\mathbb{H}_{\mathbb{R}}^{(n,m)}$. Let $\widehat{\mathbb{A}}$ be the Pontryagin dual of $\mathbb{A}$, i.e., the commutative group consisting of all unitary characters of $\mathbb{A}$. Then $\widehat{\mathbb{A}}$ is isomorphic to the additive group $\mathbb{A}$ via

$$\langle a, \widehat{a}\rangle := e^{2\pi i\, \sigma(\widehat{\mu}\,{}^t\mu + \widehat{\kappa}\kappa)}, \quad a = [0,\mu,\kappa] \in \mathbb{A},\ \widehat{a} = (\widehat{\mu},\widehat{\kappa}) \in \widehat{\mathbb{A}}. \tag{1.1.12}$$

We put

$$\mathbb{S} = \left\{ [\lambda,0,0] \in \mathbb{H}_{\mathbb{R}}^{(n,m)} \,\middle|\, \lambda \in \mathbb{R}^{(m,n)} \right\} \cong \mathbb{R}^{(m,n)}. \tag{1.1.13}$$

Then $\mathbb{S}$ acts on $\mathbb{A}$ as follows:

$$\alpha_\lambda([0,\mu,\kappa]) := [0,\mu,\kappa + \lambda\,{}^t\mu + \mu\,{}^t\lambda], \quad \alpha_\lambda = [\lambda,0,0] \in \mathbb{S}. \tag{1.1.14}$$

It is easy to see that the Heisenberg group $\left(\mathbb{H}_{\mathbb{R}}^{(n,m)}, \diamond\right)$ is isomorphic to the semidirect product $G_H := \mathbb{S} \ltimes \mathbb{A}$ of $\mathbb{S}$ and $\mathbb{A}$ whose multiplication is given by

$$(\lambda, a)\cdot(\lambda_0, a_0) = (\lambda + \lambda_0,\ a + \alpha_\lambda(a_0)), \quad \lambda,\lambda_0 \in \mathbb{S},\ a, a_0 \in \mathbb{A}.$$

On the other hand, $\mathbb{S}$ acts on $\widehat{\mathbb{A}}$ by

$$\alpha_\lambda^*(\widehat{a}) := (\widehat{\mu} + 2\widehat{\kappa}\,\lambda, \widehat{\kappa}), \quad [\lambda,0,0] \in \mathbb{S},\ \ \widehat{a} = (\widehat{\mu},\widehat{\kappa}) \in \widehat{\mathbb{A}}. \tag{1.1.15}$$

Then we have the relation $\langle \alpha_\lambda(a), \widehat{a}\rangle = \langle a, \alpha_\lambda^*(\widehat{a})\rangle$ for all $a \in \mathbb{A}$ and $\widehat{a} \in \widehat{\mathbb{A}}$.

We have three types of $\mathbb{S}$-orbits in $\widehat{\mathbb{A}}$.

TYPE I. Let $\widehat{\kappa} \in \mathrm{Sym}(m, \mathbb{R})$ be nondegenerate. The $\mathbb{S}$-orbit of $(0, \widehat{\kappa}) \in \widehat{\mathbb{A}}$ is given by

$$\widehat{\mathcal{O}}_{\widehat{\kappa}} = \left\{ (2\widehat{\kappa}\lambda, \widehat{\kappa}) \in \widehat{\mathbb{A}} \,\middle|\, \lambda \in \mathbb{R}^{(m,n)} \right\} \cong \mathbb{R}^{(m,n)}.$$

TYPE II. Let $(\widehat{\mu}, \widehat{\kappa}) \in \mathbb{R}^{(m,n)} \times \mathrm{Sym}(m, \mathbb{R})$ with $\widehat{\mu} \in \mathbb{R}^{(m,n)}$, $\widehat{\kappa} \in \mathrm{Sym}(m, \mathbb{R})$ and degenerate $\widehat{\kappa} \neq 0$. Then

$$\widehat{\mathcal{O}}_{(\widehat{\mu},\widehat{\kappa})} = \left\{ (\widehat{\mu} + 2\widehat{\kappa}\lambda, \widehat{\kappa}) \middle| \, \lambda \in \mathbb{R}^{(m,n)} \right\} \subsetneqq \mathbb{R}^{(m,n)} \times \{\widehat{\kappa}\}.$$

TYPE III. Let $\widehat{y} \in \mathbb{R}^{(m,n)}$. The $\mathbb{S}$-orbit $\widehat{\mathcal{O}}_{\widehat{y}}$ of $(\widehat{y}, 0)$ is given by

$$\widehat{\mathcal{O}}_{\widehat{y}} = \{(\widehat{y}, 0)\}.$$

We have

$$\widehat{\mathbb{A}} = \left(\bigcup_{\substack{\widehat{\kappa} \in \mathrm{Sym}(m,\mathbb{R}) \\ \widehat{\kappa} \text{ nondegenerate}}} \widehat{\mathcal{O}}_{\widehat{\kappa}} \right) \bigcup \left(\bigcup_{\widehat{y} \in \mathbb{R}^{(m,n)}} \widehat{\mathcal{O}}_{\widehat{y}} \right) \bigcup \left(\bigcup_{\substack{(\widehat{\mu},\widehat{\kappa}) \in \mathbb{R}^{(m,n)} \times \mathrm{Sym}(m,\mathbb{R}) \\ \widehat{\kappa} \neq 0 \text{ degenerate}}} \widehat{\mathcal{O}}_{(\widehat{\mu},\widehat{\kappa})} \right)$$

as a set. The stabilizer $\mathbb{S}_{\widehat{\kappa}}$ of $\mathbb{S}$ at $(0, \widehat{\kappa})$ with nondegenerate $\widehat{\kappa}$ is given by

$$\mathbb{S}_{\widehat{\kappa}} = \{0\}. \tag{1.1.16}$$

And the stabilizer $\mathbb{S}_{\widehat{y}}$ of $\mathbb{S}$ at $(\widehat{y}, 0)$ is given by

$$\mathbb{S}_{\widehat{y}} = \left\{ [\lambda, 0, 0] \,\middle|\, \lambda \in \mathbb{R}^{(m,n)} \right\} = \mathbb{S} \cong \mathbb{R}^{(m,n)}. \tag{1.1.17}$$

We put $\mathbb{C}_1^\times = \{z \in \mathbb{C} \mid |z| = 1\}$. We see that an irreducible unitary representation of $G_H \cong \mathbb{H}_{\mathbb{R}}^{(n,m)}$ is of the following form:

$$T_{\widehat{\kappa}} \overset{\text{def}}{=} \mathrm{Ind}_{\mathbb{S}_{\widehat{\kappa}} \ltimes \mathbb{A}}^{G_H} \chi_{\widehat{\kappa}} \cdot \widehat{a}(\widehat{\kappa}) = \mathrm{Ind}_{\mathbb{A}}^{G_H} \widehat{a}(\widehat{\kappa}), \quad \widehat{\kappa} \in \mathrm{Sym}(m, \mathbb{R})$$

or

$$T_{\widehat{x},\widehat{y}} \overset{\text{def}}{=} \mathrm{Ind}_{\mathbb{S}_{\widehat{y}} \ltimes \mathbb{A}}^{G_H} \chi_{\widehat{x}} \cdot \widehat{a}(\widehat{y}) = \mathrm{Ind}_{\mathbb{S} \ltimes \mathbb{A}}^{G_H} \chi_{\widehat{x}} \cdot \widehat{a}(\widehat{y}), \quad \widehat{x}, \widehat{y} \in \mathbb{R}^{(m,n)},$$

where $\chi_{\widehat{\kappa}} : \mathbb{S}_{\widehat{\kappa}} \longrightarrow \mathbb{C}_1^\times$ is a trivial unitary character of $\mathbb{S}_{\widehat{\kappa}} = \{0\}$ (cf. (1.1.16)), $\widehat{a}(\widehat{\kappa}) : \mathbb{A} \longrightarrow \mathbb{C}_1^\times$ is the unitary character of $\mathbb{A}$ defined by

$$\widehat{a}(\widehat{\kappa})([0, \mu, \kappa]) \overset{\text{def}}{=} e^{2\pi i \sigma(\widehat{\kappa}\kappa)}, \qquad [0, \mu, \kappa] \in \mathbb{A},$$

$\chi_{\widehat{x}}$ is the unitary character of $\mathbb{S}$ defined by $\chi_{\widehat{x}}(\ell) := e^{2\pi i\,\sigma(\widehat{x}^t\ell)}$ for $\ell \in \mathbb{S}$ (we note that $\mathbb{S}_{\widehat{y}} = \mathbb{S}$ by (1.1.17)) and $\widehat{a}(\widehat{y}) : \mathbb{A} \longrightarrow \mathbb{C}_1^\times$ is the unitary character of $\mathbb{A}$ defined by

$$\widehat{a}(\widehat{y})([0,\mu,\kappa]) \stackrel{\text{def}}{=} e^{2\pi\, i\,\sigma(\widehat{y}^t\mu)}, \qquad [0,\mu,\kappa] \in \mathbb{A}.$$

Therefore the unitary dual $\widehat{G}_H$ of G_H or $\mathbb{H}_{\mathbb{R}}^{(n,m)}$ is determined completely by

TYPE **I.** $\widehat{\kappa} \in \mathrm{Sym}\,(m,\mathbb{R}),\ \widehat{\kappa} \neq 0.$
TYPE **II.** $(\widehat{x},\widehat{y}) \in \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}$ with $\widehat{x},\widehat{y} \in \mathbb{R}^{(m,n)}$.

The action of the center of G_H on the representation of *type* I is nontrivial. On the other hand, the representation $\rho \in \widehat{G}_H$ of *type* II acts trivially on the center $\mathfrak{Z}$ of G_H.

1.2 Schrödinger Model

We briefly review the work of Mackey on induced representations (cf. [29]). Let G be a locally compact, separable topological group and K be a closed subgroup of G. Let σ be an irreducible unitary representation of K in a separable Hilbert space $\mathscr{H}$. Let μ be a G-invariant measure in the homogeneous space $X := K\backslash G = \{Kg \mid g \in G\}$ of the right K-cosets in G. We denote the induced representation of G from σ by

$$U_\sigma := \mathrm{Ind}_K^G\,\sigma.$$

Let $\mathscr{H}^\sigma$ be the Hilbert space consisting of all functions $\phi : G \longrightarrow \mathscr{H}$ which satisfy the following conditions:

(1) $(\phi(g), v)_{\mathscr{H}}$ is measurable with respect to dg for all $v \in \mathscr{H}$,
(2) $\phi(kg) = \sigma(k)(\phi(g))$ for all $k \in K$ and $g \in G$,
(3) $\| \phi \|^2 = \int_X \| \phi(g) \|^2\, d\mu(\dot{g}) < \infty,\ \ \dot{g} = Kg,$

where dg is a G-invariant measure on G, $(\ ,\ )_{\mathscr{H}}$ is an inner product in $\mathscr{H}$ and $\| \phi(g) \|$ is the norm in $\mathscr{H}$. The inner product $(\ ,\ )$ in $\mathscr{H}^\sigma$ is given by

$$(\phi_1,\phi_2) = \int_X (\phi_1(g),\phi_2(g))_{\mathscr{H}}\, d\mu(\dot{g}), \quad \phi_1,\phi_2 \in \mathscr{H}^\sigma.$$

Then $U_\sigma = \mathrm{Ind}_K^G\,\sigma$ is realized in the Hilbert space $\mathscr{H}^\sigma$ as follows:

$$(U_\sigma(g_0)\phi)\,(g) = \phi(gg_0), \quad g, g_0 \in G,\ \phi \in \mathscr{H}^\sigma. \tag{1.2.1}$$

It is easy to see that $\mathscr{H}^\sigma$ is isomorphic to the Hilbert space $\mathscr{H}_\sigma := L^2(X, \mu, \mathscr{H})$ of square integrable functions $f : X \longrightarrow \mathscr{H}$ with values in $\mathscr{H}$ via the formula

$$\phi_f(g) = \sigma(k_g)\,(f(\dot{g}))\,, \quad f \in \mathscr{H}_\sigma,\ g \in G, \tag{1.2.2}$$

where $\dot{g} = Kg$ and k_g is the K-component of g in the decomposition $g = k_g s_g$.

We can show easily that U_σ is realized in $\mathscr{H}_\sigma$ by

$$(U_\sigma(g_0)f)\,(\dot{g}) = \sigma(k_{s_g g_0})\,(f(\dot{g}g_0))\,, \tag{1.2.3}$$

where $g_0 \in G,\ f \in \mathscr{H}_\sigma,\ \dot{g} = Kg \in X$ and $k_{s_g g_0}$ denotes the K-component of $s_g g_0$ in the decomposition of $s_g g_0$.

If σ is a one-dimensional representation of K, U_σ is called a monomial representation.

For two fixed positive integers m and n, we put $G := H_{\mathbb{R}}^{(n,m)}$ for brevity and

$$K := \left\{(0, \mu, \kappa) \in G \mid \mu \in \mathbb{R}^{(m,n)},\ \kappa \in \mathrm{Sym}(m, \mathbb{R})\right\}. \tag{1.2.4}$$

Then K is a closed, commutative normal subgroup of G. Since $(\lambda, \mu, \kappa) = (0, \mu, \kappa + \mu\,{}^t\lambda) \circ (\lambda, 0, 0)$ for $(\lambda, \mu, \kappa) \in G$, the homogeneous space $X := K\backslash G$ is identified with $\mathbb{R}^{(m,n)}$ via

$$Kg = K \circ (\lambda, 0, 0) \longmapsto \lambda, \quad g = (\lambda, \mu, \kappa) \in G.$$

We observe that G acts on X by

$$(Kg) \cdot g_0 = K\,(\lambda + \lambda_0, 0, 0), \tag{1.2.5}$$

where $g = (\lambda, \mu, \kappa) \in G$ and $g_0 = (\lambda_0, \mu_0, \kappa_0) \in G$.

If $g = (\lambda, \mu, \kappa) \in G$, we have

$$k_g = (0, \mu, \kappa + \mu\,{}^t\lambda) \in K, \quad s_g = (\lambda, 0, 0) \tag{1.2.6}$$

in the decomposition of $g = k_g \circ s_g$. Thus if $g_0 = (\lambda_0, \mu_0, \kappa_0) \in G$, then we have

$$s_g \circ g_0 = (\lambda, 0, 0) \circ (\lambda_0, \mu_0, \kappa_0) = (\lambda + \lambda_0, \mu_0, \kappa_0 + \lambda\,{}^t\mu_0) \tag{1.2.7}$$

and so

$$k_{s_g \circ g_0} = (0, \mu_0, \kappa_0 + \mu_0\,{}^t\lambda_0 + \lambda\,{}^t\mu_0 + \mu_0\,{}^t\lambda). \tag{1.2.8}$$

For a real symmetric matrix $c = {}^t c \in \mathbb{R}^{(m,m)}$ with $c \neq 0$, we consider the one-dimensional unitary representation σ_c of K defined by

$$\sigma_c((0,\mu,\kappa)) := e^{2\pi i\,\sigma(c\kappa)}\, I, \quad (0,\mu,\kappa) \in K, \tag{1.2.9}$$

where I denotes the identity mapping. Then the induced representation $U_{\sigma_c} := \mathrm{Ind}_K^G \sigma_c$ of G induced from σ_c is realized in the Hilbert space $\mathscr{H}_{\sigma_c} = L^2(X, d\dot{g}, \mathbb{C}) \cong L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$ as follows. If $g_0 = (\lambda_0, \mu_0, \kappa_0) \in G$ and $x = Kg \in X$ with $g = (\lambda, \mu, \kappa) \in G$, then according to (1.2.3), we have

$$\left(U_{\sigma_c}(g_0) f\right)(x) = \sigma_c\left(k_{s_g \circ g_0}\right)\left(f(xg_0)\right), \quad f \in \mathscr{H}_{\sigma_c}. \tag{1.2.10}$$

It follows from (1.2.8) that

$$\left(U_{\sigma_c}(g_0) f\right)(\lambda) = e^{2\pi i \sigma\{c(\kappa_0 + \mu_0\, {}^t\lambda_0 + 2\lambda\, {}^t\mu_0)\}} f(\lambda + \lambda_0). \tag{1.2.11}$$

Here we identified $x = Kg$ (resp. $xg_0 = Kgg_0$) with λ (resp. $\lambda + \lambda_0$). The induced representation U_{σ_c} is called the Schrödinger representation of G associated with σ_c. Therefore U_{σ_c} is a monomial representation.

We denote by $\mathscr{H}^{\sigma_c}$ the Hilbert space consisting of all functions $\phi : G \longrightarrow \mathbb{C}$ which satisfy the following conditions:

(4) $\phi(g)$ is measurable with respect to dg,
(5) $\phi((0,\mu,\kappa) \circ g) = e^{2\pi i\,\sigma(c\kappa)}\phi(g)$ for all $g \in G$,
(6) $\| \phi \|^2 := \int_X |\phi(g)|^2\, d\dot{g} < \infty, \quad \dot{g} = Kg,$

where dg (resp. $d\dot{g}$) is a G-invariant measure on G (resp. $X = K\backslash G$). The inner product $(\ ,\ )$ on $\mathscr{H}^{\sigma_c}$ is given by

$$(\phi_1, \phi_2) = \int_G \phi_1(g)\,\overline{\phi_2(g)}\, dg, \quad \phi_1,\ \phi_2 \in \mathscr{H}^{\sigma_c}.$$

We observe that $\mathscr{H}_{\sigma_c} = L^2(\mathbb{R}^{(m,n)}, d\xi)$ and that the mapping $\Phi_c : \mathscr{H}_{\sigma_c} \longrightarrow \mathscr{H}^{\sigma_c}$ defined by

$$(\Phi_c(f))(g) := \phi_f(g) := e^{2\pi i \sigma\{c(\kappa + \mu\, {}^t\lambda)\}} f(\lambda) \tag{1.2.12}$$

($f \in \mathscr{H}_{\sigma_c},\ g = (\lambda, \mu, \kappa) \in G$) is an isomorphism of Hilbert spaces. The inverse $\Psi_c : \mathscr{H}^{\sigma_c} \longrightarrow \mathscr{H}_{\sigma_c}$ of Φ_c is given by

$$(\Psi_c(\phi))(\lambda) := f_\phi(\lambda) := \phi((\lambda, 0, 0)), \quad \phi \in \mathscr{H}^{\sigma_c},\ \lambda \in \mathbb{R}^{(m,n)}. \tag{1.2.13}$$

From now on, for brevity we put

$$U_c = U_{\sigma_c}, \quad \mathscr{H}_c = \mathscr{H}_{\sigma_c} \quad \text{and} \quad \mathscr{H}^c = \mathscr{H}^{\sigma_c}.$$

The Schrödinger representation U_c of G on $\mathscr{H}^c$ is given by

$$(U_c(g_0)\phi)(g) = e^{2\pi i\sigma\{c(\kappa_0+\mu_0\,{}^t\lambda_0+2\lambda\,{}^t\mu_0-2\lambda_0\,{}^t\mu)\}}\,\phi((\lambda_0,0,0)\circ g), \tag{1.2.14}$$

where $g_0 = (\lambda_0, \mu_0, \kappa_0)$, $g = (\lambda, \mu, \kappa) \in G$ and $\phi \in \mathscr{H}^c$. Using (1.2.6) and (4), Formula (1.2.14) can be expressed as follows.

$$(U_c(g_0)\phi)(g) = e^{2\pi i\sigma\{c(\kappa_0+\kappa+\mu_0\,{}^t\lambda_0+\mu\,{}^t\lambda+2\lambda\,{}^t\mu_0)\}}\,\phi((\lambda_0+\lambda,0,0)). \tag{1.2.15}$$

Theorem 1.2.1 *Let c be a positive symmetric half-integral matrix of degree m. Then the Schrödinger representation U_c of G is irreducible.*

Proof The proof can be found in [52, Theorem 3]. □

We let dU_c be the infinitesimal representation associated with the Schrödinger representation U_c. If X is an element of the Lie algebra of G, then

$$dU_c(X)f = \frac{d}{dt}\Big|_{t=0} U_c(\exp tX)f, \quad f \in \mathscr{H}_c \text{ or } \mathscr{H}^c.$$

We fix an element $\Omega \in \mathbb{H}_n$ once and for all. We let c be a positive symmetric real matrix of degree m. For each $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$, we put

$$f_{c,J}(\xi) := e^{2\pi i\,\sigma(c\,\xi\Omega\,{}^t\xi)}\,\xi^J, \quad \xi \in \mathbb{R}^{(m,n)}. \tag{1.2.16}$$

For the definitions of $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$ and ξ^J, we refer to Notations Frequently Used, p. xv. Then the set $\left\{ f_{c,J} \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ spans a dense subspace of $L^2\big(\mathbb{R}^{(m,n)}, d\xi\big) \cong \mathscr{H}_c$.

For each $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$, we put

$$\phi_{c,J}(g) = e^{2\pi i\,\sigma\{c(\kappa+\mu\,{}^t\lambda)\}}\,f_{c,J}(\lambda), \tag{1.2.17}$$

where $g = (\lambda, \mu, \kappa) \in G$. Then the set $\{\,\phi_{c,J} \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)}\,\}$ spans a dense subspace of $\mathscr{H}^c$.

1.3 Lattice Model

Let $L := \mathbb{Z}^{(m,n)} \times \mathbb{Z}^{(m,n)}$ be the lattice in the vector space $V = \mathbb{C}^{(m,n)}$. Let B be an alternating bilinear form on V such that $B(L, L) \subset \mathbb{Z}$, that is, $\mathbb{Z}$-valued on $L \times L$. The dual L_B^* of L with respect to B is defined by

$$L_B^* := \{\, v \in V \mid B(v, \ell) \in \mathbb{Z} \text{ for all } \ell \in L \,\}.$$

Then $L \subset L_B^*$. If B is nondegenerate, L_B^* is also a lattice in V, called the *dual lattice* of L. In the case where B is nondegenerate, there exist a $\mathbb{Z}$-basis $\{\xi_{11}, \xi_{12}, \ldots, \xi_{mn}, \eta_{11}, \eta_{12}, \ldots, \eta_{mn}\}$ of L and a set $\{e_{11}, e_{12}, \ldots, e_{mn}\}$ of positive integers with $e_{11}|e_{12},\ e_{12}|e_{13}, \ldots, e_{m,n-1}|e_{mn}$ such that

$$\begin{pmatrix} B(\xi_{ka}, \xi_{lb}) & B(\xi_{ka}, \eta_{lb}) \\ B(\eta_{ka}, \xi_{lb}) & B(\eta_{ka}, \eta_{lb} \end{pmatrix} = \begin{pmatrix} 0 & e \\ -e & 0 \end{pmatrix},$$

where $1 \leq k, l \leq m,\ 1 \leq a, b \leq n$ and $e := \mathrm{diag}\,(e_{11}, e_{12}, \ldots, e_{mn})$ is the diagonal matrix of degree mn with entries $e_{11}, e_{12}, \ldots, e_{mn}$. It is well known that $[L_B^* : L] = (\det e)^2 = (e_{11}e_{12} \cdots e_{mn})^2$ (cf. [18, p. 72]). The number $\det e$ is called the Pfaffian of B.

We put $G = H_{\mathbb{R}}^{(n,m)}$ for brevity. Now we consider the following subgroups of G:

$$\Gamma_L = \left\{ (\lambda, \mu, \kappa) \in G \mid (\lambda, \mu) \in L,\ \kappa \in \mathbb{R}^{(m,m)} \right\} \tag{1.3.1}$$

and

$$\Gamma_{L_B^*} = \left\{ (\lambda, \mu, \kappa) \in G \mid (\lambda, \mu) \in L_B^*,\ \kappa \in \mathbb{R}^{(m,m)} \right\}. \tag{1.3.2}$$

Then both Γ_L and $\Gamma_{L_B^*}$ are the normal subgroups of G.

Let $\mathfrak{Z} = \{(0, 0, \kappa) \mid \kappa \in \mathrm{Sym}(m, \mathbb{R})\}$ be the center of G. We put

$$\mathfrak{Z}_0 = \{ (0, 0, \kappa) \in \mathfrak{Z} \mid \kappa \in \mathrm{Sym}(m, \mathbb{R}) \text{ integral} \}. \tag{1.3.3}$$

It is easy to show that

$$\Gamma_{L_B^*} = \left\{ g \in G \mid g\gamma g^{-1}\gamma^{-1} \in \mathfrak{Z}_0 \text{ for all } \gamma \in \Gamma_L \right\}.$$

We define

$$Y_L = \left\{ \phi \in \mathrm{Hom}\,(\Gamma_L, \mathbb{C}_1^\times) \mid \phi \text{ is trivial on } \mathfrak{Z}_0 \right\} \tag{1.3.4}$$

and

$$Y_{L,S} = \left\{ \phi \in Y_L \mid \phi(\kappa) = e^{2\pi i \sigma(S\kappa)} \text{ for all } \kappa \in \mathrm{Sym}(m, \mathbb{R}) \right\} \tag{1.3.5}$$

for each symmetric real matrix S of degree m. We observe that if S is not half-integral, then $Y_L = \emptyset$ and so $Y_{L,S} = \emptyset$. It is clear that if S is symmetric half-integral, then $Y_{L,S}$ is not empty. Thus we have

$$Y_L = \bigcup_{\mathscr{M}} Y_{L,\mathscr{M}}, \tag{1.3.6}$$

where $\mathscr{M}$ runs through the set of all symmetric half-integral matrices of degree m.

Lemma 1.3.1 *Let $\mathscr{M}$ be a symmetric half-integral matrix of degree m with $\mathscr{M} \neq 0$. Then any element φ of $Y_{L,\mathscr{M}}$ is of the form $\varphi_{\mathscr{M},q}$. Here $\varphi_{\mathscr{M},q}$ is the character of Γ_L defined by*

$$\varphi_{\mathscr{M},q}((\ell,\kappa)) := e^{2\pi i\,\sigma(\mathscr{M}\kappa)} \cdot e^{\pi i\,q(\ell)}, \quad (\ell,\kappa) \in \Gamma_L, \tag{1.3.7}$$

where $q : L \longrightarrow \mathbb{R}/2\mathbb{Z} \cong [0,2)$ is a function on L satisfying the following condition:

$$q(\ell_0+\ell_1) \equiv q(\ell_0) + q(\ell_1) - 2\sigma\{\mathscr{M}(\lambda_0\,{}^t\mu_1 - \mu_0\,{}^t\lambda_1)\} \mod 2 \tag{1.3.8}$$

for all $\ell_0 = (\lambda_0,\mu_0) \in L$ and $\ell_1 = (\lambda_1,\mu_1) \in L$.

Proof (1.3.8) follows immediately from the fact that $\varphi_{\mathscr{M},q}$ is a character of Γ_L. It is obvious that any element of $Y_{L,\mathscr{M}}$ is of the form $\varphi_{\mathscr{M},q}$. □

Lemma 1.3.2 *An element of $Y_{L,0}$ is of the form $\varphi_{k,l}$ ($k,l \in \mathbb{R}^{(m,n)}$). Here $\varphi_{k,l}$ is the character of Γ_L defined by*

$$\varphi_{k,l}(\gamma) := e^{2\pi i\,\sigma(k\,{}^t\lambda + l\,{}^t\mu)}, \quad \gamma = (\lambda,\mu,\kappa) \in \Gamma_L. \tag{1.3.9}$$

Proof It is easy to prove it and so we omit the proof. □

Lemma 1.3.3 *Let $\mathscr{M}$ be a nonsingular symmetric half-integral matrix of degree m. Let $\varphi_{\mathscr{M},q_1}$ and $\varphi_{\mathscr{M},q_2}$ be the characters of Γ_L defined by (1.3.7). The character φ of Γ_L defined by $\varphi := \varphi_{\mathscr{M},q_1} \cdot \varphi_{\mathscr{M},q_2}^{-1}$ is an element of $Y_{L,0}$.*

Proof It follows from the fact that there exists an element $g = (\mathscr{M}^{-1}\lambda, \mathscr{M}^{-1}\mu, 0) \in G$ with $(\lambda,\mu) \in V$ such that

$$\varphi_{\mathscr{M},q_1}(\gamma) = \varphi_{\mathscr{M},q_2}(g\gamma g^{-1}) \quad \text{for all } \gamma \in \Gamma_L.$$

□

Let $V = \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)} \cong \mathbb{C}^{(m,n)}$. We put

$$P_{ka} = (E_{ka}, 0), \quad Q_{lb} = (0, E_{lb}),$$

where $1 \leq k,l \leq m$ and $1 \leq a,b \leq n$. Here E_{ka} (resp. E_{lb}) denotes the $m \times n$ matrix with entry 1 where the k-th (resp. the l-th) row and the a-th (resp. the b-th) column meet, all other entries 0. Then the set $\{P_{ka},\ Q_{ka}\}$ forms a basis for V. The alternating bilinear form $\mathbf{A}$ on V defined by

$$\mathbf{A}((\lambda_0,\mu_0),(\lambda,\mu)) = \sigma(\lambda_0\,{}^t\mu - \mu_0\,{}^t\lambda), \quad (\lambda_0,\mu_0),\ (\lambda,\mu) \in V.$$

is nondegenerate and $\mathbb{Z}$-valued on $L \times L$. Then we have

$$\mathbf{A}(P_{ka}, P_{lb}) = \mathbf{A}(Q_{ka}, Q_{lb}) = 0, \quad \mathbf{A}(P_{ka}, Q_{lb}) = \delta_{ab}\,\delta_{kl},$$

where $1 \le k, l \le m$ and $1 \le a, b \le n$. Therefore the elementary divisors $e_{11}, e_{12}, \ldots, e_{mn}$ of $\mathbf{A}$ are all one and L is self-dual, i.e., $L = L^*_{\mathbf{A}}$. The set

$$\{\, P_{11}, P_{12}, \ldots, P_{mn}, Q_{11}, Q_{12}, \ldots, Q_{mn} \,\}$$

forms a symplectic basis of V with respect to $\mathbf{A}$. We fix a coordinate $P_{11}, \ldots, P_{mn}, Q_{11}, \ldots, Q_{mn}$ on V.

For a unitary character $\varphi_{\mathcal{M},q}$ of Γ_L defined by (1.3.7), we let

$$\pi_{\mathcal{M},q} = \operatorname{Ind}_{\Gamma_L}^{G} \varphi_{\mathcal{M},q} \tag{1.3.10}$$

be the representation of G induced from $\varphi_{\mathcal{M},q}$. Let $\mathscr{H}_{\mathcal{M},q}$ be the Hilbert space consisting of all measurable functions $\phi : G \longrightarrow \mathbb{C}$ satisfying

(L1) $\phi(\gamma g) = \phi_{\mathcal{M},q}(\gamma)\,\phi(g)$ for all $\gamma \in \Gamma_L$ and $g \in G$.
(L2) $\|\phi\|^2_{\mathcal{M},q} = \int_{\Gamma_L \backslash G} |\phi(\bar{g})|^2\, d\bar{g} < \infty, \quad \bar{g} = \Gamma_L g$.

The induced representation $\pi_{\mathcal{M},q}$ is realized in $\mathscr{H}_{\mathcal{M},q}$ as follows:

$$\big(\pi_{\mathcal{M},q}(g_0)\phi\big)(g) = \phi(gg_0), \quad g_0, g \in G,\ \phi \in \mathscr{H}_{\mathcal{M},q}. \tag{1.3.11}$$

$\pi_{\mathcal{M},q}$ is called the lattice representation of G associated with the lattice L.

Theorem 1.3.4 *Let $\mathcal{M}$ be a positive definite, symmetric half integral matrix of degree m. Let $\varphi_{\mathcal{M}}$ be the character of Γ_L defined by $\varphi_{\mathcal{M}}((\lambda, \mu, \kappa)) := e^{2\pi i\, \sigma(\mathcal{M}\kappa)}$ for all $(\lambda, \mu, \kappa) \in \Gamma_L$. Then the representation*

$$\pi_{\mathcal{M}} := \operatorname{Ind}_{\Gamma_L}^{G} \varphi_{\mathcal{M}} \tag{1.3.12}$$

induced from the character $\varphi_{\mathcal{M}}$ is unitarily equivalent to the representation

$$\bigoplus U(\sigma_{\mathcal{M}}) := \bigoplus \operatorname{Ind}_{K}^{G} \sigma_{\mathcal{M}} \quad (\,(\det 2\mathcal{M})^n\text{-copies}\,),$$

where

$$K = \{(0, \mu, \kappa) \in G \mid \mu \in \mathbb{R}^{(m,n)},\ \kappa \in \operatorname{Sym}(m, \mathbb{R})\,\}$$

and

$$\sigma_{\mathcal{M}}((0, \mu, \kappa)) = e^{2\pi\, i\, \sigma(\mathcal{M}\kappa)} I, \quad (0, \mu, \kappa) \in K$$

is the one-dimensional unitary representation of K.

Proof We first recall that the induced representation $\pi_{\mathscr{M}}$ is realized in the Hilbert space $\mathscr{H}_{\mathscr{M}}$ consisting of all measurable functions $\phi : G \longrightarrow \mathbb{C}$ satisfying the conditions

$$\phi((\lambda_0, \mu_0, \kappa_0) \circ g) = e^{2\pi i\, \sigma(\mathscr{M}\kappa_0)}\, \phi(g), \quad (\lambda_0, \mu_0, \kappa_0) \in \Gamma_L,\ g \in G \tag{1.3.13}$$

and

$$\|\phi\|^2_{\pi,\mathscr{M}} := \int_{\Gamma_L \backslash G} |\phi(\bar{g})|^2\, d\bar{g} < \infty, \quad \bar{g} = \Gamma_L \circ g. \tag{1.3.14}$$

Now we write

$$g_0 = [\lambda_0, \mu_0, \kappa_0] \in \Gamma_L \ \ and \ \ g = [\lambda, \mu, \kappa] \in \mathbb{H}^{(n,m)}_{\mathbb{R}}.$$

For $\phi \in \mathscr{H}_{\mathscr{M}}$, we have

$$\phi(g_0 \diamond g) = \phi([\lambda_0 + \lambda, \mu_0 + \mu, \kappa_0 + \kappa + \lambda_0\, {}^t\mu + \mu\, {}^t\lambda_0]). \tag{1.3.15}$$

On the other hand, we get

$$\begin{aligned} \phi(g_0 \diamond g) &= \phi((\lambda_0, \mu_0, \kappa_0 - \mu_0\, {}^t\lambda_0) \circ g) \\ &= e^{2\pi i\, \sigma\{\mathscr{M}(\kappa_0 - \mu_0\, {}^t\lambda_0)\}}\, \phi(g) \\ &= e^{2\pi i\, \sigma(\mathscr{M}\kappa_0)}\, \phi(g) \quad (\text{because } \sigma(\mathscr{M}\mu_0\, {}^t\lambda_0) \in \mathbb{Z}). \end{aligned}$$

Thus putting $\kappa' := \kappa_0 + \lambda_0\, {}^t\mu + \mu\, {}^t\lambda_0$, we get

$$\phi([\lambda_0+\lambda, \mu_0+\mu, \kappa+\kappa']) = e^{2\pi i\, \sigma(\mathscr{M}\kappa')} \cdot e^{-4\pi i\, \sigma(\mathscr{M}\lambda_0\, {}^t\mu)}\, \phi([\lambda, \mu, \kappa]). \tag{1.3.16}$$

Putting $\lambda_0 = \kappa_0 = 0$ in (1.3.16), we have

$$\phi([\lambda, \mu + \mu_0, \kappa]) = \phi([\lambda, \mu, \kappa]) \tag{1.3.17}$$

for all $\mu_0 \in \mathbb{Z}^{(m,n)}$ and $[\lambda, \mu, \kappa] \in \mathbb{H}^{(n,m)}_{\mathbb{R}}$. Therefore if we fix λ and κ, ϕ is periodic in μ with respect to the lattice $\mathbb{Z}^{(m,n)}$ in $\mathbb{R}^{(m,n)}$. We note that

$$\phi([\lambda, \mu, \kappa]) = \phi([0, 0, \kappa] \diamond [\lambda, \mu, 0]) = e^{2\pi i\, \sigma(\mathscr{M}\kappa)}\, \phi([\lambda, \mu, 0])$$

for $[\lambda, \mu, \kappa] \in \mathbb{H}^{(n,m)}_{\mathbb{R}}$. Hence ϕ admits a Fourier expansion in μ:

$$\phi([\lambda, \mu, \kappa]) = e^{2\pi i\, \sigma(\mathscr{M}\kappa)} \sum_{N \in \mathbb{Z}^{(m,n)}} c_N(\lambda)\, e^{2\pi i\, \sigma(N\, {}^t\mu)}. \tag{1.3.18}$$

If $\lambda_0 \in \mathbb{Z}^{(m,n)}$, then we have

$$\begin{aligned}
&\phi([\lambda+\lambda_0,\mu,\kappa]) \\
&= e^{2\pi i\,\sigma(\mathscr{M}\kappa)} \sum_{N\in\mathbb{Z}^{(m,n)}} c_N(\lambda+\lambda_0)\, e^{2\pi i\,\sigma(N\,{}^t\mu)} \\
&= e^{-4\pi i\,\sigma(\mathscr{M}\lambda_0\,{}^t\mu)}\,\phi([\lambda,\mu,\kappa]) \qquad \text{(by (1.3.16))} \\
&= e^{-4\pi i\,\sigma(\mathscr{M}\lambda_0\,{}^t\mu)}\, e^{2\pi i\,\sigma(\mathscr{M}\kappa)} \sum_{N\in\mathbb{Z}^{(m,n)}} c_N(\lambda)\, e^{2\pi i\,\sigma(N\,{}^t\mu)}, \\
&= e^{2\pi i\,\sigma(\mathscr{M}\kappa)} \sum_{N\in\mathbb{Z}^{(m,n)}} c_N(\lambda)\, e^{2\pi i\,\sigma\{(N-2\mathscr{M}\lambda_0)\,{}^t\mu\}}. \quad \text{(by (1.3.18))}
\end{aligned}$$

So we get

$$\begin{aligned}
&\sum_{N\in\mathbb{Z}^{(m,n)}} c_N(\lambda+\lambda_0)\, e^{2\pi i\,\sigma(N\,{}^t\mu)} \\
&= \sum_{N\in\mathbb{Z}^{(m,n)}} c_N(\lambda)\, e^{2\pi i\,\sigma\{(N-2\mathscr{M}\lambda_0)\,{}^t\mu\}} \\
&= \sum_{N\in\mathbb{Z}^{(m,n)}} c_{N+2\mathscr{M}\lambda_0}(\lambda)\, e^{2\pi i\,\sigma(N\,{}^t\mu)}.
\end{aligned}$$

Hence we get

$$c_N(\lambda+\lambda_0) = c_{N+2\mathscr{M}\lambda_0}(\lambda) \ \text{ for all } \lambda_0 \in \mathbb{Z}^{(m,n)} \text{ and } \lambda \in \mathbb{R}^{(m,n)} \tag{1.3.19}$$

for all $\lambda_0 \in \mathbb{Z}^{(m,n)}$ and $\lambda \in \mathbb{R}^{(m,n)}$. Consequently, it is enough to know only the coefficients $c_\alpha(\lambda)$ for the representatives α in $\mathbb{Z}^{(m,n)}$ modulo $2\mathscr{M}$. It is obvious that the number of all such representatives α is $(\det 2\mathscr{M})^n$. We denote by $\mathcal{J}$ a complete system of such representatives α in $\mathbb{Z}^{(m,n)}$ modulo $2\mathscr{M}$. Then, we have

$$\begin{aligned}
&\phi([\lambda,\mu,\kappa]) \\
&= e^{2\pi i\,\sigma(\mathscr{M}\kappa)} \Big\{ \sum_{N\in\mathbb{Z}^{(m,n)}} c_{\alpha+2\mathscr{M}N}(\lambda)\, e^{2\pi i\,\sigma\{(\alpha+2\mathscr{M}N)\,{}^t\mu\}} \\
&\qquad + \sum_{N\in\mathbb{Z}^{(m,n)}} c_{\beta+2\mathscr{M}N}(\lambda)\, e^{2\pi i\sigma\{(\beta+2\mathscr{M}N)\,{}^t\mu\}} \\
&\qquad \vdots \\
&\qquad + \sum_{N\in\mathbb{Z}^{(m,n)}} c_{\gamma+2\mathscr{M}N}(\lambda)\, e^{2\pi i\,\sigma\{(\gamma+2\mathscr{M}N)\,{}^t\mu\}} \Big\},
\end{aligned}$$

where $\{\alpha, \beta, \ldots, \gamma\}$ denotes the complete system $\mathcal{J}$.

For each $\alpha \in \mathcal{J}$, we denote by $\mathscr{H}_{\mathscr{M},\alpha}$ the Hilbert space consisting of Fourier expansions

$$e^{2\pi i\,\sigma(\mathscr{M}\kappa)} \sum_{N\in\mathbb{Z}^{(m,n)}} c_{\alpha+2\mathscr{M}N}(\lambda)\, e^{2\pi i\,\sigma\{(\alpha+2\mathscr{M}N)^{t}\mu\}}, \quad (\lambda,\mu,\kappa) \in G,$$

where $c_N(\lambda)$ denotes the coefficients of the Fourier expansion (1.3.18) of $\phi \in \mathscr{H}_{\mathscr{M}}$ and ϕ runs over the set $\{\phi \in \pi_{\mathscr{M}}\}$. It is easy to see that $\mathscr{H}_{\mathscr{M},\alpha}$ is invariant under $\pi_{\mathscr{M}}$. We denote the restriction of $\pi_{\mathscr{M}}$ to $\mathscr{H}_{\mathscr{M},\alpha}$ by $\pi_{\mathscr{M},\alpha}$. Then we have

$$\pi_{\mathscr{M}} = \bigoplus_{\alpha\in\mathcal{J}} \pi_{\mathscr{M},\alpha}. \tag{1.3.20}$$

Let $\phi_\alpha \in \pi_{\mathscr{M},\alpha}$. Then for $[\lambda,\mu,\kappa] \in \mathbb{H}_{\mathbb{R}}^{(n,m)}$, we get

$$\phi_\alpha([\lambda,\mu,\kappa]) = e^{2\pi i\,\sigma(\mathscr{M}\kappa)} \sum_{N\in\mathbb{Z}^{(m,n)}} c_{\alpha+2\mathscr{M}N}(\lambda)\, e^{2\pi i\,\sigma\{(\alpha+2\mathscr{M}N)^{t}\mu\}}. \tag{1.3.21}$$

We put

$$I_\lambda = \overbrace{[0,1]\times[0,1]\times\cdots\times[0,1]}^{(m\times n)\text{-times}} \subset \left\{ [\lambda,0,0] \mid \lambda \in \mathbb{R}^{(m,n)} \right\}$$

and

$$I_\mu = \overbrace{[0,1]\times[0,1]\times\cdots\times[0,1]}^{(m\times n)\text{-times}} \subset \left\{ [0,\mu,0] \mid \mu \in \mathbb{R}^{(m,n)} \right\}.$$

Then we obtain

$$\int_{I_\mu} \phi_\alpha([\lambda,\mu,\kappa])\, e^{-2\pi i\,\sigma(\alpha^{t}\mu)}\, d\mu = e^{2\pi i\,\sigma(\mathscr{M}\kappa)}\, c_\alpha(\lambda), \quad \alpha \in \mathcal{J}. \tag{1.3.22}$$

Since $\Gamma_L\backslash G \cong I_\lambda \times I_\mu$, we get

$$\begin{aligned} &\|\phi_\alpha\|^2_{\pi,\mathscr{M},\alpha} := \|\phi_\alpha\|^2_{\pi,\mathscr{M}} \\ &= \int_{\Gamma_L\backslash G} |\phi_\alpha(\bar{g})|^2\, d\bar{g} \\ &= \int_{I_\lambda}\int_{I_\mu} |\phi_\alpha(\bar{g})|^2\, d\lambda d\mu \end{aligned}$$

$$
\begin{aligned}
&= \int_{I_\lambda \times I_\mu} \left| \sum_{N \in \mathbb{Z}^{(m,n)}} c_{\alpha + 2\mathscr{M}N}(\lambda)\, e^{2\pi i\, \sigma\{(\alpha + 2\mathscr{M}N)^{\,t}\mu\}} \right|^2 d\lambda d\mu \\
&= \int_{I_\lambda} \sum_{N \in \mathbb{Z}^{(m,n)}} |c_{\alpha + 2\mathscr{M}N}(\lambda)|^2\, d\lambda \\
&= \int_{I_\lambda} \sum_{N \in \mathbb{Z}^{(m,n)}} |c_\alpha(\lambda + N)|^2\, d\lambda \qquad (by\ (1.3.19)) \\
&= \int_{\mathbb{R}^{(m,n)}} |c_\alpha(\lambda)|^2\, d\lambda.
\end{aligned}
$$

Since $\phi_\alpha \in \pi_{\mathscr{M},\alpha}$, $\|\phi_\alpha\|_{\pi,\mathscr{M},\alpha} < \infty$ and so $c_\alpha(\lambda) \in L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$ for all $\alpha \in \mathcal{J}$.

For each $\alpha \in \mathcal{J}$, we define the mapping $\vartheta_{\mathscr{M},\alpha}$ on $L^2(\mathbb{R}^{(m,n)}, d\xi)$ by

$$
\begin{aligned}
&(\vartheta_{\mathscr{M},\alpha} f)([\lambda, \mu, \kappa]) \qquad (1.3.23) \\
&= e^{2\pi i\, \sigma(\mathscr{M}\kappa)} \sum_{N \in \mathbb{Z}^{(m,n)}} f(\lambda + N)\, e^{2\pi i\, \sigma\{(\alpha + 2\mathscr{M}N)^{\,t}\mu\}},
\end{aligned}
$$

where $f \in L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$ and $[\lambda, \mu, \kappa] \in G$.

Lemma 1.3.5 *For each $\alpha \in \mathcal{J}$, the image of $L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$ under $\vartheta_{\mathscr{M},\alpha}$ is contained in $\mathscr{H}_{\mathscr{M},\alpha}$. Moreover, the mapping $\vartheta_{\mathscr{M},\alpha}$ is a one-to-one unitary operator of $L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$ onto $\mathscr{H}_{\mathscr{M},\alpha}$ preserving the norms. In other words, the mapping*

$$
\vartheta_{\mathscr{M},\alpha} : L^2\left(\mathbb{R}^{(m,n)}, d\xi\right) \longrightarrow \mathscr{H}_{\mathscr{M},\alpha}
$$

is an isometry.

Proof We already showed that $\vartheta_{\mathscr{M},\alpha}$ preserves the norms. First, we observe that if $(\lambda_0, \mu_0, \kappa_0) \in \Gamma_L$ and $g = [\lambda, \mu, \kappa] \in \mathbb{H}_{\mathbb{R}}^{(n,m)}$,

$$
\begin{aligned}
(\lambda_0, \mu_0, \kappa_0) \circ g &= [\lambda_0, \mu_0, \kappa_0 + \mu_0\, {}^t\lambda_0] \diamond [\lambda, \mu, \kappa] \\
&= [\lambda_0 + \lambda, \mu_0 + \mu, \kappa + \kappa_0 + \mu_0\, {}^t\lambda_0 + \lambda_0\, {}^t\mu + \mu\, {}^t\lambda_0].
\end{aligned}
$$

Thus we get

$$
\begin{aligned}
&(\vartheta_{\mathscr{M},\alpha} f)((\lambda_0, \mu_0, \kappa_0) \circ g) \\
&= e^{2\pi i\, \sigma\{\mathscr{M}(\kappa + \kappa_0 + \mu_0\, {}^t\lambda_0 + \lambda_0\, {}^t\mu + \mu\, {}^t\lambda_0)\}} \\
&\quad \times \sum_{N \in \mathbb{Z}^{(m,n)}} f(\lambda + \lambda_0 + N)\, e^{2\pi i\, \sigma\{(\alpha + 2\mathscr{M}N)^{\,t}(\mu_0 + \mu)\}}
\end{aligned}
$$

$$= e^{2\pi i\,\sigma(\mathscr{M}\kappa_0)} \cdot e^{2\pi i\,\sigma(\mathscr{M}\kappa)} \cdot e^{2\pi i\,\sigma(\alpha\,{}^t\mu_0)} \sum_{N\in\mathbb{Z}^{(m,n)}} f(\lambda+N)\, e^{2\pi i\,\sigma\{(\alpha+2\mathscr{M}N)\,{}^t\mu\}}$$

$$= e^{2\pi i\,\sigma(\mathscr{M}\kappa_0)}\,(\vartheta_{\mathscr{M},\alpha} f)(g).$$

Here, in the above equalities we used the facts that $2\,\sigma(\mathscr{M}N\,{}^t\mu_0) \in \mathbb{Z}$ and $\alpha\,{}^t\mu_0 \in \mathbb{Z}$. It is easy to show that

$$\int_{\Gamma_L\backslash G} |\vartheta_{\mathscr{M},\alpha} f(\bar{g})|^2\, d\bar{g} = \int_{\mathbb{R}^{(m,n)}} |f(\lambda)|^2\, d\lambda = \|f\|_2^2 < \infty.$$

This completes the proof of Lemma 1.3.5.

Finally, it is easy to show that for each $\alpha \in \mathcal{J}$, the mapping $\vartheta_{\mathscr{M},\alpha}$ intertwines the Schrödinger representation $\left(U^{S,\mathscr{M}}, L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)\right)$ and the representation $(\pi_{\mathscr{M},\alpha}, \mathscr{H}_{\mathscr{M},\alpha})$. Therefore, by Lemma 1.3.5, for each $\alpha \in \mathcal{J}$, $\pi_{\mathscr{M},\alpha}$ is unitarily equivalent to $U(\sigma_{\mathscr{M}})$ and so $\pi_{\mathscr{M},\alpha}$ is an irreducible unitary representation of G. According to (1.3.20), the induced representation $\pi_{\mathscr{M}}$ is unitarily equivalent to

$$\bigoplus U(\sigma_{\mathscr{M}}) \quad ((\det 2\mathscr{M})^n\text{-copies}).$$

This completes the proof of Theorem 1.3.4. □

Now we state the connection between the lattice representation and theta functions. As before, we write $V = \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)} \cong \mathbb{C}^{(m,n)}$, $L = \mathbb{Z}^{(m,n)} \times \mathbb{Z}^{(m,n)}$ and $\mathscr{M}$ is a positive symmetric half-integral matrix of degree m. The function $q_{\mathscr{M}} : L \longrightarrow \mathbb{R}/2\mathbb{Z} = [0, 2)$ defined by

$$q_{\mathscr{M}}((\xi,\eta)) = 2\,\sigma(\mathscr{M}\xi\,{}^t\eta), \quad (\xi,\eta) \in L \tag{1.3.24}$$

satisfies the condition (1.3.8). We let $\varphi_{\mathscr{M},q_{\mathscr{M}}} : \Gamma_L \longrightarrow \mathbb{C}_1^\times$ be the character of Γ_L defined by

$$\varphi_{\mathscr{M},q_{\mathscr{M}}}((\ell,\kappa)) = e^{2\pi i\,\sigma(\mathscr{M}\kappa)}\, e^{\pi i\, q_{\mathscr{M}}(\ell)}, \quad (\ell,\kappa) \in \Gamma_L.$$

We denote by $\mathscr{H}_{\mathscr{M},q_{\mathscr{M}}}$ the Hilbert space consisting of measurable functions $\phi : G \longrightarrow \mathbb{C}$ which satisfy the conditions (1.3.25) and (1.3.26).

$$\phi((\ell,\kappa)\circ g) = \varphi_{\mathscr{M},q_{\mathscr{M}}}((\ell,\kappa))\,\phi(g) \tag{1.3.25}$$

for all $(\ell,\kappa) \in \Gamma_L$ and $g \in G$, and

$$\int_{\Gamma_L\backslash G} \|\phi(\dot{g})\|^2\, d\dot{g} < \infty, \quad \dot{g} = \Gamma_L \circ g. \tag{1.3.26}$$

Then the representation

$$\pi_{\mathscr{M},q_{\mathscr{M}}} = \mathrm{Ind}_{\Gamma_L}^{G} \varphi_{\mathscr{M},q_{\mathscr{M}}}$$

of G induced from the character $\varphi_{\mathscr{M},q_{\mathscr{M}}}$ is realized in $\mathscr{H}_{\mathscr{M},q_{\mathscr{M}}}$ as

$$\left(\pi_{\mathscr{M},q_{\mathscr{M}}}(g_0)\,\phi\right)(g) = \phi(gg_0), \quad g_0, g \in G,\ \phi \in \mathscr{H}_{\mathscr{M},q_{\mathscr{M}}}.$$

Let $\mathbf{H}_{\mathscr{M},q_{\mathscr{M}}}$ be the vector space consisting of measurable functions $F : V \longrightarrow \mathbb{C}$ satisfying the conditions (1.3.27) and (1.3.28).

$$F(\lambda+\xi,\mu+\eta) = e^{2\pi i\,\sigma\{\mathscr{M}(\xi\,{}^t\eta+\lambda\,{}^t\eta-\mu\,{}^t\xi)\}}\,F(\lambda,\mu), \tag{1.3.27}$$

for all $(\lambda,\mu) \in V$ and $(\xi,\eta) \in L$, and

$$\int_{L\backslash V} \|F(\dot{v})\|^2\,d\dot{v} = \int_{I_\lambda\times I_\mu} \|F(\lambda,\mu)\|^2\,d\lambda d\mu < \infty. \tag{1.3.28}$$

Given $\phi \in \mathscr{H}_{\mathscr{M},q_{\mathscr{M}}}$ and a fixed element $\Omega \in \mathbb{H}_n$, we put

$$E_\phi(\lambda,\mu) = \phi((\lambda,\mu,0)), \quad \lambda,\mu \in \mathbb{R}^{(m,n)}, \tag{1.3.29}$$

$$F_\phi(\lambda,\mu) = \phi([\lambda,\mu,0]), \quad \lambda,\mu \in \mathbb{R}^{(m,n)}, \tag{1.3.30}$$

$$F_{\Omega,\phi}(\lambda,\mu) = e^{-2\pi i\,\sigma(\mathscr{M}\lambda\Omega\,{}^t\lambda)}\,F_\phi(\lambda,\mu), \quad \lambda,\mu \in \mathbb{R}^{(m,n)}. \tag{1.3.31}$$

In addition, we put for $W = \lambda\Omega+\mu \in \mathbb{C}^{(m,n)}$,

$$\vartheta_{\Omega,\phi}(W) = \vartheta_{\Omega,\phi}(\lambda\Omega+\mu) := F_{\Omega,\phi}(\lambda,\mu). \tag{1.3.32}$$

We observe that E_ϕ, F_ϕ, $F_{\Omega,\phi}$ are functions defined on V and $\vartheta_{\Omega,\phi}$ is a function defined on $\mathbb{C}^{(m,n)}$.

Proposition 1.3.6 *If $\phi \in \mathscr{H}_{\mathscr{M},q_{\mathscr{M}}}$, $(\xi,\eta) \in L$ and $(\lambda,\mu) \in V$, then we have the formulas*

$$E_\phi(\lambda+\xi,\mu+\eta) = e^{2\pi i\,\sigma\{\mathscr{M}(\xi\,{}^t\eta+\lambda\,{}^t\eta-\mu\,{}^t\xi)\}}\,E_\phi(\lambda,\mu), \tag{1.3.33}$$

$$F_\phi(\lambda+\xi,\mu+\eta) = e^{-4\pi i\,\sigma(\mathscr{M}\xi\,{}^t\mu)}\,F_\phi(\lambda,\mu), \tag{1.3.34}$$

$$F_{\Omega,\phi}(\lambda+\xi,\mu+\eta) = e^{-2\pi i\,\sigma\{\mathscr{M}(\xi\Omega\,{}^t\xi+2\lambda\Omega\,{}^t\xi+2\mu\,{}^t\xi)\}}\,F_{\Omega,\phi}(\lambda,\mu). \tag{1.3.35}$$

If $W = \lambda\Omega + \eta \in \mathbb{C}^{(m,n)}$, *then we have*

$$\vartheta_{\Omega,\phi}(W + \xi\Omega + \eta) = e^{-2\pi i\,\sigma\{\mathscr{M}(\xi\Omega\,{}^t\xi + 2\,W\,{}^t\xi)\}}\,\vartheta_{\Omega,\phi}(W). \tag{1.3.36}$$

Moreover, F_ϕ *is an element of* $\mathbf{H}_{\mathscr{M},q_{\mathscr{M}}}$.

Proof We note that

$$(\lambda + \xi, \mu + \eta, 0) = (\xi, \eta, -\xi\,{}^t\mu + \eta\,{}^t\lambda) \circ (\lambda, \mu, 0).$$

Thus we have

$$\begin{aligned} &E_\phi(\lambda + \xi, \mu + \eta) = \phi((\lambda + \xi, \mu + \eta, 0)) \\ =&\phi((\xi, \eta, -\xi\,{}^t\mu + \eta\,{}^t\lambda) \circ (\lambda, \mu, 0)) \\ =&e^{2\pi i\,\sigma\{\mathscr{M}(\xi\,{}^t\eta + \lambda\,{}^t\eta - \mu\,{}^t\xi)\}}\,\phi((\lambda, \mu, 0)) \\ =&e^{2\pi i\,\sigma\{\mathscr{M}(\xi\,{}^t\eta + \lambda\,{}^t\eta - \mu\,{}^t\xi)\}}\,E_\phi(\lambda, \mu). \end{aligned}$$

This proves Formula (1.3.33).

We observe that

$$[\lambda + \xi, \mu + \eta, 0] = (\xi, \eta, -\xi\,{}^t\mu - \mu\,{}^t\xi - \eta\,{}^t\xi) \circ [\lambda, \mu, 0].$$

Thus we have

$$\begin{aligned} &F_\phi(\lambda + \xi, \mu + \eta) \\ =&\phi([\lambda + \xi, \mu + \eta, 0]) \\ =&e^{-2\pi i\,\sigma\{\mathscr{M}(\xi\,{}^t\mu + \mu\,{}^t\xi + \eta\,{}^t\xi)\}} \cdot e^{2\pi i\,\sigma(\mathscr{M}\xi\,{}^t\eta)}\,\phi([\lambda, \mu, 0]) \\ =&e^{-4\pi i\,\sigma(\mathscr{M}\xi\,{}^t\mu)}\,\phi([\lambda, \mu, 0]) = e^{-4\pi i\,\sigma(\mathscr{M}\xi\,{}^t\mu)}\,F_\phi(\lambda, \mu). \end{aligned}$$

This proves Formula (1.3.34).

According to (1.3.34), we have

$$\begin{aligned} &F_{\Omega,\phi}(\lambda + \xi, \mu + \eta) \\ =&e^{-2\pi i\,\sigma\{\mathscr{M}(\lambda+\xi)\Omega\,{}^t(\lambda+\xi)\}}\,F_\phi(\lambda + \xi, \mu + \eta) \\ =&e^{-2\pi i\,\sigma\{\mathscr{M}(\lambda+\xi)\Omega\,{}^t(\lambda+\xi)\}} \cdot e^{-4\pi i\,\sigma(\mathscr{M}\xi\,{}^t\mu)}\,F_\phi(\lambda, \mu) \\ =&e^{-2\pi i\,\sigma\{\mathscr{M}(\xi\Omega\,{}^t\xi + 2\lambda\Omega\,{}^t\xi + 2\mu\,{}^t\xi)\}} \cdot e^{-2\pi i\,\sigma(\mathscr{M}\lambda\Omega\,{}^t\lambda)}\,F_\phi(\lambda, \mu) \\ =&e^{-2\pi i\,\sigma\{\mathscr{M}(\xi\Omega\,{}^t\xi + 2\lambda\Omega\,{}^t\xi + 2\mu\,{}^t\xi)\}}\,F_{\Omega,\phi}(\lambda, \mu). \end{aligned}$$

This proves Formula (1.3.35). Formula (1.3.36) follows immediately from Formula (1.3.35). Indeed, if $W = \lambda\Omega + \mu$ with $\lambda, \mu \in \mathbb{R}^{(m,n)}$, we have

$$\begin{aligned}&\vartheta_{\Omega,\phi}(W + \xi\Omega + \eta) = F_{\Omega,\phi}(\lambda + \xi, \mu + \eta)\\ =&e^{-2\pi i\,\sigma\{\mathscr{M}(\xi\Omega\,{}^t\xi + 2\,(\lambda\Omega+\mu)\,{}^t\xi)\}}\,F_{\Omega,\phi}(\lambda, \mu)\\ =&e^{-2\pi i\,\sigma\{\mathscr{M}(\xi\Omega\,{}^t\xi + 2\,W\,{}^t\xi)\}}\,\vartheta_{\Omega,\phi}(W).\end{aligned}$$

□

Remark 1.3.7 The function $\vartheta_{\Omega,\phi}(W)$ is a theta function of level $2\mathscr{M}$ with respect to Ω if $\vartheta_{\Omega,\phi}$ is holomorphic. For any $\phi \in \mathscr{H}_{\mathscr{M},q_{\mathscr{M}}}$, the function $\vartheta_{\Omega,\phi}$ satisfies the transformation law (1.3.36) of a theta function. In this sense, the lattice representation $(\pi_{\mathscr{M},q_{\mathscr{M}}}, \mathscr{H}_{\mathscr{M},q_{\mathscr{M}}})$ is closely related to theta functions (cf. see Formula (1.8.1) in Sect. 1.8).

1.4 Fock Model

We consider the vector space $V := \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}$. We put

$$P_{ka} = (E_{ka}, 0), \quad Q_{lb} = (0, E_{lb}), \tag{1.4.1}$$

where $1 \le k, l \le m$ and $1 \le a, b \le n$. Then the set $\{P_{ka},\ Q_{ka}\}$ forms a basis for V. We define the alternating bilinear form $\mathbf{A} : V \times V \longrightarrow \mathbb{R}$ by

$$\mathbf{A}\,((\lambda_0, \mu_0), (\lambda, \mu)) = \sigma(\lambda_0\,{}^t\mu - \mu_0\,{}^t\lambda), \quad (\lambda_0, \mu_0),\ (\lambda, \mu) \in V. \tag{1.4.2}$$

Then we have

$$\mathbf{A}(P_{ka}, P_{lb}) = \mathbf{A}(Q_{ka}, Q_{lb}) = 0, \quad \mathbf{A}(P_{ka}, Q_{lb}) = \delta_{ab}\,\delta_{kl}, \tag{1.4.3}$$

where $1 \le k, l \le m$ and $1 \le a, b \le n$. Any element $v \in V$ can be written uniquely as

$$v = \sum_{k,a} x_{ka} P_{ka} + \sum_{l,b} y_{lb} Q_{lb}, \quad x_{ka},\ y_{lb} \in \mathbb{R}. \tag{1.4.4}$$

From now on, for brevity, we write $v = xP + y\,Q$ instead of (1.4.4). Then it is easy to see that the endomorphism $J : V \longrightarrow V$ defined by

$$J(xP + y\,Q) := -yP + x\,Q, \quad xP + y\,Q \in V \tag{1.4.5}$$

is a complex structure on V which is compatible with the alternating bilinear form $\mathbf{A}$. This means that J is an endomorphism of V satisfying the following conditions:

(J1) $J^2 = -I$ on V.
(J2) $\mathbf{A}(Jv_0, Jv) = \mathbf{A}(v_0, v)$ for all $v_0, v \in V$.
(J3) $\mathbf{A}(v, Jv) > 0$ for all $v \in V$ with $v \neq 0$.

Now we let $V_{\mathbb{C}} = V + iV$ be the complexification of V, where $i = \sqrt{-1}$. For an element $w = v_1 + i\, v_2 \in V_{\mathbb{C}}$ with $v_1, v_2 \in V$, we put

$$\overline{w} := v_1 - i\, v_2. \tag{1.4.6}$$

Let $\mathbf{A}_{\mathbb{C}}$ be the complex bilinear form on $V_{\mathbb{C}}$ extending $\mathbf{A}$ and let $J_{\mathbb{C}}$ be the complex linear map of $V_{\mathbb{C}}$ extending J. Since $J_{\mathbb{C}}^2 = -I$, $J_{\mathbb{C}}$ has only the eigenvalues $\pm i$. We denote by V^+ (resp. V^-) the eigenspace of $V_{\mathbb{C}}$ corresponding to the eigenvalues i (resp. $-i$). Thus $V_{\mathbb{C}} = V^+ + V^-$. Since

$$J_{\mathbb{C}}(P_{ka} \pm i\, Q_{ka}) = \mp i\,(P_{ka} \pm i\, Q_{ka}),$$

we have

$$V^+ = \sum_{k,a} \mathbb{C}\,(P_{ka} - i\, Q_{ka}), \quad V^- = \sum_{k,a} \mathbb{C}\,(P_{ka} + i\, Q_{ka}). \tag{1.4.7}$$

Let

$$V_* := \sum_{k,a} \mathbb{C}\, P_{ka}, \quad 1 \leq k \leq m, \quad 1 \leq a \leq n \tag{1.4.8}$$

be the subspace of $V_{\mathbb{C}}$ as a $\mathbb{C}$-vector space. It is easy to see that V_* is isomorphic to V as $\mathbb{R}$-vector spaces via the isomorphism $T : V \longrightarrow V_*$ defined by

$$T(P_{ka}) = P_{ka}, \quad T(Q_{lb}) = i\, P_{lb}. \tag{1.4.9}$$

We define the complex linear map $J_* : V_* \longrightarrow V_*$ by $J_*(P_{ka}) = i\, P_{ka}$ for $1 \leq k \leq m$, $1 \leq a \leq n$. Then J_* is compatible with J, that is, $T \circ J = J_* \circ T$. It is easily seen that there exists a unique hermitian form $\mathbf{H}$ on V_* with $\operatorname{Im}\mathbf{H} = \mathbf{A}$. Indeed, $\mathbf{H}$ is given by

$$\mathbf{H}(v, w) = \mathbf{A}(v, J_* w) + i\, \mathbf{A}(v, w), \quad v, w \in V_*. \tag{1.4.10}$$

For $v = \sum_{k,a} z_{ka} P_{ka} \in V_*$ with $z_{ka} = x_{ka} + i\, y_{ka}$ $(x_{ka}, y_{ka} \in \mathbb{R})$, for brevity we write $v = zP$. For two elements $v = zP$ and $v' = z'P$ in V_*, $\mathbf{H}(v, v') = \sum_{k,a} \overline{z_{ka}}\, z'_{ka}$. We observe that

$$V_{\mathbb{C}} = \sum_{k,a} \mathbb{C}\, P_{ka} + \sum_{l,b} \mathbb{C}\, Q_{lb} = V^+ + V^- \supset V^{\pm}.$$

For $w = z^0 P + z^1 Q \in V_{\mathbb{C}}$, we put

$$w = w^+ + w^-, \quad w^+ := z^+(P - i\,Q), \quad w^- := z^-(P + i\,Q).$$

The relations among z^0, z^1, z^+, z^- are given by

$$z^{\pm} = \frac{1}{2}(z^0 \pm i\, z^1), \quad z^0 = z^+ + z^-, \quad z^1 = i\,(z^- - z^+). \tag{1.4.11}$$

Precisely, (1.4.11) implies that

$$z^{\pm}_{ka} = \frac{1}{2}\,(z^0_{ka} \pm i\, z^1_{ka}), \quad z^0_{ka} = z^+_{ka} + z^-_{ka}, \quad z^1_{ka} = i\,(z^-_{ka} - z^+_{ka}),$$

where $1 \le k \le m$ and $1 \le a \le n$. It is easy to see that

$$\mathbf{A}_{\mathbb{C}}(w^-, w^+) = -2i \sum_{k,a} z^-_{ka} z^+_{ka} = -\frac{i}{2} \sum_{k,a} \left\{ (z^0_{ka})^2 + (z^1_{ka})^2 \right\}. \tag{1.4.12}$$

Let

$$G_{\mathbb{C}} := \left\{ (z^0, z^1, a) \;\middle|\; z^0, z^1 \in \mathbb{C}^{(m,n)}, \quad a \in \mathbb{C}^{(m,m)}, \quad a + z^1\, {}^t z^0 \text{ symmetric} \right\}$$

be the complexification of the real Heisenberg group $G := H_{\mathbb{R}}^{(n,m)}$. Analogously in the real case, the multiplication on $G_{\mathbb{C}}$ is given by (1.1.2). If $w = z^0 P + z^1 Q \overset{\text{def}}{=} \sum_{k,a} z^0_{ka} P_{ka} + \sum_{l,b} z^1_{lb} Q_{lb}$, we identify z^0, z^1 with the $m \times n$ matrices respectively:

$$z^0 := \begin{pmatrix} z^0_{11} & z^0_{12} & \cdots & z^0_{1n} \\ z^0_{21} & z^0_{22} & \cdots & z^0_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ z^0_{m1} & z^0_{m2} & \cdots & z^0_{mn} \end{pmatrix}, \quad z^1 := \begin{pmatrix} z^1_{11} & z^1_{12} & \cdots & z^1_{1n} \\ z^1_{21} & z^1_{22} & \cdots & z^1_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ z^1_{m1} & z^1_{m2} & \cdots & z^1_{mn} \end{pmatrix}.$$

That is, we identify $w = z^0 P + z^1 Q \in V_{\mathbb{C}}$ with $(z^0, z^1) \in \mathbb{C}^{(m,n)} \times \mathbb{C}^{(m,n)}$. If $w = z^0 P + z^1 Q$, $\hat{w} = \hat{z}^0 P + \hat{z}^1 Q \in V_{\mathbb{C}}$, then

$$(w, a) \circ (\widehat{w}, \hat{a}) = (w + \widehat{w}, a + \hat{a} + z^0\, {}^t \hat{z}^1 - z^1\, {}^t \hat{z}^0), \quad a, \hat{a} \in \mathbb{C}^{(m,m)}. \tag{1.4.13}$$

From now on, for brevity we put

$$R^+ := P - i\,Q, \qquad R^- := P + i\,Q. \tag{1.4.14}$$

If $w = z^+R^+ + z^-R^-,\ \widehat{w} = \hat{z}^+R^+ + \hat{z}^-R^- \in V_{\mathbb{C}}$, by an easy computation, we have

$$(w, a) \circ (\widehat{w}, \hat{a}) = (\widetilde{w}, a + \hat{a} + 2\,i\,(z^{+}\,{}^t\hat{z}^- - z^-\,{}^t\hat{z}^+)) \tag{1.4.15}$$

with

$$\widetilde{w} = (z^+ + \hat{z}^+)R^+ + (z^- + \hat{z}^-)R^-.$$

Here we identified z^+, z^- with $m \times n$ matrices

$$z^+ := \begin{pmatrix} z^+_{11} & z^+_{12} & \cdots & z^+_{1n} \\ z^+_{21} & z^+_{22} & \cdots & z^+_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ z^+_{m1} & z^+_{m2} & \cdots & z^+_{mn} \end{pmatrix}, \quad z^- := \begin{pmatrix} z^-_{11} & z^-_{12} & \cdots & z^-_{1n} \\ z^-_{21} & z^-_{22} & \cdots & z^-_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ z^-_{m1} & z^-_{m2} & \cdots & z^-_{mn} \end{pmatrix}.$$

It is easy to see that

$$P_{\mathbb{C}} := \left\{ (w^-, a) \in G_{\mathbb{C}} \mid w^- \in V^-,\ \ a \in \mathbb{C}^{(m,m)} \right\} \tag{1.4.16}$$

is a commutative subgroup of $G_{\mathbb{C}}$ and

$$G \cap P_{\mathbb{C}} = \mathfrak{Z}, \quad G_{\mathbb{C}} = G \cdot P_{\mathbb{C}},$$

where $\mathfrak{Z} := \left\{ (0, 0, \kappa) \in G \mid \kappa = {}^t\kappa \in \mathbb{R}^{(m,m)} \right\} \cong \mathrm{Sym}(m, \mathbb{R})$ is the center of G. Moreover,

$$P_{\mathbb{C}} \backslash G_{\mathbb{C}} \cong V^+ \cong \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)} \cong \mathfrak{Z} \backslash G. \tag{1.4.17}$$

For $c = {}^t c \in \mathrm{Sym}(m, \mathbb{R})$ with $c > 0$, we let $\delta_c : P_{\mathbb{C}} \longrightarrow \mathbb{C}^\times$ be a quasi-character of $P_{\mathbb{C}}$ defined by

$$\delta_c((w^-, a)) = e^{2\pi i \sigma(ca)}, \quad (w^-, a) \in P_{\mathbb{C}}. \tag{1.4.18}$$

Let

$$U^{F,c} = \mathrm{Ind}_{P_{\mathbb{C}}}^{G_{\mathbb{C}}}\, \delta_c$$

be the representation of $G_{\mathbb{C}}$ induced from a quasi-character δ_c of $P_{\mathbb{C}}$. Then the representation space $\mathscr{H}^{F,c}$ of $\mathrm{Ind}_{P_{\mathbb{C}}}^{G_{\mathbb{C}}}\, \delta_c$ consists of all holomorphic functions $\psi : G_{\mathbb{C}} \longrightarrow \mathbb{C}$ satisfying the following conditions:

(F1) $\psi((w^-, a) \circ g) = \delta_c((w^-, a))\psi(g) = e^{2\pi i\,\sigma(ca)}\,\psi(g)$
for all $(w^-, a) \in P_{\mathbb{C}}$ and $g \in G_{\mathbb{C}}$.

(F2) $\int_{\mathfrak{Z}\backslash G} |\psi(g)|^2 \, d\dot{g} < \infty$.

The inner product $\langle \ , \ \rangle_{F,c}$ on $\mathscr{H}^{F,c}$ is given by

$$\langle \psi_1, \psi_2 \rangle_{F,c} := \int_{\mathfrak{Z}\backslash G} \psi_1(g) \, \overline{\psi_2(g)} \, d\dot{g}, \quad \psi_1, \psi_2 \in \mathscr{H}^{F,c}, \ \dot{g} = \mathfrak{Z}g.$$

$U^{F,c}$ is realized by the right regular representation of $G_{\mathbb{C}}$ on $\mathscr{H}^{F,c}$:

$$\left(U^{F,c}(g_0)\psi\right)(g) = \psi(gg_0), \quad \psi \in \mathscr{H}^{F,c}, \ g_0, g \in G_{\mathbb{C}}. \tag{1.4.19}$$

Now we will show that $U^{F,c}$ is realized as a representation of G in the Fock space. The Fock space $\mathscr{H}_{F,c}$ is the Hilbert space consisting of all holomorphic functions $f : \mathbb{C}^{(m,n)} \cong V^* \longrightarrow \mathbb{C}$ satisfying the condition

$$\| f \|^2_{F,c} = \int_{\mathbb{C}^{(m,n)}} |f(W)|^2 \, e^{-2\pi\sigma(c\,W\,{}^t\overline{W})} \, dW < \infty.$$

Here V^* denotes the usual dual space of $\mathbb{C}^{(m,n)}$. The inner product $(\ , \ )_{F,c}$ on $\mathscr{H}_{F,c}$ is given by

$$(f_1, f_2)_{F,c} = \int_{\mathbb{C}^{(m,n)}} f_1(W) \, \overline{f_2(W)} \, e^{-2\pi\,\sigma(c\,W\,{}^t\overline{W})} \, dW, \quad f_1, f_2 \in \mathscr{H}_{F,c}.$$

Lemma 1.4.1 *The mapping* $\Lambda : \mathscr{H}_{F,c} \longrightarrow \mathscr{H}^{F,c}, \ \Lambda_f := \Lambda(f) \ (f \in \mathscr{H}_{F,c})$ *defined by*

$$\Lambda_f((z^0 P + z^1 Q, a)) = e^{2\pi i \, \sigma\{c(a + 2\,i\,z^-\,{}^t z^+)\}} \, f(2z^+) \tag{1.4.20}$$

is an isometry of $\mathscr{H}_{F,c}$ *onto* $\mathscr{H}^{F,c}$, *where* $2z^{\pm} = z^0 \pm i\, z^1$ (cf. (1.4.11)). *The inverse* $\Delta : \mathscr{H}^{F,c} \longrightarrow \mathscr{H}_{F,c}, \ \Delta_\psi := \Delta(\psi) \ (\psi \in \mathscr{H}^{F,c})$ *is given by*

$$\Delta_\psi(W) = \psi\left(\frac{1}{2} W R^+\right), \quad W \in \mathbb{C}^{(m,n)}, \tag{1.4.21}$$

where $R^{\pm} = P \mp i\, Q$ (cf. (1.4.14)).

Proof First we observe that for $w = z^0 P + z^1 Q = z^+ R^+ + z^- R^- \in V_{\mathbb{C}}$,

$$(w, a) = (z^- R^-, a + 2\,i\,z^-\,{}^t z^+) \circ (z^+ R^+, 0).$$

Thus if $\psi \in \mathscr{H}^{F,c}$ and $w = z^0 P + z^1 Q = z^+ R^+ + z^- R^-$, by (F1),

$$\psi((w, a)) = e^{2\pi i \, \sigma\{c(a + 2\,i\,z^-\,{}^t z^+)\}} \, \psi((z^+ R^+, 0)). \tag{1.4.22}$$

Let $W = x + i\,y \in \mathbb{C}^{(m,n)}$ with $x, y \in \mathbb{R}^{(m,n)}$. Then

$$xP + yQ = z^+R^+ + z^-R^-, \quad 2z^\pm = x \pm i\,y.$$

So $z^-\,{}^t z^+ = \frac{1}{4}\,W\,{}^t\overline{W}$. According to (1.4.22), if $\psi \in \mathscr{H}^{F,c}$, we have

$$\psi((xP + yQ, 0)) = e^{-\pi\,\sigma(c\,W\,{}^t\overline{W})}\,\psi\left(\left(\frac{1}{2}WR^+, 0\right)\right).$$

Thus we get

$$|\psi((xP + yQ, 0))|^2 = e^{-2\pi\,\sigma(c\,W\,{}^t\overline{W})}\left|\psi\left(\left(\frac{1}{2}WR^+, 0\right)\right)\right|^2.$$

Therefore

$$\int_{\mathfrak{Z}\backslash G} |\psi(\dot{g})|^2\,d\dot{g} = \int_{\mathbb{C}^{(m,n)}} e^{-2\pi\,\sigma(c\,W\,{}^t\overline{W})}\left|\Delta_\psi(W)\right|^2 dW < \infty.$$

It is easy to see that Δ is the inverse of Λ. Hence we obtain the desired results. □

Lemma 1.4.2 *The representation $U^{F,c}$ is realized as a representation of G in the Fock space $\mathscr{H}_{F,c}$ as follows. If $g = (\lambda P + \mu Q, \kappa) = (\lambda, \mu, \kappa) \in G$ and $f \in \mathscr{H}_{F,c}$, then*

$$\left(U^{F,c}(g)f\right)(W) = e^{2\pi i\,\sigma(c\,\kappa)} \cdot e^{-\pi\,\sigma\{c\,(\zeta\,{}^t\bar{\zeta} + 2\,W\,{}^t\bar{\zeta})\}}\,f(W + \zeta), \tag{1.4.23}$$

where $W \in \mathbb{C}^{(m,n)}$ and $\zeta = \lambda + i\,\mu$.

Proof For $g \in G$, $f \in \mathscr{H}_{F,c}$ and $W \in \mathbb{C}^{(m,n)}$, we have

$$\begin{aligned}
&\left(U^{F,c}(g)f\right)(W)\\
&= \left(\Delta(U^{F,c}(g)(\Lambda_f))\right)(W) = \left(U^{F,c}(g)(\Lambda_f)\right)\left(\frac{1}{2}WR^+\right)\\
&= \Lambda_f\left(\left(\frac{1}{2}WR^+, 0\right) \circ g\right) = \Lambda_f\left(\left(\frac{1}{2}W, -\frac{i}{2}W, 0\right) \circ (\lambda, \mu, \kappa)\right)\\
&= \Lambda_f\left(\left(\lambda + \frac{1}{2}W\right)P + \left(\mu - \frac{i}{2}W\right)Q, \kappa + \frac{1}{2}\,W\,{}^t\mu + \frac{i}{2}\,W\,{}^t\lambda\right)\\
&= e^{2\pi i\,\sigma\{c(\kappa + \frac{i}{2}\,W\,{}^t\bar{\zeta} + \frac{i}{2}\,\bar{\zeta}\,{}^tW + \frac{i}{2}\,\zeta\,{}^t\bar{\zeta})\}}\,f(W + \zeta) \quad (**)\\
&= e^{2\pi i\,\sigma(c\kappa)} \cdot e^{-\pi\,\sigma\{c(\zeta\,{}^t\bar{\zeta} + 2\,W\,{}^t\bar{\zeta})\}}\,f(W + \zeta),
\end{aligned}$$

where $\zeta = \lambda + i\,\mu$. In (**), we used (1.4.20) and the facts that $2\,i\,z^-\,{}^t z^+ = \frac{i}{2}(\zeta\,{}^t\overline{\zeta} + \overline{\zeta}\,{}^t W)$ and $2\,z^+ = W + \zeta$. □

Definition 1.4.3 The induced representation $U^{F,c}$ of G in the Fock space $\mathscr{H}_{F,c}$ is called the Fock representation of G.

Let $W = U + iV \in \mathbb{C}^{(m,n)}$ with $U, V \in \mathbb{R}^{(m,n)}$. If $U = (u_{ka})$, $V = (v_{lb})$ are coordinates in $\mathbb{C}^{(m,n)}$, we put

$$dU = du_{11}du_{12}\cdots du_{mn}, \quad dV = dv_{11}dv_{12}\cdots dv_{mn}$$

and $dW = dU\,dV$. And we set

$$d\mu(W) = e^{-\pi\sigma(W\,{}^t\overline{W})}\,dW. \tag{1.4.24}$$

Let f be a holomorphic function on $\mathbb{C}^{(m,n)}$. Then $f(W)$ has the Taylor expansion

$$f(z) = \sum_{J \in \mathbb{Z}_{\geq 0}^{(m,n)}} a_J W^J, \quad W = (w_{ka}) \in \mathbb{C}^{(m,n)},$$

where $J = (J_{ka}) \in \mathbb{Z}_{\geq 0}^{(m,n)}$ and $W^J := w_{11}^{J_{11}} w_{12}^{J_{12}} \cdots w_{mn}^{J_{mn}}$.

We set $|W|_\infty := \max_{k,a}(|w_{ka}|)$. Then by an easy computation, we have

$$\begin{aligned}\int_{\mathbb{C}^{(m,n)}} |f(W)|^2\,d\mu(W) &= \lim_{r\to\infty} \int_{|W|_\infty \leq r} |f(W)|^2 d\mu(W) \\ &= \lim_{r\to\infty} \sum_{J,K} a_J \overline{a_K} \int_{|W|_\infty \leq r} W^J \overline{W^K}\,d\mu(W) \\ &= \sum_J |a_J|^2 \pi^{-|J|} J!,\end{aligned}$$

where J runs over $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$. For the definitions of $|J|$ and $J!$, we refer to Notations Frequently Used, p. xv.

Let $\mathscr{H}_{m,n}$ be the Hilbert space consisting of all holomorphic functions $f : \mathbb{C}^{(m,n)} \longrightarrow \mathbb{C}$ satisfying the condition

$$\|f\|^2 = \int_{\mathbb{C}^{(m,n)}} |f(W)|^2\,d\mu(W) < \infty. \tag{1.4.25}$$

The inner product $(\ ,\ )$ on $\mathscr{H}_{m,n}$ is given by

$$(f_1, f_2) = \int_{\mathbb{C}^{(m,n)}} f_1(W)\,\overline{f_2(W)}\,d\mu(W), \quad f_1, f_2 \in \mathscr{H}_{m,n}.$$

Thus we have

Lemma 1.4.4 *Let $f \in \mathscr{H}_{m,n}$ and let $f(W) = \sum_J a_J W^J$ be the Taylor expansion of f. Then*

$$\|f\|^2 = \sum_{J \in \mathbb{Z}_{\geq 0}^{(m,n)}} |a_J|^2 \pi^{-|J|} J!.$$

For each $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$, we define the holomorphic function $\Phi_J(W)$ on $\mathbb{C}^{(m,n)}$ by

$$\Phi_J(W) := (J!)^{-\frac{1}{2}} \left(\pi^{\frac{1}{2}} W\right)^J, \quad W \in \mathbb{C}^{(m,n)}. \tag{1.4.26}$$

Then

$$(\Phi_J, \Phi_K) = \begin{cases} 1 & \text{if } J = K \\ 0 & \text{otherwise.} \end{cases} \tag{1.4.27}$$

It is easy to see that the set $\left\{ \Phi_J \,\middle|\, J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ forms a complete orthonormal system in $\mathscr{H}_{m,n}$. By the Schwarz inequality, for any $f \in \mathscr{H}_{m,n}$, we have

$$|f(W)| \leq e^{\frac{\pi}{2}\sigma(W\,{}^t\overline{W})} \|f\|, \quad W \in \mathbb{C}^{(m,n)}. \tag{1.4.28}$$

Consequently, the norm convergence in $\mathscr{H}_{m,n}$ implies the uniform convergence on any bounded subset of $\mathbb{C}^{(m,n)}$. We observe that for a fixed $W' \in \mathbb{C}^{(m,n)}$, the holomorphic function $W \longrightarrow e^{\pi\sigma(W\,{}^t\overline{W'})}$ admits the following Taylor expansion:

$$e^{\pi\sigma(W\,{}^t\overline{W'})} = \sum_{J \in \mathbb{Z}_{\geq 0}^{(m,n)}} \Phi_J(W)\, \Phi_J(\overline{W'}). \tag{1.4.29}$$

From (1.4.29), we obtain

$$\Phi_J(\overline{W'}) = (J!)^{-\frac{1}{2}} \int_{\mathbb{C}^{(m,n)}} e^{\pi\sigma(W\,{}^t\overline{W'})} \left(\pi^{\frac{1}{2}} \overline{W}\right)^J d\mu(W). \tag{1.4.30}$$

Thus if $f \in \mathscr{H}_{m,n}$, we get

$$\begin{aligned} \left(f(W),\, e^{\pi\sigma(W\,{}^t\overline{W'})}\right) &= \left(f,\, \sum_J \Phi_J(\overline{W'})\, \Phi_J(\cdot)\right) \\ &= \sum_J \Phi_J(W')\, (f, \Phi_J) \\ &= f(W'). \end{aligned}$$

Hence $e^{\pi\sigma(W\,{}^t\overline{W'})}$ is the reproducing kernel for $\mathscr{H}_{m,n}$ in the sense that for any $f \in \mathscr{H}_{m,n}$,

$$f(W) = \int_{\mathbb{C}^{(m,n)}} e^{\pi\,\sigma(W\,{}^t\overline{W'})}\, f(W')\, d\mu(W'). \tag{1.4.31}$$

We set

$$\kappa(W, W') := e^{\pi\sigma(W\,{}^t\overline{W'})}, \quad W, W' \in \mathbb{C}^{(m,n)}. \tag{1.4.32}$$

Obviously, $\kappa(W, W') = \overline{\kappa(W', W)}$. (1.4.31) may be written as

$$f(W) = \int_{\mathbb{C}^{(m,n)}} \kappa(W, W')\, f(W')\, d\mu(W'), \quad f \in \mathscr{H}_{m,n}. \tag{1.4.33}$$

Let $\mathscr{M}$ be a positive definite, symmetric half-integral matrix of degree m. We define the measure

$$d\mu_{\mathscr{M}}(W) = e^{-2\pi\,\sigma(\mathscr{M}W\,{}^t\overline{W})}\, dW. \tag{1.4.34}$$

We recall the Fock space $\mathscr{H}_{F,\mathscr{M}}$ consisting of all holomorphic functions $f : \mathbb{C}^{(m,n)} \longrightarrow \mathbb{C}$ that satisfy the condition

$$\|f\|^2_{\mathscr{M}} := \|f\|^2_{F,\mathscr{M}} := \int_{\mathbb{C}^{(m,n)}} |f(W)|^2\, d\mu_{\mathscr{M}}(W) < \infty. \tag{1.4.35}$$

The inner product $(\ ,\ )_{\mathscr{M}} := (\ ,\ )_{F,\mathscr{M}}$ on $\mathscr{H}_{F,\mathscr{M}}$ is given by

$$(f_1, f_2)_{\mathscr{M}} = \int_{\mathbb{C}^{(m,n)}} f_1(W)\, \overline{f_2(W)}\, d\mu_{\mathscr{M}}(W), \quad f_1, f_2 \in \mathscr{H}_{F,\mathscr{M}}.$$

Lemma 1.4.5 *Let $f \in \mathscr{H}_{F,\mathscr{M}}$ and let $g(W) = f\left((2\mathscr{M})^{-\frac{1}{2}} W\right)$ be the holomorphic function on $\mathbb{C}^{(m,n)}$. We let*

$$g(W) = \sum_{J \in \mathbb{Z}^{(m,n)}_{\geq 0}} a_{\mathscr{M},J}\, W^J$$

be the Taylor expansion of $g(W)$. Then we have

$$\|f\|^2_{\mathscr{M}} = (f, f)_{\mathscr{M}} = 2^{-n}(\det \mathscr{M})^{-n} \sum_{J \in \mathbb{Z}^{(m,n)}_{\geq 0}} |a_{\mathscr{M},J}|^2\, \pi^{-|J|}\, J!.$$

Proof Let $\mathscr{M}^{\frac{1}{2}}$ be the unique positive definite symmetric matrix of degree m such that $\left(\mathscr{M}^{\frac{1}{2}}\right)^2 = \mathscr{M}$. We put $\widetilde{W} := \sqrt{2}\mathscr{M}^{\frac{1}{2}}W$. Obviously, $d\widetilde{W} = 2^n (\det \mathscr{M})^n dW$. Thus for $f \in \mathscr{H}_{F,\mathscr{M}}$, we have

$$\begin{aligned}(f, f)_{\mathscr{M}} &= \int_{\mathbb{C}^{(m,n)}} |f(W)|^2 \, d\mu_{\mathscr{M}}(W) \\ &= 2^{-n} (\det \mathscr{M})^{-n} \int_{\mathbb{C}^{(m,n)}} |g(W)|^2 \, d\mu(W) \\ &= 2^{-n} (\det \mathscr{M})^{-n} \sum_{J \in \mathbb{Z}_{\geq 0}^{(m,n)}} |a_{\mathscr{M},J}|^2 \, \pi^{-|J|} \, J! \quad \text{(by Lemma 1.4.4).}\end{aligned}$$

□

For each $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$, we put

$$\Phi_{\mathscr{M},J}(W) := 2^{\frac{n}{2}} (\det \mathscr{M})^{\frac{n}{2}} (J!)^{-\frac{1}{2}} \left((2\pi\mathscr{M})^{\frac{1}{2}} W\right)^J, \quad W \in \mathbb{C}^{(m,n)}. \tag{1.4.36}$$

Lemma 1.4.6 *The set* $\left\{ \Phi_{\mathscr{M},J} \,\middle|\, J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ *is a complete orthonormal system in* $\mathscr{H}_{F,\mathscr{M}}$.

Proof For $J, K \in \mathbb{Z}_{\geq 0}^{(m,n)}$, we have

$$\begin{aligned}&\left(\Phi_{\mathscr{M},J}, \Phi_{\mathscr{M},K}\right)_{\mathscr{M}} \\ &= 2^n (\det \mathscr{M})^n (J!)^{-\frac{1}{2}} (K!)^{-\frac{1}{2}} \\ &\quad \times \int_{\mathbb{C}^{(m,n)}} \left((2\pi\mathscr{M})^{\frac{1}{2}} W\right)^J \left((2\pi\mathscr{M})^{\frac{1}{2}} \overline{W}\right)^K d\mu_{\mathscr{M}}(W) \\ &= (J!)^{-\frac{1}{2}} (K!)^{-\frac{1}{2}} \int_{\mathbb{C}^{(m,n)}} (\pi^{\frac{1}{2}} W)^J \, \overline{(\pi^{\frac{1}{2}} W)^K} \, d\mu(W) \\ &= (\Phi_J, \Phi_K).\end{aligned}$$

By (1.4.27), we have

$$(\Phi_{\mathscr{M},J}, \Phi_{\mathscr{M},K})_{\mathscr{M}} = \begin{cases} 1 & \text{if } J = K \\ 0 & \text{otherwise.} \end{cases} \tag{1.4.37}$$

We leave the proof of the completeness to the reader. □

We observe that for a fixed $W' \in \mathbb{C}^{(m,n)}$, the holomorphic function $W \mapsto e^{\pi\,\sigma(\mathscr{M} W\, {}^t\overline{W'})}$ admits the following Taylor expansion:

$$e^{\pi\,\sigma(\mathscr{M}W\,{}^t\overline{W'})} = \sum_{J\in\mathbb{Z}_{\geq 0}^{(m,n)}} \Phi_{\mathscr{M},J}(W)\,\Phi_{\mathscr{M},J}(\overline{W'}). \tag{1.4.38}$$

If $f \in \mathscr{H}_{F,\mathscr{M}}$, we have

$$\begin{aligned}\left(f(W),\ e^{\pi\,\sigma(\mathscr{M}W\,{}^t\overline{W'})}\right)_{\mathscr{M}} &= \sum_{J\in\mathbb{Z}_{\geq 0}^{(m,n)}} (f,\Phi_{\mathscr{M},J})_{\mathscr{M}}\,\Phi_{\mathscr{M},J}(W') \\ &= f(W').\end{aligned}$$

Hence $e^{\pi\,\sigma(\mathscr{M}W\,{}^t\overline{W'})}$ is the reproducing kernel for $\mathscr{H}_{F,\mathscr{M}}$ in the sense that

$$f(W) = \int_{\mathbb{C}^{(m,n)}} f(W')\,e^{\pi\,\sigma(\mathscr{M}W\,{}^t\overline{W'})}\,d\mu_{\mathscr{M}}(W'). \tag{1.4.39}$$

For $U \in \mathbb{R}^{(m,n)}$ and $W \in \mathbb{C}^{(m,n)}$, we put

$$k(U,W) := e^{2\pi\,\sigma(-U\,{}^tU + \frac{1}{2}W\,{}^tW + 2i\,U\,{}^tW)}. \tag{1.4.40}$$

Then we have the following lemma.

Lemma 1.4.7

$$\int_{\mathbb{R}^{(m,n)}} k(U,W)\,\overline{k(U,W')}\,dU = e^{2\pi\,\sigma(W\,{}^tW')}.$$

Proof We put

$$\mathcal{I}(W,W') := \int_{\mathbb{R}^{(m,n)}} k(U,W)\,\overline{k(U,W')}\,dU.$$

Then we have

$$\begin{aligned}\mathcal{I}(W,W') &= e^{\pi\,\sigma(W\,{}^tW+\overline{W'}\,{}^t\overline{W'})}\int_{\mathbb{R}^{(m,n)}} e^{-4\pi\,\sigma(U\,{}^tU)}\,e^{4\pi i\,\sigma\{U\,{}^t(W-\overline{W'})\}}\,dU \\ &= e^{\pi\,\sigma(W\,{}^tW+\overline{W'}\,{}^t\overline{W'})}\cdot\prod_{k,a}\int_{\mathbb{R}} e^{-4\pi\{u_{ka}^2 - iu_{ka}(w_{ka}-\overline{w'_{ka}})\}}\,du_{ka},\end{aligned}$$

where $W = (w_{ka})$, $W' = (w'_{ka}) \in \mathbb{C}^{(m,n)}$ and $U = (u_{ka}) \in \mathbb{R}^{(m,n)}$. It is easy to show that

$$\int_{\mathbb{R}} e^{-4\pi\{u_{ka}^2 - iu_{ka}(w_{ka}-\overline{w'_{ka}})\}}\,du_{ka} = e^{-\pi(w_{ka}-\overline{w'_{ka}})^2}.$$

Thus we get

$$\begin{aligned}\mathcal{I}(W, W') &= e^{\pi\,\sigma(W\,{}^t W+\overline{W'}\,{}^t\overline{W'})}\cdot e^{-\pi\sum_{k,a}(w_{ka}-\overline{w'_{ka}})^2}\\ &= e^{2\pi\sum_{k,a} w_{ka}\overline{w'_{ka}}}\\ &= e^{2\pi\,\sigma(W\,{}^t\overline{W'})}.\end{aligned}$$

□

For $U \in \mathbb{R}^{(m,n)}$ and $W \in \mathbb{C}^{(m,n)}$, we put

$$k_{\mathscr{M}}(U, W) := e^{2\pi\,\sigma\{\mathscr{M}(-U\,{}^t U-\frac{1}{2}\,W\,{}^t W+2\,U\,{}^t W)\}}. \tag{1.4.41}$$

Lemma 1.4.8 *Let $\mathscr{M}$ be a positive definite, symmetric half-integral matrix of degree m. Then we have*

$$k_{\mathscr{M}}(U, W) = k(\mathscr{M}^{\frac{1}{2}}U,\ -i\,\mathscr{M}^{\frac{1}{2}}W) \tag{1.4.42}$$

and

$$\int_{\mathbb{R}^{(m,n)}} k_{\mathscr{M}}(U, W)\,\overline{k_{\mathscr{M}}(U, W')}\,dU = (\det\mathscr{M})^{-\frac{n}{2}}\cdot e^{2\pi\,\sigma(\mathscr{M}W\,{}^t\overline{W'})}. \tag{1.4.43}$$

Proof Formula (1.4.42) follows immediately from a straightforward computation. We put

$$\mathcal{I}_{\mathscr{M}}(W, W') := \int_{\mathbb{R}^{(m,n)}} k_{\mathscr{M}}(U, W)\,\overline{k_{\mathscr{M}}(U, W')}\,dU.$$

Using (1.4.42), we have

$$\begin{aligned}\mathcal{I}_{\mathscr{M}}(W, W') &= \int_{\mathbb{R}^{(m,n)}} k\left(\mathscr{M}^{\frac{1}{2}}U,\ -i\mathscr{M}^{\frac{1}{2}}W\right)\overline{k\left(\mathscr{M}^{\frac{1}{2}}U, -i\mathscr{M}^{\frac{1}{2}}W'\right)}\,dU\\ &= (\det\mathscr{M})^{-\frac{n}{2}}\int_{\mathbb{R}^{(m,n)}} k\left(U, -i\mathscr{M}^{\frac{1}{2}}W\right)\overline{k\left(U, -i\mathscr{M}^{\frac{1}{2}}W'\right)}\,dU\\ &= (\det\mathscr{M})^{-\frac{n}{2}}\cdot e^{2\pi\,\sigma(\mathscr{M}W\,{}^t\overline{W'})} \qquad \text{(by Lemma 1.4.7).}\end{aligned}$$

□

We recall that the Fock representation $U^{F,\mathscr{M}}$ of the real Heisenberg group G in $\mathscr{H}_{F,\mathscr{M}}$(cf. (1.4.23)) is given by

$$\left(U^{F,\mathscr{M}}(g)f\right)(W) = e^{2\pi i\,\sigma(\mathscr{M}\kappa)}\cdot e^{-\pi\,\sigma\{\mathscr{M}(\zeta\,{}^t\bar{\zeta}+2\,W\,{}^t\bar{\zeta})\}}\,f(W+\zeta), \tag{1.4.44}$$

where $g = (\lambda, \mu, \kappa) \in G$, $f \in \mathscr{H}_{F,\mathscr{M}}$ and $\zeta = \lambda + i\,\mu \in \mathbb{C}^{(m,n)}$.

Lemma 1.4.9 *The Fock representation $U^{F,\mathscr{M}}$ of G in $\mathscr{H}_{F,\mathscr{M}}$ is unitary.*

Proof For brevity, we put $U_{g,f}(W) := \left(U^{F,\mathscr{M}}(g)f\right)(W)$ for $g = (\lambda, \mu, \kappa) \in G$ and $f \in \mathscr{H}_{F,\mathscr{M}}$. Then we have

$$
\begin{aligned}
&(U_{g,f},\, U_{g,f})_{\mathscr{M}} = \|U_{g,f}\|^2_{\mathscr{M}} \\
&= \int_{\mathbb{C}^{(m,n)}} U_{g,f}(W)\,\overline{U_{g,f}(W)}\,d\mu_{\mathscr{M}}(W) \\
&= \int_{\mathbb{C}^{(m,n)}} e^{-\pi\,\sigma\{\mathscr{M}({}^t\zeta\bar{\zeta} + 2W\,{}^t\bar{\zeta} + \bar{\zeta}\,{}^t\zeta + 2\,\overline{W}\,{}^tW + 2\,W\,{}^t\overline{W})\}}\,|f(W+\zeta)|^2\,dW \\
&= \int_{\mathbb{C}^{(m,n)}} |f(W)|^2\,d\mu_{\mathscr{M}}(W) \\
&= (f,\, f)_{\mathscr{M}} = \|f\|^2_{\mathscr{M}}.
\end{aligned}
$$

□

We recall that the Schrödinger representation $U^{S,\mathscr{M}} := U_{\sigma_{\mathscr{M}}}$ of the real Heisenberg group G in the Hilbert space $\mathscr{H}_{S,\mathscr{M}} \cong L^2\left(\mathbb{R}^{(m,n)},\, d\xi\right)$ (cf. (1.2.11)) is given by

$$
\left(U^{S,\mathscr{M}}(g)f\right)(\xi) = e^{2\pi i\,\sigma\{\mathscr{M}(\kappa + \mu\,{}^t\lambda + 2\,\mu\,{}^t\xi)\}}\,f(\xi + \lambda), \tag{1.4.45}
$$

where $g = (\lambda, \mu, \kappa) \in G$, $f \in \mathscr{H}_{S,\mathscr{M}}$ and $\xi \in \mathbb{R}^{(m,n)}$. $U^{S,\mathscr{M}}$ is called the Schrödinger representation of G of index $\mathscr{M}$. The inner product $(\ ,\ )_{S,\mathscr{M}}$ on $\mathscr{H}_{S,\mathscr{M}}$ is given by

$$
(f_1,\, f_2)_{S,\mathscr{M}} = \int_{\mathbb{R}^{(m,n)}} f_1(U)\,\overline{f_2(U)}\,dU, \quad f_1, f_2 \in \mathscr{H}_{S,\mathscr{M}}.
$$

And we define the norm $\|\ \|_{S,\mathscr{M}}$ on $\mathscr{H}_{S,\mathscr{M}}$ by

$$
\|f\|^2_{S,\mathscr{M}} = \int_{\mathbb{R}^{(m,n)}} |f(U)|^2\,dU, \quad f \in \mathscr{H}_{S,\mathscr{M}}.
$$

Theorem 1.4.10 *The Fock representation $\left(U^{F,\mathscr{M}},\, \mathscr{H}_{F,\mathscr{M}}\right)$ of G is unitarily equivalent to the Schrödinger representation $\left(U^{S,\mathscr{M}},\, \mathscr{H}_{S,\mathscr{M}}\right)$ of G of index $\mathscr{M}$. Therefore the Fock representation $U^{F,\mathscr{M}}$ is irreducible. The intertwining unitary isometry $I_{\mathscr{M}} : \mathscr{H}_{S,\mathscr{M}} \longrightarrow \mathscr{H}_{F,\mathscr{M}}$ is given by*

$$
(I_{\mathscr{M}} f)(W) = \int_{\mathbb{R}^{(m,n)}} k_{\mathscr{M}}(\xi, W)\,f(\xi)\,d\xi, \tag{1.4.46}
$$

where $f \in \mathscr{H}_{S,\mathscr{M}} = L^2\left(\mathbb{R}^{(m,n)},\, d\xi\right)$, $W \in \mathbb{C}^{(m,n)}$ *and* $k_{\mathscr{M}}(\xi, W)$ *is a function on* $\mathbb{R}^{(m,n)} \times \mathbb{C}^{(m,n)}$ *defined by (1.4.41).*

Proof For any $f \in \mathscr{H}_{S,\mathscr{M}} = L^2\left(\mathbb{R}^{(m,n)},\, d\xi\right)$, we define

$$(I_{\mathscr{M}} f)(W) = \int_{\mathbb{R}^{(m,n)}} k_{\mathscr{M}}(\xi, W)\, f(\xi)\, d\xi, \quad W \in \mathbb{C}^{(m,n)}.$$

Now we will show the following (I1), (I2) and (I3):

(I1) The image of $\mathscr{H}_{S,\mathscr{M}}$ under $I_{\mathscr{M}}$ is contained in $\mathscr{H}_{F,\mathscr{M}}$.
(I2) $I_{\mathscr{M}}$ preserves the norms, i.e., $\|f\|_{S,\mathscr{M}} = \|I_{\mathscr{M}} f\|_{\mathscr{M}}$.
(I3) $I_{\mathscr{M}}$ is a bijective operator of $\mathscr{H}_{S,\mathscr{M}}$ onto $\mathscr{H}_{F,\mathscr{M}}$.

Before we prove (I1), (I2) and (I3), we prove the following lemma.

Lemma 1.4.11 *For a fixed* $U \in \mathbb{R}^{(m,n)}$, *we consider the Taylor expansion*

$$k_{\mathscr{M}}(U, W) = \sum_{J \in \mathbb{Z}_{\geq 0}^{(m,n)}} h_J(U)\, \Phi_{\mathscr{M},J}(W), \quad W \in \mathbb{C}^{(m,n)} \tag{1.4.47}$$

of the holomorphic function $k_{\mathscr{M}}(U, \cdot)$ *on* $\mathbb{C}^{(m,n)}$. *Then the set* $\left\{ h_J \,\middle|\, J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ *forms a complete orthonormal system in* $L^2\left(\mathbb{R}^{(m,n)},\, d\xi\right)$. *Moreover, for a fixed* $W \in \mathbb{C}^{(m,n)}$, *(1.4.47) is the Fourier expansion of* $k_{\mathscr{M}}(\cdot, W)$ *with respect to this orthonormal system* $\left\{ h_J \,\middle|\, J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$.

Proof Following Igusa [18, pp. 33–34], we can prove it. The details will be left to the reader. □

If $f \in \mathscr{H}_{S,\mathscr{M}}$, then by the Schwarz inequality and Formula (1.4.43) in Lemma 1.4.8, we have

$$\begin{aligned} |\,(I_{\mathscr{M}} f)(W)| &\leq \left(\int_{\mathbb{R}^{(m,n)}} |k_{\mathscr{M}}(U, W)|^2\, dU \right)^{\frac{1}{2}} \cdot \left(\int_{\mathbb{R}^{(m,n)}} |f(U)|^2\, dU \right)^{\frac{1}{2}} \\ &= (\det \mathscr{M})^{-\frac{n}{4}} \cdot e^{\pi\, \sigma(\mathscr{M} W\, {}^t\overline{W})}\, \|f\|_{S,\mathscr{M}}. \end{aligned}$$

Thus the above integral $(I_{\mathscr{M}} f)(W)$ converges uniformly on any compact subset of $\mathbb{C}^{(m,n)}$ and hence $(I_{\mathscr{M}} f)(W)$ is holomorphic in $\mathbb{C}^{(m,n)}$. And according to Lemma 1.4.11, we get

$$\begin{aligned} (I_{\mathscr{M}} f)(W) &= \sum_{J \in \mathbb{Z}_{\geq 0}^{(m,n)}} \int_{\mathbb{R}^{(m,n)}} h_J(U)\, f(U)\, \Phi_{\mathscr{M},J}(W)\, dU \\ &= \sum_{J \in \mathbb{Z}_{\geq 0}^{(m,n)}} (h_J,\, \bar{f})_{S,\mathscr{M}}\, \Phi_{\mathscr{M},J}(W). \end{aligned}$$

Therefore we get

$$\begin{aligned}
&\|I_{\mathscr{M}} f\|^2_{F,\mathscr{M}} = \int_{\mathbb{C}^{(m,n)}} |I_{\mathscr{M}} f(W)|^2 \, d\mu_{\mathscr{M}}(W) \\
&= \sum_{J,\, K \in \mathbb{Z}^{(m,n)}_{\geq 0}} (h_J,\ \bar{f})_{S,\mathscr{M}} \cdot \overline{(h_K,\ \bar{f})} \int_{\mathbb{C}^{(m,n)}} \Phi_{\mathscr{M},J}(W)\, \overline{\Phi_{\mathscr{M},K}(W)}\, d\mu_{\mathscr{M}}(W) \\
&= \sum_{J \in \mathbb{Z}^{(m,n)}_{\geq 0}} |(h_J,\ \bar{f})_{S,\mathscr{M}}|^2 \quad \text{(by (1.4.37))} \\
&= \|f\|^2_{S,\mathscr{M}} < \infty.
\end{aligned}$$

This proves (I1) and (I2). It is easy to see that $I_{\mathscr{M}} \overline{h_J} = \Phi_{\mathscr{M},J}$ for all $J \in \mathbb{Z}^{(m,n)}_{\geq 0}$. Since the set $\left\{ \Phi_{\mathscr{M},J} \,\middle|\, J \in \mathbb{Z}^{(m,n)}_{\geq 0} \right\}$ forms a complete orthonormal system of $\mathscr{H}_{F,\mathscr{M}}$, $I_{\mathscr{M}}$ is surjective. Obviously, the injectivity of $I_{\mathscr{M}}$ follows immediately from the fact that $I_{\mathscr{M}} \overline{h_J} = \Phi_{\mathscr{M},J}$ for all $J \in \mathbb{Z}^{(m,n)}_{\geq 0}$. This proves (I3).

On the other hand, we let $f \in \mathscr{H}_{S,\mathscr{M}}$ and $g = (\lambda, \mu, \kappa) \in G$. We put $\zeta = \lambda + i\,\mu$. Then we get

$$\begin{aligned}
&\left(U^{F,\mathscr{M}}(g)(I_{\mathscr{M}} f) \right)(W) \\
&= e^{2\pi i\, \sigma(\mathscr{M}\kappa)} \cdot e^{-\pi\, \sigma\{\mathscr{M}(\zeta\, {}^t\bar{\zeta} + 2\, W\, {}^t\bar{\zeta})\}} (I_{\mathscr{M}} f)(W + \zeta) \quad \text{(by (1.4.44))} \\
&= e^{2\pi i\, \sigma(\mathscr{M}\kappa)} \cdot e^{-\pi\, \sigma\{\mathscr{M}(\zeta\, {}^t\bar{\zeta} + 2\, W\, {}^t\bar{\zeta})\}} \int_{\mathbb{R}^{(m,n)}} k_{\mathscr{M}}(U, W + \zeta)\, f(U)\, dU.
\end{aligned}$$

We define the function $A_{\mathscr{M}} : \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)} \longrightarrow \mathbb{C}$ by

$$A_{\mathscr{M}}(U, W) := \sigma \left\{ \mathscr{M} \left(-U\, {}^t U - \frac{W\, {}^t W}{2} + 2\, U\, {}^t W \right) \right\}. \tag{1.4.48}$$

Obviously, $\kappa_{\mathscr{M}}(U, W) = e^{2\pi A_{\mathscr{M}}(U,W)}$ for $U \in \mathbb{R}^{(m,n)}$ and $W \in \mathbb{C}^{(m,n)}$.

By an easy computation, we get

$$A_{\mathscr{M}}(U, W+\zeta) - A_{\mathscr{M}}(U-\lambda, W) = \sigma \left\{ \mathscr{M} \left(\frac{\zeta\, {}^t\bar{\zeta}}{2} + W\, {}^t\bar{\zeta} - i\, \lambda\, {}^t\mu + 2\, i\, U\, {}^t\mu \right) \right\}.$$

Therefore we get

$$\begin{aligned}
&k_{\mathscr{M}}(U, W + \zeta) \\
&= e^{2\pi A_{\mathscr{M}}(U-\lambda, W)} \cdot e^{2\pi\, \sigma \left\{ \mathscr{M} \left(\frac{1}{2}\, \zeta\, {}^t\bar{\zeta} + W\, {}^t\bar{\zeta} - i\, \lambda\, {}^t\mu + 2\, i\, U\, {}^t\mu \right) \right\}} \\
&= k_{\mathscr{M}}(U - \lambda, W) \cdot e^{2\pi\, \sigma \left\{ \mathscr{M} \left(\frac{1}{2}\, \zeta\, {}^t\bar{\zeta} + W\, {}^t\bar{\zeta} - i\, \lambda\, {}^t\mu + 2\, i\, U\, {}^t\mu \right) \right\}}.
\end{aligned}$$

Hence we have

$$\begin{aligned}
&\left(U^{F,\mathscr{M}}(g)(I_{\mathscr{M}} f)\right)(W) \\
&= \int_{\mathbb{R}^{(m,n)}} e^{2\pi i\,\sigma\{\mathscr{M}(\kappa+2U\,{}^t\mu-\lambda\,{}^t\mu)\}}\, k_{\mathscr{M}}(U-\lambda, W)\, f(U)\, dU \\
&= \int_{\mathbb{R}^{(m,n)}} e^{2\pi i\,\sigma\{\mathscr{M}(\kappa+2\lambda\,{}^t\mu+2U\,{}^t\mu-\lambda\,{}^t\mu)\}}\, k_{\mathscr{M}}(U, W)\, f(U+\lambda)\, dU \\
&= \int_{\mathbb{R}^{(m,n)}} e^{2\pi i\,\sigma\{\mathscr{M}(\kappa+2U\,{}^t\mu+\lambda\,{}^t\mu)\}}\, k_{\mathscr{M}}(U, W)\, f(U+\lambda)\, dU \\
&= \int_{\mathbb{R}^{(m,n)}} k_{\mathscr{M}}(U, W)\, \left(U^{S,\mathscr{M}}(g) f\right)(U)\, dU \qquad \text{(by (1.4.45))} \\
&= \left(I_{\mathscr{M}}\left(U^{S,\mathscr{M}}(g) f\right)\right)(W).
\end{aligned}$$

So far we proved that $U^{F,\mathscr{M}} \circ I_{\mathscr{M}} = I_{\mathscr{M}} \circ U^{S,\mathscr{M}}(g)$ for all $g \in G$. That is, the unitary isometry $I_{\mathscr{M}}$ of $\mathscr{H}_{S,\mathscr{M}}$ onto $\mathscr{H}_{F,\mathscr{M}}$ is the intertwining operator. This completes the proof. □

1.5 Hermite Operators

We recall the Schrödinger representation U_c of $G := H_{\mathbb{R}}^{(n,m)}$ induced from σ_c (cf. (1.2.11)). We consider the special case when $c = I_m$ is the identity matrix of degree m. Then it is easy to see that

$$\begin{aligned}
dU_{I_m}(D^0_{kl})\, f(\xi) &= 2\pi\, i\, \delta_{kl}\, f(\xi), \\
dU_{I_m}(D_{ka})\, f(\xi) &= \frac{\partial f(\xi)}{\partial \xi_{ka}}, \\
dU_{I_m}(\widehat{D}_{lb})\, f(\xi) &= 4\pi\, i\, \xi_{lb}\, f(\xi),
\end{aligned}$$

where $f \in \mathcal{S}(\mathbb{R}^{(m,n)})$ or $C^\infty(\mathbb{R}^{(m,n)})$, the Schwartz space and $\xi_{11}, \ldots, \xi_{mn}$ are the coordinates of ξ. In Lemma 1.1.2 in Sect. 1.1, we put

$$\begin{aligned}
Z^0_{kl} &:= -i\, D^0_{kl}, \qquad 1 \le k \le l \le m, \qquad i = \sqrt{-1}, \\
Y^+_{ka} &:= \frac{1}{2}\,(D_{ka} + i\, \widehat{D}_{ka}), \quad 1 \le k \le m,\ 1 \le a \le n, \\
Y^-_{lb} &:= \frac{1}{2}\,(D_{lb} - i\, \widehat{D}_{lb}), \quad 1 \le l \le m,\ 1 \le b \le n.
\end{aligned}$$

We set

$$A_{ka}^{+} := dU_{I_m}(Y_{ka}^{+}) = \frac{1}{2}\, dU_{I_m}(D_{ka}) + \frac{i}{2}\, dU_{I_m}(\widehat{D}_{ka}), \tag{1.5.1}$$

$$A_{lb}^{-} := dU_{I_m}(Y_{lb}^{-}) = \frac{1}{2}\, dU_{I_m}(D_{lb}) - \frac{i}{2}\, dU_{I_m}(\widehat{D}_{lb})) \tag{1.5.2}$$

and

$$C_{kl} := dU_{I_m}(Z_{kl}^{0}) = -i\, dU_{I_m}(D_{kl}^{0}). \tag{1.5.3}$$

By Lemma 1.1.2, we have

$$\begin{aligned}
&[A_{ka}^{+}, A_{lb}^{-}] = \delta_{ab}\, C_{kl},\\
&[A_{ka}^{+}, A_{lb}^{+}] = [A_{ka}^{-}, A_{lb}^{-}] = 0,\\
&[C_{kl}, C_{mn}] = [C_{kl}, A_{ma}^{+}] = [C_{kl}, A_{ma}^{-}] = 0.
\end{aligned}$$

In particular, we have

$$[A_{ka}^{+}, A_{ka}^{-}] = 2\pi \cdot \mathrm{Id}, \qquad 1 \le k \le m, \quad 1 \le a \le n. \tag{1.5.4}$$

We note that A_{ka}^{+} and A_{lb}^{-} act on the Schwartz space $\mathcal{S}(\mathbb{R}^{(m,n)})$ or $C^{\infty}(\mathbb{R}^{(m,n)})$ as the following linear differential operators:

$$A_{ka}^{+} = \frac{1}{2}\left(\frac{\partial}{\partial \xi_{ka}} - 4\pi\, \xi_{ka}\right) \tag{1.5.5}$$

and

$$A_{lb}^{-} = \frac{1}{2}\left(\frac{\partial}{\partial \xi_{lb}} - 4\pi\, \xi_{lb}\right), \tag{1.5.6}$$

where $1 \le k, l \le m$ and $1 \le a, b \le n$. The differential operators A_{ka}^{+} and A_{lb}^{-} are called the creation operator of an energy quantum and the annihilation operator of an energy quantum respectively. It is easy to see that the adjoint of A_{ka}^{-} is $-A_{ka}^{+}$.

We start with the ground state $f_0(\xi)$ given by

$$f_0(\xi) = (\sqrt{2})^{mn} e^{-2\pi \sum_{k=1}^{m} \sum_{a=1}^{n} \xi_{ka}^2}. \tag{1.5.7}$$

By an easy computation, we have

$$\langle f_0, f_0 \rangle = 1, \quad A_{ka}^{-}(f_0) = 0 \tag{1.5.8}$$

for all $1 \le k \le m$ and $1 \le a \le n$. This means that f_0 is a unit vector in $L^2(\mathbb{R}^{(m,n)}, d\xi)$ which is annihilated by the annihilation operator $A^-_{ka} : \mathcal{S}(\mathbb{R}^{(m,n)}) \longrightarrow \mathcal{S}(\mathbb{R}^{(m,n)})$. For the definitions of $\mathbb{Z}^{(m,n)}_{\ge 0}$, $|J|$, $J \pm \epsilon_{ka}$, $J!$ and ξ^J, we refer to Notations Frequently Used, p. xv. For any $J \in \mathbb{Z}^{(m,n)}_{\ge 0}$, we define

$$f_J(\xi) := (A^+)^J f_0(\xi) := (A^+_{11})^{J_{11}} \cdots (A^+_{ka})^{J_{ka}} \cdots (A^+_{mn})^{J_{mn}} f_0(\xi). \tag{1.5.9}$$

We give a lexicographic ordering on $\mathbb{Z}^{(m,n)}_{\ge 0}$. That is, for $J, K \in \mathbb{Z}^{(m,n)}_{\ge 0}$, $J < K$ if and only if $J_{11} = K_{11}, \ldots, J_{ij} = K_{ij}, J_{i,j+1} < K_{i,j+1}, \ldots$.

Lemma 1.5.1 *For each k, a with $1 \le k \le m$ and $1 \le a \le n$, we have*

$$A^-_{ka}(f_J) = -2\pi\, J_{ka}\, f_{J-\epsilon_{ka}}. \tag{1.5.10}$$

Proof We prove this by induction on J. If $J = (0, \ldots, 0)$, (1.5.10) holds. Suppose (1.5.10) holds for J. For $\tilde{J} = J + \epsilon_{ka}$,

$$\begin{aligned} A^-_{ka}\left(f_{J+\epsilon_{ka}}\right) &= A^-_{ka} \circ A^+_{ka}(f_J) \\ &= \left(A^+_{ka} \circ A^-_{ka} - [A^+_{ka}, A^-_{ka}]\right)(f_J) \\ &= A^+_{ka}(-2\pi\, J_{ka}\, f_{J-\epsilon_{ka}}) - 2\pi\, f_J \\ &= -2\pi\, J_{ka}\, f_J - 2\pi\, f_J \\ &= -2\pi\,(J_{ka} + 1)\, f_J. \end{aligned}$$

This completes the proof. □

Lemma 1.5.2

$$\langle f_J, f_K \rangle = \begin{cases} (2\pi)^J\, J! & \text{if } J = K \\ 0 & \text{otherwise.} \end{cases}$$

Proof If $J > K$, we have

$$\begin{aligned} \langle\, f_J, f_K \,\rangle &= \langle\, (A^+)^J f_0, (A^+)^K f_0 \,\rangle \\ &= (-1)^J \langle\, f_0, (A^-)^J \circ (A^+)^K f_0 \,\rangle \\ &= 0 \qquad \text{(by Lemma 1.5.1).} \end{aligned}$$

In the case when $J < K$, $\langle\, f_J, f_K \,\rangle = \langle\, f_K, f_J \,\rangle = 0$. In the case when $J = K$, we prove the above identity by induction on J. If $J = (0, 0, \ldots, 0)$, then $\langle\, f_0, f_0 \,\rangle = 1$.

Assume that $(f_J, f_J) = (2\pi)^J J!$. Then according to (1.5.4) and Lemma 1.5.1, we have

$$\begin{aligned}\langle f_{J+\epsilon_{ka}}, f_{J+\epsilon_{ka}} \rangle &= \langle A^+_{ka}(f_J), A^+_{ka}(f_J) \rangle \\ &= -\langle f_J, A^-_{ka} \circ A^+_{ka}(f_J) \rangle \\ &= -\langle f_J, (A^+_{ka} \circ A^-_{ka} - [A^+_{ka}, A^-_{ka}]) f_J \rangle \\ &= -\langle f_J, -2\pi J_{ka} f_J - 2\pi f_J \rangle \\ &= 2\pi (J_{ka} + 1) \langle f_J, f_J \rangle \\ &= (2\pi)^{J+\epsilon_{ka}} (J + \epsilon_{ka})!.\end{aligned}$$

□

We define the normalized function $h_J \in \mathcal{S}(\mathbb{R}^{(m,n)})$ by

$$h_J := \left(\frac{1}{\sqrt{2\pi}}\right)^J (J!)^{-1/2} f_J, \quad J \in \mathbb{Z}^{(m,n)}_{\geq 0}. \tag{1.5.11}$$

Lemma 1.5.3 *For each $J \in \mathbb{Z}^{(m,n)}_{\geq 0}$ and all $k, a \in \mathbb{Z}$ with $1 \leq k \leq m$ and $1 \leq a \leq n$, we have*

$$A^+_{ka}(h_J) = \{2\pi (J_{ka} + 1)\}^{1/2} h_{J+\epsilon_{ka}} \tag{1.5.12}$$

and

$$A^-_{ka}(h_J) = -(2\pi J_{ka})^{1/2} h_{J-\epsilon_{ka}}. \tag{1.5.13}$$

Proof According to (1.5.9), we have

$$\begin{aligned}A^+_{ka}(h_J) &= \left(\frac{1}{\sqrt{2\pi}}\right)^J (J!)^{-1/2} f_{J+\epsilon_{ka}} \\ &= (2\pi)^{1/2} (J_{ka} + 1)^{1/2} h_{J+\epsilon_{ka}} \\ &= \{2\pi (J_{ka} + 1)\}^{1/2} h_{J+\epsilon_{ka}}.\end{aligned}$$

According to Lemma 1.5.1, we have

$$\begin{aligned} A_{ka}^{-}(h_J) &= \left(\frac{1}{\sqrt{2\pi}}\right)^J (J!)^{-1/2} A_{ka}^{-}(f_J) \\ &= \left(\frac{1}{\sqrt{2\pi}}\right)^J (J!)^{-1/2} (-2\pi) J_{ka} f_{J-\epsilon_{ka}} \\ &= -(2\,\pi\, J_{ka})^{1/2} \left(\frac{1}{\sqrt{2\pi}}\right)^{J-\epsilon_{ka}} \{(J-\epsilon_{ka})!\}^{-1/2} f_{J-\epsilon_{ka}} \\ &= -(2\,\pi\, J_{ka})^{1/2} h_{J-\epsilon_{ka}}. \end{aligned}$$

□

Lemma 1.5.4 *For each $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$ and $k, a \in \mathbb{Z}^+$ with $1 \leq k \leq m$ and $1 \leq a \leq n$, we have*

$$A_{ka}^{+} \circ A_{ka}^{-}(h_J) = -2\,\pi\, J_{ka} h_J, \tag{1.5.14}$$

$$A_{ka}^{-} \circ A_{ka}^{+}(h_J) = -2\,\pi\, (J_{ka}+1)\, h_J. \tag{1.5.15}$$

Proof It follows immediately from (1.5.12) and (1.5.13):

$$\begin{aligned} A_{ka}^{+} \circ A_{ka}^{-}(h_J) &= -(2\,\pi\, J_{ka})^{1/2}\, A_{ka}^{+}(h_{J-\epsilon_{ka}}) \\ &= -(2\,\pi\, J_{ka})^{1/2}\, (2\pi J_{ka})^{1/2}\, h_J \\ &= -(2\,\pi\, J_{ka})\, h_J, \\ A_{ka}^{-} \circ A_{ka}^{+}(h_J) &= \{2\,\pi\, (J_{ka}+1)\}^{1/2}\, A_{ka}^{-}(h_{J+\epsilon_{ka}}) \\ &= \{2\,\pi\, (J_{ka}+1)\}^{1/2}\, (-1)\, \{2\pi (J_{ka}+1)\}^{1/2}\, h_J \\ &= -2\pi (J_{ka}+1)\, h_J. \end{aligned}$$

□

The linear differential operators

$$A_{ka}^{+} \circ A_{ka}^{-} = \frac{1}{4}\left(\frac{\partial^2}{\partial \xi_{ka}^2} - 16\,\pi^2\, \xi_{ka}^2 + 4\,\pi\right)$$

and

$$A_{ka}^{-} \circ A_{ka}^{+} = \frac{1}{4}\left(\frac{\partial^2}{\partial \xi_{ka}^2} - 16\,\pi^2\, \xi_{ka}^2 - 4\,\pi\right)$$

are called the number operators for the family $\{h_J \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)}\}$. Now we consider the so-called Hermite differential operator

$$H_{ka} := -2\left(A_{ka}^{+} \circ A_{ka}^{-} + A_{ka}^{-} \circ A_{ka}^{+}\right) = -\frac{\partial^2}{\partial \xi_{ka}^2} + 16\,\pi^2\,\xi_{ka}^2.$$

H_{ka} is also called the Schrödinger Hamiltonian for the harmonic oscillator system in quantum mechanics. Obviously, we have

$$H_{ka}(h_J) = 8\,\pi\,\left(J_{ka} + \frac{1}{2}\right) h_J, \qquad J \in \mathbb{Z}_{\geq 0}^{(m,n)}. \tag{1.5.16}$$

Thus the $\left\{ h_J \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ is the set of normalized eigenfunctions of all Hermite operators H_{ka} with eigenvalues $\left\{8\,\pi\,(J_{ka} + \frac{1}{2}) \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)}\right\}$. In other words, each h_J $(J \in \mathbb{Z}_{\geq 0}^{(m,n)})$ is the harmonic oscillator wave function with equidistant energies $\left\{8\,\pi\,(J_{ka} + \frac{1}{2}) \mid 1 \leq k \leq m,\ 1 \leq a \leq n \right\}$ in natural units. The Hermite operator H_{ka} acts on the Schwartz space $\mathcal{S}(\mathbb{R}^{(m,n)}) \subset L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)$ and is self-adjoint.

Lemma 1.5.5 *For each $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$ and $k, a \in \mathbb{Z}$ with $1 \leq k \leq m$ and $1 \leq a \leq n$,*

$$h_J(-\xi) = (-1)^J\, h_J(\xi), \tag{1.5.17}$$

$$\left(\frac{\partial}{\partial \xi_{ka}} - 4\,\pi\,\xi_{ka}\right) h_J(\xi) = 2\,\{2\,\pi\,(J_{ka}+1)\}^{1/2}\, h_{J+\epsilon_{ka}}(\xi), \tag{1.5.18}$$

$$\widehat{h_J} = (-i)^J\, h_J, \tag{1.5.19}$$

$$CF(h_J) = i^J\, h_J. \tag{1.5.20}$$

Thus $\widehat{h_J}$ and $CF(h_J)$ satisfy the differential equation (1.5.18). Here $\widehat{f}(\eta)$ denotes the Fourier transform of $f(\xi)$ on $\mathbb{R}^{(m,n)}$ defined by

$$\widehat{f}(\eta) := \int_{\mathbb{R}^{(m,n)}} f(\xi)\, e^{-2\,\pi\, i\, \langle \xi, \eta \rangle}\, d\xi, \quad \eta \in \mathbb{R}^{(m,n)}$$

and $CF(f)$ denotes the Fourier cotransform of f on $\mathbb{R}^{(m,n)}$ defined by

$$CF(f)(\xi) := \int_{\mathbb{R}^{(m,n)}} f(\eta)\, e^{2\,\pi\, i\, \langle \eta, \xi \rangle}\, d\eta, \quad \xi \in \mathbb{R}^{(m,n)}.$$

Proof (1.5.17) is obvious. (1.5.18) follows immediately from (1.5.5) and (1.5.12). (1.5.19) and (1.5.20) follow from a simple computation. □

For $\xi = (\xi_{ka}) \in \mathbb{R}^{(m,n)}$, we briefly put $|\xi|^2 := \sum_{k=1}^{m} \sum_{a=1}^{n} \xi_{ka}^2$. We define the functions $P_J \left(J \in \mathbb{Z}_{\geq 0}^{(m,n)}\right)$ by

$$h_J(\xi) := P_J(\xi)\, e^{-2\pi |\xi|^2}, \qquad \xi \in \mathbb{R}^{(m,n)}. \tag{1.5.21}$$

Indeed, it will be proved later that $P_J(\xi)$ are the Hermite polynomials of degree $J = (J_{11}, \ldots, J_{mn})$ normalized in such a way that they form an orthonormal family in $L^2\big(\mathbb{R}^{(m,n)}, e^{-4\pi|\xi|^2} d\xi\big)$.

Lemma 1.5.6 *For each $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$ and $k, a \in \mathbb{Z}^+$ with $1 \leq k \leq m$, and $1 \leq a \leq n$, we have*

$$\frac{\partial P_J(\xi)}{\partial \xi_{ka}} - 8\pi\, \xi_{ka}\, P_J(\xi) - 2\{2\pi\,(J_{ka}+1)\}^{1/2}\, P_{J+\epsilon_{ka}}(\xi) = 0 \tag{1.5.22}$$

and

$$\frac{\partial P_{J+\epsilon_{ka}}(\xi)}{\partial \xi_{ka}} + 2\{2\pi\,(J_{ka}+1)\}^{1/2}\, P_J(\xi) = 0. \tag{1.5.23}$$

Proof (1.5.22) follows from (1.5.18). (1.5.23) follows from (1.5.6) and (1.5.13). □

Differentiating (1.5.22) with respect to ξ_{ka}, and then using (1.5.23), we see that $P_J(\xi)$ satisfies the so-called Hermite equation,

$$\frac{\partial^2 P_J(\xi)}{\partial \xi_{ka}^2} - 8\pi\, \xi_{ka}\, \frac{\partial P_J(\xi)}{\partial \xi_{ka}} + 8\pi\, J_{ka}\, P_J(\xi) = 0, \tag{1.5.24}$$

where $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$, $1 \leq k \leq m$ and $1 \leq a \leq n$. We set $\partial_{ka} := \frac{\partial}{\partial \xi_{ka}}$. Then (1.5.24) becomes

$$\partial_{ka}^2 P_J(\xi) - 8\pi\, \xi_{ka}\, \partial_{ka}\, P_J(\xi) + 8\pi J_{ka} P_J(\xi) = 0.$$

Differentiating (1.5.18) with respect to ξ_{ka}, we obtain

$$\begin{aligned} &\partial_{ka}^2 h_J(\xi) - 4\pi\, \xi_{ka}\, \partial_{ka} h_J(\xi) - 4\pi\, h_J(\xi) \\ &\quad - 2\{2\pi\,(J_{ka}+1)\}^{1/2}\, \partial_{ka} h_{J+\epsilon_{ka}}(\xi) = 0. \end{aligned} \tag{1.5.25}$$

By the way, according to (1.5.23), we have

$$\begin{aligned} \partial_{ka} h_{J+\epsilon_{ka}}(\xi) &= \partial_{ka} P_{J+\epsilon_{ka}}(\xi)\, e^{-2\pi|\xi|^2} - 4\pi\, \xi_{ka}\, P_{J+\epsilon_{ka}}(\xi)\, e^{-2\pi|\xi|^2} \\ &= -2\{2\pi\,(J_{ka}+1)\}^{1/2}\, h_J(\xi) - 4\pi\, \xi_{ka}\, h_{J+\epsilon_{ka}}(\xi). \end{aligned}$$

If we substitute this relation into (1.5.25), we obtain

$$\partial_{ka}^2 h_J(\xi) - 16\pi^2 \xi_{ka}^2 h_J(\xi) = -8\pi \left(J_{ka} + \frac{1}{2} \right) h_J(\xi). \tag{1.5.26}$$

Theorem 1.5.7 *The set* $\left\{ h_J \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ *of normalized Hermitian function in* $\mathcal{S}(\mathbb{R}^{(m,n)})$ *forms an orthonormal basis of* $L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$. *These* h_J *are eigenfunctions of the Fourier transform and the Fourier cotransform with eigenvalues* $(-i)^J$ *and* i^J *respectively.*

Proof If X is a left-invariant vector field on G, we set, for brevity

$$U(X) := dU_{I_m}(X).$$

We will prove that the set

$$\left\{ U\left(\exp_G \left(\sum_{k,a} x_{ka} D_{ka} + \sum_{l,b} y_{lb} \widehat{D}_{lb} \right) \right)(f_0) \;\middle|\; x_{ka}, y_{lb} \in \mathbb{R} \right\}$$

$(1 \leq k, l \leq m,\ 1 \leq a, b \leq n)$ is contained in the closed vector subspace of $L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$ spanned by the set $\left\{ h_J \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ and the subspace generated by the above set is invariant under the action of U. Since the Schrödinger representation $\left(U_{I_m}, L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)\right)$ is irreducible, we conclude that the set $\left\{ h_J \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ is a *complete* orthonormal basis for $L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$.

According to the commutation relations among D_{kl}^0, D_{ka}, $\widehat{D}_{lb}$ (cf. Lemma 1.1.1) and the fact that $U(D_{kl}^0) f = 2\pi i \delta_{kl} f$ for all $f \in \mathcal{S}(\mathbb{R}^{(m,n)})$, it suffices to prove the case $m = 1$ and $n = 1$. We put $D^0 := D_{11}^0$, $D := D_{11}$ and $\widehat{D} := \widehat{D}_{11}$. In other words, it remains to prove that the set

$$\left\{ U(\exp_G(xD + y\widehat{D}))(f_0) \;\middle|\; x, y \in \mathbb{R} \right\}$$

is contained in the closed vector subspace of $L^2(\mathbb{R}, d\xi)$ spanned by the set $\{h_j \mid j = 0, 1, 2, \ldots\}$.

First we note that by (1.5.1) and (1.5.2)

$$A^+ = \frac{1}{2}\left(U(D) + i\, U(\widehat{D}) \right) \quad \text{and} \quad A^- = \frac{1}{2}\left(U(D) - i\, U(\widehat{D}) \right).$$

For the present, we fix real numbers $x, y \in \mathbb{R}$. We put $z = x + i\, y \in \mathbb{C}$. It is obvious that $U(xD + y\widehat{D}) = \bar{z}A^+ + zA^-$. For all integers $k \geq 0,\ \ell \geq 0$ with $0 \leq k \leq \ell$,

We define the complex numbers $c_{k\ell}$ by

$$U(xD + y\widehat{D})^{\ell}(f_0) = \sum_{k=0}^{\ell} c_{k,\ell} f_k.$$

By the fact that $A^-(f_0) = 0$ and by (1.5.10), we have

$$U(xD + y\widehat{D})^{\ell+1}(f_0) = (\bar{z}A^+ + zA^-)\left(\sum_{k=0}^{\ell} c_{k,\ell} f_k\right)$$

$$= \sum_{k=0}^{\ell} c_{k,\ell}\,(\bar{z} f_{k+1} - 2\pi\, k\, z\, f_{k-1}).$$

Thus we get the recurrence formula

$$c_{k,\ell+1} = \bar{z} c_{k-1,\ell} - 2\pi\,(k+1)\, z\, c_{k+1,\ell}, \qquad 1 \le k \le \ell - 1.$$

Let $z = |z|\, e^{2\pi i \varphi}$ with $\varphi \in [0, 1)$ for $z \neq 0$. We put

$$d_{k,\ell} := \left(|z|^{-1/2}\, e^{\pi i \varphi}\right)^{\ell+k} \left((2\pi\,|z|)^{-1/2}\, e^{-\pi i\,(\varphi - \frac{1}{2})}\right)^{\ell-k} c_{k,\ell}.$$

Then we have the recurrence formula

$$d_{k,\ell+1} = d_{k-1,\ell} + (k+1)\, d_{k+1,\ell}, \qquad 1 \le k \le \ell - 1.$$

For $1 \le k \le \ell - 1$, we put

$$b_{k,\ell} := d_{\ell-k,\ell}.$$

Then we get the recurrence formula

$$b_{k,\ell} = b_{k,\ell-1} + (\ell - k + 1)\, b_{k-2,\ell-1}, \qquad 2 \le k \le \ell - 1.$$

If the starting value is $b_{0,0}$ and we define $b_{k,0} = 0$ for $k \ge 1$, then we get

$$b_{2p+1,\ell} = 0 \qquad \text{for } 0 \le p \le \frac{1}{2}(\ell - 1)$$

and

$$b_{2p,\ell} = \frac{\ell!}{2^p\, p!\,(\ell - 2p)!} \quad \text{for } 0 \le p \le \frac{1}{2}\ell.$$

Thus we obtain

$$
\begin{aligned}
&U(xD+y\widehat{D})^{\ell}(f_0)=\sum_{k=0}^{\ell} c_{k,\ell}\, f_k\\
&=\sum_{k=0}^{\ell}\left(|z|^{1/2}\, e^{-\pi i\varphi}\right)^{\ell+k}\left((2\,\pi\,|z|)^{1/2}\, e^{\pi\, i\,(\varphi-\frac{1}{2})}\right)^{\ell-k} d_{k,\ell}\, f_k\\
&=\sum_{k=0}^{\ell}\left(|z|^{1/2}\, e^{-\pi i\varphi}\right)^{2\ell-k}\left((2\,\pi\,|z|)^{1/2}\, e^{\pi\, i\,(\varphi-\frac{1}{2})}\right)^{k} b_{k,\ell}\, f_{\ell-k}\\
&=\sum_{p=0}^{[\frac{\ell}{2}]}\left(|z|^{1/2}\, e^{-\pi\, i\,\varphi}\right)^{2\ell-2p}\left((2\,\pi\,|z|)^{1/2}\, e^{\pi\, i\,(\varphi-\frac{1}{2})}\right)^{2p} b_{2p,\ell}\, f_{\ell-2p}\\
&=\sum_{p=0}^{[\frac{\ell}{2}]}\bar{z}^{\ell-p}(-2\,\pi\, z)^{p}\,\frac{\ell!}{2^p\, p!\,(\ell-2p)!}\, f_{\ell-2p}\\
&=\sum_{p=0}^{[\frac{\ell}{2}]}\bar{z}^{\ell-p}(-\pi\, z)^{p}\,\frac{\ell!}{p!\,(\ell-2p)!}\, f_{\ell-2p}.
\end{aligned}
$$

Therefore we get

$$
\begin{aligned}
&e^{U(xD+y\widehat{D})}(f_0)\\
&=\sum_{\ell=0}^{\infty}\frac{1}{\ell!}\, U(xD+y\widehat{D})^{\ell}(f_0)\\
&=\sum_{\ell=0}^{\infty}\sum_{p=0}^{[\frac{\ell}{2}]}\frac{1}{\ell!}\,\bar{z}^{\ell-p}\,(-\pi z)^{p}\,\frac{\ell!}{p!\,(\ell-2p)!}\, f_{\ell-2p}\\
&=\sum_{\ell=0}^{\infty}\sum_{p=0}^{[\frac{\ell}{2}]}\frac{1}{p!\,(\ell-2p)!}\,(-\pi|z|^2)^{p}\,\bar{z}^{\ell-2p}\, f_{\ell-2p}\\
&=\sum_{k=0}^{\infty}\left\{\sum_{p=0}^{\infty}\frac{1}{p!}\,(-\pi\,|z|^2)^{p}\right\}\frac{\bar{z}^{k}}{k!}\, f_k\\
&=e^{-\pi|z|^2}e^{\bar{z}A^{+}}(f_0)\\
&=e^{-\pi|z|^2}\sum_{k=0}^{\infty}\frac{(\sqrt{2\pi}\,\bar{z})^{k}}{(k!)^{1/2}}\, h_k.
\end{aligned}
$$

Therefore $U(\exp_G(xD + y\widehat{D}))(f_0)$ belongs to the closed subspace of $L^2(\mathbb{R}, d\xi)$ spanned by the set $\{h_j \mid j = 0, 1, 2, \dots\}$. The latter part of the theorem follows immediately from (1.5.19) and (1.5.20). This completes the proof. □

Corollary 1.5.8 *The set* $\left\{ P_J \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ *of Hermite polynomials forms an orthonormal basis for the* L^2*-space* $L^2\left(\mathbb{R}^{(m,n)}, e^{-4\pi|\xi|^2} d\xi\right)$.

Proof The proof follows immediately from Theorem 1.5.7 and (1.5.21). □

1.6 Harmonic Analysis on the Quotient Space of $H_{\mathbb{R}}^{(n,m)}$

We fix an element $\Omega \in \mathbb{H}_n$ once and for all. Let $\mathcal{M}$ be a positive symmetric half-integral matrix of degree m. Let $L^2\left(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathcal{M}}\right)$ be the L^2-space of $\mathbb{R}^{(m,n)}$ with respect to the measure

$$d\xi_{\Omega,\mathcal{M}} = e^{\pi i \sigma\{\mathcal{M}\xi(\Omega - \overline{\Omega})^t \xi\}} d\xi.$$

It is easy to show that the transformation $f(\xi) \longmapsto e^{\pi i \sigma\{\mathcal{M}\xi\Omega^t\xi\}} f(\xi)$ of $L^2\left(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathcal{M}}\right)$ into $L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$ is an isomorphism. Since the set $\left\{\xi^J \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)}\right\}$ is a basis of $L^2\left(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathcal{M}}\right)$, the set

$$\left\{ e^{\pi i \sigma\{\mathcal{M}\xi\Omega^t\xi\}} \xi^J \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$$

is a basis of $L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$. For the definitions of $\mathbb{Z}_{\geq 0}^{(m,n)}$ and ξ^J, we refer to Notations Frequently Used, p. xv. We observe that there exists a canonical bijection A from the cosets $\mathcal{T} := \mathbb{Z}^{(m,n)}/(2\mathcal{M})\mathbb{Z}^{(m,n)}$ to $(2\mathcal{M})^{-1}\mathbb{Z}^{(m,n)}/\mathbb{Z}^{(m,n)}$. We denote by A_α the image of $\alpha \in \mathcal{T}$ under the bijection A.

For brevity, we put $G = H_{\mathbb{R}}^{(n,m)}$. For each $A_\alpha \in \mathcal{T}$ and each $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$, we define a function $\Phi_J^{(\mathcal{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}(\Omega|\cdot)$ on G by

$$\begin{aligned} &\Phi_J^{(\mathcal{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}(\Omega|(\lambda, \mu, \kappa)) \\ &\stackrel{\text{def}}{=} e^{2\pi i \sigma\{\mathcal{M}(\kappa - \lambda^t\mu)\}} \sum_{N \in \mathbb{Z}^{(m,n)}} (\lambda + N + A_\alpha)^J \\ &\times e^{2\pi i \sigma\{\mathcal{M}((\lambda+N+A_\alpha)\Omega^t(\lambda+N+A_\alpha) + 2(\lambda+N+A_\alpha)^t\mu)\}}, \end{aligned} \tag{1.6.1}$$

where $(\lambda, \mu, \kappa) \in G$. We let $\Gamma_G = H_{\mathbb{Z}}^{(n,m)}$ be the discrete subgroup of G consisting of integral elements. That is,

$$\Gamma_G = \{(\lambda, \mu, \kappa) \in G \mid \lambda, \mu, \kappa \text{ integral}\}.$$

According to [52, Proposition 1.3], the function $\Phi_J^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix} \big(\Omega|(\lambda, \mu, \kappa)\big)$ satisfies the transformation behaviour

$$\Phi_J^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix} \big(\Omega|\gamma \circ g\big) = \Phi_J^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix} (\Omega|g) \tag{1.6.2}$$

holds for all $\gamma \in \Gamma_G$ and $g \in G$. Thus the functions

$$\Phi_J^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix} (\Omega|(\lambda, \mu, \kappa)) \ \left(J \in \mathbb{Z}_{\geq 0}^{(m,n)}\right)$$

are real analytic functions on the quotient space $\Gamma_G \backslash G$. Let $H_\Omega^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}$ be the completion of the vector space spanned by

$$\Phi_J^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix} \big(\Omega|(\lambda, \mu, \kappa)\big) \ \left(J \in \mathbb{Z}_{\geq 0}^{(m,n)}\right)$$

and let $\overline{H_\Omega^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}}$ be the complex conjugate of $H_\Omega^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}$.

Let $L^2(\Gamma_G \backslash G)$ be the L^2-space of the quotient space $\Gamma_G \backslash G$ with respect to the invariant measure

$$d\lambda_{11} \cdots d\lambda_{m,n-1} d\lambda_{mn} d\mu_{11} \cdots d\mu_{m,n-1} d\mu_{mn} d\kappa_{11} d\kappa_{12} \cdots d\kappa_{mm}.$$

Let ρ be the right regular representation of G on the Hilbert space $L^2(\Gamma_G \backslash G)$ given by

$$(\rho(g_0)\phi)(g) := \phi(gg_0), \quad g_0, g \in G, \quad \phi \in L^2(\Gamma_G \backslash G).$$

In [52], the author proved that the subspaces $H_\Omega^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}$ and $\overline{H_\Omega^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}}$ are irreducible invariant subspaces of $L^2(\Gamma_G \backslash G)$ with respect to ρ and the decomposi-

tion of the right regular representation ρ is given by

$$L^2(\Gamma_G\backslash G) = \bigoplus_{\mathscr{M},\alpha} H_\Omega^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix} \bigoplus \left(\overline{\bigoplus_{\mathscr{M},\alpha} H_\Omega^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}}\right)$$
$$\bigoplus \left(\bigoplus_c R(c)\right) \bigoplus \left(\bigoplus_{k,\ell\in\mathbb{Z}^{(m,n)}} \mathbb{C}\, e^{2\pi i\,\sigma(k^t\lambda+\ell^t\mu)}\right),$$

where $\mathscr{M}$ (respectively c) runs over the set of all positive symmetric half integral matrices of degree m (respectively the set of all half-integral nonzero matrices of degree m which are neither positive nor negative definite), $R(c)$ is the sum of irreducible representations π_c which occur in ρ and A_α runs over a complete system of representatives of the cosets $(2\mathscr{M})^{-1}\mathbb{Z}^{(m,n)}/\mathbb{Z}^{(m,n)}$.

Lemma 1.6.1 *The transform of* $L^2\big(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathscr{M}}\big)$ *onto* $H_\Omega^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}$ *given by*

$$\xi^J \longmapsto \Phi_J^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}\big(\Omega|(\lambda,\mu,\kappa)\big) \tag{1.6.3}$$

is an isomorphism of Hilbert spaces.

Proof For the proof, we refer to [52, Lemma 3.2]. □

We write

$$f_{\Omega,J}^{(\mathscr{M})}(\xi) := e^{2\pi i\,\sigma(\mathscr{M}\xi\Omega^t\xi)}\xi^J, \quad J \in \mathbb{Z}_{\geq 0}^{(m,n)}. \tag{1.6.4}$$

We let $\Phi_{\Omega,\alpha}^{(\mathscr{M})}$ be the transform of $L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)$ onto $H_\Omega^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}$ defined by

$$\Phi_{\Omega,\alpha}^{(\mathscr{M})}\left(f_{\Omega,J}^{(\mathscr{M})}\right) := \Phi_J^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}\big(\Omega|(\lambda,\mu,\kappa)\big). \tag{1.6.5}$$

Then $\Phi_{\Omega,\alpha}^{(\mathscr{M})}$ is an isometry of $L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)$ onto $H_\Omega^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}$ such that

$$U^{S,\mathscr{M}}\left((\lambda,\mu,\kappa)\right)\circ\Phi_{\Omega,\alpha}^{(\mathscr{M})} = \Phi_{\Omega,\alpha}^{(\mathscr{M})}\circ U^{S,\mathscr{M}}\big((\lambda,-\mu,-\kappa)\big),$$

where $U^{S,\mathscr{M}}$ is the Schrödinger representation of G defined by (1.4.45).

Let $\Delta_{\Omega,\mathscr{M}}$ be the isometry of $L^2\left(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathscr{M}}\right)$ onto $L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$ defined by

$$(\Delta_{\Omega,\mathscr{M}} f)(\xi) := e^{\pi i\, \sigma\{\mathscr{M}\xi\Omega\, {}^t\xi\}} f(\xi). \tag{1.6.6}$$

We define the unitary representation $U_\Omega^{S,\mathscr{M}}$ of G on $L^2\left(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathscr{M}}\right)$ by

$$\left(U_\Omega^{S,\mathscr{M}}(g) f\right)(\xi) := \Delta_{\Omega,\mathscr{M}}^{-1}\left(\left(U^{S,\mathscr{M}}(g)(\Delta_{\Omega,\mathscr{M}} f)\right)(\xi)\right), \tag{1.6.7}$$

where $f \in L^2\left(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathscr{M}}\right)$ and $\xi \in \mathbb{R}^{(m,n)}$.

Now we write the image of $f_{\Omega,J}^{(\mathscr{M})} \in L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$ under $\vartheta_{\mathscr{M},\alpha}$ (cf. (1.3.23)) explicitly:

$$\begin{aligned}
&\left(\vartheta_{\mathscr{M},\alpha} f_{\Omega,J}^{(\mathscr{M})}\right)((\lambda,\mu,\kappa)) \\
&= \left(\vartheta_{\mathscr{M},\alpha} f_{\Omega,J}^{(\mathscr{M})}\right)([\lambda,\mu,\kappa+\mu\, {}^t\lambda]) \\
&= e^{2\pi i\, \sigma\{\mathscr{M}(\kappa+\mu\, {}^t\lambda)\}} \sum_{N\in\mathbb{Z}^{(m,n)}} e^{2\pi i\, \sigma\{\mathscr{M}((\lambda+N)\Omega\, {}^t(\lambda+N)+2\, N\, {}^t\mu)\}} (\lambda+N)^J \\
&= e^{2\pi i\, \sigma\{\mathscr{M}(\kappa-\lambda\, {}^t\mu)+\alpha\, {}^t\mu\}} \sum_{N\in\mathbb{Z}^{(m,n)}} e^{2\pi i\, \sigma\{\mathscr{M}((\lambda+N)\Omega\, {}^t(\lambda+N)+2\,(\lambda+N)\, {}^t\mu)\}} (\lambda+N)^J.
\end{aligned}$$

In particular, if $\alpha = 0$, $\kappa = 0$ and $J = 0$, then we have

$$\begin{aligned}
&\left(\vartheta_{\mathscr{M},0} f_{\Omega,0}^{(\mathscr{M})}\right)((\lambda,\mu,0)) \\
&= e^{-2\pi i\, \sigma(\mathscr{M}\lambda\, {}^t\mu)} \sum_{N\in\mathbb{Z}^{(m,n)}} e^{2\pi i\, \sigma\{\mathscr{M}((\lambda+N)\Omega\, {}^t(\lambda+N)+2\,(\lambda+N)\, {}^t\mu)\}} \\
&= e^{2\pi\, i\, \sigma\{\mathscr{M}(\lambda\,\Omega\, {}^t\lambda+\lambda\, {}^t\mu)\}} \sum_{N\in\mathbb{Z}^{(m,n)}} e^{2\pi i\, \sigma\{\mathscr{M}(N\Omega\, {}^t N+2\,(\lambda\Omega+\mu)^t N)\}} \\
&= e^{2\pi i\, \sigma\{\mathscr{M}(\lambda\,\Omega\, {}^t\lambda+\lambda\, {}^t\mu)\}}\, \vartheta^{(2\mathscr{M})}\begin{bmatrix}0\\0\end{bmatrix}(\Omega, \lambda\Omega+\mu).
\end{aligned}$$

Therefore we obtain:

Proposition 1.6.2 *Let $\mathscr{M}$ be a positive symmetric half-integral matrix of degree m. Let $\alpha \in \mathcal{T}$ and $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$. Then we have*

$$\begin{aligned}
&\left(\vartheta_{\mathscr{M},\alpha} f_{\Omega,J}^{(\mathscr{M})}\right)((\lambda,\mu,\kappa)) \\
&= e^{2\pi i\, \sigma\{\mathscr{M}((\kappa-\lambda\, {}^t\mu)+\alpha\, {}^t\mu)\}} \sum_{N\in\mathbb{Z}^{(m,n)}} e^{2\pi i\, \sigma\{\mathscr{M}((\lambda+N)\Omega\, {}^t(\lambda+N)+2(\lambda+N)\, {}^t\mu)\}} (\lambda+N)^J.
\end{aligned}$$

In particular,

$$\left(\vartheta_{\mathscr{M},0} f_{\Omega,0}^{(\mathscr{M})}\right)((\lambda,\mu,0)) = e^{2\pi i\,\sigma\{\mathscr{M}(\lambda\Omega\,{}^t\lambda+\lambda\,{}^t\mu)\}}\,\vartheta^{(2\mathscr{M})}\begin{bmatrix}0\\0\end{bmatrix}(\Omega,\lambda\,\Omega+\mu).$$

It is easy to see that the following diagrams are commutative:

$$\begin{array}{ccc}
L^2\left(\mathbb{R}^{(m,n)},d\xi\right) & \xrightarrow{U^{S,\mathscr{M}}(g)} & L^2\left(\mathbb{R}^{(m,n)},d\xi\right)\\
\Big\downarrow{\scriptstyle \vartheta_{\mathscr{M},\alpha}} & & \Big\downarrow{\scriptstyle \vartheta_{\mathscr{M},\alpha}}\\
\mathscr{H}_{\mathscr{M},\alpha} & & \mathscr{H}_{\mathscr{M},\alpha}\\
\| & \xrightarrow[\pi_{\mathscr{M},\alpha}(g)]{} & \|\\
\pi_{\mathscr{M},\alpha} & & \pi_{\mathscr{M},\alpha}
\end{array}$$

[diagram 1.6.1]

$$\begin{array}{ccc}
L^2\left(\mathbb{R}^{(m,n)},d\xi_{\Omega,\mathscr{M}}\right) & \xrightarrow{U_{\Omega}^{S,\mathscr{M}}(g)} & L^2\left(\mathbb{R}^{(m,n)},d\xi_{\Omega,\mathscr{M}}\right)\\
\Big\downarrow{\scriptstyle \Delta_{\Omega,\mathscr{M}}} & & \Big\downarrow{\scriptstyle \Delta_{\Omega,\mathscr{M}}}\\
L^2\left(\mathbb{R}^{(m,n)},d\xi\right) & \xrightarrow[U^{S,\mathscr{M}}(g)]{} & L^2\left(\mathbb{R}^{(m,n)},d\xi\right)
\end{array}$$

[diagram 1.6.2]

$$\begin{array}{ccc}
L^2\left(\mathbb{R}^{(m,n)},d\xi\right) & \xrightarrow{U^{S,\mathscr{M}}(g)} & L^2\left(\mathbb{R}^{(m,n)},d\xi\right)\\
\Big\downarrow{\scriptstyle I_{\mathscr{M}}} & & \Big\downarrow{\scriptstyle I_{\mathscr{M}}}\\
\mathscr{H}_{F,\mathscr{M}} & \xrightarrow[U^{F,\mathscr{M}}(g)]{} & \mathscr{H}_{F,\mathscr{M}}
\end{array}$$

[diagram 1.6.3]

$$\begin{array}{ccc}
L^2\left(\mathbb{R}^{(m,n)},d\xi\right) & \xrightarrow{U^{S,\mathscr{M}}(g)} & L^2\left(\mathbb{R}^{(m,n)},d\xi\right)\\
\Big\downarrow{\scriptstyle \Phi_{\Omega,\alpha}^{(\mathscr{M})}} & & \Big\downarrow{\scriptstyle \Phi_{\Omega,\alpha}^{(\mathscr{M})}}\\
H_{\Omega}^{(\mathscr{M})}\begin{bmatrix}A_\alpha\\0\end{bmatrix} & \xrightarrow[\rho_{\mathscr{M}}(g)]{} & H_{\Omega}^{(\mathscr{M})}\begin{bmatrix}A_\alpha\\0\end{bmatrix}
\end{array}$$

[diagram 1.6.4]

Here $g \in G$ and $\rho_{\mathscr{M}}$ denotes the restriction of the right regular representation ρ to $H_\Omega^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}$. We know that the mapping $\vartheta_{\mathscr{M},\alpha}$, $\Delta_{\Omega,\mathscr{M}}$, $I_{\mathscr{M}}$ (cf. (1.4.46)) and $\Phi_{\Omega,\alpha}^{(\mathscr{M})}$ are all the isomorphisms preserving the norms. Hence we have:

Theorem 1.6.3 *For each $\alpha \in \mathcal{T}$, $\Omega \in \mathbb{H}_n$ and $\mathscr{M}$ a positive symmetric half-integral matrix of degree m, the following five representations are unitarily equivalent to each other via the intertwining operators $\vartheta_{\mathscr{M},\alpha}$, $I_{\mathscr{M}}$, $\Phi_{\Omega,\alpha}^{(\mathscr{M})}$ and $\Delta_{\Omega,\mathscr{M}}$.*

(1) *the Schrödinger representation* $\left(U^{S,\mathscr{M}}, L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)\right)$,
(2) *the lattice representation* $(\pi_{\mathscr{M},\alpha}, \mathscr{H}_{\mathscr{M},\alpha})$,
(3) *the Fock representation* $\left(U^{F,\mathscr{M}}, \mathscr{H}_{F,\mathscr{M}}\right)$,
(4) *the representation* $\left(\rho_{\mathscr{M}}, H_\Omega^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}\right)$ *and*
(5) *the representation* $\left(U_\Omega^{S,\mathscr{M}}, L^2\left(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathscr{M}}\right)\right)$.

Remark 1.6.4 The multiplicity of the Schrödinger representation $U^{S,\mathscr{M}}$ of G in $\left(\rho, L^2(\Gamma_G\backslash G)\right)$ is $(\det 2\mathscr{M})^n$.

We refer to [52] for details. Theorem 1.6.3 may be pictured as follows:

$$\begin{array}{ccccc} & & \mathscr{H}_{F,\mathscr{M}} & & \\ & & \uparrow{\scriptstyle I_{\mathscr{M}}} & & \\ L^2\left(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathscr{M}}\right) & \xrightarrow{\Delta_{\mathscr{M},\alpha}} & L^2\left(\mathbb{R}^{(m,n)}, d\xi\right) & \xrightarrow{\vartheta_{\mathscr{M},\alpha}} & \mathscr{H}_{\mathscr{M},\alpha} \\ & & \downarrow{\scriptstyle \Phi_{\Omega,\alpha}^{(\mathscr{M})}} & & \\ & & H_\Omega^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix} & & \end{array}$$

[diagram 1.6.5]

Finally, we describe explicitly the orthonormal bases of

$$L^2\left(\mathbb{R}^{(m,n)}, d\xi\right),\ L^2\left(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathscr{M}}\right),\ \mathscr{H}_{\mathscr{M},\alpha},\ \mathscr{H}_{F,\mathscr{M}} \text{ and } H_\Omega^{(\mathscr{M})}\begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}$$

respectively.

In the previous section, we proved that the family of the functions

$$h_J(\xi) = \left(\frac{1}{\sqrt{2\pi}}\right)^J (J!)^{-1/2} f_J(\xi), \quad J \in \mathbb{Z}_{\geq 0}^{(m,n)}$$

forms an orthonormal basis of $L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)$. Therefore the set

$$\left\{ e^{-\pi\, i\, \sigma(\mathscr{M}\xi\Omega^{\,t}\xi)}\, h_J(\xi) \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$$

forms an orthonormal basis for $L^2\big(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathscr{M}}\big)$. For each $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$, the set of functions

$$\begin{aligned}
&\vartheta_{\mathscr{M},\alpha,J}(\lambda,\mu,\kappa):\\
&= (\vartheta_{\mathscr{M},\alpha} h_J)(\lambda,\mu,\kappa)\\
&= e^{2\pi i\, \sigma\{\mathscr{M}(\kappa+\mu\,{}^t\lambda+\alpha\,{}^t\mu)\}} \sum_{N\in\mathbb{Z}^{(m,n)}} h_J(\lambda+N)\, e^{4\pi i\, \sigma(\mathscr{M}N\,{}^t\mu)}, \quad J \in \mathbb{Z}_{\geq 0}^{(m,n)}
\end{aligned}$$

forms an orthonormal basis for $\mathscr{H}_{\mathscr{M},\alpha}$. For each $J \in \mathbb{Z}_{\geq 0}^{(m,n)}$, we define the function

$$\begin{aligned}
&H_J^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix} \big(\Omega|(\lambda,\mu,\kappa)\big) \stackrel{\text{def}}{=} 2^{\frac{mn}{4}-\frac{|J|}{2}}\, (J!)^{-1/2}\, (\det 2\mathscr{M})^{n/4} (\det \operatorname{Im}\Omega)^{m/4}\\
&\times e^{\pi i\, \sigma(\mathscr{M}(\kappa-\lambda\,{}^t\mu))} \sum_{N\in\mathbb{Z}^{(m,n)}} H_J\left(\sqrt{2\pi}\, (2\mathscr{M})^{1/2}\, (\lambda+N+A_\alpha)(\operatorname{Im}\Omega)^{1/2}\right)\\
&\times e^{\pi i\, \sigma\{\mathscr{M}((\lambda+N+A_\alpha)\Omega^{\,t}(\lambda+N+A_\alpha)+2\,(\lambda+N+A_\alpha)^{\,t}\mu)\}},
\end{aligned}$$

where $H_J(\xi)$ is the Hermite polynomial on $\mathbb{R}^{(m,n)}$ in several variables defined by

$$H_J(\xi) := H_{J_{11}}(\xi_{11}) H_{J_{12}}(\xi_{12}) \cdots H_{J_{mn}}(\xi_{mn}).$$

It was proved in [56] that the functions $H_J^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix} \big(\Omega|(\lambda,\mu,\kappa)\big) \quad \left(J \in \mathbb{Z}_{\geq 0}^{(m,n)}\right)$ form an orthonormal basis for $H_\Omega^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}$. We have:

Theorem 1.6.5 *We have an orthonormal basis for each of the above representations:*

(1) *The set* $\left\{ h_J \;\middle|\; J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ *forms an orthonormal basis for* $L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)$.

(2) *The set* $\left\{ e^{-\pi i\, \sigma(\mathscr{M}\xi\Omega^{\,t}\xi)}\, h_J \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ *forms an orthonormal basis for* $L^2\big(\mathbb{R}^{(m,n)}, d\xi_{\Omega,\mathscr{M}}\big)$.

(3) *The set* $\left\{ \vartheta_{\mathscr{M},\alpha,J} \;\middle|\; J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ *forms an orthonormal basis for* $\mathscr{H}_{\mathscr{M},\alpha}$.

(4) *The set* $\left\{ \Phi_{\mathscr{M},J} \mid J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ *(cf. (1.4.36)) forms an orthonormal basis for* $\mathscr{H}_{F,\mathscr{M}}$.

(5) *The set* $\left\{ H_J^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix} \big(\Omega|(\lambda, \mu, \kappa)\big) \,\Big|\, J \in \mathbb{Z}_{\geq 0}^{(m,n)} \right\}$ *forms an orthonormal basis for* $H_\Omega^{(\mathscr{M})} \begin{bmatrix} A_\alpha \\ 0 \end{bmatrix}$.

1.7 Coadjoint Orbits of the Heisenberg Group $H_{\mathbb{R}}^{(n,m)}$

In this section, we find the coadjoint orbits of $H_{\mathbb{R}}^{(n,m)}$ and describe the connection between the coadjoint orbits of $H_{\mathbb{R}}^{(n,m)}$ and the irreducible unitary representations of $H_{\mathbb{R}}^{(n,m)}$ following the work on the orbit method done by A. Kirillov.

First we briefly outline the results of Kirillov on unitary representations of a nilpotent real Lie group. We refer to [21–23] for more details.

Let G be a simply connected real Lie group with its Lie algebra $\mathfrak{g}$. Let $\mathrm{Ad}_G : G \longrightarrow GL(\mathfrak{g})$ be the adjoint representation of G. That is, for each $g \in G$, $\mathrm{Ad}_G(g)$ is the differential map of I_g at the identity element e, where $I_g : G \longrightarrow G$ is the conjugation by g given by

$$I_g(x) := gxg^{-1} \quad \text{for all } x \in G.$$

Let $\mathfrak{g}^*$ be the dual space of the vector space $\mathfrak{g}$. Let $\mathrm{Ad}_G^* : G \longrightarrow GL(\mathfrak{g}^*)$ be the contragredient of the adjoint representation Ad_G. We call Ad_G^* the *coadjoint representation* of G. For each $\ell \in \mathfrak{g}^*$, we define the alternating bilinear form B_ℓ on $\mathfrak{g}$ by

$$B_\ell(X, Y) = \ell([X, Y]) \stackrel{\text{def}}{=} \langle \ell, [X, Y] \rangle \quad \text{for all } X, Y \in \mathfrak{g}. \tag{1.7.1}$$

Definition 1.7.1

(1) A Lie subalgebra $\mathfrak{h}$ of $\mathfrak{g}$ is said to be subordinate to $\ell \in \mathfrak{g}^*$ if $\mathfrak{h}$ forms a totally isotropic vector subspace of $\mathfrak{g}$ relative to the alternating bilinear form B_ℓ defined by (1.7.1), i.e., $B_\ell|_{\mathfrak{h}\times\mathfrak{h}} = 0$.

(2) A Lie subalgebra $\mathfrak{h}$ of $\mathfrak{g}$ is called a polarization of $\mathfrak{g}$ for ℓ if $\mathfrak{h}$ is maximal among the totally isotropic vector subspace of $\mathfrak{g}$ relative to B_ℓ. In other words, if $\mathfrak{f}$ is a vector subspace of $\mathfrak{g}$ such that $\mathfrak{h} \subset \mathfrak{f}$ and $B_\ell|_{\mathfrak{f}\times\mathfrak{f}} = 0$, then we have $\mathfrak{f} = \mathfrak{h}$.

(3) Let $\ell \in \mathfrak{g}^*$ and let $\mathfrak{h}$ be a polarization of $\mathfrak{g}$ for ℓ. We let H the simply connected closed subgroup of G corresponding to the Lie subalgebra $\mathfrak{h}$. We define the unitary character $\chi_{\ell,\mathfrak{h}}$ of H by

$$\chi_{\ell,\mathfrak{h}}(\exp_H(X)) = e^{2\pi i \langle \ell, X \rangle} \quad \text{for all } X \in \mathfrak{h}, \;\; i = \sqrt{-1}, \tag{1.7.2}$$

where $\exp_H : \mathfrak{h} \longrightarrow H$ denotes the exponential map of $\mathfrak{h}$ to H. It is known that $\exp_H$ is surjective.

Dixmier and Kirillov [7, 21] proved the following important theorem.

Theorem 1.7.2 *A simply connected real nilpotent Lie group G is monomial, that is, each irreducible unitary representation of G can be unitarily induced by a unitary character of some closed subgroup of G.*

Now we may state Theorem 1.7.2 explicitly.

Theorem 1.7.3 *Let G be a simply connected real nilpotent Lie group with its Lie algebra $\mathfrak{g}$. Assume that there is given an irreducible unitary representation π of G. Then there exist an element $\ell \in \mathfrak{g}^*$ and a polarization $\mathfrak{h}$ of $\mathfrak{g}$ for ℓ such that $\pi \cong \mathrm{Ind}_H^G \chi_{\ell,\mathfrak{h}}$, where $\chi_{\ell,\mathfrak{h}}$ is the unitary character of $H = \exp(\mathfrak{h})$ defined by (1.7.2). Here $\cong$ means the unitary equivalence.*

Definition 1.7.4 Let $(\mathscr{H}, \langle\ ,\ \rangle)$ be a separable Hilbert space with inner product $\langle\ ,\ \rangle$. Let $\{e_\alpha \mid \alpha \in \mathbb{I}\}$ be an orthonormal basis of $\mathscr{H}$ and $A : \mathscr{H} \longrightarrow \mathscr{H}$ be a bounded linear operator on $\mathscr{H}$. The trace of A is denoted by $\mathrm{Tr}(A)$ and defined as

$$\mathrm{Tr}(A) \stackrel{\mathrm{def}}{=} \sum_{\alpha \in \mathbb{I}} \langle Ae_\alpha, e_\alpha \rangle.$$

Here we observe that $\mathrm{Tr}(A)$ is independent of the choice of an orthonormal basis of $\mathscr{H}$. A bounded operator $B : \mathscr{H} \longrightarrow \mathscr{H}$ is said to be trace class if

$$\mathrm{Tr}(|B|) < \infty,$$

where $|B| :\stackrel{\mathrm{def}}{=} (B^*B)^{1/2}$ denotes the positive definite Hermitian square root of B^*B. The trace norm of a trace class operator T is defined to be

$$\|T\|_1 \stackrel{\mathrm{def}}{=} \mathrm{Tr}(|T|).$$

One can show that the trace norm is a norm on the space $\mathbb{B}(\mathscr{H})$ of all trace class operators on $\mathscr{H}$ and that $\mathbb{B}(\mathscr{H})$ with the trace norm becomes a Banach space.

Definition 1.7.5 Let $(\pi, \mathscr{H})$ be a unitary representation of a real Lie group G with representation space $\mathscr{H}$, where $\mathscr{H}$ is a separable Hilbert space. The unitary representation $(\pi, \mathscr{H})$ is called trace class if for each compactly supported smooth function $f \in C_c^\infty(G)$ on G, the operator $\pi(f) : \mathscr{H} \longrightarrow \mathscr{H}$ defined by

$$\pi(f)(v) \stackrel{\mathrm{def}}{=} \int_G f(g)\pi(g)v dg$$

is trace class, where dg is a Haar measure on G.

Theorem 1.7.6 *Let G be a simply connected real nilpotent Lie group with its Lie algebra $\mathfrak{g}$. If $\ell \in \mathfrak{g}^*$, there exists a polarization $\mathfrak{h}$ of $\mathfrak{g}$ for ℓ such that the monomial representation $\mathrm{Ind}_H^G \chi_{\ell,\mathfrak{h}}$ is irreducible and trace class. If ℓ' is an element of $\mathfrak{g}^*$ which belongs to the coadjoint orbit $\mathrm{Ad}_G^*(G)\ell$ and $\mathfrak{h}'$ is a polarization of $\mathfrak{g}$ for ℓ', then the monomial representations $\mathrm{Ind}_H^G \chi_{\ell,\mathfrak{h}}$ and $\mathrm{Ind}_{H'}^G \chi_{\ell',\mathfrak{h}'}$ are unitarily equivalent. Here H and H' are simply connected closed subgroups of G corresponding to the Lie subalgebras $\mathfrak{h}$ and $\mathfrak{h}'$ respectively. Conversely, if $\mathfrak{h}$ and $\mathfrak{h}'$ are polarizations of $\mathfrak{g}$ for $\ell \in \mathfrak{g}^*$ and $\ell' \in \mathfrak{g}^*$ respectively such that the monomial representations $\mathrm{Ind}_H^G \chi_{\ell,\mathfrak{h}}$ and $\mathrm{Ind}_{H'}^G \chi_{\ell',\mathfrak{h}'}$ are unitarily equivalent, then ℓ and ℓ' belong to the same coadjoint G-orbit in $\mathfrak{g}^*$. Finally, for each irreducible unitary representation π of G, there exists a unique coadjoint G-orbit Ω in $\mathfrak{g}^*$ such that for any linear form $\ell \in \Omega$ and each polarization $\mathfrak{h}$ of $\mathfrak{g}$ for ℓ, the representations π and $\mathrm{Ind}_H^G \chi_{\ell,\mathfrak{h}}$ are unitarily equivalent. Any irreducible unitary representation of G is trace class.*

Kirillov [21] proved Theorem 1.7.3 and Theorem 1.7.6.

Remark 1.7.7

(a) The bijection of the orbit space $\mathfrak{g}^*/G$ of coadjoint G-orbits in $\mathfrak{g}^*$ onto the unitary dual $\widehat{G}$ of G given by Theorem 1.7.6 is called the Kirillov correspondence of G. It provides a parametrization of $\widehat{G}$ by means of the coadjoint G-orbit space.

(b) Let G be a connected simply connected nilpotent real Lie group with its Lie algebra $\mathfrak{g}$. Then the Kirillov correspondence

$$\widehat{G} \longrightarrow \mathcal{O}(G) = \mathfrak{g}^*/G$$

is a homeomorphism.

Theorem 1.7.8 *Let G be a connected simply connected real nilpotent Lie group with its Lie algebra $\mathfrak{g}$. Let $\mathfrak{h}$ be a subalgebra of $\mathfrak{g}$. Let $p : \mathfrak{g}^* \longrightarrow \mathfrak{h}^*$ be the natural projection. Let H be the simply connected subgroup of G with its Lie algebra $\mathfrak{h}$. Then the following (i), (ii) and (iii) hold:*

(i) Let π be an irreducible unitary representation of G corresponding to a coadjoint G-orbit $\Omega \subset \mathfrak{g}^*$ via the Kirillov correspondence. Then the restriction $\mathrm{Res}_H^G \pi$ of π to H decomposes into the direct integral of irreducible representations of H corresponding to a coadjoint H-orbit $\omega (\subset \mathfrak{h}^*)$ such that $\omega \subset p(\Omega)$.

(ii) Let τ be an irreducible unitary representation of H corresponding to a coadjoint H-orbit $\omega \subset \mathfrak{h}^*$. Then the induced representation $\mathrm{Ind}_H^G \tau$ decomposes into the direct integral of irreducible representations π_Ω of G corresponding to coadjoint G-orbits $\Omega (\subset \mathfrak{g}^*)$ such that $p(\Omega) \supset \omega$.

(iii) Let π_1 and π_2 be the irreducible unitary representation of G corresponding to coadjoint G-orbits Ω_1 and Ω_2 respectively. Then the tensor product $\pi_1 \otimes \pi_2$ decomposes into the direct integral of irreducible representations π_Ω of G corresponding to coadjoint G-orbits $\Omega (\subset \mathfrak{g}^*)$ such that $\Omega \subset \Omega_1 + \Omega_2$.

For more details on Theorem 1.7.8, we refer to [21, 23] and [22, §15, Theorem 1, p. 249].

Theorem 1.7.9 *Let G be a connected simply connected real nilpotent Lie group with its Lie algebra $\mathfrak{g}$. Let π be an irreducible unitary representation of G corresponding to a coadjoint G-orbit $\Omega \subset \mathfrak{g}^*$. Then the character χ_π is a distribution on $\mathcal{S}(G)$ and its Fourier transform coincides with the canonical measure on Ω given by the symplectic structure. Here $\mathcal{S}(G)$ denotes the Schwartz space of rapidly decreasing functions on G.*

For more details on Theorem 1.7.9, we refer to [21] and [22, pp. 251–258].

Now we find the coadjoint orbits of the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ and describe the connection between the coadjoint orbits and the unitary dual of $H_{\mathbb{R}}^{(n,m)}$.

For brevity, we let $G := H_{\mathbb{R}}^{(n,m)}$ as before. Let $\mathfrak{g}$ be the Lie algebra of G and let $\mathfrak{g}^*$ be the dual space of $\mathfrak{g}$. We recall that $\mathrm{Sym}(m, \mathbb{R})$ denotes the space of all $m \times m$ real symmetric matrices (see Notations Frequently Used, p. xiv). We observe that $\mathfrak{g}$ can be regarded as the subalgebra consisting of all $2(m+n) \times 2(m+n)$ real matrices of the form

$$X(\alpha, \beta, \gamma) := \begin{pmatrix} 0 & 0 & 0 & {}^t\beta \\ \alpha & 0 & \beta & \gamma \\ 0 & 0 & 0 & -{}^t\alpha \\ 0 & 0 & 0 & 0 \end{pmatrix}, \ \alpha, \beta \in \mathbb{R}^{(m,n)}, \ \gamma \in \mathrm{Sym}(m, \mathbb{R})$$

in the Lie algebra $\mathfrak{sp}(2(m+n), \mathbb{R})$ of the symplectic group $Sp(2(m+n), \mathbb{R})$. An easy computation yields

$$[X(\alpha, \beta, \gamma), \ X(\delta, \epsilon, \xi)] = X(0, 0, \alpha\,{}^t\epsilon + \epsilon\,{}^t\alpha - \beta\,{}^t\delta - \delta\,{}^t\beta).$$

The dual space $\mathfrak{g}^*$ of $\mathfrak{g}$ can be identified with the vector space consisting of all $2(m+n) \times 2(m+n)$ real matrices of the form

$$F(a, b, c) := \begin{pmatrix} 0 & {}^ta & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & {}^tb & 0 & 0 \\ b & c & -a & 0 \end{pmatrix}, \ a, b \in \mathbb{R}^{(m,n)}, \ c \in \mathrm{Sym}(m, \mathbb{R})$$

so that

$$\begin{aligned} \langle F(a, b, c), X(\alpha, \beta, \gamma) \rangle : &= \sigma(F(a, b, c)\, X(\alpha, \beta, \gamma)) \\ &= 2\,\sigma({}^t\alpha\, a + {}^tb\, \beta) + \sigma(c\, \gamma). \end{aligned} \tag{1.7.3}$$

The adjoint representation Ad_G of G is given by $\mathrm{Ad}_G(g)X = gXg^{-1}$ for $g \in G$ and $X \in \mathfrak{g}$. For $g \in G$ and $F \in \mathfrak{g}^*$, gFg^{-1} is not of the form $F(a, b, c)$. We denote

by $(gFg^{-1})_*$ the

$$\begin{pmatrix} 0 & * & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & * & 0 & 0 \\ * & * & * & 0 \end{pmatrix} - \text{part}$$

of the matrix gFg^{-1}. Then it is easy to see that the coadjoint representation $\mathrm{Ad}_G^* : G \longrightarrow GL(\mathfrak{g}^*)$ is given by $\mathrm{Ad}_G^*(g)F = (gFg^{-1})_*$, where $g \in G$ and $F \in \mathfrak{g}^*$. More precisely,

$$\mathrm{Ad}_G^*(g)F(a,b,c) = F(a + c\mu, b - c\lambda, c), \tag{1.7.4}$$

where $g = (\lambda, \mu, \kappa) \in G$. So the coadjoint G-orbit $\Omega_{a,b}$ at $F(a,b,0) \in \mathfrak{g}^*$ is given by

$$\Omega_{a,b} = \mathrm{Ad}_G^*(G)\, F(a,b,0) = \{F(a,b,0)\}, \text{ a single point.} \tag{1.7.5}$$

And for any $a, b \in \mathbb{R}^{(m,n)}$ and $c \in \mathrm{Sym}(m,\mathbb{R})$ with $1 \le k = \mathrm{rank}\ c \le m$, the coadjoint G-orbit $\Omega_{a,b,c,k}$ at $F(a,b,c) \in \mathfrak{g}^*$ is given by

$$\Omega_{a,b,c,k} = \left\{ F(a + c\mu, b - c\lambda, c) \,\middle|\, \lambda, \mu \in \mathbb{R}^{(m,n)} \right\} \cong \mathbb{R}^{(k,n)} \times \mathbb{R}^{(k,n)}. \tag{1.7.6}$$

Therefore the coadjoint G-orbits in $\mathfrak{g}^*$ fall into two classes:

(**I**) The single points $\left\{ \Omega_{a,b} \,\middle|\, a, b \in \mathbb{R}^{(m,n)} \right\}$ located in the plane $c = 0$.

(**II**) The affine planes $\left\{ \Omega_{a,b,c,k} \,\middle|\, a, b \in \mathbb{R}^{(m,n)},\ c \in \mathrm{Sym}(m,\mathbb{R}),\ 1 \le \mathrm{rank}\ c = k \le m \right\}$.

In other words, the orbit space $\mathcal{O}(G)$ of coadjoint orbits is parametrized by

$$\begin{cases} a, b \in \mathbb{R}^{(m,n)},\ c \in \mathrm{Sym}(m,\mathbb{R}),\ 1 \le k = \mathrm{rank}\, c \le m; \\ (a,b)-\text{plane} \cong \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}. \end{cases}$$

Definition 1.7.10

(a) The single point coadjoint orbits of the type $\Omega_{a,b}$ are said to be the extremely degenerate G-orbits in $\mathfrak{g}^*$.

(b) The flat coadjoint orbits of the type $\Omega_{a,b,c,k}$ with $1 \le k = \mathrm{rank}\, c < m$ are said to be the $(m-k)$-degenerate G-orbits in $\mathfrak{g}^*$.

(c) The flat coadjoint orbits of the type $\Omega_{a,b,c,m}$ with $\mathrm{rank}\, c = m$ are said to be the nondegenerate G-orbits in $\mathfrak{g}^*$.

Since G is a connected and simply connected 2-step nilpotent Lie group, according to A. Kirillov (cf. [21] or [22, Theorem 1, p. 249]), the unitary dual $\widehat{G}$

of G is given by

$$\widehat{G} = \coprod_{a,b,c,k} \Omega_{a,b,c,k} \coprod \left(\mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)} \right), \tag{1.7.7}$$

where $\coprod$ denotes the disjoint union, a and b run over $\mathbb{R}^{(m,n)}$ and c ($1 \leq k = \operatorname{rank} c \leq m$) runs over $\mathrm{Sym}(m, \mathbb{R})$. We observe that $\Omega_{a,b,c,k} \cong \mathbb{R}^{(k,n)} \times \mathbb{R}^{(k,n)}$. The topology of $\widehat{G}$ may be described as follows. The topology on $\{\Omega_{a,b,c,k} \mid a, b \in \mathbb{R}^{(m,n)},\ 0 \neq c \in \mathrm{Sym}(m, \mathbb{R}),\ 1 \leq k \leq m\}$ is the usual topology of the Euclidean space and the topology on $\{F(a, b, 0) \mid a, b \in \mathbb{R}^{(m,n)}\}$ is the usual Euclidean topology. But a sequence on $\{\Omega_{a,b,c,k} \mid a, b \in \mathbb{R}^{(m,n)},\ 0 \neq c \in \mathrm{Sym}(m, \mathbb{R}),\ 1 \leq k \leq m\}$ which converges to 0 in the usual topology converges to the whole Euclidean space $\mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)} \times \mathrm{Sym}(m, \mathbb{R})$ in the topology of $\widehat{G}$. This is just the quotient topology on $\mathfrak{g}^*/G$ so that algebraically and topologically $\widehat{G} = \mathfrak{g}^*/G$.

It is well known that each coadjoint orbit is a symplectic manifold. We will state this fact in detail. For the present, we fix an element F of $\mathfrak{g}^*$ once and for all. We consider the alternating $\mathbb{R}$-bilinear form $\mathbf{B}_F$ on $\mathfrak{g}$ defined by

$$\mathbf{B}_F(X, Y) \stackrel{\text{def}}{=} \langle F, [X, Y] \rangle = \langle \mathrm{ad}_{\mathfrak{g}}^*(Y)F, X \rangle, \quad X, Y \in \mathfrak{g}, \tag{1.7.8}$$

where $\mathrm{ad}_{\mathfrak{g}}^* : \mathfrak{g} \longrightarrow \mathrm{End}(\mathfrak{g}^*)$ denotes the differential of the coadjoint representation $\mathrm{Ad}_G^* : G \longrightarrow GL(\mathfrak{g}^*)$. More precisely, if $F = F(a, b, c)$, $X = X(\alpha, \beta, \gamma)$, and $Y = X(\delta, \epsilon, \xi)$, then

$$\mathbf{B}_F(X, Y) = \sigma\{c\,({}^t\alpha\epsilon + {}^t\epsilon\alpha - {}^t\beta\delta - {}^t\delta\beta)\}. \tag{1.7.9}$$

For $F \in \mathfrak{g}^*$, we let

$$G_F = \left\{ g \in G \mid \mathrm{Ad}_G^*(g)F = F \right\}$$

be the stabilizer of the coadjoint action Ad^* of G on $\mathfrak{g}^*$ at F. Since G_F is a closed subgroup of G, G_F is a Lie subgroup of G. We denote by $\mathfrak{g}_F$ the Lie subalgebra of $\mathfrak{g}$ corresponding to G_F. Then it is easy to show that

$$\mathfrak{g}_F = \operatorname{rad} \mathbf{B}_F = \left\{ X \in \mathfrak{g} \mid \mathrm{ad}_{\mathfrak{g}}^*(X)F = 0 \right\}. \tag{1.7.10}$$

Here $\operatorname{rad} \mathbf{B}_F$ denotes the radical of $\mathbf{B}_F$ in $\mathfrak{g}$. We let $\dot{\mathbf{B}}_F$ be the non-degenerate alternating $\mathbb{R}$-bilinear form on the quotient vector space $\mathfrak{g}/\operatorname{rad} \mathbf{B}_F$ induced from $\mathbf{B}_F$. Since we may identify the tangent space of the coadjoint orbit $\Omega_F \cong G/G_F$ with $\mathfrak{g}/\mathfrak{g}_F = \mathfrak{g}/\operatorname{rad} \mathbf{B}_F$, we see that the tangent space of Ω_F at F is a symplectic vector space with respect to the symplectic form $\dot{\mathbf{B}}_F$.

Now we are ready to prove that the coadjoint orbit $\Omega_F = \mathrm{Ad}_G^*(G)F$ is a symplectic manifold. We denote by $\widetilde{X}$ the smooth vector field on $\mathfrak{g}^*$ associated with

$X \in \mathfrak{g}$. That means that for each $\ell \in \mathfrak{g}^*$, we have

$$\widetilde{X}(\ell) = \mathrm{ad}^*_{\mathfrak{g}}(X)\ \ell. \tag{1.7.11}$$

We define the differential 2-form B_{Ω_F} on Ω_F by

$$B_{\Omega_F}(\widetilde{X}, \widetilde{Y}) = B_{\Omega_F}(\mathrm{ad}^*_{\mathfrak{g}}(X)F, \mathrm{ad}^*_{\mathfrak{g}}(Y)F) := \mathbf{B}_F(X, Y), \tag{1.7.12}$$

where $X, Y \in \mathfrak{g}$.

Lemma 1.7.11 *B_{Ω_F} is non-degenerate.*

Proof Let $\widetilde{X}$ be the smooth vector field on $\mathfrak{g}^*$ associated with $X \in \mathfrak{g}$ such that $B_{\Omega_F}(\widetilde{X}, \widetilde{Y}) = 0$ for all $\widetilde{Y}$ with $Y \in \mathfrak{g}$. Since $B_{\Omega_F}(\widetilde{X}, \widetilde{Y}) = \mathbf{B}_F(X, Y) = 0$ for all $Y \in \mathfrak{g},\ X \in \mathfrak{g}_F$. Thus $\widetilde{X} = 0$. Hence B_{Ω_F} is non-degenerate. □

Lemma 1.7.12 *B_{Ω_F} is closed.*

Proof If $\widetilde{X_1}$, $\widetilde{X_2}$, and $\widetilde{X_3}$ are three smooth vector fields on $\mathfrak{g}^*$ associated with $X_1,\ X_2,\ X_3 \in \mathfrak{g}$, then

$$\begin{aligned}
&dB_{\Omega_F}(\widetilde{X_1}, \widetilde{X_2}, \widetilde{X_3}) \\
&= \widetilde{X_1}(B_{\Omega_F}(\widetilde{X_2}, \widetilde{X_3})) - \widetilde{X_2}(B_{\Omega_F}(\widetilde{X_1}, \widetilde{X_3})) + \widetilde{X_3}(B_{\Omega_F}(\widetilde{X_1}, \widetilde{X_2})) \\
&\quad - B_{\Omega_F}([\widetilde{X_1}, \widetilde{X_2}], \widetilde{X_3}) + B_{\Omega_F}([\widetilde{X_1}, \widetilde{X_3}], \widetilde{X_2}) - B_{\Omega_F}([\widetilde{X_2}, \widetilde{X_3}], \widetilde{X_1}) \\
&= -\langle F, [[X_1, X_2], X_3] + [[X_2, X_3], X_1] + [[X_3, X_1], X_2]\rangle \\
&= 0 \qquad \text{(by the Jacobi identity)}.
\end{aligned}$$

Therefore B_{Ω_F} is closed. □

In summary, (Ω_F, B_{Ω_F}) is a symplectic manifold of dimension $2\,k\,n$ $(1 \le k \le m)$ or 0.

In order to describe the irreducible unitary representations of G corresponding to the coadjoint orbits under the Kirillov correspondence, we have to determine the polarizations of $\mathfrak{g}$ for the linear forms $F \in \mathfrak{g}^*$.

Case I. $F = F(a, b, 0),\ a, b \in \mathbb{R}^{(m,n)}$; the extremely degenerate case.

According to (1.7.5), $\Omega_F = \Omega_{a,b} = \{F(a, b, 0)\}$ is a single point. It follows from (1.7.9) that $\mathbf{B}_F(X, Y) = 0$ for all $X, Y \in \mathfrak{g}$. Thus $\mathfrak{g}$ is the unique polarization of $\mathfrak{g}$ for F. The Kirillov correspondence says that the irreducible unitary representation $\pi_{a,b}$ of G corresponding to the coadjoint orbit $\Omega_{a,b}$ is given by

$$\pi_{a,b}(\exp X(\alpha, \beta, \gamma)) = e^{2\pi i \langle F, X(\alpha,\beta,\gamma)\rangle} = e^{4\pi i\, \sigma({}^t a\alpha + {}^t b\beta)}. \tag{1.7.13}$$

That is, $\pi_{a,b}$ is a one-dimensional extremely degenerate representation of G.

Case II. $F = F(a, b, c)$, $a, b \in \mathbb{R}^{(m,n)}$, $c \in \mathrm{Sym}(m, \mathbb{R})$ with $1 \leq \mathrm{rank}\, c \leq m$. According to (1.7.6),

$$\Omega_F = \Omega_{a,b,c,k} = \left\{ F(a + c\,\mu, b - c\,\lambda, c) \mid \lambda, \mu \in \mathbb{R}^{(m,n)} \right\}.$$

By (1.7.9), we see that

$$\mathfrak{k} = \left\{ X(0, \beta, \gamma) \,\middle|\, \beta \in \mathbb{R}^{(m,n)},\ \gamma \in \mathrm{Sym}(m, \mathbb{R}) \right\} \tag{1.7.14}$$

is a polarization of $\mathfrak{g}$ for F, i.e.,$\mathfrak{k}$ is a Lie subalgebra of $\mathfrak{g}$ subordinate to $F \in \mathfrak{g}^*$ which is maximal among the totally isotropic vector subspaces of $\mathfrak{g}$ relative to the alternating $\mathbb{R}$-bilinear form $\mathbf{B}_F$ (cf. Definition 1.7.1). Let K be the simply connected Lie subgroup of G corresponding to the Lie subalgebra $\mathfrak{k}$ of $\mathfrak{g}$. We let

$$\chi_{a,b,c,k;\mathfrak{k}} : K \longrightarrow \mathbb{C}_1^\times$$

be the unitary character of K defined by

$$\chi_{a,b,c,k;\mathfrak{k}}(\exp X(0, \beta, \gamma)) = e^{2\pi i \langle F, X(0,\beta,\gamma)\rangle} = e^{2\pi i\, \sigma(c\,\gamma + 2\,\beta\,{}^t b)}. \tag{1.7.15}$$

The Kirillov correspondence says that the irreducible unitary representation $\pi_{a,b,c,k;\mathfrak{k}}$ of G corresponding to the coadjoint orbit $\Omega_F = \Omega_{a,b,c,k}$ is given by

$$\pi_{a,b,c,k;\mathfrak{k}} = \mathrm{Ind}_K^G\, \chi_{a,b,c,k;\mathfrak{k}}. \tag{1.7.16}$$

For $a, b, \widetilde{a}, \widetilde{b} \in \mathbb{R}^{(m,n)}$, we have

$$\pi_{a,b,c,k;\mathfrak{k}}(\exp X(0, 0, \gamma)) = \pi_{\widetilde{a},\widetilde{b},c,k;\mathfrak{k}}(\exp X(0, 0, \gamma))$$

for all $\gamma \in \mathrm{Sym}(m, \mathbb{R})$. According to Theorem 1.7.6, $\pi_{a,b,c,k;\mathfrak{k}}$ is unitarily equivalent to $\pi_{\widetilde{a},\widetilde{b},c,k;\mathfrak{k}}$ for all $a, b, \widetilde{a}, \widetilde{b} \in \mathbb{R}^{(m,n)}$. So we denote the equivalence class of $\pi_{a,b,c,k;\mathfrak{k}}$ by $\pi_{c,k;\mathfrak{k}}$. According to Kirillov's theorem (cf. [21]), we know that the induced representation $\pi_{c,k;\mathfrak{k}}$ is, up to equivalence, independent of the choice of a polarization of $\mathfrak{g}$ for F. Thus we denote the equivalence class of $\pi_{c,k;\mathfrak{k}}$ by $\pi_{c,k}$. We see that $\pi_{c,k}$ is realized on the representation space $L^2(\mathbb{R}^{(m,n)}, d\xi)$ as follows:

$$(\pi_{c,k}(g) f)(\xi) = e^{2\pi i\, \sigma\{c(\kappa + \mu\,{}^t\lambda + 2\,\xi\,{}^t\mu)\}} f(\xi + \lambda), \tag{1.7.17}$$

where $g = (\lambda, \mu, \kappa) \in G$ and $\xi \in \mathbb{R}^{(m,n)}$. Using the fact that

$$\exp X(\alpha, \beta, \gamma) = \left(\alpha, \beta, \gamma + \frac{1}{2}\left(\alpha\,{}^t\beta - \beta\,{}^t\alpha\right)\right),$$

we see that $\pi_{c,k}$ is nothing but the Schrödinger representation $U_c = U(\sigma_c)$ of G induced from the one-dimensional unitary representation σ_c of K given by $\sigma_c((0,\mu,\kappa)) = e^{2\pi i\,\sigma(c\,\kappa)} I$ (cf. (1.2.9) and (1.2.11)). We note that $\pi_{c,k}$ is the representation of G with central character $\chi_{c,k} : \mathfrak{Z} \longrightarrow \mathbb{C}_1^\times$ given by $\chi_{c,k}((0,0,\kappa)) = e^{2\pi i\,\sigma(c\,\kappa)}$. Here $\mathfrak{Z} = \{(0,0,\kappa) \mid \kappa \in \mathrm{Sym}(m,\mathbb{R})\}$ denotes the center of G.
It is well known that the monomial representation $\left(\pi_{c,k}, L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)\right)$ of G extends to an operator of trace class

$$\pi_{c,k}^1(\phi) : L^2\left(\mathbb{R}^{(m,n)}, d\xi\right) \longrightarrow L^2\left(\mathbb{R}^{(m,n)}, d\xi\right) \tag{1.7.18}$$

for all $\phi \in C_c^\infty(G)$. Here $C_c^\infty(G)$ is the vector space of all smooth functions on G with compact support. We let $C_c^\infty(\mathfrak{g})$ and $C(\mathfrak{g}^*)$ the vector space of all smooth functions on $\mathfrak{g}$ with compact support and the vector space of all continuous functions on $\mathfrak{g}^*$ respectively. If $f \in C_c^\infty(\mathfrak{g})$, we define the Fourier cotransform

$$\mathcal{C}F_{\mathfrak{g}} : C_c^\infty(\mathfrak{g}) \longrightarrow C(\mathfrak{g}^*)$$

by

$$\left(\mathcal{C}F_{\mathfrak{g}}(f)\right)(F') := \int_{\mathfrak{g}} f(X)\, e^{2\pi i\,\langle F', X\rangle} dX, \tag{1.7.19}$$

where $F' \in \mathfrak{g}^*$ and dX denotes the usual Lebesgue measure on $\mathfrak{g}$. According to A. Kirillov (cf. [21]), there exists a measure β on the coadjoint orbit

$$\Omega_{a,b,c,k} \approx \mathbb{R}^{(k,n)} \times \mathbb{R}^{(k,n)}$$

$$(a, b \in \mathbb{R}^{(m,n)},\ c \in \mathrm{Sym}(m,\mathbb{R}),\ \mathrm{rank}\, c = k,\ 1 \le k \le m)$$

which is invariant under the coadjoint action of G such that

$$\mathrm{tr}\, \pi_{c,k}^1(\phi) = \int_{\Omega_c} \mathcal{C}F_{\mathfrak{g}}(\phi \circ \exp)(F') d\beta(F') \tag{1.7.20}$$

holds for all test functions $\phi \in C_c^\infty(G)$, where exp denotes the exponential mapping of $\mathfrak{g}$ onto G. We recall that

$$\pi_{c,k}^1(\phi)(f) := \int_G \phi(x)\, \left(\pi_{c,k}(x) f\right) dx,$$

where $\phi \in C_c^\infty(G)$ and $f \in L^2(\mathbb{R}^{(m,n)}, d\xi)$. By the Plancherel theorem, the mapping

$$S(G/\mathfrak{Z}) \ni \varphi \longmapsto \pi_{c,k}^1(\varphi) \in \mathrm{TC}(L^2(\mathbb{R}^{(m,n)}, d\xi))$$

extends to a unitary isometry

$$\pi_{c,k}^2 : L^2(G/\mathfrak{Z}, \chi_{c,k}) \longrightarrow \mathrm{HS}\big(L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)\big) \tag{1.7.21}$$

of the representation space $L^2(G/\mathfrak{Z}, \chi_{c,k})$ of $\mathrm{Ind}_{\mathfrak{Z}}^{G} \chi_{c,k}$ onto the complex Hilbert space $\mathrm{HS}\big(L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)\big)$ consisting of all Hilbert–Schmidt operators on $L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)$, where $S(G/\mathfrak{Z})$ is the Schwartz space of all infinitely differentiable complex-valued functions on $G/\mathfrak{Z} \cong \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}$ that are rapidly decreasing at infinity and $\mathrm{TC}\big(L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)\big)$ denotes the complex vector space of all continuous $\mathbb{C}$-linear mappings of $L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)$ into itself which are trace class.

In summary, we have the following result.

Theorem 1.7.13 *For $F = F(a, b, 0) \in \mathfrak{g}^*$, the irreducible unitary representation $\pi_{a,b}$ of G corresponding to the coadjoint orbit $\Omega_F = \Omega_{a,b}$ under the Kirillov correspondence is an extremely degenerate representation of G given by*

$$\pi_{a,b}\big(\exp X(\alpha, \beta, \gamma)\big) = e^{4\pi i\, \sigma({}^t a\alpha + {}^t b\beta)}.$$

On the other hand, for $F = F(a, b, c) \in \mathfrak{g}^*$ with $a, b \in \mathbb{R}^{(m,n)}$, $c \in \mathrm{Sym}(m, \mathbb{R})$ with $1 \leq k = \mathrm{rank}\, c \leq m$, the irreducible unitary representation $\big(\pi_{c,k}, L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)\big)$ of G corresponding to the coadjoint orbit $\Omega_{a,b,c,k}$ under the Kirillov correspondence is unitarily equivalent to the Schrödinger representation $\big(U_c, L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)\big)$ and this representation $\pi_{c,k}$ is square integrable modulo its center $\mathfrak{Z}$. For all test functions $\phi \in C_c^\infty(G)$, the character formula

$$\mathrm{Tr}\left(\pi_{c,k}^2(\phi)\right) = \mathcal{C}(\phi, c) \int_{\mathrm{Sym}(m,\mathbb{R})} \phi(0, 0, \kappa)\, e^{2\pi i\, \sigma(c\kappa)} d\kappa$$

holds for some constant $\mathcal{C}(\phi, c)$ depending on ϕ and c, where $d\kappa$ is the Lebesgue measure on the Euclidean space $\mathrm{Sym}(m, \mathbb{R})$.

Now we consider the subgroup K of G (cf. (1.2.4)) given by

$$K := \big\{\, (0, \mu, \kappa) \in G \;\big|\; \mu \in \mathbb{R}^{(m,n)},\ \kappa \in \mathrm{Sym}(m, \mathbb{R}) \,\big\}.$$

The Lie algebra $\mathfrak{k}$ of K is given by (1.7.14). The dual space $\mathfrak{k}^*$ of $\mathfrak{k}$ may be identified with the space

$$\big\{F(0, b, c) \;\big|\; b \in \mathbb{R}^{(m,n)},\ c \in \mathrm{Sym}(m, \mathbb{R})\big\}.$$

We let $\mathrm{Ad}_K^* : K \longrightarrow GL(\mathfrak{k}^*)$ be the coadjoint representation of K on $\mathfrak{k}^*$. The coadjoint K-orbit $\omega_{b,c,k}$ at $F(0,b,c) \in \mathfrak{k}^*$ with $k = \operatorname{rank} c$ and $1 \leq k \leq m$ is given by

$$\omega_{b,c,k} = \mathrm{Ad}_K^*(K)\, F(0,b,c) = \{F(c\mu, b, c) \mid \mu \in \mathbb{R}^{(m,n)}\}. \tag{1.7.22}$$

Since K is a commutative group, $[\mathfrak{k}, \mathfrak{k}] = 0$ and so the alternating $\mathbb{R}$-bilinear form $\mathbf{B}_F$ on $\mathfrak{k}$ associated with $F := F(0,b,c)$ identically vanishes on $\mathfrak{k} \times \mathfrak{k}$ (cf. (1.7.8)). $\mathfrak{k}$ is the unique polarization of $\mathfrak{k}$ for $F = F(0,b,c)$. The Kirillov correspondence says that the irreducible unitary representation $\chi_{b,c}$ of K corresponding to the coadjoint orbit $\omega_{b,c}$ is given by

$$\chi_{b,c}\big(\exp X(0,\beta,\gamma)\big) = e^{2\pi i \langle F(0,b,c), X(0,\beta,\gamma)\rangle} = e^{2\pi i\, \sigma(2\, b\, {}^t\beta + c\gamma)} \tag{1.7.23}$$

or

$$\chi_{b,c}((0,\mu,\kappa)) = e^{2\pi i\, \sigma(2\, b\, {}^t\mu + c\kappa)}, \quad (0,\mu,\kappa) \in K. \tag{1.7.24}$$

For $0 \neq c \in \mathrm{Sym}(m,\mathbb{R})$ with $1 \leq k = \operatorname{rank} c \leq m$, we let $\pi_{c,k}$ be the Schrödinger representation of G given by (1.7.17). We know that $\pi_{c,k}$ is the irreducible unitary representation of G corresponding to the coadjoint orbit

$$\Omega_{a,b,c,k} = \mathrm{Ad}_G^*(G)\, F(a,b,c) = \left\{ F(a + c\mu, b - c\lambda, c) \mid a, b \in \mathbb{R}^{(m,n)} \right\}.$$

Let $p : \mathfrak{g}^* \longrightarrow \mathfrak{k}^*$ be the natural projection defined by $p(F(a,b,c)) = F(0,b,c)$. Obviously, we have

$$p(\Omega_{a,b,c,k}) = \left\{ F(0,b,c) \,\middle|\, b \in \mathbb{R}^{(m,n)} \right\} = \bigcup_{b \in \mathbb{R}^{(m,n)}} \omega_{b,c,k}.$$

According to Kirillov's theorem (cf. [22, Theorem 1, p. 249]), the restriction $\pi_{c,k}|_K$ of $\pi_{c,k}$ to K is the direct integral of all one-dimensional representations $\chi_{b,c}$ of K ($b \in \mathbb{R}^{(m,n)}$). Conversely, we let $\chi_{b,c}$ be the element of $\widehat{K}$ corresponding to the coadjoint orbit $\omega_{b,c,k}$ of K. The induced representation $\mathrm{Ind}_K^G \chi_{b,c}$ is nothing but the Schrödinger representation $\pi_{c,k}$. The coadjoint orbit $\Omega_{a,b,c,k}$ of G is the only coadjoint orbit such that $\Omega_{a,b,c,k} \cap p^{-1}(\omega_{b,c,k})$ is nonempty.

1.8 Realization of Theta Functions as Functions on the Heisenberg Group

In this section, we identify theta functions with smooth functions on the Heisenberg group $(\mathbb{H}_{\mathbb{R}}^{(n,m)}, \diamond)$ satisfying some conditions (see Theorem 1.8.17).

We fix an element $\Omega \in \mathbb{H}_n$ once and for all. From now on, we put $i = \sqrt{-1}$. Let $\mathscr{M}$ be a positive definite, symmetric integral matrix of degree m. A holomorphic function $f : \mathbb{C}^{(m,n)} \longrightarrow \mathbb{C}$ satisfying the following equation,

$$f(W + \xi\Omega + \eta) = e^{-\pi i\, \sigma\{\mathscr{M}(\xi\Omega\, {}^t\xi + 2\xi\, {}^tW)\}} f(W), \quad W \in \mathbb{C}^{(m,n)}, \tag{1.8.1}$$

for all $\xi,\ \eta \in \mathbb{Z}^{(m,n)}$ is called a theta function of level $\mathscr{M}$ with respect to Ω. The set $R^{\Omega}_{\mathscr{M}}$ of all theta functions of level $\mathscr{M}$ with respect to Ω is a complex vector space of dimension $(\det \mathscr{M})^n$ with a basis consisting of theta functions

$$\vartheta^{(\mathscr{M})} \begin{bmatrix} A \\ 0 \end{bmatrix} (\Omega, W) := \sum_{N \in \mathbb{Z}^{(m,n)}} e^{\pi i\, \sigma\{\mathscr{M}((N+A)\Omega\, {}^t(N+A) + 2\, W\, {}^t(N+A))\}}, \tag{1.8.2}$$

where A runs over a complete system of the cosets $\mathscr{M}^{-1}\mathbb{Z}^{(m,n)}/\mathbb{Z}^{(m,n)}$.

Definition 1.8.1 Let S be a positive definite, symmetric real matrix of degree m and let $A, B \in \mathbb{R}^{(m,n)}$. We define the theta function

$$\vartheta^{(S)} \begin{bmatrix} A \\ B \end{bmatrix} (\Omega, W) = \sum_{N \in \mathbb{Z}^{(m,n)}} e^{\pi i\, \sigma\{S((N+A)\Omega\, {}^t(N+A) + 2\,(W+B)\, {}^t(N+A))\}} \tag{1.8.3}$$

with characteristic (A, B) converging normally on $\mathbb{H}_n \times \mathbb{C}^{(m,n)}$.

We have a general definition of theta functions.

Definition 1.8.2 Let V be a complex vector space and let $L \subset V$ be a lattice of V. A theta function on V relative to L is a nonzero holomorphic function ϑ on V satisfying the following condition:

$$\vartheta(W + \xi) = e^{2\pi i(Q_\xi(W) + c_\xi)} \vartheta(W),$$

where Q_ξ is a $\mathbb{C}$-linear form on V and c_ξ is an element of $\mathbb{C}$, for every $W \in V$ and $\xi \in L$.

If ϑ is a theta function on V relative to L, then the mapping $J_\vartheta : L \times V \longrightarrow \mathbb{C}^*$ defined by

$$J_\vartheta(\xi, W) := e^{2\pi i(Q_\xi(W) + c_\xi)}, \quad \xi \in L, \quad W \in V$$

is easily seen to be an automorphic factor. This means that J_ϑ satisfies the following condition,

$$J_\vartheta(\xi_1 + \xi_2, W) = J_\vartheta(\xi_1, W + \xi_2)\, J_\vartheta(\xi_2, W),$$

for all $\xi_1, \xi_2 \in L$ and $W \in V$. We observe that for all $\xi_1, \xi_2 \in L$ and $W \in V$,

$$Q_{\xi_1+\xi_2}(W) + c_{\xi_1+\xi_2} \equiv Q_{\xi_1}(W + \xi_2) + Q_{\xi_2}(W) + c_{\xi_1} + c_{\xi_2} \mod \mathbb{Z}.$$

J_ϑ is called the automorphic factor of the theta function ϑ on V relative to L.

Definition 1.8.3 Let V be a complex vector space. A mapping $(W_1, W_2) \longmapsto Q(W_1, W_2)$, $W_1, W_2 \in V$ is called a quasi-Hermitian form on $V \times V$ if it is $\mathbb{C}$-linear in W_1, $\mathbb{R}$-linear in W_2, and if

$$A_Q(W_1, W_2) = \frac{1}{2i} \{Q(W_1, W_2) - Q(W_2, W_1)\}$$

is in $\mathbb{R}$ for every W_1, W_2 in V (cf. [18, p. 64]).

Definition 1.8.4 A continuous mapping f of a locally compact continuous group Y to $\mathbb{C}_1^\times$ such that

$$(y_1, y_2) \longmapsto \frac{f(y_1 + y_2)}{f(y_1) f(y_2)}, \qquad y_1, y_2 \in Y$$

is a bicharacter of Y, i.e., a character of Y in y_1 and y_2, is called a second degree character of Y. Here $\mathbb{C}_1^\times = \{z \in \mathbb{C} \mid |z| = 1\}$ (see Notations Frequently Used, p. xiii). See [18, p. 66].

Definition 1.8.5 Suppose X is a finite dimensional real vector space, L is a lattice in X and $A : X \times X \longrightarrow \mathbb{R}$ an alternating form on $X \times X$ which is integral valued on $L \times L$. Let B be a bilinear real form on $X \times X$ which is integral valued on $L \times L$ and satisfies the equation

$$A(x, y) = B(x, y) - B(y, x) \qquad \text{for all } x, y \in X.$$

A mapping $\psi : L \longrightarrow \mathbb{C}_1^\times$ of L to $\mathbb{C}_1^\times$ is said to be associated with A if the following second degree character by

$$\chi(\xi) = \psi(\xi)\, e^{\pi i B(\xi, \xi)}, \quad \xi \in L$$

is a character of L. Let

$$X_0 \stackrel{\text{def}}{=} \{x_0 \in X \mid A(x_0, y) = 0 \text{ for all } y \in X\}$$

be the subspace of X over $\mathbb{R}$. We put $L_0 \stackrel{\text{def}}{=} L \cap X_0$. A second degree character $\varphi : L \longrightarrow \mathbb{C}_1^\times$ is said to be strongly associated with A if $\varphi(\xi + \xi_0) = \varphi(\xi)$ for all $\xi \in L$ and all $\xi_0 \in L$. We refer to [18, p. 67].

Theorem 1.8.6 (Igusa [18, p. 67]) *Let $J : L \times V \longrightarrow \mathbb{C}^\times$ be the automorphic factor of a theta function ϑ on V relative to L. Then there exists a unique triple*

(Q, ℓ, ψ) *such that*

$$J(\xi, W) = e^{\pi\{Q(W,\xi)+\frac{1}{2}Q(\xi,\xi)+2i\,\ell(\xi)\}}\psi(\xi), \quad \xi \in L, \ W \in V, \tag{1.8.4}$$

where

(1) Q *is a quasi-Hermitian form on* $V \times V$,
(2) *the Hermitian form* $H := \mathrm{Her}\,(Q)$ *defined by*

$$H(W_1, W_2) = \frac{1}{2i}\left\{Q(iW_1, W_2) - Q(W_1, iW_2)\right\}, \quad W_1, \ W_2 \in V$$

is a Riemann form with respect to L, *that is,* $H = {}^t\overline{H} > 0$ *and* $(\mathrm{Im}\,H)(L \times L) \subset \mathbb{Z}$,
(3) $\ell : V \longrightarrow \mathbb{C}$ *is a* $\mathbb{C}$*-linear form on* V,
(4) ψ *is a second degree character of* L *which is associated with* $A := \mathrm{Im}\,H$,
(5) ψ *is strongly associated with* A.

Remark 1.8.7 The condition (4) means that $\psi : L \longrightarrow \mathbb{C}_1^\times$ is a semi-character of L satisfying the functional equation

$$\psi(\xi_1 + \xi_2) = e^{\pi i A(\xi_1,\xi_2)}\psi(\xi_1)\psi(\xi_2), \quad \xi_1, \xi_2 \in L. \tag{$*$}$$

Definition 1.8.8 A theta function with the automorphic factor of the form (1.8.4) is called a theta function of type (Q, ℓ, ψ). We denote by $L(Q, \ell, \psi)$ the union of theta functions of type (Q, ℓ, ψ) and the constant 0. A theta function of type (Q, ℓ, ψ) is said to be normalized if $\mathrm{Sym}\,Q = 0$ and $\ell = 0$. Here $\mathrm{Sym}\,Q : V \times V \longrightarrow \mathbb{C}$ is a symmetric $\mathbb{C}$-linear form on $V \times V$ defined by

$$(\mathrm{Sym}\,Q)(z, w) = \frac{1}{2i}\left\{Q(iz, w) + Q(z, iw)\right\}, \quad z, w \in V.$$

We observe that $Q = \mathrm{Her}\,Q + \mathrm{Sym}\,Q$. We note that $\mathrm{Sym}\,Q = 0$ if and only if $Q = \mathrm{Her}\,Q = H$. We denote by $Th(H, \psi, L)$ the union of the set of all normalized theta functions of type $(H, 0, \psi)$ and the constant 0. It is easily seen that if $\vartheta \in Th(H, \psi, L)$, for all $W \in V$, $\xi \in L$, we have

$$\vartheta(W + \xi) = e^{\pi H(W+\frac{1}{2}\xi,\xi)}\psi(\xi)\,\vartheta(W). \tag{1.8.5}$$

Theorem 1.8.9 *Let* S *be a positive definite, symmetric integral matrix of degree* m *and let* A, B *be two* $m \times n$ *real matrices. Then for* $\Omega \in \mathbb{H}_n$ *and* $W \in \mathbb{C}^{(m,n)}$, *we have*

$$(\theta.1) \qquad \vartheta^{(S)}\begin{bmatrix} A \\ B \end{bmatrix}(\Omega, -W) = \vartheta^{(S)}\begin{bmatrix} -A \\ -B \end{bmatrix}(\Omega, W),$$

$$(\theta.2)\qquad \vartheta^{(S)}\begin{bmatrix}A\\B\end{bmatrix}(\Omega, W+\lambda\Omega+\mu)$$
$$= e^{-\pi i\,\sigma\{S(\lambda\Omega\,{}^t\lambda+2\,(W+\mu)\,{}^t\lambda)\}}\,e^{-2\pi i\,\sigma(SB\,{}^t\lambda)}\cdot\vartheta^{(S)}\begin{bmatrix}A+\lambda\\B+\mu\end{bmatrix}(\Omega, W)$$
for all $\lambda, \mu \in \mathbb{R}^{(m,n)}$.

$$(\theta.3)\qquad \vartheta^{(S)}\begin{bmatrix}A\\B\end{bmatrix}(\Omega, W)$$
$$= e^{\pi i\,\sigma\{S(A\Omega\,{}^tA)+2\,(W+B)\,{}^tA)\}}\,\vartheta^{(S)}\begin{bmatrix}0\\0\end{bmatrix}(\Omega, W+A\Omega+B).$$

Moreover, if S is a positive definite, symmetric integral matrix of degree m, we have

$$(\theta.4)\qquad \vartheta^{(S)}\begin{bmatrix}A+\xi\\B+\eta\end{bmatrix}(\Omega, W) = e^{2\pi i\,\sigma(SA\,{}^t\eta)}\,\vartheta^{(S)}\begin{bmatrix}A\\B\end{bmatrix}(\Omega, W).$$
for all $\xi, \eta \in \mathbb{Z}^{(m,n)}$.

$$(\theta.5)\qquad \vartheta^{(S)}\begin{bmatrix}A\\B\end{bmatrix}(\Omega, W+\xi\Omega+\eta)$$
$$= e^{-\pi i\,\sigma\{S(\xi\Omega\,{}^t\xi+2\,W\,{}^t\xi)\}}\cdot e^{2\pi i\,\sigma\{S(A\,{}^t\eta-B\,{}^t\xi)\}}\cdot\vartheta^{(S)}\begin{bmatrix}A\\B\end{bmatrix}(\Omega, W)$$
for all $\xi, \eta \in \mathbb{Z}^{(m,n)}$.

Proof $(\theta.1)$ follows immediately from Formula (1.8.3). $(\theta.2)$ follows immediately from the relation

$$\begin{aligned}&(N+A)\Omega\,{}^t(N+A)+2(W+\lambda\Omega+\mu+B)\,{}^t(N+A)\\ =&(N+A+\lambda)\,\Omega\,{}^t(N+A+\lambda)+2(W+\mu+B)\,{}^t(N+A+\lambda)\\ &-(N+A)\Omega\,{}^t\lambda+\lambda\Omega\,{}^t(N+A)-\lambda\Omega\,{}^t\lambda-2(W+\mu+B)\,{}^t\lambda.\end{aligned}$$

If we put $A = B = 0$ and replace λ, μ by A, B in $(\theta.2)$, then we obtain $(\theta.3)$. For $\xi, \eta \in \mathbb{Z}^{(m,n)}$, we have

$$\begin{aligned}&\vartheta^{(S)}\begin{bmatrix}A+\xi\\B+\eta\end{bmatrix}(\Omega, W)\\ =&\sum_{N\in\mathbb{Z}^{(m,n)}} e^{\pi i\,\sigma\{S((A+N+\xi)\Omega\,{}^t(A+N+\xi)+2\,(W+B)\,{}^t(A+N+\xi))\}}\\ &\qquad\times e^{2\pi i\,\sigma\{S\eta\,{}^t(N+\xi)\}}\cdot e^{2\pi i\,\sigma(S\,{}^t\eta A)}\\ =&\ e^{2\pi i\,\sigma(SA\,{}^t\eta)}\cdot\vartheta^{(S)}\begin{bmatrix}A\\B\end{bmatrix}(\Omega, W).\end{aligned}$$

Here in the last equality we used the fact that $\sigma(S\eta\,{}^t(N+\xi)) \in \mathbb{Z}$ because S is integral. $(\theta.5)$ follows from $(\theta.2)$, $(\theta.4)$ and the fact that $\sigma(S\eta\,{}^t\xi)$ is integral. □

For a positive definite, symmetric real matrix S of degree m, $\Omega \in \mathbb{H}_n$ and $A, B \in \mathbb{R}^{(m,n)}$, we put

$$\chi_{S,\Omega,A,B}(\xi\Omega+\eta) := \chi_{S,\Omega,A,B}(\xi,\eta) := e^{2\pi i\,\sigma\{S(A\,{}^t\eta - B\,{}^t\xi)\}}, \tag{1.8.6}$$

where $\xi, \eta \in \mathbb{Z}^{(m,n)}$.

We define

$$q_{S,\Omega}(W) = \frac{1}{2}\,\sigma\left(SW(\Omega-\overline{\Omega})^{-1}\,{}^tW\right), \quad W \in \mathbb{C}^{(m,n)} \tag{1.8.7}$$

and also define

$$H_{S,\Omega}(W_1,W_2) = 2i\,\sigma\left(SW_1(\Omega-\overline{\Omega})^{-1}\,{}^t\overline{W}_2\right), \quad W_1, W_2 \in \mathbb{C}^{(m,n)}. \tag{1.8.8}$$

It is easy to check that $H_{S,\Omega}$ is a positive definite Hermitian form on $\mathbb{C}^{(m,n)}$.

Lemma 1.8.10 *For $W \in \mathbb{C}^{(m,n)}$ and $\ell \in \mathbb{Z}^{(m,n)}\Omega + \mathbb{Z}^{(m,n)}$, we have*

$$q_{S,\Omega}(W+\ell) = q_{S,\Omega}(W) + q_{S,\Omega}(\ell) + \sigma\left(S\ell(\Omega-\overline{\Omega})^{-1}\,{}^tW\right) \tag{1.8.9}$$

and

$$\begin{aligned} H_{S,\Omega}\left(W+\frac{\ell}{2},\ell\right) = \sigma\left(S\left(W+\tfrac{\ell}{2}\right)(\operatorname{Im}\Omega)^{-1}\,{}^t\ell\right) \\ -2i\,\sigma\left(S\left(W+\tfrac{\ell}{2}\right)\,{}^t\xi\right), \end{aligned} \tag{1.8.10}$$

where $\ell = \xi\Omega + \eta$, $\xi, \eta \in \mathbb{Z}^{(m,n)}$.

Proof It follows immediately from a straightforward computation. □

Lemma 1.8.11 *Let S be a positive definite, symmetric integral matrix of degree m. For $\Omega \in \mathbb{H}_n$, we let $L_\Omega := \mathbb{Z}^{(m,n)}\Omega + \mathbb{Z}^{(m,n)}$ be the lattice in $\mathbb{C}^{(m,n)}$. We define the mapping $\psi_{S,\Omega} : L_\Omega \longrightarrow \mathbb{C}_1^*$ by*

$$\psi_{S,\Omega}(\xi\Omega+\eta) = e^{\pi i\,\sigma(S\eta\,{}^t\xi)}, \quad \xi,\eta \in \mathbb{Z}^{(m,n)}. \tag{1.8.11}$$

Then:

(a) *$\psi_{S,\Omega}$ is a second-degree character of L_Ω associated with* $\operatorname{Im} H_{S,\Omega}$.
(b) *$\psi_{S,\Omega}\cdot\chi_{S,\Omega,A,B}$ is a second-degree character of L_Ω associated with* $\operatorname{Im} H_{S,\Omega}$.

Proof

(a) We fix $\ell = \xi\Omega + \eta \in L_\Omega$ with $\xi, \eta \in \mathbb{Z}^{(m,n)}$. We define $f_\ell : L_\Omega \longrightarrow \mathbb{C}_1^\times$ by

$$f_\ell(\ell_1) := \frac{\psi_{S,\Omega}(\ell_1 + \ell)}{\psi_{S,\Omega}(\ell_1)\,\psi_{S,\Omega}(\ell)}, \quad \ell_1 \in L_\Omega.$$

It is easy to see that f_ℓ is a character of L_Ω and hence to see that the map from $L_\Omega \times L_\Omega$ to $\mathbb{C}_1^*$ defined by

$$(\ell_1, \ell_2) \longmapsto \frac{\psi_{S,\Omega}(\ell_1 + \ell_2)}{\psi_{S,\Omega}(\ell_1)\,\psi_{S,\Omega}(\ell_2)}$$

is a bicharacter of L_Ω, i.e., a character of L_Ω in ℓ_1 and ℓ_2. Hence $\psi_{S,\Omega}$ is a second degree character of L_Ω. In order to show that $\psi_{S,\Omega}$ is associated with $H_{S,\Omega}$, it is enough to prove that

$$\psi_{S,\Omega}(\ell_1 + \ell_2) = e^{\pi i A_{S,\Omega}(\ell_1,\ell_2)}\,\psi_{S,\Omega}(\ell_1)\,\psi_{S,\Omega}(\ell_2) \tag{1.8.12}$$

for all $\ell_1, \ell_2 \in L_\Omega$. Here $A_{S,\Omega}$ denotes the imaginary part of the positive hermitian form $H_{S,\Omega}$. By an easy computation, we have

$$A_{S,\Omega}(\ell_1, \ell_2) = \sigma\{S(\xi_1\,{}^t\eta_2 - \eta_1\,{}^t\xi_2)\}, \tag{1.8.13}$$

where $\ell_i = \xi_i\Omega + \eta_i \in L_\Omega\,(1 \le i \le 2)$. Hence (1.8.12) follows immediately from (1.8.13).

(b) We fix $\ell = \xi\Omega + \eta \in L_\Omega$ with $\xi, \eta \in \mathbb{Z}^{(m,n)}$. We put $\widetilde{\psi}_{S,\Omega,A,B} := \psi_{S,\Omega} \cdot \chi_{S,\Omega,A,B}$. Then the map $\tilde{f}_\ell : L_\Omega \longrightarrow \mathbb{C}_1^*$ defined by

$$\tilde{f}_\ell(\ell_1) = \frac{\widetilde{\psi}_{S,\Omega,A,B}(\ell_1 + \ell)}{\widetilde{\psi}_{S,\Omega,A,B}(\ell_1)\,\widetilde{\psi}_{S,\Omega,A,B}(\ell)} \quad \ell_1 \in L_\Omega$$

is a character of L_Ω. So $\widetilde{\psi}_{S,\Omega,A,B}$ is a second degree character of L_Ω. In order to show that $\widetilde{\psi}_{S,\Omega,A,B}$ is associated with $A_{S,\Omega}$, it suffices to prove that

$$\widetilde{\psi}_{S,\Omega,A,B}(\ell_1 + \ell_2) = e^{\pi i\,A_{S,\Omega}(\ell_1,\ell_2)}\,\widetilde{\psi}_{S,\Omega,A,B}(\ell_1)\,\widetilde{\psi}_{S,\Omega,A,B}(\ell_2) \tag{1.8.14}$$

for all $\ell_1, \ell_2 \in L_\Omega$. An easy calculation yields (1.8.14).

□

Theorem 1.8.12 *We assume that S is a positive definite, symmetric integral matrix of degree m. Let $\Omega \in \mathbb{H}_n$. We denote by R_S^Ω the vector space of all holomorphic functions $f : \mathbb{C}^{(m,n)} \longrightarrow \mathbb{C}$ satisfying the transformation behaviour*

$$f(W + \xi\Omega + \eta) = e^{-\pi i\,\sigma\{S(\xi\Omega\,{}^t\xi + 2\xi\,{}^tW)\}}\,f(W), \quad W \in \mathbb{C}^{(m,n)}$$

for all $\xi, \eta \in \mathbb{Z}^{(m,n)}$. *Then the mapping*

$$\Theta : R_S^{\Omega} \longrightarrow Th(H_{S,\Omega}, \psi_{S,\Omega}, L_{\Omega})$$

defined by

$$(\Theta(F))(W) := e^{2\pi i\, q_{S,\Omega}(W)} F(W), \quad F \in R_S^{\Omega},\ W \in \mathbb{C}^{(m,n)}$$

is an isomorphism of vector spaces, where L_{Ω} *and* $\psi_{S,\Omega}$ *are the same as those in Lemma 1.8.11.*

Proof First of all, we will show the image $\Theta(R_S^{\Omega})$ is contained in $Th(H_{S,\Omega}, \psi_{S,\Omega}, L_{\Omega})$. If $F \in R_S^{\Omega}$, $W \in \mathbb{C}^{(m,n)}$ and $\ell = \xi\Omega + \eta \in L_{\Omega}$, then we have

$$\begin{aligned}
\Theta(F)(W+\ell) &= e^{2\pi i\, q_{S,\Omega}(W+\ell)}\, F(W+\ell) \\
&= e^{2\pi i \{q_{S,\Omega}(W) + q_{S,\Omega}(\ell) + \sigma(Sl(\Omega - \overline{\Omega}^{-1}\, {}^t W)\}} \\
&\quad \times e^{-\pi i\, \sigma\{S(\xi\Omega\, {}^t\xi + 2\, W\, {}^t\xi)\}}\, F(W) \qquad (\text{by Lemma } 1.8.10) \\
&= e^{2\pi i\, \sigma\{S(W + \frac{\ell}{2})(\Omega - \overline{\Omega})^{-1}\, {}^t\ell\}} \\
&\quad \times e^{-\pi i\, \sigma\{S(\xi\Omega\, {}^t\xi + 2\, W\, {}^t\xi)\}} \cdot \Theta(F)(W) \\
&= e^{\pi H_{S,\Omega}(W + \frac{\ell}{2}, \ell)} \cdot e^{-\pi i\, \sigma(S\eta\, {}^t\xi)}\, \Theta(F)(W) \\
&= e^{\pi H_{S,\Omega}(W + \frac{\ell}{2}, \ell)}\, \psi_{S,\Omega}(\ell)\, \Theta(F)(W).
\end{aligned}$$

Thus $\Theta(F)$ is contained in the set $Th(H_{S,\Omega}, \psi_{S,\Omega}, L_{\Omega})$. It is easy to see that the mapping Θ is an isomorphism. □

Proposition 1.8.13 *Let* S *be as above in Theorem 1.8.12 and* $A, B \in \mathbb{R}^{(m,n)}$. *We denote by* $R_{S,A,B}^{\Omega}$ *the union of the set of all theta functions with characteristic* (A, B) *with respect to* S *and* Ω *and the constant* 0. *Then we have an isomorphism*

$$R_{S,A,B}^{\Omega} \cong Th(H_{S,\Omega}, \psi_{S,\Omega} \cdot \chi_{S,\Omega,A,B}, L_{\Omega}).$$

Proof First we observe that $\psi_{S,\Omega} \cdot \chi_{S,\Omega,A,B}$ is a second degree character of L_{Ω} associated with $A_{S,\Omega}$ (cf. Lemma 1.8.11 (b)). In a similar way in the proof of Theorem 1.8.12, using $(\theta.5)$, we can show that the mapping

$$\Theta_{A,B}(f)(W) := e^{2\pi i\, q_{S,\Omega}(W)}\, f(W), \quad f \in R_{S,A,B}^{\Omega},\ W \in \mathbb{C}^{(m,n)}$$

has its image in $Th(H_{S,\Omega}, \psi_{S,\Omega} \cdot \chi_{S,\Omega,A,B}, L_{\Omega})$. □

Proposition 1.8.14 *Let S be as above in Theorem 1.8.12 and let $A, B \in \mathbb{R}^{(m,n)}$. Then we have an isomorphism*

$$Th(H_{S,\Omega}, \psi_{S,\Omega}, L_\Omega) \cong Th(H_{S,\Omega}, \psi_{S,\Omega} \cdot \chi_{S,\Omega,A,B}, L_\Omega).$$

Proof The proof follows from the fact that the dimension of the complex vector space $Th(H_{S,\Omega}, \psi_{S,\Omega}, L_\Omega)$ is equal to that of $Th(H_{S,\Omega}, \psi_{S,\Omega} \cdot \chi_{S,\Omega,A,B}, L_\Omega)$. It is well known that the dimension of $Th(H_{S,\Omega}, \psi_{S,\Omega}, L_\Omega)$ is equal to the Pfaffian of $A_{S,\Omega}$ relative to L_Ω (cf. [18, p. 72]). □

Remark 1.8.15 From Theorem 1.8.12, Proposition 1.8.13 and Proposition 1.8.14, R_S^Ω is isomorphic to $R_{S,A,B}^\Omega$ for any $A, B \in \mathbb{R}^{(m,n)}$.

Now as before, we fix an element $\Omega \in \mathbb{H}_n$ and let S be a positive symmetric integral matrix of degree m. Let

$$L \stackrel{\text{def}}{=} \mathbb{Z}^{(m,n)} \times \mathbb{Z}^{(m,n)} \subset \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)} \cong \mathbb{C}^{(m,n)}$$

be the lattice in $\mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}$. Then the lattice L acts on $\mathbb{C}^{(m,n)}$ freely by

$$(\xi,\ \eta) \cdot W = W + \xi\Omega + \eta, \quad \xi, \eta \in \mathbb{Z}^{(m,n)}, \quad W \in \mathbb{C}^{(m,n)}.$$

Lemma 1.8.16 *Let $A, B \in \mathbb{R}^{(m,n)}$. Let $J_{S,\Omega,A,B} : L \times \mathbb{C}^{(m,n)} \longrightarrow \mathbb{C}^\times$ be the mapping defined by*

$$J_{S,\Omega,A,B}(\ell, W) := e^{\pi i\, \sigma\{S(\xi\Omega\, {}^t\xi + 2\, W\, {}^t\xi)\}} \cdot e^{-2\pi i\, \sigma\{S(A\, {}^t\eta - B\, {}^t\xi)\}}, \tag{1.8.15}$$

where $\ell = (\xi, \eta) \in L$ and $W \in \mathbb{C}^{(m,n)}$. Then $J_{S,\Omega,A,B}$ is an automorphic factor for the lattice L.

Proof For brevity, we write $J := J_{S,\Omega,A,B}$. For any two elements $\ell_i = (\xi_i,\ \eta_i)\ (i = 1, 2)$ of L and $W \in \mathbb{C}^{(m,n)}$, we must show that

$$J(\ell_1 + \ell_2,\ W) = J(\ell_1,\ \ell_2 \cdot W)\, J(\ell_2, W). \tag{1.8.16}$$

Using the fact that $\sigma(2S\eta_2\, {}^t\xi_1)$ is an even integer, an easy computation yields (1.8.16). □

The Heisenberg group $(\mathbb{H}_{\mathbb{R}}^{(n,m)}, \diamond)$ (see Formula (1.1.8)) with multiplication $\diamond$ acts on $\mathbb{C}^{(m,n)}$ by

$$[\lambda_0, \mu_0, \kappa_0] \cdot (\lambda\Omega + \mu) := (\lambda_0 + \lambda)\Omega + (\mu_0 + \mu), \quad \lambda, \mu \in \mathbb{R}^{(m,n)}.$$

Since the center $\mathfrak{Z} = \left\{ [0, 0, \kappa] \mid \kappa = {}^t\kappa \in \mathbb{R}^{(m,m)} \right\}$ of $\mathbb{H}_{\mathbb{R}}^{(n,m)}$ is the stabilizer of $\mathbb{H}_{\mathbb{R}}^{(n,m)}$ at 0, the homogeneous space $\mathbb{H}_{\mathbb{R}}^{(n,m)}/\mathfrak{Z}$ is identified with $\mathbb{C}^{(m,n)}$ via

$$[\lambda, \mu, \kappa] \cdot \mathfrak{Z} \longmapsto [\lambda, \mu, \kappa] \cdot 0 = \lambda\Omega + \mu.$$

Thus the automorphic factor $J_{S,\Omega,A,B} : L \times \mathbb{C}^{(m,n)} \longrightarrow GL(1, \mathbb{C})$ for the lattice L may be lifted to the automorphic factor $\tilde{J}_{S,\Omega,A,B} : \mathbb{H}_{\mathbb{R}}^{(n,m)} \times \mathbb{C}^{(m,n)} \longrightarrow \mathbb{C}^\times$ defined by

$$\tilde{J}_{S,\Omega,A,B}(g_0, W) = e^{\pi i\, \sigma\{S(\lambda\Omega\, {}^t\lambda + 2\, W\, {}^t\lambda + \kappa)\}} \cdot e^{-\pi i\, \sigma\{S(A\, {}^t\mu - B\, {}^t\lambda)\}}, \tag{1.8.17}$$

where $g_0 = [\lambda, \mu, \kappa] \in \mathbb{H}_{\mathbb{R}}^{(n,m)}$.

We denote by $\mathcal{A}_{S,\Omega}$ the complex vector space consisting of $\mathbb{C}$-valued smooth functions φ on $\mathbb{H}_{\mathbb{R}}^{(n,m)}$ satisfying the following conditions:

(*a*) $\varphi([\xi, \eta, 0] \diamond g_0) = \varphi(g_0)$ for all $\xi, \eta \in \mathbb{Z}^{(m,n)}$ and $g_0 \in \mathbb{H}_{\mathbb{R}}^{(n,m)}$,
(*b*) $\varphi(g_0 \diamond [0, 0, \kappa]) = e^{\pi i\, \sigma(S\kappa)}\, \varphi(g_0)$ for all $\kappa \in \mathrm{Sym}(m, \mathbb{R})$ and $g_0 \in \mathbb{H}_{\mathbb{R}}^{(n,m)}$,
(*c*) $\left(\mathfrak{L}_{X_{ka}} - \sum_{b=1}^{n} \Omega_{ab} \mathfrak{L}_{\widehat{X}_{kb}} \right)\varphi = 0$ for all $1 \le k \le m$ and $1 \le a \le n$.

Here, $\Omega = (\Omega_{ab}) \in \mathbb{H}_n$ and if X is an element of the Lie algebra of $\mathbb{H}_{\mathbb{R}}^{(n,m)}$,

$$\left(\mathfrak{L}_X \varphi \right)(g_0) = \frac{d}{dt}\bigg|_{t=0} \varphi(g_0 \diamond \exp t X), \quad g_0 \in \mathbb{H}_{\mathbb{R}}^{(n,m)}.$$

Theorem 1.8.17 *Let S and Ω be as before. Then the vector space R_S^Ω is isomorphic to the vector space $\mathcal{A}_{S,\Omega}$ via the mapping*

$$f \longmapsto \varphi_f(g_0) := \tilde{J}_{S,\Omega,0,0}(g_0, 0)\, f(g_0 \cdot 0),$$

where $g_0 \in \mathbb{H}_{\mathbb{R}}^{(n,m)}$ and $f \in R_S^\Omega$. The inverse of the above isomorphism is given by

$$\varphi \longmapsto f_\varphi(W) := \tilde{J}_{S,\Omega,0,0}(g_0, 0)^{-1}\, \varphi(g_0), \quad \varphi \in \mathcal{A}_{S,\Omega},$$

where $W = g_0 \cdot 0$. This definition does not depend on the choice of g_0 with $W = g_0 \cdot 0$.

Proof For brevity, we write $\tilde{J} := \tilde{J}_{S,\Omega,0,0}$. If $\gamma = [\xi, \eta, 0] \in \mathbb{H}_{\mathbb{R}}^{(n,m)}$ with $\xi, \eta \in \mathbb{Z}^{(m,n)}$, we have for all $g_0 \in H_{\mathbb{R}}^{(n,m)}$

$$\begin{aligned}
\varphi_f(\gamma \diamond g_0) &= \tilde{J}(\gamma \diamond g_0,\ 0)\ f((\gamma \diamond g_0) \cdot 0) \\
&= \tilde{J}(\gamma,\ g_0 \cdot 0)\ \tilde{J}(g_0,\ 0)\ f(g_0 \cdot 0 + \xi\,\Omega + \eta) \\
&= \tilde{J}(\gamma,\ g_0 \cdot 0)\ \tilde{J}(g_0, 0)\ J((\xi, \eta),\ g_0 \cdot 0)^{-1}\ f(g_0 \cdot 0) \\
&= \tilde{J}(\gamma, 0)\ f(g_0 \cdot 0) \\
&= \varphi_f(g_0).
\end{aligned}$$

And if $\kappa = {}^t\kappa \in \mathbb{R}^{(m,m)}$, we have

$$\begin{aligned}
\varphi_f(g_0 \diamond [0, 0, \kappa]) &= \tilde{J}(g_0 \diamond [0, 0, \kappa],\ 0)\ f((g_0 \diamond [0, 0, \kappa]) \cdot 0) \\
&= \tilde{J}(g_0,\ [0, 0, \kappa] \cdot 0)\ \tilde{J}([0, 0, \kappa],\ 0)\ f(g_0 \cdot 0) \\
&= e^{\pi i\, \sigma(S\kappa)}\ \tilde{J}(g_0, 0)\ f(g_0 \cdot 0) \\
&= e^{\pi\, i\, \sigma(S\kappa)}\ \varphi_f(g_0).
\end{aligned}$$

We introduce a system of complex coordinates on $\mathbb{C}^{(m,n)}$ with respect to Ω:

$$W = \lambda\,\Omega + \mu, \quad \overline{W} = \lambda\,\overline{\Omega} + \mu, \quad \lambda, \mu \text{ real}.$$

We set

$$dW = \begin{pmatrix} dW_{11} & dW_{12} & \dots & dW_{1n} \\ dW_{21} & dW_{22} & \dots & dW_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ dW_{m1} & dW_{m2} & \dots & dW_{mn} \end{pmatrix}, \quad \frac{\partial}{\partial W} = \begin{pmatrix} \frac{\partial}{\partial W_{11}} & \frac{\partial}{\partial W_{21}} & \cdots & \frac{\partial}{\partial W_{m1}} \\ \frac{\partial}{\partial W_{12}} & \frac{\partial}{\partial W_{22}} & \cdots & \frac{\partial}{\partial W_{m2}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial W_{1n}} & \frac{\partial}{\partial W_{2n}} & \cdots & \frac{\partial}{\partial W_{mn}} \end{pmatrix}.$$

Then an easy computation yields

$$\begin{aligned}
\frac{\partial}{\partial \lambda} &= \Omega \frac{\partial}{\partial W} + \overline{\Omega} \frac{\partial}{\partial \overline{W}}, \\
\frac{\partial}{\partial \mu} &= \frac{\partial}{\partial W} + \frac{\partial}{\partial \overline{W}}.
\end{aligned}$$

Thus we obtain the following:

$$\frac{\partial}{\partial \overline{W}} = \frac{i}{2}\ (\operatorname{Im}\Omega)^{-1} \left(\frac{\partial}{\partial \lambda} - \Omega \frac{\partial}{\partial \mu} \right). \tag{1.8.18}$$

Since f is holomorphic, according to (1.8.18), f satisfies the conditions

$$\left(\frac{\partial}{\partial \lambda_{ka}} - \sum_{b=1}^{n} \Omega_{ab} \frac{\partial}{\partial \mu_{kb}} \right) f(W) = 0, \quad 1 \le k \le m,\ 1 \le a \le n. \tag{1.8.19}$$

Conversely, if a smooth function on $\mathbb{C}^{(m,n)}$ satisfies the condition (1.8.19), it is holomorphic.

In order to prove that φ_f satisfies the condition (c), we first compute $\mathfrak{L}_{X_{ka}}\varphi_f$ and $\mathfrak{L}_{\widehat{X}_{lb}}\varphi_f$ for $1 \le k, l \le m$ and $1 \le a, b \le n$. If $g = [\lambda, \mu, \kappa] \in \mathbb{H}_{\mathbb{R}}^{(n,m)}$ and $S = (s_{kl})$,

$$\begin{aligned}
\left(\mathfrak{L}_{X_{ka}} \varphi_f \right) &= \frac{d}{dt}\bigg|_{t=0} \varphi_f(g \diamond \exp t X_{ka}) \\
&= \frac{d}{dt}\bigg|_{t=0} \varphi_f([\lambda, \mu, \kappa] \diamond [tE_{ka}, 0, 0]) \\
&= \frac{d}{dt}\bigg|_{t=0} \tilde{J}([\lambda + tE_{ka}, \mu, \kappa],\ 0)\, f((\lambda + tE_{ka})\Omega + \mu) \\
&= \frac{d}{dt}\bigg|_{t=0} e^{\pi i\, \sigma\{S(\lambda + tE_{ka})\Omega\, {}^t(\lambda + tE_{ka})\}}\, e^{\pi i\, \sigma(S\kappa)}\, f((\lambda + tE_{ka})\Omega + \mu) \\
&= e^{\pi i\, \sigma\{S(\kappa + \lambda\Omega\, {}^t\lambda)\}} \left\{ 2\pi i \left(\sum_{b=1}^{n} \sum_{l=1}^{m} s_{kl} \Omega_{ab} \lambda_{ab} \right) + \frac{\partial}{\partial \lambda_{ka}} \right\} f(W).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\left(\mathfrak{L}_{\widehat{X}_{lb}} \varphi_f \right)(g) &= \frac{d}{dt}\bigg|_{t=0} \varphi_f(g \diamond \exp t \widehat{X}_{lb}) \\
&= \frac{d}{dt}\bigg|_{t=0} \varphi_f([\lambda, \mu, \kappa] \diamond [0, tE_{lb}, 0]) \\
&= \frac{d}{dt}\bigg|_{t=0} \varphi_f([\lambda, \mu + tE_{lb}, \kappa + t\lambda\, {}^t E_{lb} + tE_{lb}\, {}^t\lambda]) \\
&= e^{\pi i\, \sigma\{S(\kappa + \lambda\Omega\, {}^t\lambda)\}} \frac{d}{dt}\bigg|_{t=0} e^{2\pi i t\, \sigma(S\lambda\, {}^t E_{lb})}\, f(\lambda\Omega + (\mu + tE_{lb})) \\
&= e^{\pi i\, \sigma\{S(\kappa + \lambda\Omega\, {}^t\lambda)\}} \left\{ 2\pi i \left(\sum_{p=1}^{m} s_{lp} \lambda_{pb} \right) + \frac{\partial}{\partial \mu_{lb}} \right\} f(W).
\end{aligned}$$

Thus

$$\left(\mathfrak{L}_{X_{ka}} - \sum_{b=1}^{n} \Omega_{ab}\, \mathfrak{L}_{\widehat{X}_{kb}}\right)\varphi_f(g)$$

$$= e^{\pi i\, \sigma\{S(\kappa+\lambda\Omega\,{}^t\lambda)\}} \left\{ \frac{\partial}{\partial \lambda_{ka}} - \sum_{b=1}^{n} \Omega_{ab} \frac{\partial}{\partial \mu_{kb}} \right\} f(W) = 0.$$

This completes the proof. □

1.9 Characterization of Theta Functions as Sections of a Line Bundle over an Abelian Variety

In this section, we will show that a theta function is characterized as a global section of a holomorphic line bundle over a complex abelian variety.

First of all we briefly review the theory of line bundles over an abelian variety. Let $A = \mathbb{C}^g/L$ be a complex abelian variety of dimension g equipped with an ample line bundle over A, where L is a lattice of rank $2g$ in $\mathbb{C}^g$. Then A is a projective variety which is a connected, compact commutative complex Lie group. We define $\mathbb{C}_1^\times := \{\xi \in \mathbb{C} \mid |\xi| = 1\}$. For an integer k with $0 \le k \le 2g$, we denote by $H^k(A)$ the space of all harmonic k-forms on A. Since $A \cong (\mathbb{C}_1^\times)^{2g}$ topologically, $\dim_{\mathbb{C}} H^k(A) = \binom{2g}{k}$ with $0 \le k \le 2g$. The Hodge decomposition of A is given

$$H^g(A, \mathbb{C}) \cong \bigoplus_{p+q=g} H^p(A, \Omega^q),$$

where Ω^p is the sheaf of holomorphic p-forms on A. We have canonical isomorphisms

$$H^r(A, \mathbb{Z}) \cong \{\text{group of alternating } r-\text{forms } L \times \cdots \times L \longrightarrow \mathbb{Z}\}.$$

See [33, pp. 3–4] for details.

Definition 1.9.1 Let $A = \mathbb{C}^g/L$ be an abelian variety with a lattice L in $\mathbb{C}^g$. A Hermitian form H on $\mathbb{C}^g$ is called a Riemann form for A if it satisfies the following conditions (R1) and (R2):

(R1) H is nondegenerate;
(R2) The imaginary part $E = \operatorname{Im} H$ of H is integral valued on $L \times L$.

Let $A = \mathbb{C}^g/L$ be an abelian variety and let H be a Riemann form for A. A map $\chi : L \longrightarrow T$ is said to be a semi–character of L with respect to $E = \operatorname{Im} H$ if it

satisfies the following condition

$$\chi(\alpha+\beta) = \chi(\alpha)\,\chi(\beta)\exp\{i\,\pi\,E(\alpha,\beta)\}, \quad \alpha,\beta \in L. \tag{1.9.1}$$

For any Riemann form H on $\mathbb{C}^g$, a semi-character χ_H satisfying the condition (1.9.1) always exists uniquely (cf. [33, p. 20]). The mapping $J_{H,\chi_H} : L \times \mathbb{C}^g \longrightarrow GL(1,\mathbb{C})$ defined by

$$J_{H,\chi_H}(\alpha,z) \overset{\text{def}}{=} \chi_H(\alpha)\exp\left\{\pi H(z,\alpha)+\frac{\pi}{2}H(\alpha,\alpha)\right\}, \quad \alpha \in L,\ z \in \mathbb{C}^g \tag{1.9.2}$$

satisfies the following equation

$$J_{H,\chi_H}(\alpha+\beta,z) = J_{H,\chi_H}(\alpha,\beta+z)J_{H,\chi_H}(\beta,z)$$

for all $\alpha, \beta \in L$ and $z \in \mathbb{C}^g$. In other words, J_{H,χ_H} is an automorphic factor of rank one with respect to the lattice L in $\mathbb{C}^g$. Let $L(H,\chi_H)$ be the holomorphic line bundle over A defined by the automorphic factor J_{H,χ_H} for L. More precisely, $L(H,\chi_H)$ is expressed as follows. We first observe that the lattice L acts on $\mathbb{C}^g \times \mathbb{C}$ by

$$\rho_\alpha(z,\xi) := (z+\alpha,\, J_{H,\chi_H}(\alpha,z)\,\xi), \tag{1.9.3}$$

where $\alpha \in L$, $z \in \mathbb{C}^g$ and $\xi \in \mathbb{C}$. $L(H,\chi_H)$ is the quotient of the trivial line bundle $\mathbb{C}^g \times \mathbb{C}$ over $\mathbb{C}^g$ for the action of L given by (1.9.3). The global sections of $L(H,\chi_H)$ are essentially the global sections θ of the trivial line bundle $\mathbb{C}^g \times \mathbb{C}$ over $\mathbb{C}^g$ (i.e. holomorphic functions θ on $\mathbb{C}^g$) which are invariant under the action (1.9.3) of L on $\mathbb{C}^g \times \mathbb{C}$, that is, which satisfies the functional equation

$$\theta(z+\alpha) = J_{H,\chi_H}(\alpha,z)\,\theta(z), \quad z \in \mathbb{C}^g,\ \alpha \in L. \tag{1.9.4}$$

In other words, a function θ is a theta function for H and χ_H. We can say that the cohomology group $H^0(A, L(H,\chi_H))$ is the space of theta functions for H and χ_H. According to [33, Proposition, p. 26], we have

$$\dim\, H^0(A, L(H,\chi_H)) = \sqrt{\det E}. \tag{1.9.5}$$

According to the Lefschetz theorem ([33, pp. 29–30]), if H is a positive definite Riemann form for A, the line bundle $L(H,\chi_H)$ is ample, that is, the space of holomorphic sections $L(H,\chi_H)^{\otimes n}$ $(n \geq 3)$ gives an embedding of A as a closed complex submanifold into a projective space $\mathbb{P}^N$. Therefore A is a smooth projective variety of dimension g.

Without difficulty we can show that there is an isomorphism of holomorphic line bundles over A

$$L(H_1,\chi_1)\otimes L(H_2,\chi_2) \cong L(H_1+H_2,\chi_1\chi_2).$$

Therefore the set of all $L(H, \chi)$ forms a group under a tensor product. We note that two holomorphic line bundles $L(H, \chi)$ and $L(\widetilde{H}, \widetilde{\chi})$ over A are isomorphic if and only if their automorphic factors $J_{H,\chi}$ and $J_{\widetilde{H},\widetilde{\chi}}$ are holomorphically equivalent, i.e., there exists a holomorphic mapping $h : \mathbb{C}^g \longrightarrow GL(1, \mathbb{C})$ such that

$$h(z + \alpha) J_{H,\chi}(\alpha, z) = h(z) J_{\widetilde{H},\widetilde{\chi}}(\alpha, z)$$

for all $\alpha \in L$ and $z \in \mathbb{C}^g$.

We denote by $\mathrm{Pic}(A)$ the group of all holomorphic line bundles over an abelian variety $A = \mathbb{C}^g/L$ whose multiplication is the tensor product. Let $\mathrm{Pic}^0(A)$ be the group of holomorphic line bundles over an abelian variety A with zero Chern class. In fact, $\mathrm{Pic}^0(A)$ is an abelian variety called the dual abelian variety.

Now we have the famous theorem.

Appell–Humbert Theorem *Any holomorphic line bundle over an abelian variety A is isomorphic to the line bundle $L(H, \chi)$ for a uniquely determined Riemann form H for A and a uniquely determined semi-character χ.*

Proof The proof of the above theorem can be found in [33, pp. 20–22]. □

We fix an element $\Omega \in \mathbb{H}_n$ and a positive definite integral symmetric matrix S of degree m. Let

$$L_\Omega \stackrel{\text{def}}{=} \mathbb{Z}^{(m,n)}\Omega + \mathbb{Z}^{(m,n)}$$

be the lattice in $\mathbb{C}^{(m,n)}$. We define the mapping $J_{\Omega,S} : L_\Omega \times \mathbb{C}^{(m,n)} \longrightarrow GL(1, \mathbb{C})$ by

$$J_{\Omega,S}(\alpha, W) \stackrel{\text{def}}{=} e^{-\pi i \sigma\{S(\lambda\Omega\, {}^t\lambda + 2\, W\, {}^t\lambda)\}}, \tag{1.9.6}$$

where $\alpha = \lambda\,\Omega + \mu$ with $\lambda, \mu \in \mathbb{Z}^{(m,n)}$ and $W \in \mathbb{C}^{(m,n)}$. Then $J_{\Omega,S}$ is an automorphic factor for L_Ω on $\mathbb{C}^{(m,n)}$. Indeed, for any two elements $\alpha = \lambda\,\Omega + \mu$ and $\beta = \xi\,\Omega + \eta$ in L_Ω with $\lambda, \mu, \xi, \eta \in \mathbb{Z}^{(m,n)}$, we have

$$J_{\Omega,S}(\alpha + \beta, W) = e^{-\pi i \sigma\{S((\lambda+\xi)\Omega\, {}^t(\lambda+\xi) + 2\, W\, {}^t(\lambda+\xi))\}}$$

and

$$\begin{aligned} &J_{\Omega,S}(\alpha, W + \beta) J_{\Omega,S}(\beta, W) \\ =&e^{-\pi i \sigma\{S(\lambda\Omega\, {}^t\lambda + 2\,(W+\beta)\, {}^t\lambda)\}} \cdot e^{-\pi i \sigma\{S(\xi\Omega\, {}^t\xi + 2\, W\, {}^t\xi)\}}. \end{aligned}$$

Since $\sigma(S\eta\, {}^t\lambda)$ is an integer, for any $\alpha, \beta \in L_\Omega$ and $W \in \mathbb{C}^{(m,n)}$, we get

$$J_{\Omega,S}(\alpha + \beta, W) = J_{\Omega,S}(\alpha, W + \beta)\, J_{\Omega,S}(\beta, W).$$

L_Ω acts on $\mathbb{C}^{(m,n)} \times \mathbb{C}$ by

$$\rho_\alpha(W, \xi) \stackrel{\text{def}}{=} (W + \alpha,\, J_{\Omega,S}(\alpha, W)\, \xi), \tag{1.9.7}$$

where $\alpha = \lambda\, \Omega + \mu \in L_\Omega$, $W \in \mathbb{C}^{(m,n)}$ and $\xi \in \mathbb{C}$. Let

$$\mathfrak{L}_{\Omega,S} \stackrel{\text{def}}{=} (\mathbb{C}^{(m,n)} \times \mathbb{C})/L_\Omega$$

be the quotient space of the trivial line bundle over $\mathbb{C}^{(m,n)}$ by L_Ω. Then $\mathfrak{L}_{\Omega,S}$ is a holomorphic line bundle over the complex abelian variety $A_\Omega := \mathbb{C}^{(m,n)}/L_\Omega$ of dimension mn. Let $\vartheta(W)$ be a global section of $\mathfrak{L}_{\Omega,S}$. Then

$$(W, \vartheta(W)) \sim (W + \alpha, \vartheta(W + \alpha)) \qquad \text{for all } \alpha \in L_\Omega.$$

According to (1.9.7), for any $\alpha = \lambda\, \Omega + \mu$ and $W \in \mathbb{C}^{(m,n)}$, we get

$$\vartheta(W + \alpha) = \vartheta(W + \lambda\, \Omega + \mu) = J_{\Omega,S}(\alpha, W)\vartheta(W).$$

Therefore $\vartheta(W)$ is a theta function of level S with respect to Ω (see Formula (1.8.1)).

Conversely, if $\theta : \mathbb{C}^{(m,n)} \longrightarrow \mathbb{C}$ is a theta function of level S with respect to Ω, then θ satisfies the following equation:

$$\theta(W + \lambda\Omega + \mu) = e^{-\pi\, i\, \sigma\{S(\lambda\Omega\, {}^t\lambda + 2\, W\, {}^t\lambda)\}}\theta(W),$$

where $W \in \mathbb{C}^{(m,n)}$ and $\lambda\Omega + \mu \in L_\Omega$ with $\lambda, \mu \in \mathbb{Z}^{(m,n)}$. In other words,

$$\theta(W + \alpha) = J_{\Omega,S}(\alpha, W)\theta(W)$$

for all $W \in \mathbb{C}^{(m,n)}$ and $\alpha \in L_\Omega$. Therefore $\theta(W)$ is a global section of the holomorphic line bundle $\mathfrak{L}_{\Omega,S}$ over the abelian variety A_Ω.

We fix an element $\Omega \in \mathbb{H}_n$ and a positive definite integral symmetric matrix S of degree m. Let

$$A_\Omega = \mathbb{C}^{(m,n)}/L_\Omega$$

be the complex abelian variety, where $L_\Omega = \mathbb{Z}^{(m,n)}\Omega + \mathbb{Z}^{(m,n)}$ is a lattice in $\mathbb{C}^{(m,n)}$.

Lemma 1.9.2 *The Hermitian form $H_{S,\Omega}$ defined by (1.8.8) is a positive definite Riemann form for A_Ω* (see Definition 1.9.1).

Proof For brevity, we put $H := H_{S,\Omega}$. It is easily seen that H is a positive definite Hermitian form on $\mathbb{C}^{(m,n)}$. Let $\mathbb{S}$ and $\mathbb{A}$ be the real part of H and the imaginary part

of H respectively. Then

$$H(W_1, W_2) = \mathbb{S}(W_1, W_2) + i\,\mathbb{A}(W_1, W_2) = \mathbb{A}(i\,W_1, W_2) + i\,\mathbb{A}(W_1, W_2),$$

where $W_1, W_2 \in \mathbb{C}^{(m,n)}$. If $\alpha = \lambda\,\Omega + \mu,\ \beta = \xi\,\Omega + \eta \in L_\Omega$ with $\lambda, \mu, \xi, \eta \in \mathbb{Z}^{(m,n)}$, then

$$\mathbb{A}(\alpha, \beta) = \sigma\{S(\lambda\,{}^t\eta - \mu\,{}^t\xi)\}$$

is integral valued on $L_\Omega \times L_\Omega$. Thus H is a Riemann form for A_Ω. □

Let χ_H be the semi-character of L_Ω with respect to $\mathbb{A}$. Such a χ_H is determined uniquely by L_Ω and $\mathbb{A}$ (cf. [33, p. 20]). We define the mapping $J_H : L_\Omega \times \mathbb{C}^{(m,n)} \longrightarrow GL(1, \mathbb{C})$ by

$$J_H(\alpha, W) \stackrel{\text{def}}{=} \chi_H(\alpha) \exp\left\{\pi H(W, \alpha) + \frac{\pi}{2} H(\alpha, \alpha)\right\}, \tag{1.9.8}$$

where $\alpha \in L_\Omega$ and $W \in \mathbb{C}^{(m,n)}$. Then J_H is an automorphic factor with respect to L_Ω, that is, J_H satisfies the following equation,

$$J_H(\alpha + \beta, W) = J_H(\alpha, W + \beta)\, J_H(\beta, W),$$

for all $\alpha, \beta \in L_\Omega$ and $W \in \mathbb{C}^{(m,n)}$. We see that the lattice L_Ω acts on the trivial line bundle $\mathbb{C}^{(m,n)} \times \mathbb{C}$ over $\mathbb{C}^{(m,n)}$ by

$$\alpha \cdot (W, \xi) := (W + \alpha, J_H(\alpha, W)\,\xi), \tag{1.9.9}$$

where $\alpha \in L_\Omega,\ W \in \mathbb{C}^{(m,n)}$ and $\xi \in \mathbb{C}$. Let $\mathfrak{L}(H, \chi_H)$ be the holomorphic line bundle over A_Ω determined by the action (1.9.9), that is, the quotient of the trivial line bundle $\mathbb{C}^{(m,n)} \times \mathbb{C}$ over $\mathbb{C}^{(m,n)}$ for the action (1.9.9) of L_Ω. If Θ is a global section of $\mathfrak{L}(H, \chi_H)$, then

$$\Theta(W + \alpha) = J_H(\alpha, W)\,\Theta(W), \quad \alpha \in L_\Omega,\ W \in \mathbb{C}^{(m,n)}.$$

Thus Θ is a theta function for H and χ_H. Since H is a positive definite Riemann form by Lemma 1.9.2, according to the Lefschetz theorem (cf. [33, pp. 29–30]), the line bundle $\mathfrak{L}(H, \chi_H)$ is ample, that is, the space of holomorphic sections $\mathfrak{L}(H, \chi_H)^{\otimes n}$ $(n \geq 3)$ gives an embedding of A_Ω as a closed complex submanifold into a projective space $\mathbb{P}^N$.

Remark 1.9.3 We refer to [51] for details on holomorphic vector bundles of rank r with $r > 1$ over an abelian variety.

Chapter 2
The Schrödinger–Weil Representation

2.1 The Symplectic Group

We recall that

$$Sp(2n, \mathbb{R}) = \{M \in \mathbb{R}^{(2n,2n)} \mid {}^tM J_n M = J_n\}$$

is the symplectic group of degree n, where

$$J_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

If $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$ with $A, B, C, D \in \mathbb{R}^{(n,n)}$, then

$$ {}^tAD - {}^tCB = I_n, \qquad {}^tAC = {}^tCA, \qquad {}^tBD = {}^tDB. \tag{2.1.1}$$

We note that $Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$. The inverse of $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$ is

$$M^{-1} = J_n^{-1}\, {}^tM\, J_n = \begin{pmatrix} {}^tD & -{}^tB \\ -{}^tC & {}^tA \end{pmatrix}.$$

Since $J_n^{-1} = -J_n$ and ${}^tM^{-1} J_n M^{-1} = J_n$ with $M \in Sp(2n, \mathbb{R})$, we see that

$${}^tM^{-1} J_n^{-1} M^{-1} = J_n^{-1}, \quad \text{that is,} \quad M J_n\, {}^tM = J_n.$$

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Thus if $M \in Sp(2n, \mathbb{R})$, then ${}^tM \in Sp(2n, \mathbb{R})$. If $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$, then

$$A\,{}^tD - B\,{}^tC = I_n, \qquad A\,{}^tB = B\,{}^tA, \qquad C\,{}^tD = D\,{}^tC. \tag{2.1.2}$$

Lemma 2.1.1 *Let* $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$ *and* $\Omega \in \mathbb{H}_n$. *Then:*

(a) $C\Omega + D$ *is nonsingular.*
(b) $(A\Omega + B)(C\Omega + D)^{-1}$ *is an element of* $\mathbb{H}_n$.

Proof Let $\Omega = X + i\,Y \in \mathbb{H}_n$ with $X, Y \in \mathbb{R}^{(n,n)}$ and $Y > 0$. Then

$${}^t\begin{pmatrix} \Omega \\ I_n \end{pmatrix} J_n \begin{pmatrix} \Omega \\ I_n \end{pmatrix} = 0 \tag{2.1.3}$$

and

$${}^t\begin{pmatrix} \Omega \\ I_n \end{pmatrix} J_n \overline{\begin{pmatrix} \Omega \\ I_n \end{pmatrix}} = 2\,i\,Y. \tag{2.1.4}$$

We set

$$S = A\Omega + B \qquad \text{and} \qquad T = C\Omega + D.$$

By (2.1.3), we have

$$\begin{aligned} {}^t\begin{pmatrix} S \\ T \end{pmatrix} J_n \begin{pmatrix} S \\ T \end{pmatrix} &= {}^t\left\{ M \begin{pmatrix} \Omega \\ I_n \end{pmatrix} \right\} J_n \left\{ M \begin{pmatrix} \Omega \\ I_n \end{pmatrix} \right\} \\ &= {}^t\begin{pmatrix} \Omega \\ I_n \end{pmatrix} {}^tM\, J_n M \begin{pmatrix} \Omega \\ I_n \end{pmatrix} \\ &= {}^t\begin{pmatrix} \Omega \\ I_n \end{pmatrix} J_n \begin{pmatrix} \Omega \\ I_n \end{pmatrix} = 0. \end{aligned}$$

By (2.1.4), we have

$$\frac{1}{2i}\,{}^t\begin{pmatrix} S \\ T \end{pmatrix} J_n \overline{\begin{pmatrix} S \\ T \end{pmatrix}} = \frac{1}{2i}\,{}^t\begin{pmatrix} \Omega \\ I_n \end{pmatrix} J_n \overline{\begin{pmatrix} \Omega \\ I_n \end{pmatrix}} = Y > 0.$$

Thus we have

$${}^tST - {}^tT\overline{S} = 0 \qquad \text{and} \qquad \frac{1}{2\,i}\left({}^tS\overline{T} - {}^tT\overline{S}\right) = Y > 0. \tag{2.1.5}$$

Assume $Tv = (C\Omega + D)v = 0$ for some $v \in \mathbb{C}^n$. Then $\overline{T}\overline{v} = 0$, ${}^t v\, {}^t T = 0$ and hence

$$\frac{1}{2i}\, {}^t v\big({}^t S\overline{T} - {}^t T\overline{S}\big)\overline{v} = 0.$$

By (2.1.5), $v = 0$ and so $T = C\Omega + D$ is nonsingular. This proves the statement (a).

We set

$$\Omega_* = (A\Omega + B)(C\Omega + D)^{-1} = ST^{-1}.$$

By (2.1.5), we have $\Omega_* = {}^t\Omega_*$ and

$$\begin{aligned}\operatorname{Im}\Omega_* &= \frac{1}{2i}\big(\Omega_* - \overline{\Omega}_*\big) = \frac{1}{2i}\big({}^t\Omega_* - \overline{\Omega}_*\big)\\ &= \frac{1}{2i}\, {}^t T^{-1}\big({}^t S\overline{T} - {}^t T\overline{S}\big)\,\overline{T}^{-1}\\ &= {}^t T^{-1} Y\, \overline{T}^{-1} > 0.\end{aligned}$$

Therefore $\Omega_* \in \mathbb{H}_n$. This completes the proof of the statement (b). □

Lemma 2.1.2 *The symplectic group $Sp(2n, \mathbb{R})$ acts on the Siegel upper half plane $\mathbb{H}_n$ transitively by*

$$M \cdot \Omega = (A\Omega + B)(C\Omega + D)^{-1}, \tag{2.1.6}$$

where $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$ *and* $\Omega \in \mathbb{H}_n$.

Proof Let $\Omega = X + iY \in \mathbb{H}_n$ with $X, Y \in \mathrm{Sym}(n, \mathbb{R})$ and $Y > 0$. It suffices to show that there exists an element $M \in Sp(2n, \mathbb{R})$ such that $M \cdot (iI_n) = \Omega$. We choose $Q \in GL(n, \mathbb{R})$ such that ${}^t Q\, Q = Y$. We take

$$M = \begin{pmatrix} I_n & X \\ 0 & I_n \end{pmatrix}\begin{pmatrix} {}^t Q & 0 \\ 0 & Q^{-1} \end{pmatrix}.$$

According to (2.1.2), $M \in Sp(2n, \mathbb{R})$. Clearly, $M \cdot (iI_n) = X + iY = \Omega$. □

It is known (cf. [10, pp. 322–328] or [20, p. 10]) that $Sp(2n, \mathbb{R})$ is generated by the following elements:

$$t_b = \begin{pmatrix} I_n & b \\ 0 & I_n \end{pmatrix} \text{ with } b = {}^t b \in \mathbb{R}^{(n,n)},$$

$$d_a = \begin{pmatrix} {}^t a & 0 \\ 0 & a^{-1} \end{pmatrix} \text{ with } a \in GL(n, \mathbb{R}),$$

$$\sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}.$$

Thus if $M \in Sp(2n, \mathbb{R})$, $\det M = 1$.

A subgroup Γ is said to be *discrete* if $\Gamma \cap K$ is finite for any compact subset K of $Sp(2n, \mathbb{R})$.

Theorem 2.1.3 *A discrete subgroup Γ of $Sp(2n, \mathbb{R})$ acts properly discontinuously on $\mathbb{H}_n$, that is, for any two compact subsets C_1, C_2 of $\mathbb{H}_n$, the set*

$$\{\, \gamma \in \Gamma \mid \gamma \cdot C_1 \cap C_2 \neq \emptyset \,\}$$

is finite.

Proof We can show that the mapping

$$p : Sp(2n, \mathbb{R}) \longrightarrow \mathbb{H}_n, \quad M \longmapsto M \cdot (i I_n), \quad M \in Sp(2n, \mathbb{R})$$

is *proper*, i.e., for any compact subset $X \subset \mathbb{H}_n$, $p^{-1}(X)$ is compact in $Sp(2n, \mathbb{R})$ (cf. [10, pp. 28–29]). Suppose X_1 and X_2 are two compact subsets of $\mathbb{H}_n$. Then $Z_1 = p^{-1}(X_1)$ and $Z_2 = p^{-1}(X_2)$ are compact in $Sp(2n, \mathbb{R})$. Since the image of $Z_2 \times Z_1$ under the continuous mapping $(M_2, M_1) \longmapsto M_2 M_1^{-1}$ is compact, the set

$$Z_2 Z_1^{-1} = \{\, M_2 M_1^{-1} \mid M_1 \in Z_1,\ M_2 \in Z_2 \,\}$$

is compact. It remains to show that $\{\gamma \in \Gamma \mid \gamma \cdot X_1 \cap X_2 \neq \emptyset\}$ is finite. If $\gamma \in \Gamma$ such that $\gamma \cdot X_1 \cap X_2 \neq \emptyset$, then

$$\gamma \cdot \Omega_1 = M_2 \cdot (i I_n) \in X_2 \quad \text{for some } \Omega_1 \in X_1 \text{ and } M_2 \in Z_2.$$

Since $(\gamma^{-1} M_2) \cdot (i I_n) = \Omega_1 \in X_1$, we have $\gamma^{-1} M_2 \in Z_1$, that is, $M_2^{-1} \gamma \in Z_1^{-1}$. Therefore $\gamma \in M_2 Z_1^{-1} \subset Z_2 Z_1^{-1}$. Since $\Gamma \cap Z_2 Z_1^{-1}$ is finite, the set $\{\gamma \in \Gamma \mid \gamma \cdot X_1 \cap X_2 \neq \emptyset\}$ is finite. □

By Theorem 2.1.3, the Siegel modular group $\Gamma_n = Sp(2n, \mathbb{Z})$ acts properly discontinuously on $\mathbb{H}_n$. Therefore the stabilizer $(\Gamma_n)_\Omega$ of $\Omega \in \mathbb{H}_n$ given by

$$(\Gamma_n)_\Omega = \{\, \gamma \in \Gamma_n \mid \gamma \cdot \Omega = \Omega \,\}$$

is a finite subgroup of Γ_n.

Let q be a positive integer. The set

$$\Gamma_n(q) = \{M \in \Gamma_n \mid M \equiv I_{2n} \bmod q\}$$

is a normal subgroup of Γ_n because it is the kernel of the homomorphism $\Gamma_n \longrightarrow Sp(2n, \mathbb{Z}/q\mathbb{Z})$ defined by $\gamma \longrightarrow \gamma \bmod q$. It is called the *principal congruence subgroup* of *level* q. We have $\Gamma_n(1) = \Gamma_n$. A subgroup Γ of $Sp(2n, \mathbb{R})$ which contains $\Gamma_n(\ell)$ for some positive integer ℓ as a subgroup of finite index is called a *modular group*. A subgroup Γ of Γ_n which contains $\Gamma_n(\ell)$ for some positive integer ℓ as a subgroup of finite index is called a *congruence subgroup* of Γ_n.

Let Ω_1 and Ω_2 be two points of $\mathbb{H}_n$ and $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$. We write $\Omega_i^* = M \cdot \Omega_i$ $(i = 1, 2)$. Then by the symplectic conditions (2.1.1) and (2.1.2), we have

$$\Omega_2^* - \Omega_1^* = {}^t(C\Omega_2 + D)^{-1}(\Omega_2 - \Omega_1)(C\Omega_1 + D)^{-1} \tag{2.1.7}$$

and

$$\Omega_2^* - \overline{\Omega_1^*} = {}^t(C\Omega_2 + D)^{-1}\big(\Omega_2 - \overline{\Omega_1}\big)(C\overline{\Omega_1} + D)^{-1}. \tag{2.1.8}$$

Let $\Omega = X + iY \in \mathbb{H}_n$ with $X, Y \in \mathbb{R}^{(n,n)}$. If $\Omega^* = M \cdot \Omega$, then we write $\Omega^* = X^* + iY^*$ with $X^*, Y^* \in \mathbb{R}^{(n,n)}$. Then by (2.1.8),

$$\Omega^* - \overline{\Omega^*} = {}^t(C\Omega + D)^{-1}(\Omega - \overline{\Omega})(C\overline{\Omega} + D)^{-1} \tag{2.1.9}$$

and hence

$$Y^* = {}^t(C\Omega + D)^{-1}Y(C\overline{\Omega} + D)^{-1}. \tag{2.1.10}$$

Therefore we obtain

$$\det Y^* = \det Y \cdot |\det(C\Omega + D)|^{-2}. \tag{2.1.11}$$

And

$$\begin{aligned} d\Omega^* &= d(M\cdot\Omega) = d\big\{(A\Omega + B)(C\Omega + D)^{-1}\big\} \\ &= A\,d\Omega\,(C\Omega + D)^{-1} - (A\Omega + B)(C\Omega + D)^{-1}C\,d\Omega\,(C\Omega + D)^{-1} \\ &= {}^t(C\Omega + D)^{-1}\big\{\Omega({}^tCA - {}^tAC) + ({}^tDA - {}^tBC)\big\}d\Omega\,(C\Omega + D)^{-1} \\ &= {}^t(C\Omega + D)^{-1}d\Omega\,(C\Omega + D)^{-1}. \end{aligned}$$

Thus we have

$$d\Omega^* = {}^t(C\Omega + D)^{-1}\,d\Omega\,(C\Omega + D)^{-1}. \tag{2.1.12}$$

By Formulas (2.1.10) and (2.1.12),

$$ds^2 = \sigma(Y^{-1}d\Omega\,Y^{-1}d\overline{\Omega}) \tag{2.1.13}$$

is invariant under the action (2.1.6) of $Sp(2n,\mathbb{R})$. For $\Omega = i\, I_n$,

$$ds^2 = \sum_{i=1} \left(dx_{ii}^2 + dy_{ii}^2\right) + 2 \sum_{1\le i\le j\le n} \left(dx_{ij}^2 + dy_{ij}^2\right).$$

Since $Sp(2n,\mathbb{R})$ acts on $\mathbb{H}_n$ transitively, ds^2 is an $Sp(2n,\mathbb{R})$-invariant Riemannian metric on $\mathbb{H}_n$.

The tangent space $T_\Omega(\mathbb{H}_n)$ of $\mathbb{H}_n$ at Ω is identified with the vector space $\mathrm{Sym}(n,\mathbb{C})$ consisting of all $n \times n$ symmetric complex matrices. By (2.1.12), the differential

$$dM_\Omega : T_\Omega(\mathbb{H}_n) \longrightarrow T_{M\cdot\Omega}(\mathbb{H}_n)$$

of the symplectic transformation M at Ω is given by

$$dM_\Omega(W) = {}^t(C\Omega + D)^{-1}\, W\, (C\Omega + D)^{-1}, \quad W \in \mathrm{Sym}(n,\mathbb{C}). \tag{2.1.14}$$

We can see that the Jacobian of the symplectic transformation $M \in Sp(2n,\mathbb{R})$ is given by

$$\frac{\partial(\Omega^*)}{\partial(\Omega)} = \det(C\Omega + D)^{-(n+1)},$$

where $\Omega^* = M\cdot\Omega$ with $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$.

For $\Omega = (\omega_{ij}) \in \mathbb{H}_n$, we write $\Omega = X + i\, Y$ with $X = (x_{ij})$, $Y = (y_{ij})$ real and $d\Omega = (d\omega_{ij})$. We put

$$\frac{\partial}{\partial\Omega} = \left(\frac{1+\delta_{ij}}{2}\frac{\partial}{\partial\omega_{ij}}\right) \qquad \text{and} \qquad \frac{\partial}{\partial\overline{\Omega}} = \left(\frac{1+\delta_{ij}}{2}\frac{\partial}{\partial\overline{\omega}_{ij}}\right).$$

H. Maaß [27, 28] proved that the Laplacian of the Siegel metric ds^2 (cf. (2.1.13)) is given by

$$\Delta = 4\sigma\left(Y\,{}^t\!\left(Y\frac{\partial}{\partial\overline{\Omega}}\right)\frac{\partial}{\partial\Omega}\right). \tag{2.1.15}$$

And

$$dv_n(\Omega) = (\det Y)^{-(n+1)} \prod_{1\le i\le j\le n} dx_{ij} \prod_{1\le i\le j\le n} dy_{ij} \tag{2.1.16}$$

is a $Sp(2n,\mathbb{R})$-invariant volume element on $\mathbb{H}_n$ (cf. [44, p. 130]).

Theorem 2.1.4 (Siegel [43])

(1) There exists exactly one geodesic joining two arbitrary points Ω_0, Ω_1 in $\mathbb{H}_n$. Let $R(\Omega_0, \Omega_1)$ be the cross-ratio defined by

$$R(\Omega_0, \Omega_1) = (\Omega_0 - \Omega_1)(\Omega_0 - \overline{\Omega}_1)^{-1}(\overline{\Omega}_0 - \overline{\Omega}_1)(\overline{\Omega}_0 - \Omega_1)^{-1}. \tag{2.1.17}$$

For brevity, we put $R_* = R(\Omega_0, \Omega_1)$. Then the symplectic length $\rho(\Omega_0, \Omega_1)$ of the geodesic joining Ω_0 and Ω_1 is given by

$$\rho(\Omega_0, \Omega_1)^2 = \sigma\left(\left(\log \frac{1 + R_*^{\frac{1}{2}}}{1 - R_*^{\frac{1}{2}}}\right)^2\right), \tag{2.1.18}$$

where

$$\left(\log \frac{1 + R_*^{\frac{1}{2}}}{1 - R_*^{\frac{1}{2}}}\right)^2 = 4\, R_* \left(\sum_{k=0}^{\infty} \frac{R_*^k}{2k+1}\right)^2.$$

(2) For $M \in Sp(2n, \mathbb{R})$, we set

$$\widetilde{\Omega}_0 = M \cdot \Omega_0 \quad \text{and} \quad \widetilde{\Omega}_1 = M \cdot \Omega_1.$$

Then $R(\Omega_1, \Omega_0)$ and $R(\widetilde{\Omega}_1, \widetilde{\Omega}_0)$ have the same eigenvalues.

(3) All geodesics are symplectic images of the special geodesics

$$\alpha(t) = i\, \mathrm{diag}(a_1^t, a_2^t, \ldots, a_n^t), \tag{2.1.19}$$

where $a_1, a_2, \ldots, a_n$ are arbitrary positive real numbers satisfying the condition

$$\sum_{k=1}^{n} (\log a_k)^2 = 1.$$

The proof of the above theorem can be found in [43, pp. 289–293].

Let

$$\mathbb{D}_n = \left\{ W \in \mathbb{C}^{(n,n)} \mid W = {}^t W,\ I_n - W\overline{W} > 0 \right\}$$

be the generalized unit disk of degree n. The Cayley transform $\Psi : \mathbb{D}_n \longrightarrow \mathbb{H}_n$ defined by

$$\Psi(W) = i\,(I_n + W)(I_n - W)^{-1}, \quad W \in \mathbb{D}_n \tag{2.1.20}$$

is a biholomorphic mapping of $\mathbb{D}_n$ onto $\mathbb{H}_n$ which gives the bounded realization of $\mathbb{H}_n$ by $\mathbb{D}_n$ (cf. [43]). A. Korányi and J. Wolf [25] gave a realization of a bounded symmetric domain as a Siegel domain of the third kind by investigating a generalized Cayley transform of a bounded symmetric domain that generalizes the Cayley transform Ψ of $\mathbb{D}_n$.

Let

$$T = \frac{1}{\sqrt{2}} \begin{pmatrix} I_n & I_n \\ iI_n & -iI_n \end{pmatrix} \tag{2.1.21}$$

be the $2n \times 2n$ matrix represented by Ψ. Then

$$T^{-1} Sp(2n, \mathbb{R})\, T = \left\{ \begin{pmatrix} P & Q \\ \overline{Q} & \overline{P} \end{pmatrix} \Big| \, {}^t P\overline{P} - {}^t\overline{Q}Q = I_n,\ {}^t P\overline{Q} = {}^t\overline{Q}P \right\}. \tag{2.1.22}$$

Indeed, if $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$, then

$$T^{-1} M T = \begin{pmatrix} P & Q \\ \overline{Q} & \overline{P} \end{pmatrix}, \tag{2.1.23}$$

where

$$P = \frac{1}{2} \left\{ (A + D) + i\,(B - C) \right\} \tag{2.1.24}$$

and

$$Q = \frac{1}{2} \left\{ (A - D) - i\,(B + C) \right\}. \tag{2.1.25}$$

For brevity, we set

$$G_* = T^{-1} Sp(2n, \mathbb{R}) T.$$

Then G_* is a subgroup of $SU(n, n)$, where

$$SU(n, n) = \left\{ h \in \mathbb{C}^{(n,n)} \,\Big|\, {}^t h I_{n,n} \overline{h} = I_{n,n} \right\}, \quad I_{n,n} = \begin{pmatrix} I_n & 0 \\ 0 & -I_n \end{pmatrix}.$$

In the case $n = 1$, we observe that

$$T^{-1} Sp(2, \mathbb{R}) T = T^{-1} SL_2(\mathbb{R}) T = SU(1, 1).$$

If $n > 1$, then G_* is a *proper* subgroup of $SU(n,n)$. In fact, since ${}^tT J_n T = -i J_n$, we get

$$G_* = \left\{ h \in SU(n,n) \,\middle|\, {}^t h J_n h = J_n \right\} = SU(n,n) \cap Sp(2n, \mathbb{C}), \tag{2.1.26}$$

where

$$Sp(2n, \mathbb{C}) = \left\{ \alpha \in \mathbb{C}^{(2n,2n)} \,\middle|\, {}^t\alpha \, J_n \, \alpha = J_n \right\}.$$

Let

$$P^+ = \left\{ \begin{pmatrix} I_n & Z \\ 0 & I_n \end{pmatrix} \,\middle|\, Z = {}^t Z \in \mathbb{C}^{(n,n)} \right\}$$

be the P^+-part of the complexification of $G_* \subset SU(n,n)$. We note that the Harish-Chandra decomposition of an element $\begin{pmatrix} P & Q \\ \overline{Q} & \overline{P} \end{pmatrix}$ in G_* is

$$\begin{pmatrix} P & Q \\ \overline{Q} & \overline{P} \end{pmatrix} = \begin{pmatrix} I_n & Q\overline{P}^{-1} \\ 0 & I_n \end{pmatrix} \begin{pmatrix} P - Q\overline{P}^{-1}\overline{Q} & 0 \\ 0 & \overline{P} \end{pmatrix} \begin{pmatrix} I_n & 0 \\ \overline{P}^{-1}\overline{Q} & I_n \end{pmatrix}.$$

For more details, we refer to [24, p. 155]. Thus the P^+-component of the following element,

$$\begin{pmatrix} P & Q \\ \overline{Q} & \overline{P} \end{pmatrix} \cdot \begin{pmatrix} I_n & W \\ 0 & I_n \end{pmatrix}, \quad W \in \mathbb{D}_n$$

of the complexification of G_* is given by

$$\begin{pmatrix} I_n & (PW + Q)(\overline{Q}W + \overline{P})^{-1} \\ 0 & I_n \end{pmatrix}. \tag{2.1.27}$$

We note that $Q\overline{P}^{-1} \in \mathbb{D}_n$. We get the Harish-Chandra embedding of $\mathbb{D}_n$ into P^+ (cf. [24, p. 155] or [40, pp. 58–59]). Therefore we see that G_* acts on $\mathbb{D}_n$ transitively by

$$\begin{pmatrix} P & Q \\ \overline{Q} & \overline{P} \end{pmatrix} \cdot W = (PW + Q)(\overline{Q}W + \overline{P})^{-1}, \tag{2.1.28}$$

where $\begin{pmatrix} P & Q \\ \overline{Q} & \overline{P} \end{pmatrix} \in G_*$ and $W \in \mathbb{D}_n$. The isotropy subgroup K_* of G_* at the origin o is given by

$$K_* = \left\{ \begin{pmatrix} P & 0 \\ 0 & \overline{P} \end{pmatrix} \;\middle|\; P \in U(n) \right\}.$$

Thus G_*/K_* is biholomorphic to $\mathbb{D}_n$. It is known that the action (2.1.6) is compatible with the action (2.1.28) via the Cayley transform Ψ (cf. (2.1.20)). In other words, if $M \in Sp(2n, \mathbb{R})$ and $W \in \mathbb{D}_n$, then

$$M \cdot \Psi(W) = \Psi(M_* \cdot W), \tag{2.1.29}$$

where $M_* = T^{-1} M T \in G_*$.

For $W = (w_{ij}) \in \mathbb{D}_n$, we write $dW = (dw_{ij})$ and $d\overline{W} = (d\overline{w}_{ij})$. We put

$$\frac{\partial}{\partial W} = \left(\frac{1+\delta_{ij}}{2} \frac{\partial}{\partial w_{ij}} \right) \qquad \text{and} \qquad \frac{\partial}{\partial \overline{W}} = \left(\frac{1+\delta_{ij}}{2} \frac{\partial}{\partial \overline{w}_{ij}} \right).$$

Using the Cayley transform $\Psi : \mathbb{D}_n \longrightarrow \mathbb{H}_n$, Siegel showed (cf. [43]) that

$$ds_*^2 = 4\,\sigma\left((I_n - W\overline{W})^{-1} dW\, (I_n - \overline{W}W)^{-1} d\overline{W} \right) \tag{2.1.30}$$

is a G_*-invariant Riemannian metric on $\mathbb{D}_n$ and Maaß [27] showed that its Laplacian is given by

$$\Delta_* = \sigma\left((I_n - W\overline{W}) \,{}^t\!\left((I_n - W\overline{W}) \frac{\partial}{\partial \overline{W}} \right) \frac{\partial}{\partial W} \right). \tag{2.1.31}$$

Now we discuss the differential operators on $\mathbb{H}_n$ invariant under the action (2.1.6). The isotropy subgroup K at $i\,I_n$ for the action (2.1.6) is a maximal compact subgroup given by

$$K = \left\{ \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \;\middle|\; A\,{}^t\!A + B\,{}^t\!B = I_n,\ A\,{}^t\!B = B\,{}^t\!A,\ A, B \in \mathbb{R}^{(n,n)} \right\}.$$

Let $\mathfrak{k}$ be the Lie algebra of K. Then the Lie algebra $\mathfrak{sp}(2n, \mathbb{R})$ of $Sp(2n, \mathbb{R})$ has a Cartan decomposition $\mathfrak{sp}(2n, \mathbb{R}) = \mathfrak{k} \oplus \mathfrak{p}$, where

$$\mathfrak{p} = \left\{ \begin{pmatrix} X & Y \\ Y & -X \end{pmatrix} \;\middle|\; X = {}^t\!X,\ Y = {}^t\!Y,\ X, Y \in \mathbb{R}^{(n,n)} \right\}.$$

The subspace $\mathfrak{p}$ of $\mathfrak{sp}(2n, \mathbb{R})$ may be regarded as the tangent space of $\mathbb{H}_n$ at $i\,I_n$. The adjoint representation of $Sp(2n, \mathbb{R})$ on $\mathfrak{sp}(2n, \mathbb{R})$ induces the action of K on $\mathfrak{p}$

given by

$$k \cdot Z = kZ\,{}^t k, \quad k \in K,\ Z \in \mathfrak{p}. \tag{2.1.32}$$

Let $\mathbb{T}_n$ be the vector space of $n \times n$ symmetric complex matrices. We let $\psi : \mathfrak{p} \longrightarrow \mathbb{T}_n$ be the map defined by

$$\psi\left(\begin{pmatrix} X & Y \\ Y & -X \end{pmatrix}\right) = X + iY, \quad \begin{pmatrix} X & Y \\ Y & -X \end{pmatrix} \in \mathfrak{p}. \tag{2.1.33}$$

We let $\delta : K \longrightarrow U(n)$ be the isomorphism defined by

$$\delta\left(\begin{pmatrix} A & -B \\ B & A \end{pmatrix}\right) = A + iB, \quad \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \in K, \tag{2.1.34}$$

where $U(n)$ denotes the unitary group of degree n. We identify $\mathfrak{p}$ (resp. K) with $\mathbb{T}_n$ (resp. $U(n)$) through the map ψ (resp. δ). We consider the action of $U(n)$ on $\mathbb{T}_n$ defined by

$$h \cdot Z = hZ\,{}^t h, \quad h \in U(n),\ Z \in \mathbb{T}_n. \tag{2.1.35}$$

Then the adjoint action (2.1.32) of K on $\mathfrak{p}$ is compatible with the action (2.1.35) of $U(n)$ on $\mathbb{T}_n$ through the map ψ. Precisely for any $k \in K$ and $\omega \in \mathfrak{p}$, we get

$$\psi(k\,\omega\,{}^t k) = \delta(k)\,\psi(\omega)\,{}^t\delta(k). \tag{2.1.36}$$

The action (2.1.35) induces the action of $U(n)$ on the polynomial algebra $\mathrm{Pol}(\mathbb{T}_n)$ and the symmetric algebra $S(\mathbb{T}_n)$ respectively. We denote by $\mathrm{Pol}(\mathbb{T}_n)^{U(n)}\Big(\text{resp. } S(\mathbb{T}_n)^{U(n)}\Big)$ the subalgebra of $\mathrm{Pol}(\mathbb{T}_n)$ $\Big(\text{resp. } S(\mathbb{T}_n)\Big)$ consisting of $U(n)$-invariants. The following inner product $(\ ,\ )$ on $\mathbb{T}_n$ defined by

$$(Z, W) = \sigma\big(Z\,\overline{W}\big), \quad Z, W \in \mathbb{T}_n$$

gives an isomorphism as vector spaces

$$\mathbb{T}_n \cong \mathbb{T}_n^*, \quad Z \mapsto f_Z, \quad Z \in \mathbb{T}_n, \tag{2.1.37}$$

where $\mathbb{T}_n^*$ denotes the dual space of $\mathbb{T}_n$ and f_Z is the linear functional on $\mathbb{T}_n$ defined by

$$f_Z(W) = (W, Z), \quad W \in \mathbb{T}_n.$$

It is known that there is a canonical linear bijection of $S(\mathbb{T}_n)^{U(n)}$ onto the algebra $\mathbb{D}(\mathbb{H}_n)$ of differential operators on $\mathbb{H}_n$ invariant under the action (2.1.6)

of $Sp(2n, \mathbb{R})$. Identifying $\mathbb{T}_n$ with $\mathbb{T}_n^*$ by the above isomorphism (2.1.37), we get a canonical linear bijection

$$\Phi : \mathrm{Pol}(\mathbb{T}_n)^{U(n)} \longrightarrow \mathbb{D}(\mathbb{H}_n) \tag{2.1.38}$$

of $\mathrm{Pol}(\mathbb{T}_n)^{U(n)}$ onto $\mathbb{D}(\mathbb{H}_n)$. The map Φ is described explicitly as follows. Similarly, the action (2.1.32) induces the action of K on the polynomial algebra $\mathrm{Pol}(\mathfrak{p})$ and $S(\mathfrak{p})$ respectively. Through the map ψ, the subalgebra $\mathrm{Pol}(\mathfrak{p})^K$ of $\mathrm{Pol}(\mathfrak{p})$ consisting of K-invariants is isomorphic to $\mathrm{Pol}(\mathbb{T}_n)^{U(n)}$. We put $N = n(n+1)$. Let $\{\xi_\alpha \mid 1 \le \alpha \le N\}$ be a basis of $\mathfrak{p}$. If $P \in \mathrm{Pol}(\mathfrak{p})^K$, then

$$\Big(\Phi(P)f\Big)(gK) = \left[P\left(\frac{\partial}{\partial t_\alpha}\right) f\left(g \exp\left(\sum_{\alpha=1}^{N} t_\alpha \xi_\alpha\right) K\right)\right]_{(t_\alpha)=0}, \tag{2.1.39}$$

where $f \in C^\infty(\mathbb{H}_n)$ and $g \in Sp(2n, \mathbb{R})$. We refer to [15, 16] for more details. In general, it is hard to express $\Phi(P)$ explicitly for a polynomial $P \in \mathrm{Pol}(\mathfrak{p})^K$.

According to the work of Harish-Chandra [13, 14], the algebra $\mathbb{D}(\mathbb{H}_n)$ is generated by n algebraically independent generators and is isomorphic to the commutative ring $\mathbb{C}[x_1, \cdots, x_n]$ with n indeterminates. We note that n is the real rank of $Sp(2n, \mathbb{R})$. Let $\mathfrak{sp}(2n, \mathbb{C})$ be the complexification of $\mathfrak{sp}(2n, \mathbb{R})$. It is known that $\mathbb{D}(\mathbb{H}_n)$ is a surjective image of the center of the universal enveloping algebra of $\mathfrak{sp}(2n, \mathbb{C})$ (cf. [42]).

2.2 The Universal Covering Group of the Symplectic Group

In this section, we describe the universal covering group of $Sp(2n, \mathbb{R})$ using the so-called Maslov index.

Let (V, B) be a symplectic (real) vector space of dimension $2n$ with a non-degenerate alternating form B on V. A subspace L of (V, B) such that $B(x, y) = 0$ for all $x, y \in L$ is said to be totally isotropic. For a subspace L of (V, B), we will denote by $L^\perp$ the orthogonal complement of L in V relative to B, i.e.,

$$L^\perp = \big\{x \in V \mid B(x, y) = 0 \quad \text{for all } y \in L\big\}.$$

If L is a subspace of (V, B) such that $L = L^\perp$, then L is called a Lagrangian subspace of (V, B). If L is a totally isotropic subspace of V such that $B(x, L) = 0$ implies $x \in L$, then L is said to be maximally totally isotropic. We note that if L is a Lagrangian subspace of (V, B), then $\dim L = n$ because $\dim L + \dim L^\perp = 2n$.

Let $Sp(B)$ be the symplectic group defined by

$$Sp(B) = \big\{g \in GL(V) \mid B(gx, gy) = B(x, y) \quad \text{for all } x, y \in V\big\}.$$

Definition 2.2.1 Let L_1, L_2, L_3 be three Lagrangian subspaces of V. The integer $\tau(L_1, L_2, L_3)$ is defined to be the signature of the quadratic form $Q(x_1 + x_2 + x_3)$ on the $3n$-dimensional vector space $L_1 \oplus L_2 \oplus L_3$ defined by

$$Q(x_1 + x_2 + x_3) := B(x_1, x_2) + B(x_2, x_3) + B(x_3, x_1),$$

where $x_1 \in L_1$, $x_2 \in L_2$ and $x_3 \in L_3$. The integer $\tau(L_1, L_2, L_3)$ is called the Maslov index of (V, B). Here the signature of Q is defined as follows: In a certain basis of $L_1 \oplus L_2 \oplus L_3$, the matrix of Q is diagonal and contains p times the coefficient $+1$, q times the coefficient -1; the signature of Q is $p - q$.

Lemma 2.2.2 *Let L_1, L_2, L_3 be three Lagrangian subspaces of (V, B). Then we have the following properties:*

(1) *The Maslov index is $Sp(2n, \mathbb{R})$-invariant, i.e., for any $g \in Sp(2n, \mathbb{R})$, we have $\tau(gL_1, gL_2, gL_3) = \tau(L_1, L_2, L_3)$.*
(2) $\tau(L_1, L_2, L_3) = -\tau(L_2, L_1, L_3) = -\tau(L_1, L_3, L_2)$.

Proof It follows immediately from the definition. □

For a sequence $(L_1, L_2, \ldots, L_k)$ with $k \geq 4$ of Lagrangian subspaces of (V, B), we define the generalized Maslov index $\tau(L_1, L_2, \ldots, L_k)$ by

$$\tau(L_1, L_2, \ldots, L_k) = \tau(L_1, L_2, L_3) + \tau(L_1, L_3, L_4) + \cdots + \tau(L_1, L_{k-1}, L_k).$$

Proposition 2.2.3

(1) *The Maslov index $\tau(L_1, L_2, \ldots, L_k)$ is invariant under the action of the symplectic group $Sp(B)$, and its value is unchanged under circular permutation.*
(2) *For any Lagrangian subspace $L_1, L_2, L_3, L_1', L_2', L_3'$, we have*

$$\begin{aligned}\tau(L_1', L_2', L_3') = \tau(L_1, L_2, L_3) + \tau(L_1', L_2', L_2, L_1) + \tau(L_2', L_3', L_3, L_2) \\ + \tau(L_3', L_1', L_1, L_3).\end{aligned}$$

Proof See [26, pp. 45–46]. □

Let Λ be the space of all Lagrangian subspaces of (V, B). Then Λ may be regarded as a closed submanifold of the Grassmannian manifold of all n-dimensional subspaces in $\mathbb{R}^{2n}$. We define

$$\widetilde{\Lambda} := \Lambda \times \mathbb{Z} = \big\{ (L, u) \,\big|\, L \in \Lambda,\ u \in \mathbb{Z} \big\}.$$

We fix a Lagrangian subspace L_0 of (V, B). Let $(L_1, u_1) \in \widetilde{\Lambda}$ and let $\mathscr{U}$ be a neighborhood of L_1. Let L_2 be a Lagrangian subspace of V transverse to L_1. We define

$$U(L_1, u_1; \mathscr{U}, L_2) := \big\{ (L, u) \,\big|\, L \in \mathscr{U},\ u = u_1 + \tau(L, L_0, L_1, L_2) \big\}.$$

It is proved in [26, Proposition 1.9.5] that the set of all such $U(L_1, u_1; \mathscr{U}, L_2)$'s form a neighborhood for a topology on $\widetilde{\Lambda}$. Let $\pi : \widetilde{\Lambda} \longrightarrow \Lambda$ be the projection defined by $\pi(L, u) = L$. Clearly, π is a continuous map and hence $\widetilde{\Lambda}$ is a covering of Λ.

Let L_* be a fixed element of Λ. We define the group

$$\widetilde{Sp(B)}_* := Sp(B) \times \mathbb{Z} \tag{2.2.1}$$

equipped with the multiplication law

$$(g_1, n_1) \cdot (g_2, n_2) = \big(g_1 g_2, n_1 + n_2 + \tau(L_*, g_1 L_*, g_1 g_2 L_*)\big),$$

where $g_1, g_2 \in Sp(B)$ and $n_1, n_2 \in \mathbb{Z}$. Then it is easy to see that $\widetilde{Sp(B)}_*$ acts on $\widetilde{\Lambda}$ by

$$(g, n) \cdot (L, u) = \big(gL, n + u + \tau(L_*, gL_*, gL)\big),$$

where $g \in Sp(B)$, $n, u \in \mathbb{Z}$ and $L \in \Lambda$.

Let L_2 be a Lagrangian subspace of (V, B) transverse to L_* and $\mathscr{E}$ be a neighborhood of e in $Sp(B)$, where e is the identity element of $Sp(B)$. We define

$$\mathscr{W}(\mathscr{E}, L_2) := \big\{ \big(g, -\tau(gL_*, L_2, L_*)\big) \,\big|\, g \in \mathscr{E} \big\}.$$

Then the set of all such $\mathscr{W}(\mathscr{E}, L_2)$'s form a fundamental system of neighborhoods of $(e, 0)$ on $\widetilde{Sp(B)}_*$. Therefore $\widetilde{Sp(B)}_*$ has the structure of a topological group. It is easily seen that $\widetilde{Sp(B)}_*$ acts on $\widetilde{\Lambda}$ continuously.

Definition 2.2.4 An *oriented* vector space of dimension n is a pair (W, ε), where W is a real vector space of dimension n and ε is an orientation of W, i.e., a connected component of $\Lambda^n W - \{0\}$. If (W, ε) is an oriented vector space, we define the oriented dual vector space $(W, \varepsilon)^* = (W^*, \varepsilon^*)$ of (W, ε) by choosing ε^* such that $\langle x, y\rangle > 0$ for $x \in \varepsilon^*$ and $y \in \varepsilon$. If (W_1, ε_1) and (W_2, ε_2) are two oriented vector spaces of dimension n and A is a linear invertible map from W_1 to W_2, we define the sign of the determinant of A denoted by $\delta(A) = \pm 1$, by the condition

$$\big(\Lambda^n A\big)\, \varepsilon_1 = c\, \delta(A)\, \varepsilon_2 \quad \text{with } c > 0.$$

Let L and M be two Lagrangian subspaces of a symplectic vector space (V, B). We define $g_{M,L} : L \longrightarrow M^*$ by $\langle\, g_{M,L}(x), y\rangle = B(x, y)$ for all $x \in L$ and $y \in M$. Here M^* denotes the dual vector space of M. Let (L_1, ε_1) and (L_2, ε_2) be two oriented Lagrangian subspaces of (V, B) which are transverse. Then $g_{L_2,L_1} : (L_1, \varepsilon_1) \longrightarrow (L_2, \varepsilon_2)^*$ is invertible. We define

$$\xi\big((L_1, \varepsilon_1), (L_2, \varepsilon_2)\big) := \delta(g_{L_2,L_1}).$$

This depends only on the relative orientation of (L_1, ε_1) and (L_2, ε_2). More generally if L_1 and L_2 are not transverse, we define (L_1, ε_1) and (L_2, ε_2) as follows: Let ε be an orientation of $H = L_1 \cap L_2$. Then ε defines an orientation $\widetilde{\varepsilon}_i$ $(i = 1, 2)$ on L_i/H by $\widetilde{\varepsilon}_i \wedge \varepsilon = \varepsilon_i$. Since L_1/H and L_2/H are two transverse subspaces of $(L_1 + L_2)/H = H^{\perp}/H$, we can define

$$\xi\big((L_1, \varepsilon_1), (L_2, \varepsilon_2)\big) := \xi\big((L_1/H, \widetilde{\varepsilon}_1), (L_2/H, \widetilde{\varepsilon}_2)\big).$$

We observe that this is independent of the choice of the orientation ε of H because $\widetilde{\varepsilon}_1$ and $\widetilde{\varepsilon}_2$ change simultaneously if we change ε to $-\varepsilon$.

If $L_1 = L_2$, we define

$$\xi\big((L_1, \varepsilon_1), (L_2, \varepsilon_2)\big) := \begin{cases} 1 & \text{if } \varepsilon_1 = \varepsilon_2, \\ -1 & \text{if } \varepsilon_1 \neq \varepsilon_2. \end{cases}$$

Definition 2.2.5 Let (L_1, ε_1) and (L_2, ε_2) be two oriented Lagrangian subspaces of a symplectic vector space (V, B). We define

$$s\big((L_1, \varepsilon_1), (L_2, \varepsilon_2)\big) := \big(\sqrt{-1}\big)^{n-\dim(L_1 \cap L_2)}\, \xi\big((L_1, \varepsilon_1), (L_2, \varepsilon_2)\big).$$

Definition 2.2.6 Let L be a Lagrangian subspace of a symplectic vector space (V, B). We choose an orientation L^+ on L. We define the map $s_L : Sp(B) \longrightarrow \mathbb{C}$ by

$$s_L(g) := s(L^+, gL^+), \quad g \in Sp(B).$$

This is well-defined because $s_L(g)$ is independent of the choice of the orientation on L.

We define the map $\tilde{s}_* : \widetilde{Sp(B)}_* \longrightarrow \mathbb{C}$ by

$$\tilde{s}_*(g, n) := e^{\frac{\pi n i}{2}}\, s_{L_*}(g), \quad g \in Sp(B),\ n \in \mathbb{Z}.$$

Lemma 2.2.7 *$\tilde{s}_*(g, n)$ is a character of $\widetilde{Sp(B)}_*$ with values in $\mathbb{Z}/4\mathbb{Z}$.*

Proof The proof can be found in [26, p. 72]. □

Lion and Vergne (cf. [26, p. 94]) proved that the kernel of $\tilde{s}_*$ is the universal covering group of $Sp(B)$ and the fundamental group $\pi_1(Sp(B))$ of $Sp(B)$ is isomorphic to $\mathbb{Z}$. Therefore $\widetilde{Sp(B)}_*$ is the union of four connected components such that each of them is simply connected.

We now consider the group

$$Sp(B, L_*) := Sp(B) \times \mathbb{C}_1^{\times}$$

equipped with the multiplication law

$$(g_1, t_1) \bullet (g_2, t_2) = \big(g_1 g_2,\ t_1 t_2\, c_*(g_1, g_2)^{-1} \big),$$

where $g_1, g_2 \in Sp(B),\ t_1, t_2 \in \mathbb{C}_1^\times$ and

$$c_*(g_1, g_2) := e^{-\frac{\pi i}{4}\tau(L_*, g_1 L_*, g_1 g_2 L_*)}.$$

We define the mapping $\varphi : \widetilde{Sp(B)}_* \longrightarrow Sp(B, L_*)$ by

$$\varphi(g, n) := \big(g, e^{\frac{\pi n i}{4}}\big), \quad g \in Sp(B),\ n \in \mathbb{Z}.$$

We refer to (2.2.1) for the definition of $\widetilde{Sp(B)}_*$. Then φ is a group homomorphism. Indeed, if (g_1, n_1) and (g_2, n_2) are two elements in $\widetilde{Sp(B)}_*$, then we have

$$\begin{aligned}
\varphi((g_1, n_1) \cdot (g_2, n_2)) &= \varphi(g_1 g_2, n_1 + n_2 + \tau(L_*, g_1 L_*, g_1 g_2 L_*)) \\
&= \Big(g_1 g_2, e^{\frac{\pi i}{4}\{n_1 + n_2 + \tau(L_*, g_1 L_*, g_1 g_2 L_*)\}}\Big) \\
&= \Big(g_1 g_2, e^{\frac{\pi n_1 i}{4}}\, e^{\frac{\pi n_2 i}{4}}\, e^{\frac{\pi i}{4}\tau(L_*, g_1 L_*, g_1 g_2 L_*)}\Big) \\
&= \Big(g_1 g_2, e^{\frac{\pi n_1 i}{4}}\, e^{\frac{\pi n_2 i}{4}}\, c_*(g_1, g_2)^{-1}\Big) \\
&= \Big(g_1, e^{\frac{\pi n_1 i}{4}}\Big) \bullet \Big(g_2, e^{\frac{\pi n_2 i}{4}}\Big) \\
&= \varphi((g_1, n_1)) \bullet \varphi((g_2, n_2)).
\end{aligned}$$

We define

$$Mp(B)_* := \big\{ (g, t) \in Sp(B, L_*) \,\big|\, t^2 = s_{L_*}(g)^{-1} \big\}. \tag{2.2.2}$$

2.3 The Weil Representation

We recall that for a real symmetric positive definite matrix $c \in \mathbb{R}^{(m,m)}$, the Schrödinger representation U_c of $H_{\mathbb{R}}^{(n,m)}$ is defined by Formula (1.2.11). We refer to Formula (1.4.45). For convenience, we rewrite Formula (1.2.11):

(1.2.11) $$\big(U_c(g_0) f\big)(x) = e^{2\pi i \sigma(c(\kappa_0 + \mu_0\, {}^t\lambda_0 + 2\, x\, {}^t\mu_0))} f(x + \lambda_0),$$

for $g_0 = (\lambda_0, \mu_0, \kappa_0) \in H_{\mathbb{R}}^{(n,m)},\ x \in \mathbb{R}^{(m,n)}$ and $f \in L^2\big(\mathbb{R}^{(m,n)}\big)$.

We let

$$G^J = Sp(2n, \mathbb{R}) \ltimes H_{\mathbb{R}}^{(n,m)} \quad \text{(semi-direct product)}$$

be the Jacobi group endowed with the following multiplication law:

$$\Big(M, (\lambda, \mu, \kappa)\Big)\Big(M', (\lambda', \mu', \kappa')\Big) = \Big(MM', (\widetilde{\lambda}+\lambda', \widetilde{\mu}+\mu', \kappa+\kappa'+\widetilde{\lambda}\,{}^t\mu' - \widetilde{\mu}\,{}^t\lambda')\Big),$$

with $M, M' \in Sp(2n, \mathbb{R})$, (λ, μ, κ), $(\lambda', \mu', \kappa') \in H_{\mathbb{R}}^{(n,m)}$ and $(\widetilde{\lambda}, \widetilde{\mu}) = (\lambda, \mu)M'$. Then $Sp(2n, \mathbb{R})$ acts on $H_{\mathbb{R}}^{(n,m)}$ by conjugation inside G^J,

$$M \star (\lambda, \mu, \kappa) = M(\lambda, \mu, \kappa)M^{-1} = (\lambda_*, \mu_*, \kappa), \tag{2.3.1}$$

where $M \in Sp(2n, \mathbb{R})$, $(\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$ and $(\lambda_*, \mu_*) = (\lambda, \mu)M^{-1}$.

We fix an element $M \in Sp(2n, \mathbb{R})$. We consider the mapping U_c^M of $H_{\mathbb{R}}^{(n,m)}$ into $\mathrm{Aut}\big(L^2\big(\mathbb{R}^{(m,n)}\big)\big)$ defined by

$$U_c^M(g) = U_c(M \star g) = U_c(MgM^{-1}), \qquad g \in H_{\mathbb{R}}^{(n,m)}. \tag{2.3.2}$$

Lemma 2.3.1 *U_c^M is an irreducible representation of $H_{\mathbb{R}}^{(n,m)}$ on $L^2\big(\mathbb{R}^{(m,n)}\big)$ such that*

$$U_c^M\big((0, 0, \kappa)\big) = U_c\big((0, 0, \kappa)\big) \qquad \text{for all } \kappa \in \mathrm{Sym}(m, \mathbb{R}).$$

Thus U_c^M is unitarily equivalent to U_c.

Proof If $g_1, g_2 \in H_{\mathbb{R}}^{(n,m)}$, then

$$\begin{aligned} U_c^M(g_1g_2) &= U_c(M \star (g_1g_2)) = U_c\big(M(g_1g_2)M^{-1}\big) \\ &= U_c\big((Mg_1M^{-1})(Mg_2M^{-1})\big) \\ &= U_c\big(Mg_1M^{-1}\big)\, U_c\big(Mg_2M^{-1}\big) \\ &= U_c^M(g_1)\, U_c^M(g_2). \end{aligned}$$

Thus U_c^M is a representation of $H_{\mathbb{R}}^{(n,m)}$. The irreducibility of U_c^M follows immediately from that of U_c. It is easily seen that

$$U_c^M\big((0, 0, \kappa)\big) = U_c\big(M \star (0, 0, \kappa)\big) = U_c\big((0, 0, \kappa)\big)$$

for all $\kappa \in \mathrm{Sym}(m, \mathbb{R})$. Therefore it follows from the Stone–von Neumann theorem that U_c^M is unitarily equivalent to U_c. □

Since U_c^M is unitarily equivalent to U_c, there exists a unitary operator $R_c(M) : L^2\big(\mathbb{R}^{(m,n)}\big) \longrightarrow L^2\big(\mathbb{R}^{(m,n)}\big)$ such that $U_c^M(g)\, R_c(M) = R_c(M)\, U_c(g)$ for all $g \in H_{\mathbb{R}}^{(n,m)}$. For convenience, we take $R_c(I_{2n}) = I_c$, where I_c is the identity operator on $L^2\big(\mathbb{R}^{(m,n)}\big)$. We observe that $R_c(M)$ is determined uniquely up to a scalar of modulus one. For any two elements M_1, M_2 of $Sp(2n, \mathbb{R})$, the unitary operator $R_c(M_2)^{-1} R_c(M_1)^{-1} R_c(M_1 M_2)$ commutes with U_c. Indeed, for any element $g \in H_{\mathbb{R}}^{(n,m)}$, we have

$$\begin{aligned}
&U_c(g)\, R_c(M_2)^{-1}\, R_c(M_1)^{-1}\, R_c(M_1M_2)\\
&= R_c(M_2)^{-1} U_c^{M_2}(g)\, R_c(M_1)^{-1}\, R_c(M_1M_2)\\
&= R_c(M_2)^{-1} U_c(M_2 \star g)\, R_c(M_1)^{-1}\, R_c(M_1M_2)\\
&= R_c(M_2)^{-1}\, R_c(M_1)^{-1}\, U_c^{M_1}(M_2 \star g)\, R_c(M_1M_2)\\
&= R_c(M_2)^{-1}\, R_c(M_1)^{-1}\, U_c\big((M_1M_2) \star g\big)\, R_c(M_1M_2)\\
&= R_c(M_2)^{-1}\, R_c(M_1)^{-1}\, U_c^{M_1M_2}(g)\, R_c(M_1M_2)\\
&= R_c(M_2)^{-1}\, R_c(M_1)^{-1}\, R_c(M_1M_2)\, U_c(g).
\end{aligned}$$

According to Schur's lemma, we obtain a map $\alpha_c : Sp(2n, \mathbb{R}) \times Sp(2n, \mathbb{R}) \longrightarrow \mathbb{C}_1^\times$ satisfying the condition

$$R_c(M_1M_2) = \alpha_c(M_1, M_2)\, R_c(M_1)\, R_c(M_2) \tag{2.3.3}$$

for all $M_1, M_2 \in Sp(2n, \mathbb{R})$. Thus R_c is a projective representation of $Sp(2n, \mathbb{R})$ with its multiplier α_c.

Lemma 2.3.2 *The map α_c satisfies the cocycle condition*

$$\alpha_c(M_1M_2, M_3)\, \alpha_c(M_1, M_2) = \alpha_c(M_1, M_2M_3)\, \alpha_c(M_2, M_3) \tag{2.3.4}$$

for all $M_1, M_2, M_3 \in Sp(2n, \mathbb{R})$.

Proof Let $M_1, M_2, M_3 \in Sp(2n, \mathbb{R})$. Then according to Formula (2.3.3),

$$\begin{aligned}
&R_c\big((M_1M_2)M_3\big)\\
&= \alpha_c(M_1M_2, M_3)\, R_c(M_1M_2)\, R_c(M_3)\\
&= \alpha_c(M_1M_2, M_3)\, \alpha_c(M_1, M_2)\, R_c(M_1)\, R_c(M_2)\, R_c(M_3)
\end{aligned}$$

and

$$\begin{aligned} & R_c\big(M_1(M_2M_3)\big) \\ &= \alpha_c(M_1, M_2M_3)\, R_c(M_1)\, R_c(M_2M_3) \\ &= \alpha_c(M_1, M_2M_3)\, \alpha_c(M_2, M_3)\, R_c(M_1)\, R_c(M_2)\, R_c(M_3). \end{aligned}$$

Hence we obtain the cocycle condition (2.3.4). □

The cocycle α_c yields the central extension $Sp(2n,\mathbb{R})_*$ of $Sp(2n,\mathbb{R})$ by $\mathbb{C}_1^\times$. The group $Sp(2n,\mathbb{R})_*$ is the set $Sp(2n,\mathbb{R})\times\mathbb{C}_1^\times$ with the following group multiplication,

$$(M_1, t_1)\cdot(M_2, t_2) = \big(M_1M_2,\, t_1t_2\,\alpha_c(M_1, M_2)^{-1}\big), \tag{2.3.5}$$

for all $M_1, M_2 \in Sp(2n,\mathbb{R})$ and $t_1, t_2 \in \mathbb{C}_1^\times$. We see that the map $\widetilde{R}_c : Sp(2n,\mathbb{R})_* \longrightarrow \mathrm{Aut}\big(L^2\big(\mathbb{R}^{(m,n)}\big)\big)$ defined by

$$\widetilde{R}_c(M, t) = t\, R_c(M), \qquad M \in Sp(2n,\mathbb{R}),\ t \in \mathbb{C}_1^\times \tag{2.3.6}$$

is a true representation of $Sp(2n,\mathbb{R})_*$. As in [26, pp. 70–71, Section 1.7], we can define the map $s_c : Sp(2n,\mathbb{R}) \longrightarrow \mathbb{C}_1^\times$ satisfying the relation

$$\alpha_c(M_1, M_2)^2 = s_c(M_1)^{-1}s_c(M_2)^{-1}s_c(M_1M_2)$$

for all $M_1, M_2 \in Sp(2n,\mathbb{R})$. For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$ and $\Omega \in \mathbb{H}_n$, we put

$$J(M,\Omega) \overset{\text{def}}{=} \det(C\Omega + D).$$

According to [38, p. 401], we may take

$$s_c(M) = |J(M, i\,I_n)|\, J(M, i\,I_n)^{-1}, \quad M \in Sp(2n,\mathbb{R}).$$

The following subset,

$$Mp(2n,\mathbb{R}) = \Big\{(M, t) \in Sp(2n,\mathbb{R})_* \mid t^2 = s_c(M)^{-1}\Big\}, \tag{2.3.7}$$

is a subgroup of $Sp(2n,\mathbb{R})_*$ that is called the metaplectic group. We can show that $Mp(2n,\mathbb{R})$ is a two-fold covering group of $Sp(2n,\mathbb{R})$. The restriction ω_c of $\widetilde{R}_c$ to $Mp(2n,\mathbb{R})$ is a true representation of $Mp(2n,\mathbb{R})$ which is called the Weil representation of $Sp(2n,\mathbb{R})$.

Now we describe the action of ω_c explicitly. It is known that $Sp(2n,\mathbb{R})$ is generated by the following elements:

$$t_b = \begin{pmatrix} I_n & b \\ 0 & I_n \end{pmatrix} \text{ with } b = {}^t b \in \mathbb{R}^{(n,n)},$$

$$d_a = \begin{pmatrix} {}^t a & 0 \\ 0 & a^{-1} \end{pmatrix} \text{ with } a \in GL(n,\mathbb{R}),$$

$$\sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}.$$

Theorem 2.3.3 *The actions of R_c on the generators t_b, d_a and σ_n of $Sp(2n,\mathbb{R})$ are given by*

$$(R_c(t_b)f)(x) = e^{2\pi i \sigma(c\,x\,b\,{}^t x)} f(x), \tag{2.3.8}$$

$$(R_c(d_a)f)(x) = (\det a)^{\frac{m}{2}} f(x\,{}^t a), \tag{2.3.9}$$

$$(R_c(\sigma_n)f)(x) = \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}} \times \int_{\mathbb{R}^{(m,n)}} f(y) e^{-4\pi i\, \sigma(c\,y\,{}^t x)}\, dy, \tag{2.3.10}$$

where $f \in L^2\big(\mathbb{R}^{(m,n)}\big)$ and $x \in \mathbb{R}^{(m,n)}$. Here the function

$$h(a) \overset{\text{def}}{=} (\det a)^{\frac{1}{2}}, \quad a \in GL(n,\mathbb{R}) \tag{2.3.11}$$

is the smooth function on $GL(n,\mathbb{R})$ determined uniquely by the following properties (i) and (ii):

(i) $h(a)^2 = \det a$ *for all* $a \in GL(n,\mathbb{R})$;
(ii) $h(a) = \sqrt{\det a}$ *for all* $a \in GL(n,\mathbb{R})$ *with* $\det a > 0$.

For a positive integer m, we define

$$(\det a)^{\frac{m}{2}} \overset{\text{def}}{=} \left\{(\det a)^{\frac{1}{2}}\right\}^m, \qquad a \in GL(n,\mathbb{R}). \tag{2.3.12}$$

Proof Let $g = (\lambda,\mu,\kappa) \in H_{\mathbb{R}}^{(n,m)}$, $x \in \mathbb{R}^{(m,n)}$ and $f \in L^2\big(\mathbb{R}^{(m,n)}\big)$. For each $t_b \in Sp(2n,\mathbb{R})$ with $b = {}^t b \in \mathbb{R}^{(n,n)}$, we put

$$\big(T_c(t_b)f\big)(x) = e^{2\pi i \sigma(c\,x\,b\,{}^t x)} f(x) \qquad \text{for all } f \in L^2\big(\mathbb{R}^{(m,n)}\big).$$

Then we have

$$\begin{aligned}
&\big(T_c(t_b)U_c(g)f\big)(x)\\
&= e^{2\pi i\,\sigma(c\,x\,b\,{}^t x)}\,\big(U_c(g)f\big)(x)\\
&= e^{2\pi i\,\sigma(c\,x\,b\,{}^t x)}\cdot e^{2\pi i\,\sigma(c(\kappa+\mu\,{}^t\lambda+2x\,{}^t\mu))}\,f(x+\lambda)\\
&= e^{2\pi i\,\sigma(c(\kappa+\mu\,{}^t\lambda+2x\,{}^t\mu+x\,b\,{}^t x))}\,f(x+\lambda).
\end{aligned}$$

Since $t_b \star (\lambda, \mu, \kappa) = t_b(\lambda, \mu, \kappa)\, t_b^{-1} = (\lambda, -\lambda\, b + \mu, \kappa)$, we obtain

$$\begin{aligned}
&\big(U_c^{t_b}(g)\; T_c(t_b)\, f\big)(x)\\
&= \big(U_c(t_b \star g)\; T_c(t_b)\, f\big)(x)\\
&= \big(U_c(\lambda, -\lambda\, b + \mu, \kappa)\; T_c(t_b)\, f\big)(x)\\
&= e^{2\pi i\,\sigma(c(\kappa+(-\lambda\, b+\mu)\,{}^t\lambda+2x\,{}^t(-\lambda b+\mu)))}\big(T_c(t_b)\, f\big)(x+\lambda)\\
&= e^{2\pi i\,\sigma(c(\kappa+(-\lambda\, b+\mu)\,{}^t\lambda+2x\,{}^t(-\lambda b+\mu)))}\cdot e^{2\pi i\,\sigma(c(x+\lambda)b\,{}^t(x+\lambda))}\,f(x+\lambda).\\
&= e^{2\pi i\,\sigma(c(\kappa+\mu\,{}^t\lambda+2x\,{}^t\mu+x\,b\,{}^t x))}\,f(x+\lambda).
\end{aligned}$$

Therefore

$$T_c(t_b)\, U_c(g) f = U_c^{t_b}(g)\; T_c(t_b) f$$

for all $b = {}^t b \in \mathbb{R}^{(n,n)}$, $g \in H_{\mathbb{R}}^{(n,m)}$ and $f \in L^2\big(\mathbb{R}^{(m,n)}\big)$.

Since $T_c(t_0) = T_c(I_{2n}) = I_c = R_c(I_{2n})$, we see that

$$R_c(t_b) = T_c(t_b) \qquad \text{for all } b = {}^t b \in \mathbb{R}^{(n,n)}.$$

We recall that I_c is the identity operator on $L^2\big(\mathbb{R}^{(m,n)}\big)$.

On the other hand, for each $a \in GL(n, \mathbb{R})$ and $f \in L^2\big(\mathbb{R}^{(m,n)}\big)$, we put

$$\big(A_c(d_a) f\big)(x) = (\det a)^{\frac{m}{2}}\; f(x\,{}^t a).$$

Then we have

$$\begin{aligned}
&\big(A_c(d_a)U_c(g)f\big)(x)\\
&= \big(\det a\big)^{\frac{m}{2}}\,\big(U_c(g)f\big)(x\,{}^t a)\\
&= \big(\det a\big)^{\frac{m}{2}}\; e^{2\pi i\,\sigma(c(\kappa+\mu\,{}^t\lambda+2x\,{}^t a\,{}^t\mu))}\,f(x\,{}^t a+\lambda).
\end{aligned}$$

Since $d_a \star (\lambda, \mu, \kappa) = d_a(\lambda, \mu, \kappa)\, d_a^{-1} = (\lambda\, {}^t a^{-1}, \mu\, a, \kappa)$,

$$\begin{aligned}
&\big(U_c^{d_a}(g)\, A_c(d_a) f\big)(x) \\
&= \big(U_c(d_a \star g)\, A_c(d_a) f\big)(x) \\
&= \big(U_c(\lambda\, {}^t a^{-1}, \mu\, a, \kappa)\, A_c(d_a) f\big)(x) \\
&= e^{2\pi i\, \sigma(c(\kappa + (\mu\, a)\, {}^t(\lambda\, {}^t a^{-1}) + 2\, x\, {}^t(\mu\, a)))}\, \big(A_c(d_a) f\big)\big(x + \lambda\, {}^t a^{-1}\big) \\
&= \big(\det a\big)^{\frac{m}{2}}\, e^{2\pi i\, \sigma(c(\kappa + \mu\, {}^t\lambda + 2\, x\, {}^t a\, {}^t\mu))}\, f(x\, {}^t a + \lambda).
\end{aligned}$$

Thus

$$A_c(d_a) U_c(g) f = U_c^{d_a}(g)\, A_c(d_a) f$$

for all $a \in GL(n, \mathbb{R})$, $g \in H_{\mathbb{R}}^{(n,m)}$ and $f \in L^2\big(\mathbb{R}^{(m,n)}\big)$.

Since $A_c(d_{I_n}) = I_c = R_c(d_{I_n})$, we obtain $R_c(d_a) = A_c(d_a)$ for all $a \in GL(n, \mathbb{R})$.

Finally, we put

$$\big(B_c(\sigma_n) f\big)(x) = \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} f(y)\, e^{-4\pi i\, \sigma(c\, y\, {}^t x)}\, dy$$

for all $f \in L^2\big(\mathbb{R}^{(m,n)}\big)$. Here $\left(\frac{2}{i}\right)^{\frac{mn}{2}}$ must be understood as follows. We observe that $\left(\frac{2}{i}\right)^{\frac{mn}{2}} = (-2\, i)^{\frac{mn}{2}} = 2^{\frac{mn}{2}}\, (-i)^{\frac{mn}{2}}$. We define

$$(-i)^{\frac{mn}{2}} \stackrel{\text{def}}{=} e^{\frac{mn}{2} \ln(-i)},$$

where we choose the principal branch $\operatorname{Ln} z$ of $\ln z$ which is a continuous function on a connected open subset of $\mathbb{C}$ such $\operatorname{Ln} z$ is a logarithm of z for all $z \in U$. Or since

$$\begin{aligned}
-i &= |-i| \left(\cos\left(\frac{3}{2}\pi + 2k\pi\right) + i\, \sin\left(\frac{3}{2}\pi + 2k\pi\right)\right) \\
&\qquad k = 0, \pm 1, \pm 2, \ldots,
\end{aligned}$$

we have

$$\begin{aligned}
(-i)^{\frac{1}{2}} &= \cos\left(\frac{3}{4}\pi + k\pi\right) + i\, \sin\left(\frac{3}{4}\pi + k\pi\right) \\
&= \pm\left(\frac{\sqrt{2}}{2} - i\, \frac{\sqrt{2}}{2}\right), \quad k = 0, \pm 1, \pm 2, \ldots
\end{aligned}$$

We choose the principal square root $\frac{\sqrt{2}}{2} - i\,\frac{\sqrt{2}}{2}$ of $-i$, which is one of the eighth roots of unity. We define

$$(-i)^{\frac{mn}{2}} \stackrel{\text{def}}{=} \left\{(-i)^{\frac{1}{2}}\right\}^{mn}.$$

Thus we have

$$\left(\frac{2}{i}\right)^{\frac{mn}{2}} = 2^{\frac{mn}{2}} \left\{(-i)^{\frac{1}{2}}\right\}^{mn}.$$

$$\begin{aligned}
&\big(B_c(\sigma_n)U_c(g)f\big)(x) \\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} \big(U_c(g)f\big)(y)\, e^{-4\pi i\ \sigma(c\,y\,{}^t x)}\, dy \\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} e^{2\pi i\,\sigma(c(\kappa+\mu\,{}^t\lambda+2\,y\,{}^t\mu))} \cdot e^{-4\pi i\ \sigma(c\,y\,{}^t x)}\, f(y+\lambda)\, dy \\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}}\, e^{2\pi i\,\sigma(c(\kappa+\mu\,{}^t\lambda))} \int_{\mathbb{R}^{(m,n)}} e^{4\pi i\ \sigma(c\,y\,{}^t(\mu-x))}\, f(y+\lambda)\, dy \\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}}\, e^{2\pi i\,\sigma(c(\kappa+\mu\,{}^t\lambda))} \int_{\mathbb{R}^{(m,n)}} e^{4\pi i\ \sigma(c\,(\widetilde{y}-\lambda)\,{}^t(\mu-x))}\, f(\widetilde{y})\, d\widetilde{y} \\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}}\, e^{2\pi i\,\sigma(c(\kappa+\mu\,{}^t\lambda))} \cdot e^{-4\pi i\,\sigma(c\,\lambda\,{}^t(\mu-x))} \\
&\quad \times \int_{\mathbb{R}^{(m,n)}} f(y)\, e^{4\pi i\ \sigma(c\,y\,{}^t(\mu-x))}\, dy \\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}}\, e^{2\pi i\,\sigma(c(\kappa-\lambda\,{}^t\mu+2\,x\,{}^t\lambda))} \int_{\mathbb{R}^{(m,n)}} f(y)\, e^{-4\pi i\ \sigma(c\,y\,{}^t(x-\mu))}\, dy.
\end{aligned}$$

Since $\sigma_n \star (\lambda, \mu, \kappa) = \sigma_n(\lambda, \mu, \kappa)\,\sigma_n^{-1} = (-\mu, \lambda, \kappa)$ for $g = (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$, we obtain

$$\begin{aligned}
&\big(U_c^{\sigma_n}(g)\ B_c(\sigma_n)f\big)(x) \\
&= \big(U_c(\sigma_n \star g)\ B_c(\sigma_n)f\big)(x) \\
&= \big(U_c(-\mu, \lambda, \kappa)\ B_c(\sigma_n)f\big)(x) \\
&= e^{2\pi i\,\sigma(c(\kappa-\lambda\,{}^t\mu+2\,x\,{}^t\lambda))}\ \big(B_c(\sigma_n)f\big)(x-\mu) \\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}}\, e^{2\pi i\,\sigma(c(\kappa-\lambda\,{}^t\mu+2\,x\,{}^t\lambda))} \int_{\mathbb{R}^{(m,n)}} f(y)\, e^{-4\pi i\ \sigma(c\,y\,{}^t(x-\mu))}\, dy.
\end{aligned}$$

Therefore

$$B_c(\sigma_n)U_c(g)f = U_c^{\sigma_n}(g)\, B_c(\sigma_n)f$$

for all $g \in H_{\mathbb{R}}^{(n,m)}$ and $f \in L^2\big(\mathbb{R}^{(m,n)}\big)$. We note that we can take

$$R_c(\sigma_n) = B_c(\sigma_n).$$

Hence we complete the proof of the above theorem. □

Now we review some properties of ω_c. The Weil representation ω_c is not an irreducible representation of $Mp(2n,\mathbb{R})$. In [20], Kashiwara and Vergne found an explicit decomposition of ω_c into irreducibles. First we observe that the orthogonal group $O(m)$ acts on $L^2\big(\mathbb{R}^{(m,n)}\big)$ by

$$(\alpha \cdot f)(x) \;=\; f(\alpha^{-1}x), \qquad \alpha \in O(m),\ x \in \mathbb{R}^{(m,n)},\ f \in L^2\big(\mathbb{R}^{(m,n)}\big).$$

This action commutes with ω_c. For each irreducible representation (σ, V_σ) of $O(m)$, we let $L^2\big(\mathbb{R}^{(m,n)};\sigma\big)$ be the space of all V_σ-valued square integrable functions $f : \mathbb{R}^{(m,n)} \longrightarrow V_\sigma$ satisfying the condition

$$f(\alpha^{-1}x) = \sigma(\alpha^{-1})f(x) \qquad \text{for all } \alpha \in O(m),\ x \in \mathbb{R}^{(m,n)}.$$

Let $\omega_c(\sigma)$ be the restriction of ω_c to the subspace $L^2\big(\mathbb{R}^{(m,n)};\sigma\big)$ of $L^2\big(\mathbb{R}^{(m,n)}\big)$. We denote by $\widehat{O(m)}$ the unitary dual of $O(m)$. In other words, $\widehat{O(m)}$ is the set of all equivalence classes of irreducible representations of $O(m)$. Let

$$\Sigma_m := \left\{ \sigma \in \widehat{O(m)} \;\middle|\; L^2\big(\mathbb{R}^{(m,n)};\sigma\big) \neq 0 \right\}.$$

Kashiwara and Vergne proved that for any $\sigma \in \Sigma_m$, the representation $\omega_c(\sigma)$ is an irreducible unitary representation of $Mp(2n,\mathbb{R})$ on $L^2\big(\mathbb{R}^{(m,n)};\sigma\big)$ and that ω_c is decomposed into irreducibles as follows:

$$\omega_c \;=\; \bigoplus_{\sigma\in\Sigma_m} (\dim V_\sigma)\, \omega_c(\sigma).$$

We realize $\omega_c(\sigma)$ in the space of vector-valued holomorphic functions on $\mathbb{H}_n$. We note that $\mathbb{H}_n$ is biholomorphic to the Hermitian symmetric Einstein-Kähler manifold $Sp(2n,\mathbb{R})/K$ with $K := U(n)$ via the map

$$Sp(2n,\mathbb{R})/K \;\longrightarrow\; \mathbb{H}_n, \qquad gK \mapsto g\cdot(i\,I_n),\quad M \in Sp(2n,\mathbb{R}).$$

Let $\widehat{K}$ be the unitary dual of K. For any $(\tau, V_\tau) \in \widehat{K}$, we let $\mathscr{O}(\mathbb{H}_n, V_\tau)$ be the space of V_τ-valued holomorphic functions on $\mathbb{H}_n$. Let T_τ be the representation of

$Mp(2n, \mathbb{R})$ on $\mathscr{O}(\mathbb{H}_n, V_\tau)$ defined by

$$\big(T_\tau(M)f\big)(\Omega) := \tau({}^t(C\Omega + D))f(M^{-1} \cdot \Omega), \tag{2.3.13}$$

where $M^{-1} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$, $f \in \mathscr{O}(\mathbb{H}_n, V_\tau)$ and $\Omega \in \mathbb{H}_n$. Here τ can be extended uniquely to a representation of the complexification $GL(n, \mathbb{C})$ of K. If v_τ is the highest weight vector of τ, then $\Phi_\tau(\Omega) := \tau(\Omega + iI_n)v_\tau$ is the highest weight vector of T_τ. It can be shown that T_τ is an irreducible representation of $Sp(2n, \mathbb{R})$ with highest weight vector Φ_τ.

Definition 2.3.4 A polynomial $f : \mathbb{R}^{(m,n)} \longrightarrow \mathbb{C}$ is called pluriharmonic if

$$\sum_{k=1}^{m} \frac{\partial^2 f}{\partial x_{ki}\, \partial x_{kj}} = 0 \qquad \text{for all } 1 \le i, j \le n.$$

Let $\mathfrak{H}$ be the space of all pluriharmonic polynomials on $\mathbb{R}^{(m,n)}$. Then $O(m) \times GL(n, \mathbb{R})$ acts on $\mathfrak{H}$ by

$$\big((\alpha, a) \cdot P\big)(x) = P(\alpha^{-1} x a),$$

where $\alpha \in O(m)$, $a \in GL(n, \mathbb{R})$, $P \in \mathfrak{H}$ and $x \in \mathbb{R}^{(m,n)}$. For $(\sigma, V_\sigma) \in \Sigma_m$, we let $\mathfrak{H}(\sigma)$ be the space of all V_σ-valued pluriharmonic polynomials $P : \mathbb{R}^{(m,n)} \longrightarrow V_\sigma$ such that

$$P(\alpha x) = \sigma(\alpha^{-1})^{-1} P(x) \quad \text{for all } \alpha \in O(m) \text{ and } x \in \mathbb{R}^{(m,n)}.$$

Let $\tau(\sigma)$ be the representation of $GL(n, \mathbb{R})$ on $\mathfrak{H}(\sigma)$ defined by

$$\big(\tau(\sigma)(a)P\big)(x) = P(xa) \quad \text{for all } a \in GL(n, \mathbb{R}) \text{ and } P \in \mathfrak{H}(\sigma).$$

For $\sigma \in \Sigma_m$, we see that $\mathfrak{H}(\sigma) \neq 0$ and $\tau(\sigma)$ is an irreducible finite-dimensional representation of $GL(n, \mathbb{R})$ on $\mathfrak{H}(\sigma)$. They proved that the mapping $\sigma \longrightarrow \tau(\sigma)$ is an injection from Σ_m into $\widehat{GL(n, \mathbb{R})}$ and

$$\mathfrak{H} = \bigoplus_{\sigma \in \Sigma_m} \tau(\sigma) \otimes \sigma^* = \bigoplus_{\sigma \in \Sigma_m} \mathfrak{H}(\sigma) \otimes \sigma^*$$

as $O(m) \times GL(n, \mathbb{R})$-module.

Let $\sigma \in \Sigma_m$. We assume that $P : \mathbb{R}^{(m,n)} \longrightarrow \mathrm{Hom}_{\mathbb{C}}(V_{\tau(\sigma)}, V_\sigma)$ is a $\mathrm{Hom}_{\mathbb{C}}(V_{\tau(\sigma)}, V_\sigma)$-valued pluriharmonic polynomial on $\mathbb{R}^{(m,n)}$ satisfying the conditions

$$\text{(A)} \qquad P(\alpha x) = \sigma(\alpha^{-1})^{-1} P(x) \qquad \text{for all } \alpha \in O(m) \text{ and } x \in \mathbb{R}^{(m,n)}$$

and

$$\text{(B)} \qquad P(xa) \;=\; P(x)\left(\tau(\sigma)\otimes \det{}^{\frac{m}{2}}\right)(a) \qquad \text{for all } a \in GL(n,\mathbb{R}).$$

The unitary operator

$$\mathscr{F}_\sigma : L^2\big(\mathbb{R}^{(m,n)};\sigma\big) \longrightarrow \mathscr{O}\big(\mathbb{H}_n, V_{\tau(\sigma)}\big)$$

defined by

$$(\mathscr{F}_\sigma f)(\Omega) := \int_{\mathbb{R}^{(m,n)}} e^{\pi\, i\, \sigma(x\,\Omega\,{}^t x)}\, P(x)^* f(x)\, dx,$$

($f \in L^2\big(\mathbb{R}^{(m,n)};\sigma\big)$, $\Omega \in \mathbb{H}_n$) intertwines $\omega_c(\sigma)$ with $T_{\tau(\sigma)\otimes\det^{-\frac{m}{2}}}$. Here the function $\chi : GL(n,\mathbb{R}) \longrightarrow \mathbb{C}^\times$ defined by

$$\chi(a) \overset{\text{def}}{=} \det{}^{-\frac{1}{2}}(a), \qquad a \in GL(n,\mathbb{R})$$

is the one-dimensional representation of $GL(n,\mathbb{R})$ determined uniquely by the following properties (i) and (ii):

(i) $\chi(a)^2 = (\det a)^{-1}$ for all $a \in GL(n,\mathbb{R})$;
(ii) $\chi(a) = (\sqrt{\det a})^{-1}$ for all $a \in GL(n,\mathbb{R})$ with $\det a > 0$.

For a positive integer m, we define the one-dimensional representation $\det^{-\frac{m}{2}}$ of $GL(n,\mathbb{R})$ by

$$\det{}^{-\frac{m}{2}} \overset{\text{def}}{=} \left(\det{}^{-\frac{1}{2}}\right)^m.$$

The norm on $\mathscr{O}(\mathbb{H}_n, V_{\tau(\sigma)})$ is defined as follows: Let $\langle\ ,\ \rangle_\sigma : V_\sigma \times V_\sigma \longrightarrow \mathbb{C}$ be an inner product on V_σ such that

$$\langle \sigma(\alpha)u, \sigma(\alpha)v\rangle_\sigma = \langle u, v\rangle_\sigma$$

for all $\alpha \in O(m)$ and $u, v \in V_\sigma$. For an element $F \in \mathscr{O}(\mathbb{H}_n, V_{\tau(\sigma)})$, the norm on $\mathscr{O}(\mathbb{H}_n, V_{\tau(\sigma)})$ is given by

$$\|F\|^2_{\mathbb{H}_n,\tau(\sigma)} = \int_{\mathbb{H}_n}\int_{\mathbb{R}^{(m,n)}} \langle F(\Omega)(x), F(\Omega)(x)\rangle_\sigma\, dx\, dv_n(\Omega),$$

where dx is the Lebesgue measure on $\mathbb{R}^{(m,n)}$ and $dv_n(\Omega)$ is a $Sp(2n,\mathbb{R})$-invariant volume element on $\mathbb{H}_n$ (cf. (2.1.16)).

2.4 Covariant Maps for the Weil Representation

Let c be a symmetric positive definite real matrix of degree m. We define the map $\mathscr{F}^{(c)} : \mathbb{H}_n \longrightarrow L^2\big(\mathbb{R}^{(m,n)}\big)$ by

$$\mathscr{F}^{(c)}(\Omega)(x) := e^{2\pi i \sigma(c\, x\, \Omega\, {}^t x)}, \qquad \Omega \in \mathbb{H}_n,\ x \in \mathbb{R}^{(m,n)}. \tag{2.4.1}$$

We recall that $Sp(2n, \mathbb{R})$ acts on $\mathbb{H}_n$ transitively by

$$M \cdot \Omega = (A\Omega + B)(C\Omega + D)^{-1}, \quad \Omega \in \mathbb{H}_n, \tag{2.4.2}$$

where $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$. Put $J_1(M, \Omega) = \det(C\Omega + D)^{\frac{1}{2}}$ for $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$ and $\Omega \in \mathbb{H}_n$. We know that for a fixed element $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$, the function

$$p(\Omega) \overset{\text{def}}{=} \det(C\Omega + D)^{\frac{1}{2}}, \quad \Omega \in \mathbb{H}_n$$

is a holomorphic function on $\mathbb{H}_n$ determined uniquely by the properties (a) and (b):

(a) $p(\Omega)^2 = \det(C\Omega + D)$ for all $\Omega \in \mathbb{H}_n$.
(b) $p(\Omega)$ takes the principal square root of $\det(C\Omega + D)$.

Then $J_1 : Sp(2n, \mathbb{R}) \times \mathbb{H}_n \longrightarrow GL(1, \mathbb{C})$ is the function on $Sp(2n, \mathbb{R}) \times \mathbb{H}_n$ satisfying the following properties (1) and (2):

(1) $\{J_1(M, \Omega)\}^2 = \det(C\Omega + D)$ for all $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$ and $\mathbb{H}_n$.

(2) If $\sigma = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix} \in Sp(2n, \mathbb{R})$ and $Y \in \mathrm{Sym}(n, \mathbb{R})$ with $Y > 0$, we take

$$J_1(\sigma, i\, Y) = \{\det(i\, Y)\}^{\frac{1}{2}} = i^{\frac{n}{2}} \det Y = \left(\frac{\sqrt{2}}{2} + i\,\frac{\sqrt{2}}{2}\right)^n \det Y,$$

where $\frac{\sqrt{2}}{2} + i\,\frac{\sqrt{2}}{2}$ is the principal square root of i and we put $i^{\frac{n}{2}} = \left(i^{\frac{1}{2}}\right)^n$ for any positive integer n. For any positive integer m, we define

$$J_m(M, \Omega) \overset{\text{def}}{=} \{J_1(M, \Omega)\}^m \text{ for all } M \in Sp(2n, \mathbb{R}) \text{ and } \Omega \in \mathbb{H}_n. \tag{2.4.3}$$

Theorem 2.4.1 *The map $\mathscr{F}^{(c)} : \mathbb{H}_n \longrightarrow L^2\big(\mathbb{R}^{(m,n)}\big)$ defined by Formula (2.4.1) is a covariant map for the Weil representation ω_c of $Mp(2n, \mathbb{R})$ with respect to the*

function J_m defined by Formula (2.4.3). In other words, $\mathscr{F}^{(c)}$ satisfies the following covariant relation,

$$R_c(M)\mathscr{F}^{(c)}(\Omega) = J_m(M,\Omega)^{-1}\,\mathscr{F}^{(c)}(M\cdot\Omega), \tag{2.4.4}$$

for all $M \in Sp(2n,\mathbb{R})$ and $\Omega \in \mathbb{H}_n$. We recall that

$$\omega_c\big((M,t)\big) = t\,R_c(M) \qquad \text{(cf. (2.3.6))}$$

for all $(M,t) \in Mp(2n,\mathbb{R})$ with $M \in Sp(2n,\mathbb{R})$ and $t \in \mathbb{C}_1^\times$.

Proof For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$ and $\Omega \in \mathbb{H}_n$, we put

$$\Omega_* = M\cdot\Omega = (A\Omega + B)(C\Omega + D)^{-1}. \tag{2.4.5}$$

In this section, we use the notations t_b, d_a and σ_n from Sect. 2.3. It suffices to prove the covariance relation (2.4.4) for the generators t_b ($b = {}^t b \in \mathbb{R}^{(n,n)}$), d_a ($a \in GL(n,\mathbb{R})$) and σ_n of $Sp(2n,\mathbb{R})$.

Case I $M = t_b$ with $b = {}^t b \in \mathbb{R}^{(n,n)}$.

In this case, we have

$$\Omega_* = \Omega + b.$$

By Formula (2.3.8) in Theorem 2.3.3,

$$\begin{aligned} &\Big(R_c(M)\mathscr{F}^{(c)}(\Omega)\Big)(x) \\ &= \Big(R_c(t_b)\mathscr{F}^{(c)}(\Omega)\Big)(x) \\ &= e^{2\pi i\sigma(c\,x\,b\,{}^t x)}\,\mathscr{F}^{(c)}(\Omega)(x). \end{aligned}$$

On the other hand, according to Formula (2.4.3),

$$\begin{aligned} &J_m(M,\Omega)^{-1}\mathscr{F}^{(c)}(M\cdot\Omega)(x) \\ &= \mathscr{F}^{(c)}(\Omega+b)(x) \\ &= e^{2\pi i\sigma(c\,x(\Omega+b)\,{}^t x))} \\ &= e^{2\pi i\sigma(c\,x\,b\,{}^t x)}\,\mathscr{F}^{(c)}(\Omega)(x). \end{aligned}$$

Thus

$$R_c(t_b)\mathscr{F}^{(c)}(\Omega) = J_m(t_b,\Omega)^{-1}\mathscr{F}^{(c)}(t_b\cdot\Omega)$$

for all $b = {}^t b \in \mathbb{R}^{(n,n)}$ and $\Omega \in \mathbb{H}_n$. Therefore we proved the covariance relation (2.4.4) in the case $M = t_b$ with $b = {}^t b \in \mathbb{R}^{(n,n)}$.

Case II $M = d_a = \begin{pmatrix} {}^t a & 0 \\ 0 & a^{-1} \end{pmatrix}$ with $a \in GL(n, \mathbb{R})$.

In this case, we have

$$\Omega_* = {}^t a \, \Omega \, a.$$

By Formula (2.3.9) in Theorem 2.3.3,

$$\begin{aligned} &\left(R_c(M)\mathscr{F}^{(c)}(\Omega)\right)(x) \\ &= (\det a)^{\frac{m}{2}} \ \mathscr{F}^{(c)}(\Omega)(x\,{}^t a) \\ &= (\det a)^{\frac{m}{2}} \ e^{2\pi i \sigma(c\, x\, {}^t a\, \Omega\, a\, {}^t x)}. \end{aligned}$$

On the other hand, according to Formula (2.4.3),

$$\begin{aligned} &J_m(M, \Omega)^{-1} \mathscr{F}^{(c)}(M \cdot \Omega)(x) \\ &= \left(\det\left(a^{-1}\right)\right)^{-\frac{m}{2}} \ \mathscr{F}^{(c)}\left({}^t a\, \Omega\, a\right)(x) \\ &= (\det a)^{\frac{m}{2}} \ e^{2\pi i \sigma(c\, x\, {}^t a\, \Omega\, a\, {}^t x)}. \end{aligned}$$

Thus

$$R_c(d_a)\mathscr{F}^{(c)}(\Omega) = J_m(d_a, \Omega)^{-1} \mathscr{F}^{(c)}(d_a \cdot \Omega)$$

for all $a \in GL(n, \mathbb{R})$ and $\Omega \in \mathbb{H}_n$. Therefore we proved the covariance relation (2.4.4) in the case $M = d_a$ with $d_a \in GL(n, \mathbb{R})$.

Case III $M = \sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$.

In this case, we have

$$\Omega_* = -\Omega^{-1} \qquad \text{and} \qquad J_m(M, \Omega) = (\det \Omega)^{\frac{m}{2}}.$$

In order to prove the covariance relation (2.4.4), we need the following useful lemma.

Lemma 2.4.2 *For a fixed element $\Omega \in \mathbb{H}_n$ and a fixed element $Z \in \mathbb{C}^{(m,n)}$, we obtain the following property:*

$$\begin{aligned} &\int_{\mathbb{R}^{(m,n)}} e^{\pi\, i\, \sigma(x\,\Omega\,{}^t x + 2\,x\,{}^t Z)}\, dx_{11} \cdots dx_{mn} \qquad (2.4.6)\\ &= \left(\det \frac{\Omega}{i}\right)^{-\frac{m}{2}} e^{-\pi\, i\, \sigma(Z\,\Omega^{-1}\,{}^t Z)}, \end{aligned}$$

where $x = (x_{ij}) \in \mathbb{R}^{(m,n)}$.

Proof of Lemma 2.4.2. By a simple computation, we see that

$$e^{\pi i\, \sigma(x\Omega\,{}^t x + 2x\,{}^t Z)} = e^{-\pi i\, \sigma(Z\Omega^{-1}\,{}^t Z)} \cdot e^{\pi i\, \sigma\{(x+Z\Omega^{-1})\Omega\,{}^t(x+Z\Omega^{-1})\}}.$$

We observe that the real Jacobi group $Sp(2n, \mathbb{R}) \ltimes H_{\mathbb{R}}^{(n,m)}$ acts on $\mathbb{H}_n \times \mathbb{C}^{(m,n)}$ holomorphically and transitively by

$$\big(M, (\lambda, \mu, \kappa)\big) \cdot (\Omega, Z) = \big(M \cdot \Omega, (Z + \lambda\,\Omega + \mu)(C\Omega + D)^{-1}\big), \qquad (2.4.7)$$

where $M \in Sp(2n, \mathbb{R})$, $(\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$, $\Omega \in \mathbb{H}_n$ and $Z \in \mathbb{C}^{(m,n)}$. So we may put

$$\Omega = i\, A\,{}^t A, \quad Z = i\, V, \quad A \in \mathbb{R}^{(n,n)}, \quad V = (v_{ij}) \in \mathbb{R}^{(m,n)}.$$

Then we obtain

$$\begin{aligned} &\int_{\mathbb{R}^{(m,n)}} e^{\pi i\, \sigma(x\Omega\,{}^t x + 2x\,{}^t Z)}\, dx_{11} \cdots dx_{mn}\\ &= e^{-\pi i\, \sigma(Z\Omega^{-1}\,{}^t Z)} \int_{\mathbb{R}^{(m,n)}} e^{\pi i\, \sigma[\{x + i\,V(iA\,{}^tA)^{-1}\}(iA\,{}^tA)\,{}^t\{x + i\,V(iA\,{}^tA)^{-1}\}]}\, dx_{11} \cdots dx_{mn}\\ &= e^{-\pi i\, \sigma(Z\Omega^{-1}\,{}^t Z)} \int_{\mathbb{R}^{(m,n)}} e^{\pi i\, \sigma[\{x + V(A\,{}^tA)^{-1}\}A\,{}^tA\,{}^t\{x + V(A\,{}^tA)^{-1}\}]}\, dx_{11} \cdots dx_{mn}\\ &= e^{-\pi i\, \sigma(Z\Omega^{-1}\,{}^t Z)} \int_{\mathbb{R}^{(m,n)}} e^{-\pi\, \sigma\{(uA)\,{}^t(uA)\}}\, du_{11} \cdots du_{mn}\\ &\qquad \big(\text{Put } u = x + V(A\,{}^tA)^{-1} = (u_{ij})\big)\\ &= e^{-\pi i\, \sigma(Z\Omega^{-1}\,{}^t Z)} \int_{\mathbb{R}^{(m,n)}} e^{-\pi\, \sigma(w\,{}^t w)} (\det A)^{-m}\, dw_{11} \cdots dw_{mn}\\ &\qquad \big(\text{Put } w = uA = (w_{ij})\big) \end{aligned}$$

$$
\begin{aligned}
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\,(\det A)^{-m}\cdot\left(\prod_{i=1}^{m}\prod_{j=1}^{n}\int_{\mathbb{R}} e^{-\pi\,w_{ij}^2}\,dw_{ij}\right)\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\,(\det A)^{-m}\quad\left(\text{because }\int_{\mathbb{R}} e^{-\pi\,w_{ij}^2}\,dw_{ij}=1\quad\text{for all } i,\,j\right)\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\,\left(\det\left(A\,{}^tA\right)\right)^{-\frac{m}{2}}\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\,\left(\det\left(\frac{\Omega}{i}\right)\right)^{-\frac{m}{2}}.
\end{aligned}
$$

This completes the proof of Lemma 2.4.2. □

According to Formula (2.3.10) in Theorem 2.3.3,

$$
\begin{aligned}
&\left(R_c(\sigma_n)\mathscr{F}^{(c)}(\Omega)\right)(x)\\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}}(\det c)^{\frac{n}{2}}\int_{\mathbb{R}^{(m,n)}}\mathscr{F}^{(c)}(\Omega)(y)\,e^{-4\pi\,i\,\sigma(c\,y\,{}^t x)}\,dy\\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}}(\det c)^{\frac{n}{2}}\int_{\mathbb{R}^{(m,n)}} e^{2\pi\,i\,\sigma(c\,y\,\Omega\,{}^t y)}\cdot e^{-4\pi\,i\,\sigma(c\,y\,{}^t x)}\,dy\\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}}(\det c)^{\frac{n}{2}}\int_{\mathbb{R}^{(m,n)}} e^{\pi\,i\,\sigma\{c\,(y\,(2\,\Omega)\,{}^t y+2\,y\,{}^t(-2\,x))\}}\,dy.
\end{aligned}
$$

If we substitute $u = c^{1/2}\,y$, then $du = \left(\det c\right)^{\frac{n}{2}}\,dy$. Therefore according to Lemma 2.4.2, we obtain

$$
\begin{aligned}
&\left(R_c(\sigma_n)\mathscr{F}^{(c)}(\Omega)\right)(x)\\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}}(\det c)^{\frac{n}{2}}\int_{\mathbb{R}^{(m,n)}} e^{\pi\,i\,\sigma(u\,(2\,\Omega)\,{}^t u+2\,c^{1/2}\,u\,{}^t(-2\,x))}\,(\det c)^{-\frac{n}{2}}\,du\\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}}\int_{\mathbb{R}^{(m,n)}} e^{\pi\,i\,\sigma(u\,(2\,\Omega)\,{}^t u+2\,u\,{}^t(-2\,c^{1/2}\,x))}\,du\\
&= \left(\frac{2}{i}\right)^{\frac{mn}{2}}\left(\det\frac{2\,\Omega}{i}\right)^{-\frac{m}{2}}\,e^{-\pi\,i\,\sigma((-2\,c^{1/2}\,x)\,(2\,\Omega)^{-1}\,{}^t(-2\,c^{1/2}\,x))}\\
&= \left(\det\Omega\right)^{-\frac{m}{2}}\,e^{-2\pi\,i\,\sigma(c\,x\,\Omega^{-1}\,{}^t x)}.
\end{aligned}
$$

On the other hand, according to Formula (2.4.3),

$$
J_m(M,\,\Omega)^{-1}\mathscr{F}^{(c)}(M\cdot\Omega)(x)
$$

$$= J_m(\sigma, \Omega)^{-1}\, \mathscr{F}^{(c)}\big(-\Omega^{-1}\big)(x)$$
$$= \big(\det \Omega\big)^{-\frac{m}{2}}\, e^{2\pi i \sigma(c\, x\, (-\Omega^{-1})^t x)}$$
$$= \big(\det \Omega\big)^{-\frac{m}{2}}\, e^{-2\pi i \sigma(c\, x\, \Omega^{-1\, t} x)}.$$

So we see that

$$R_c(\sigma_n)\mathscr{F}^{(c)}(\Omega) = J_m(\sigma_n, \Omega)^{-1}\, \mathscr{F}^{(c)}(\sigma_n \cdot \Omega). \tag{2.4.8}$$

Therefore the covariance relation (2.4.4) holds for the case $\sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$.

We see that if the covariance relation (2.4.4) holds for M_1, M_2 in $Sp(2n, \mathbb{R})$, then it holds for $M_1 M_2$. Finally, we complete the proof. □

Now we can give another realization of the metaplectic group $Mp(2n, \mathbb{R})$ that was dealt with in Sect. 2.3.

Proposition 2.4.3 *Let $(U_c, \mathscr{H}_c)$ be the Schrödinger representation of the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ defined by Formula (1.2.11) with the model $\mathscr{H}_c = L^2\big(\mathbb{R}^{(m,n)}, d\xi\big)$. We denote by $U(\mathscr{H}_c)$ the group of all unitary isomorphisms of $\mathscr{H}_c$. Let $\widetilde{Mp(c)}$ be the set of all $R \in U(\mathscr{H}_c)$ such that*

$$R\, U_c(g) = U_c(M \star g)R = U_c(MgM^{-1})R$$

for all $g \in H_{\mathbb{R}}^{(n,m)}$ and for some $M \in Sp(2n, \mathbb{R})$. Then for a given element $R \in \widetilde{Mp(c)}$, the corresponding $M \in Sp(2n, \mathbb{R})$ is determined uniquely, denoted by $M = \nu_c(R)$. Moreover, there is an exact sequence of groups

$$1 \longrightarrow \mathbb{C}_1^{\times} \longrightarrow \widetilde{Mp(c)} \xrightarrow{\nu_c} Sp(2n, \mathbb{R}) \longrightarrow 1. \tag{2.4.9}$$

Proof First of all we observe that $\widetilde{Mp(c)}$ is a subgroup of $U(\mathscr{H}_c)$. Let $R \in \widetilde{Mp(c)}$, and $M_1, M_2 \in Sp(2n, \mathbb{R})$ such that

$$R\, U_c(g) = U_c(M_1 \star g)\, R = U_c(M_2 \star g)\, R \qquad \text{for all } g \in H_{\mathbb{R}}^{(n,m)}.$$

Then $U_c(M_1 \star g) = U_c(M_2 \star g)$ for all $g \in H_{\mathbb{R}}^{(n,m)}$. According to Formula (1.2.11), $(M_1^{-1}M_2)g = g(M_1^{-1}M_2)$ for all $g \in H_{\mathbb{R}}^{(n,m)}$. Thus $M_1 = M_2$. It follows that the map $\nu_c : \widetilde{Mp(c)} \longrightarrow Sp(2n, \mathbb{R})$ is well defined. It is easily checked that ν_c is a group homomorphism. The kernel of ν_c is given by

$$\ker \nu_c = \Big\{ R \in U(\mathscr{H}_c) \;\Big|\; R\, U_c(g) = U_c(g)\, R \quad \text{for all } g \in H_{\mathbb{R}}^{(n,m)} \Big\}.$$

Since U_c is irreducible and unitary, according to Schur's lemma, $\ker \nu_c = \mathbb{C}_1^\times$. The surjectivity of ν_c follows from the arguments in Sect. 2.3. □

According to Theorem 2.3.3, $R_c(t_b)$, $R_c(d_a)$ and $R_c(\sigma_n)$ are members of $\widetilde{Mp(c)}$ sitting above the generators t_b, d_a and σ_n of $Sp(2n,\mathbb{R})$ respectively. That is, $\nu_c(R_c(t_b)) = t_b$, $\nu_c(R_c(d_a)) = d_a$ and $\nu_c(R_c(\sigma_n)) = \sigma_n$.

Theorem 2.4.4 *Let $P \in \widetilde{Mp(c)}$ and $\nu_c(P) = M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$. Then for any $\Omega \in \mathbb{H}_n$,*

$$P\mathscr{F}^{(c)}(\Omega) = B_c(P;\Omega)\mathscr{F}^{(c)}(M\cdot\Omega),$$

where $B_c(P;\Omega)$ is, up to a scalar of absolute one, a branch of the holomorphic function $\left\{\det(C\Omega+D)^{\frac{1}{2}}\right\}^{-m}$ on $\mathbb{H}_n$.

Proof Let $\mathbb{G}_1$ be the subgroup of $\widetilde{Mp(c)}$ consisting of all $P \in \widetilde{Mp(c)}$ such that

$$P\mathscr{F}^{(c)}(\Omega) = c_P\,\mathscr{F}^{(c)}\big(\nu_c(P)\cdot\Omega\big) \quad \text{for all } \Omega \in \mathbb{H}_n,$$

where c_P is a constant depending only on P. For $P \in \mathbb{G}_1$, we write

$$P\mathscr{F}^{(c)}(\Omega) = B_c(P;\Omega)\,\mathscr{F}^{(c)}\big(\nu_c(P)\cdot\Omega\big) \quad \text{for all } \Omega \in \mathbb{H}_n.$$

Let $\mathbb{G}_2$ be the set of all $P \in \mathbb{G}_1$ satisfying the following conditions ($\mathbb{G}1$) and ($\mathbb{G}2$):

($\mathbb{G}1$) $B_c(P;\Omega)$ is continuous in $\Omega \in \mathbb{H}_n$.
($\mathbb{G}2$) $\{B_c(P;\Omega)\}^2\,|\det(C\Omega+D)|^m$ is independent of Ω with values in $\mathbb{C}_1^\times$ for $\nu_c(P) = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$.

It is easily checked that for $P, Q \in \mathbb{G}_2$,

$$B_c(QP;\Omega) = B_c(P;\Omega)\,B_c(Q;\nu_c(P)\cdot\Omega) \qquad \text{for all } \Omega \in \mathbb{H}_n. \tag{2.4.10}$$

Indeed, we get

$$\begin{aligned}
&(QP)\mathscr{F}^{(c)}(\Omega)\\
&= Q\big(P\mathscr{F}^{(c)}(\Omega)\big)\\
&= B_c(P;\Omega)\left(Q\big(\mathscr{F}^{(c)}(\nu_c(P)\cdot\Omega)\big)\right)\\
&= B_c(P;\Omega)\,B_c(Q;\nu_c(P)\cdot\Omega)\,\mathscr{F}^{(c)}\big(\nu_c(Q)\cdot(\nu_c(P)\cdot\Omega)\big)\\
&= B_c(P;\Omega)\,B_c(Q;\nu_c(P)\cdot\Omega)\,\mathscr{F}^{(c)}\big(\nu_c(QP)\cdot\Omega\big).
\end{aligned}$$

By Formula (2.4.9) together with the fact that $J(M,\Omega) := \det(C\Omega + D)$ for $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$ and $\Omega \in \mathbb{H}_n$ is an automorphic factor, we see that $\mathbb{G}_2$ is a subgroup of $\mathbb{G}_1$. We observe that $R_c(t_b),\ R_c(d_a),\ R_c(\sigma_n)$ in Theorem 2.3.3 and $\alpha \in \mathbb{C}_1^\times$ generate the group $\widetilde{Mp(c)}$. We shall show that $R_c(t_b),\ R_c(d_a),\ R_c(\sigma_n)$ and $\alpha \in \mathbb{C}_1^\times$ belong to $\mathbb{G}_2$. Then $\mathbb{G}_1 = \mathbb{G}_2 = \widetilde{Mp(c)}$. This implies the proof of the theorem.

Now we shall prove that $R_c(t_b),\ R_c(d_a),\ R_c(\sigma_n)$ and $\alpha \in \mathbb{C}_1^\times$ belong to $\mathbb{G}_2$. For brevity we put $F_c(P;\Omega) = \{B_c(P;\Omega)\}^2\,|\det(C\Omega + D)|^m$ for $\nu_c(P) = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$ with $P \in \widetilde{Mp(c)}$.

Case I $P = \alpha \in \mathbb{C}_1^\times \subset \widetilde{Mp(c)}$.

In this case, we obtain

$$P\mathscr{F}^{(c)}(\Omega) = \alpha\,\mathscr{F}^{(c)}(\Omega).$$

So we get $B_c(P;\Omega) = \alpha$ and $F_c(P;\Omega) = \alpha^2$. Thus $\alpha \in \mathbb{G}_2$.

Case II $P = R_c(t_b)$ with $t_b = \begin{pmatrix} I_n & b \\ 0 & I_n \end{pmatrix} \in Sp(2n,\mathbb{R})$.

In this case, according to Formula (2.3.8), we obtain

$$\begin{aligned} P\mathscr{F}^{(c)}(\Omega) &= e^{2\pi i\,\sigma(c\,x b\,{}^t x)}\,\mathscr{F}^{(c)}(\Omega)(x) \\ &= e^{2\pi i\,\sigma\{c\,x(\Omega+b)\,{}^t x\}} \\ &= \mathscr{F}^{(c)}(\Omega + b)(x) = \mathscr{F}^{(c)}(t_b \cdot \Omega)(x) \\ &= \mathscr{F}^{(c)}\big(\nu_c(R_c(t_b)) \cdot \Omega\big)(x). \end{aligned}$$

We get $B_c(P;\Omega) = 1$ and $F_c(P;\Omega) = 1$. Thus $R_c(t_b) \in \mathbb{G}_2$.

Case III $P = R_c(d_a)$ with $d_a = \begin{pmatrix} {}^t a & 0 \\ 0 & a^{-1} \end{pmatrix} \in Sp(2n,\mathbb{R})$.

In this case, according to Formula (2.3.9), we obtain

$$\begin{aligned} P\mathscr{F}^{(c)}(\Omega) &= (\det a)^{\frac{m}{2}}\,\mathscr{F}^{(c)}(\Omega)(x\,{}^t a) \\ &= (\det a)^{\frac{m}{2}}\,e^{2\pi i\,\sigma\{c\,x({}^t a\,\Omega\,a)\,{}^t x\}} \\ &= (\det a)^{\frac{m}{2}}\,\mathscr{F}^{(c)}(d_a \cdot \Omega)(x) \\ &= (\det a)^{\frac{m}{2}}\,\mathscr{F}^{(c)}\big(\nu_c(R_c(d_a)) \cdot \Omega\big)(x). \end{aligned}$$

We get $B_c(P;\Omega) = (\det a)^{\frac{m}{2}}$ and $F_c(P;\Omega) = 1$. Thus $R_c(d_a) \in \mathbb{G}_2$.

Case IV $P = R_c(\sigma_n)$ with $\sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix} \in Sp(2n, \mathbb{R})$.

In this case, according to Formula (2.3.10), we obtain

$$\begin{aligned} P\mathscr{F}^{(c)}(\Omega) &= \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} \mathscr{F}^{(c)}(\Omega)(y)\, e^{-4\pi i\, \sigma(c\, y\, {}^t x)}\, dy \\ &= \left(\frac{2}{i}\right)^{\frac{mn}{2}} (\det c)^{\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} e^{2\pi i\, \sigma\{c\,(y\,\Omega\, {}^t y - 2\, y {}^t x)\}}\, dy \\ &= (\det \Omega)^{-\frac{m}{2}}\, e^{-2\pi i\, \sigma(c\, x\, \Omega^{-1}\, {}^t x)} \qquad \text{(by Lemma 2.4.2)} \\ &= (\det \Omega)^{-\frac{m}{2}}\, \mathscr{F}^{(c)}(-\Omega^{-1})(x) \\ &= (\det \Omega)^{-\frac{m}{2}}\, \mathscr{F}^{(c)}\big(\nu_c(R_c(\sigma_n)) \cdot \Omega\big)(x). \end{aligned}$$

We get $B_c(P; \Omega) = (\det \Omega)^{-\frac{m}{2}}$ with $B_c(P; i\, I_n) = i^{-\frac{mn}{2}}$, and $F_c(P; \Omega) = i^{-\frac{mn}{2}}$. Thus $R_c(\sigma_n) \in \mathbb{G}_2$. Hence we complete the proof. □

Definition 2.4.5 Let $\chi_c : \widetilde{Mp(c)} \longrightarrow \mathbb{C}$ be the map defined by

$$\chi_c(P) = \det(C\Omega + D)^m \big\{B_c(P; \Omega)\big\}^2, \quad P \in \widetilde{Mp(c)},$$

where $\nu_c(P) = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$. According to Theorem 2.4.4, the image of χ_c is contained in $\mathbb{C}_1^\times$ and $\chi_c : \widetilde{Mp(c)} \longrightarrow \mathbb{C}_1^\times$ is a character of $\widetilde{Mp(c)}$. Furthermore, we have

$$\chi_c(\alpha) = \alpha^2 \quad \text{for any } \alpha \in \mathbb{C}_1^\times \subset \widetilde{Mp(c)}.$$

We denote by $Mp(2n, \mathbb{R})_c$ the kernel of χ_c. We call $Mp(2n, \mathbb{R})_c$ the metaplectic group attached to U_c.

We let

$$m_\diamond : \widetilde{Mp(c)} \times \widetilde{Mp(c)} \longrightarrow \widetilde{Mp(c)}$$

be the multiplication map and let

$$\Phi_{[c]} : \widetilde{Mp(c)} \times \mathbb{H}_n \longrightarrow \mathbb{C}^\times$$

be the map defined by

$$\Phi_{[c]}(P, \Omega) := B_c(P; \Omega), \quad P \in \widetilde{Mp(c)},\ \Omega \in \mathbb{H}_n.$$

We provide $\widetilde{Mp(c)}$ with the weakest topology such that the following three maps,

$$\nu_c : \widetilde{Mp(c)} \longrightarrow Sp(2n,\mathbb{R}), \quad m_\diamond : \widetilde{Mp(c)} \times \widetilde{Mp(c)} \longrightarrow \widetilde{Mp(c)},$$
$$\Phi_{[c]} : \widetilde{Mp(c)} \times \mathbb{H}_n \longrightarrow \mathbb{C}^\times,$$

are all continuous.

Then we have the following properties.

Lemma 2.4.6 *$\widetilde{Mp(c)}$ is a Hausdorff space on the above weakest topology.*

Proof Fix an element $\Omega_0 \in \mathbb{H}_n$. Let $\eta : \widetilde{Mp(c)} \longrightarrow Sp(2n,\mathbb{R}) \times \mathbb{C}^\times$ by

$$\eta(P) := \big(\nu_c(P), B_c(P;\Omega_0)\big), \quad P \in \widetilde{Mp(c)}. \tag{2.4.11}$$

Then by the weak topology on $\widetilde{Mp(c)}$, η is continuous. If $P, Q \in \widetilde{Mp(c)}$ such that $\eta(P) = \eta(Q)$, then $\nu_c(P) = \nu_c(Q)$ and $B_c(P;\Omega_0) = B_c(Q;\Omega_0)$. $QP^{-1} = \alpha \in \mathbb{C}_1^*$ because $\nu_c(QP^{-1}) = 1$. Thus $Q = \alpha P$. By assumption,

$$B_c(Q;\Omega_0) = B_c(\alpha P;\Omega_0) = \alpha B_c(P;\Omega_0) = B_c(P;\Omega_0).$$

Therefore $\alpha = 1$, that is, $P = Q$. This implies that η is one-to-one.

Let $f : Sp(2n,\mathbb{R}) \times \mathbb{C}^\times \longrightarrow \mathbb{C}^\times$ be the map defined by

$$f(M,\alpha) := \alpha^2 \{\det(C\Omega_0 + D)\}^m, \tag{2.4.12}$$

where $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$ and $\alpha \in \mathbb{C}^\times$. By Theorem 2.4.4, $\eta\big(\widetilde{Mp(c)}\big) = f^{-1}(\mathbb{C}_1^\times)$. Since $\frac{\partial f}{\partial \alpha} \neq 0$ and $\mathbb{C}_1^\times$ is a submanifold of $\mathbb{C}^\times$, we see that $\eta\big(\widetilde{Mp(c)}\big)$ is a submanifold of $Sp(2n,\mathbb{R}) \times \mathbb{C}^\times$. Therefore $\eta\big(\widetilde{Mp(c)}\big)$ is Hausdorff because $Sp(2n,\mathbb{R})$ and $\mathbb{C}^\times$ are Hausdorff. □

Lemma 2.4.7 *Let $h : \widetilde{Mp(c)} \longrightarrow Sp(2n,\mathbb{R}) \times \mathbb{C}_1^\times$ be the map defined by*

$$h(P) := \big(\nu_c(P), \chi_c(P)\big), \quad P \in \widetilde{Mp(c)}.$$

Then the map h defines a connected double covering of the Lie group $Sp(2n,\mathbb{R}) \times \mathbb{C}_1^\times$, and hence gives $\widetilde{Mp(c)}$ the structure of a Lie group.

Proof We note that h is continuous. We see that

$$\ker h = \ker \nu_c \cap \ker \chi_c = \mathbb{C}_1^\times \cap \ker \chi_c = \ker\big(\chi_c|_{\mathbb{C}_1^\times}\big) = \{\pm 1\}.$$

Let $h_* : Sp(2n,\mathbb{R}) \times \mathbb{C}^\times \longrightarrow Sp(2n,\mathbb{R}) \times \mathbb{C}^\times$ be the map defined by

$$h_*(M,\alpha) := (M,\, f(M,\alpha)), \quad M \in Sp(2n,\mathbb{R}),\ \alpha \in \mathbb{C}^\times, \tag{2.4.13}$$

where f is the map defined by (2.4.12). Then $h = h_* \circ \eta$, where η is the map defined by (2.4.11). Clearly, h_* is a double covering projection. Since $h_*^{-1}\big(Sp(2n,\mathbb{R}) \times \mathbb{C}_1^\times\big) = \eta\big(\widetilde{Mp(c)}\big)$, the restriction $h_{*,\eta}$ of h_* to $\eta\big(\widetilde{Mp(c)}\big)$ is a double covering

$$h_{*,\eta} : \widetilde{Mp(c)} \longrightarrow Sp(2n,\mathbb{R}) \times \mathbb{C}_1^\times$$

of the manifold $Sp(2n,\mathbb{R}) \times \mathbb{C}_1^\times$. It only remains to prove that $\widetilde{Mp(c)}$ is connected. Since $Sp(2n,\mathbb{R})$ and $\mathbb{C}_1^\times$ are connected, according to the exact sequence (2.4.9), $\widetilde{Mp(c)}$ is connected. □

Proposition 2.4.8 *$Mp(2n,\mathbb{R})_c$ is a closed connected subgroup of $\widetilde{Mp(c)}$ and $q_c : Mp(2n,\mathbb{R})_c \longrightarrow Sp(2n,\mathbb{R})$ is a double covering projection with $\ker q_c = \{\pm 1\}$, where q_c is the restriction of ν_c to $Mp(2n,\mathbb{R})_c$.*

Proof By Lemma 2.4.7, q_c is a double covering projection of $Sp(2n,\mathbb{R})$. It only remains to prove that $Mp(2n,\mathbb{R})_c$ is connected. The stabilizer at $i\,I_n$ under the action (2.1.6) of $Sp(2n,\mathbb{R})$ is given by

$$\left\{ \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \in Sp(2n,\mathbb{R}) \;\middle|\; A\,{}^tA + B\,{}^tB = I_n,\ A\,{}^tB = B\,{}^tA \right\},$$

which is isomorphic to $U(n)$ via $\begin{pmatrix} A & B \\ -B & A \end{pmatrix} \longmapsto A + i\,B$. The map

$$Mp(2n,\mathbb{R})_c \longrightarrow \mathbb{H}_n, \quad P \longmapsto q_c(P)\cdot(i\,I_n), \quad P \in Mp(2n,\mathbb{R})_c$$

gives the coset space of $Mp(2n,\mathbb{R})_c$ with respect to $q_c^{-1}(U(n))$, i.e.,

$$Mp(2n,\mathbb{R})_c / q_c^{-1}(U(n)) = \mathbb{H}_n.$$

For $\Omega = i\,I_n$, the map $B_{c;i\,I_n} : q_c^{-1}(U(n)) \longrightarrow \mathbb{C}_1^\times$ defined by

$$B_{c;i\,I_n}(P) := B_c(P; i\,I_n), \quad P \in q_c^{-1}(U(n)) \tag{2.4.14}$$

is a continuous character. If $P \in q_c^{-1}(U(n))$ with $q_c(P) = \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \in U(n)$, then

$$\big\{B_{c;i\,I_n}(P)\big\}^2 = \big\{B_c(P; i\,I_n)\big\}^2 = \big\{\det(A - i\,B)\big\}^{-m}.$$

We define the map $\det^*_{c,m} : U(n) \longrightarrow \mathbb{C}_1^\times$ by

$$\det{}^*_{c,m}\begin{pmatrix} A & B \\ -B & A \end{pmatrix} := \{\det(A - i\,B)\}^{-m}, \quad \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \in U(n). \tag{2.4.15}$$

and the map $Sq : \mathbb{C}_1^\times \longrightarrow \mathbb{C}_1^\times$ by $Sq(\alpha) = \alpha^2$ with $\alpha \in \mathbb{C}_1^\times$. The we have the following commutative diagram:

$$\begin{array}{ccc} q_c^{-1}(U(n)) & \xrightarrow{B_{c;iI_n}} & \mathbb{C}_1^\times \\ {\scriptstyle q_c}\downarrow & & \downarrow{\scriptstyle Sq} \\ U(n) & \xrightarrow{\det^*_{c,m}} & \mathbb{C}_1^\times \end{array}$$

[diagram 2.4.1]

Thus $q_c^{-1}(U(n))$ along with its topology is the fibre product of $\det^*_{c,m}$ and Sq. Since $U(n)$ and $\mathbb{C}_1^\times$ are connected, $q_c^{-1}(U(n))$ is connected. □

Corollary 2.4.9 *The exact sequence*

$$1 \longrightarrow \{\pm 1\} \longrightarrow Mp(2n,\mathbb{R})_c \xrightarrow{q_c} Sp(2n,\mathbb{R}) \longrightarrow 1 \tag{2.4.16}$$

is non-split and $[Mp(2n,\mathbb{R})_c, Mp(2n,\mathbb{R})_c] = Mp(2n,\mathbb{R})_c$.

Proof Embed $U(1)$ into $U(n)$ via $z \mapsto \mathrm{diag}(z, 1, 1, \cdots, 1)$, and embed $U(n)$ into $Sp(2n,\mathbb{R})$ via

$$U(n) \ni A + iB \longmapsto \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \in Sp(2n,\mathbb{R}) \quad \text{with } A, B \in \mathbb{R}^{(n,n)}.$$

So $U(1) \subset U(n) \subset Sp(2n,\mathbb{R})$. According to the commutative diagram in the proof of Proposition 2.4.8, the exact sequence

$$1 \longrightarrow \{\pm 1\} \longrightarrow q_c^{-1}(U(1)) \xrightarrow{q_c} U(1) \longrightarrow 1$$

can be identified to

$$1 \longrightarrow \{\pm 1\} \longrightarrow \mathbb{C}_1^\times \xrightarrow{Sq} \mathbb{C}_1^\times \longrightarrow 1. \tag{2.4.17}$$

If we restrict the exact sequence (2.4.17) to the torsion subgroups, then we get the non-split exact sequence

$$1 \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow \mathbb{Q}/\mathbb{Z} \xrightarrow{m_2} \mathbb{Q}/\mathbb{Z} \longrightarrow 1, \tag{2.4.18}$$

where $m_2 : \mathbb{Q}/\mathbb{Z} \longrightarrow \mathbb{Q}/\mathbb{Z}$ is the map defined by $m_2(x) = 2x$ for $x \in \mathbb{Q}/\mathbb{Z}$. Thus the exact sequence (2.4.16) is non-split.

For brevity, we put $Mp_{(c)} := Mp(2n, \mathbb{R})_c$. Since $[Sp(2n, \mathbb{R}), Sp(2n, \mathbb{R})] = Sp(2n, \mathbb{R})$, $[Mp_{(c)}, Mp_{(c)}]$ sits in the exact sequence

$$1 \longrightarrow \{\pm 1\} \cap [Mp_{(c)}, Mp_{(c)}] \longrightarrow [Mp_{(c)}, Mp_{(c)}] \longrightarrow Sp(2n, \mathbb{R}) \longrightarrow 1. \tag{2.4.19}$$

Assume $\{\pm 1\} \cap [Mp_{(c)}, Mp_{(c)}]$ is trivial. Then according to the above exact sequence (2.4.19), we have an isomorphism $\phi : Sp(2n, \mathbb{R}) \longrightarrow [Mp_{(c)}, Mp_{(c)}] \neq Mp_{(c)}$. Thus the exact sequence (2.4.16) is split because $q_c \circ \phi$ is the identity map. This contradicts the fact that the exact sequence (2.4.16) is non-split. Hence we obtain

$$\{\pm 1\} \cap [Mp_{(c)}, Mp_{(c)}] = \{\pm 1\} \qquad \text{and} \qquad [Mp_{(c)}, Mp_{(c)}] = Mp_{(c)}.$$

□

Corollary 2.4.10 *For a fixed element $\Omega \in \mathbb{H}_n$, we let*

$$U(\Omega) = \left\{ M \in Sp(2n, \mathbb{R}) \,\middle|\, M \cdot \Omega = \Omega \right\}.$$

Let $M_\Omega \in Sp(2n, \mathbb{R})$ such that $\Omega = M \cdot (i\, I_n)$. Then $U(\Omega) = M_\Omega\, U(n) M_\Omega^{-1}$. If $P \in q_c^{-1}(U(\Omega))$ such that $q_c(P) = M_\Omega\, q_c(P_0) M_\Omega^{-1}$ with $P_0 \in q_c^{-1}(U(n))$, then

$$\left\{B_c(P; \Omega)\right\}^2 = \det{}^*_{c,m}\big(q_c(P_0)\big),$$

where $\det^*_{c,m} : U(n) \longrightarrow \mathbb{C}_1^\times$ is the map defined by Formula (2.4.15).

Proof The case $\Omega = i\, I_n$ has already been proved before. We note that $U(i\, I_n) = U(n)$. For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ and $\Omega \in \mathbb{H}_n$, we put $J(M, \Omega) = \det(C\Omega + D)$. By definition, if $P \in q_c^{-1}(U(\Omega))$ such that $q_c(P) = M_\Omega\, q_c(P_0) M_\Omega^{-1}$ with $P_0 \in q_c^{-1}(U(n))$ and $q_c(P_0) = \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \in U(n)$, then

$$\begin{aligned}
\left\{B_c(P; \Omega)\right\}^2 &= J(q_c(P), \Omega)^{-m} = \left\{J\big(M_\Omega\, q_c(P_0) M_\Omega^{-1}, \Omega\big)\right\}^{-m} \\
&= \left\{J(M_\Omega\, q_c(P_0), i\, I_n)\, J(M_\Omega^{-1}, \Omega)\right\}^{-m} \\
&= \left\{J(M_\Omega, i\, I_n)\, J(M_\Omega^{-1}, \Omega)\, J(q_c(P_0), i\, I_n)\right\}^{-m} \\
&= \left\{J(q_c(P_0), i\, I_n)\right\}^{-m} \\
&= \left\{\det(A - iB)\right\}^{-m} = \det{}^*_{c,m}\big(q_c(P_0)\big).
\end{aligned}$$

□

2.5 Transformation Formula of Theta Series with Quadratic Forms

In this section, we review the theta series of several types.

Definition 2.5.1 A symmetric integral matrix S of degree m is said to be even if ${}^t\xi\, S\, \xi \equiv 0 \bmod 2$ for all $\xi \in \mathbb{Z}^{(m,1)}$. The level q of an even symmetric nonsingular matrix S is defined to be the smallest positive integer such that qS^{-1} is even.

It is well known that if S is a positive definite even integral matrix of degree m such that $\det S = 1$, then m is divisible by 8 (cf. [10, 0.4 Satz, p. 16]).

Definition 2.5.2 For a symmetric integral matrix T of degree n and a symmetric integral matrix S of degree m, we define

$$A(S, T) := \sharp\{\, \xi \in \mathbb{Z}^{(m,n)} \mid {}^t\xi\, S\, \xi \;=\; T\,\}.$$

We observe that if S is positive definite, $A(S, T)$ is finite.

Lemma 2.5.3 *Let S_1 and S_2 be two positive definite $m \times m$ symmetric integral matrices. Assume $A(S_1, S_2) \neq 0$. Then S_1 and S_2 are equivalent, that is, ${}^tUS_1U = S_2$ for some $U \in GL(m, \mathbb{Z})$ if and only if $A(S_1, T) = A(S_2, T)$ for any positive integer n and symmetric integral $n \times n$ matrices T.*

Proof Suppose S_1 and S_2 are equivalent, i.e, ${}^tUS_1U = S_2$ for some $U \in GL(m, \mathbb{Z})$. For any positive integer n and any symmetric integral $n \times n$ matrices T, we let

$$\widetilde{A}(S_i, T) \stackrel{\text{def}}{=} \left\{ \xi \in \mathbb{Z}^{(m,n)} \mid {}^t\xi\, S_i\, \xi = T \right\}, \quad i = 1, 2.$$

We define the mapping

$$\Phi_U : \widetilde{A}(S_1, T) \longrightarrow \widetilde{A}(S_2, T)$$

by

$$\Phi_U(\xi) := U^{-1}\xi, \quad \xi \in \widetilde{A}(S_1, T).$$

If $\xi \in \widetilde{A}(S_1, T)$, then

$$T = {}^t\xi\, S_1\, \xi = {}^t(U^{-1}\xi)\, S_2\, (U^{-1}\xi).$$

Thus $U^{-1}\xi \in \widetilde{A}(S_2, T)$. The inverse $\Psi_U = \Phi_U$ of Φ_U is given by

$$\Psi_U(\eta) = U\eta, \quad \eta \in \widetilde{A}(S_2, T).$$

Therefore

$$A(S_1, T) = \sharp(\widetilde{A}(S_1, T)) = \sharp(\widetilde{A}(S_2, T)) = A(S_2, T).$$

The converse is obvious. Since $A(S_1, S_2) \neq 0$, we see that S_1 is equivalent to S_2. □

Let S be a positive definite integral matrix of degree m. We define the theta series $\vartheta_S : \mathbb{H}_n \longrightarrow \mathbb{C}$ by

$$\vartheta_S(\Omega) = \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{\pi i \sigma(S\xi\, \Omega\, {}^t\xi)}, \quad \Omega \in \mathbb{H}_n. \tag{2.5.1}$$

Then $\vartheta_S(\Omega)$ is a holomorphic function on $\mathbb{H}_n$. We see that

$$\vartheta_S(\Omega) = \sum_{T = {}^tT \geq 0} A(S, T)\, e^{\pi i \sigma(T\Omega)},$$

where T runs over the set of all semipositive symmetric integral matrices of degree n.

Theorem 2.5.4 *Let S be a positive definite symmetric integral matrix of degree m. Then $\vartheta_S(\Omega)$ satisfies the transformation formula*

$$\vartheta_{S^{-1}}(-\Omega^{-1}) = (\det S)^{\frac{n}{2}} \left(\det \frac{\Omega}{i}\right)^{\frac{m}{2}} \vartheta_S(\Omega) \qquad \text{for all } \Omega \in \mathbb{H}_n. \tag{2.5.2}$$

Here we put

$$\left(\det \frac{\Omega}{i}\right)^{\frac{m}{2}} = \left\{\left(\det \frac{\Omega}{i}\right)^{\frac{1}{2}}\right\}^m, \qquad \Omega \in \mathbb{H}_n.$$

for a positive integer m with the function $h : \mathbb{H}_n \longrightarrow \mathbb{C}$

$$h(\Omega) = \left(\det \frac{\Omega}{i}\right)^{\frac{1}{2}}, \qquad \Omega \in \mathbb{H}_n$$

determined uniquely by the following properties:

(a) $h^2(\Omega) = \left(\det \frac{\Omega}{i}\right), \quad \Omega \in \mathbb{H}_n,$

(b) $h(i\, Y) = (\det Y)^{\frac{1}{2}}$ for any positive definite symmetric real matrix Y

of degree n.

Proof For a fixed element $\Omega \in \mathbb{H}_n$, we define $f : \mathbb{R}^{(m,n)} \longrightarrow \mathbb{C}$ by

$$f(x) = \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{\pi i \sigma(S(\xi+x)\,\Omega\,{}^t(\xi+x))}, \quad x \in \mathbb{R}^{(m,n)}. \tag{2.5.3}$$

We observe that f is well defined because the sum of the right-hand side of (2.5.3) converges absolutely. It is clear that if $x = (x_{ij})$ is a coordinate in $\mathbb{R}^{(m,n)}$, then f is periodic in x_{ij} with period 1. That is,

$$f(x+\alpha) = f(x) \qquad \text{for all } \alpha \in \mathbb{Z}^{(m,n)}.$$

Thus f has the Fourier series

$$f(x) = \sum_{\alpha \in \mathbb{Z}^{(m,n)}} c_\alpha\, e^{2\pi i \sigma(x\,{}^t\alpha)}, \quad x \in \mathbb{R}^{(m,n)}, \tag{2.5.4}$$

where

$$\begin{aligned}
c_\alpha &= \int_0^1 \cdots \int_0^1 f(y)\, e^{-2\pi i \sigma(y\,{}^t\alpha)}\, dy \\
&= \int_0^1 \cdots \int_0^1 \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{\pi i \sigma\{S(\xi+y)\,\Omega\,{}^t(\xi+y)\}} \cdot e^{-2\pi i \sigma(y\,{}^t\alpha)}\, dy \\
&= \int_{\mathbb{R}^{(m,n)}} e^{\pi i \sigma(S\,y\,\Omega\,{}^t y - 2\,y\,{}^t\alpha)}\, dy \\
&= \int_{\mathbb{R}^{(m,n)}} e^{\pi i \sigma\{S(y\,\Omega\,{}^t y - 2\,y\,{}^t(S^{-1}\alpha))\}}\, dy \\
&= (\det S)^{-\frac{n}{2}} \left(\det \frac{\Omega}{i}\right)^{-\frac{m}{2}} e^{-\pi i \sigma(S^{-1}\alpha\,\Omega^{-1}\,{}^t\alpha)} \quad \text{(by Lemma 2.4.2).}
\end{aligned}$$

According to Formulas (2.5.3) and (2.5.4),

$$\begin{aligned}
\vartheta_S(\Omega) = f(0) &= \sum_{\alpha \in \mathbb{Z}^{(m,n)}} c_\alpha \\
&= (\det S)^{-\frac{n}{2}} \left(\det \frac{\Omega}{i}\right)^{-\frac{m}{2}} e^{\pi i \sigma\{S^{-1}\alpha\,(-\Omega^{-1})\,{}^t\alpha\}} \\
&= (\det S)^{-\frac{n}{2}} \left(\det \frac{\Omega}{i}\right)^{-\frac{m}{2}} \vartheta_{S^{-1}}(-\Omega^{-1}).
\end{aligned}$$

Consequently, we obtain Formula (2.5.2). □

Let S be a positive definite even integral symmetric matrix of degree m. Let A and B be $m \times n$ rational matrices. We define the theta series

$$\vartheta_{S;A,B}(\Omega) = \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{\pi i \sigma\{S(\xi+\frac{1}{2}A)\,\Omega\,{}^t(\xi+\frac{1}{2}A)+{}^tB\,\xi\}}.$$

Theorem 2.5.5 *Let S be an positive definite even integral symmetric matrix of degree m. Let A and B be $m \times n$ rational matrices. Then $\vartheta_{S;A,B}(\Omega)$ satisfies the transformation formula*

$$\vartheta_{S^{-1};A,B}(-\Omega^{-1}) = e^{-\frac{1}{2}\pi i \sigma({}^tAB)} (\det S)^{\frac{n}{2}} \left(\det \frac{\Omega}{i}\right)^{\frac{m}{2}} \vartheta_{S;B,-A}(\Omega) \tag{2.5.5}$$

for all $\Omega \in \mathbb{H}_n$.

Proof Following the argument of the proof of Theorem 2.5.4, we can obtain Formula (2.5.5). We leave the details to the reader. □

Definition 2.5.6 A holomorphic function $f : \mathbb{H}_n \longrightarrow \mathbb{C}$ is called a Siegel modular form of weight $k \in \mathbb{Z}$ if it satisfies the following properties:

(1) $f(M\cdot\Omega) = \det(C\Omega + D)^k f(\Omega)$ for $\gamma = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \Gamma_n$.

(2) f is bounded in the domain $Y \geq Y_0 > 0$ with $\Omega = X + iY$, X, Y real.

We will give some examples of Siegel modular forms using the so-called thetanullwerte. For $a, b \in \mathbb{Z}^n$, we consider the thetanullwerte

$$\vartheta(\Omega\,; a, b) = \sum_{\xi \in \mathbb{Z}^n} e^{\pi i \sigma\{{}^t(\xi+\frac{1}{2}a)\,\Omega\,(\xi+\frac{1}{2}a)+{}^tb\,\xi\}}, \quad \Omega \in \mathbb{H}_n. \tag{2.5.6}$$

Lemma 2.5.7 *Let $a, b \in \mathbb{Z}^n$. Then $\vartheta(\Omega\,; a, b)$ satisfies the following properties:*

(a) $\vartheta(\Omega\,; a, b_1) = \vartheta(\Omega\,; a, b_2)$ *if* $b_1 \equiv b_2$ *mod 2.*
(b) *If* $\widetilde{a} \in \mathbb{Z}^n$, *then* $\vartheta(\Omega\,; a + 2\widetilde{a}, b) = (-1)^{{}^tb\widetilde{a}}\,\vartheta(\Omega\,; a, b)$.
(c) $\vartheta(\Omega\,; a, b) = (-1)^{{}^tab}\,\vartheta(\Omega\,; a, b)$.
(d) $\vartheta(\Omega\,; a, b) = 0$ *if* ${}^tab \not\equiv 0$ *mod 2.*

Proof (a) follows from a direct computation. If we put $\xi_* = \xi + \widetilde{a}$,

$$\begin{aligned}
&\vartheta(\Omega\,; a + 2\widetilde{a}, b) \\
&= \sum_{\xi \in \mathbb{Z}^n} e^{\pi i \sigma\{{}^t(\xi+\frac{1}{2}a+\widetilde{a})\,\Omega\,(\xi+\frac{1}{2}a+\widetilde{a})+{}^tb\,\xi\}} \\
&= \sum_{\xi_* \in \mathbb{Z}^n} e^{\pi i \sigma\{{}^t(\xi_*+\frac{1}{2}a)\,\Omega\,(\xi_*+\frac{1}{2}a)+{}^tb(\xi_*-\widetilde{a})\}} \\
&= e^{-\pi i\,{}^tb\widetilde{a}}\,\vartheta(\Omega\,; a, b).
\end{aligned}$$

Therefore we get the formula (b). If we substitute ξ into $-\xi - a$, we obtain the formula (c). (d) follows immediately from the formula (c). □

A pair $\{a, b\}$ with $a, b \in \{0, 1\}^n$ is called a theta characteristic. A theta characteristic $\{a, b\}$ is said to be even (resp. odd) if ${}^t ab$ is even (resp. odd). By induction on n, we can show that the number of even theta characteristics is $(2^n + 1)\, 2^{n-1}$.

Let $\gamma = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \Gamma_n$ and let $\{a, b\}$ be a theta characteristic. We define

$$\gamma \diamond \begin{pmatrix} a \\ b \end{pmatrix} :\equiv \begin{pmatrix} D & -C \\ -B & A \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} (C\,{}^t D)_0 \\ (A\,{}^t B)_0 \end{pmatrix} \bmod 2, \tag{2.5.7}$$

where T_0 is the column vector determined by the diagonal entries of an $n \times n$ matrix T.

Theorem 2.5.8

(1) *The Siegel modular group Γ_n acts on the set $\mathscr{C}$ of theta characteristics by*

$$\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \gamma \diamond \begin{pmatrix} a \\ b \end{pmatrix}, \quad \gamma \in \Gamma_n,\ \{a, b\} \in \mathscr{C}.$$

(2) *The sign $(-1)^{{}^t ab}$ of the theta characteristic $\{a, b\}$ is invariant under the action (2.5.7) of Γ_n.*

(3) *Γ_n acts transitively on the set $\mathscr{C}^e$ of all even theta characteristics*

(4) *If $\gamma = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \Gamma_n$, $\Omega \in \mathbb{H}_n$ and $\{a, b\} \in \mathscr{C}$, then we have*

$$\vartheta^2(\gamma \cdot \Omega\,; a, b) = \nu(\gamma)\, \det(C\Omega + D)\, \vartheta^2\big(\Omega\,; \widetilde{a}, \widetilde{b}\big), \tag{2.5.8}$$

where

(a) $\nu(\gamma)^4 = 1$,

(b) $\begin{pmatrix} \widetilde{a} \\ \widetilde{b} \end{pmatrix} = \gamma \diamond \begin{pmatrix} a \\ b \end{pmatrix}$.

Proof By a direct computation, we prove the statement (a). It suffices to show the invariance of the sign of $(-1)^{{}^t ab}$ under the generators $t_S = \begin{pmatrix} I_n & S \\ 0 & I_n \end{pmatrix}$ with $S = {}^t S \in \mathbb{Z}^{(n,n)}$ and J_n. By a simple computation,

$$\begin{pmatrix} I_n & S \\ 0 & I_n \end{pmatrix} \diamond \begin{pmatrix} a \\ b \end{pmatrix} \equiv \begin{pmatrix} a \\ b - Sa + S_0 \end{pmatrix} \bmod 2,$$

$$J_n \diamond \begin{pmatrix} a \\ b \end{pmatrix} \equiv \begin{pmatrix} b \\ -a \end{pmatrix} \pmod 2, \qquad J_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Therefore it is obvious that the sign of $(-1)^{{}^t ab}$ is invariant under the actions of t_S and J_n. In order to prove the transitivity of Γ_n on $\mathscr{C}^e$, first of all we have to prove the fact that given an even characteristic $\{a, b\} \in \mathscr{C}^e$, there exists an element $\gamma = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \Gamma_n$ such that

$$\gamma \diamond \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}, \quad i.e., \quad a \equiv (C\,{}^t D)_0, \ b \equiv (A\,{}^t B)_0 \bmod 2.$$

We decompose

$$a = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \quad a_1 \in \mathbb{Z}, \ a_2 \in \mathbb{Z}^{n-1},$$

$$b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad b_1 \in \mathbb{Z}, \ b_2 \in \mathbb{Z}^{n-1}.$$

Case 1. $a_1 b_1 = 0$

Then $\{a_1, b_1\}$ is even and also $\{a_2, b_2\}$ is even. By induction on n, we can find $\gamma \in \Gamma_n$ such that $\gamma \diamond \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$.

Case 2. $a_1 = b_1 = 1$

Since ${}^t ab$ is even, there exists an index ν with $2 \le \nu \le n$ such that $a_\nu = b_\nu = 1$. Therefore we can find a symmetric integral matrix $S = {}^t S \in \mathbb{Z}^{(n,n)}$ so that

$$\begin{pmatrix} I_n & S \\ 0 & I_n \end{pmatrix} \diamond \begin{pmatrix} a \\ b \end{pmatrix} \equiv \begin{pmatrix} a \\ b - S\,a + S_0 \end{pmatrix} \bmod 2$$

is an even theta characteristic satisfying the assumption of Case 1.

According to Case 1 and Case 2, we see that Γ_n acts on $\mathscr{C}^e$ transitively.

The transformation formula (2.5.8) for the generator J_n follows from the formula (2.5.5) with $S = 1$ and $m = 1$. For a generator t_S with $S = {}^t S \in \mathbb{Z}^{(n,n)}$, it is easy to see that

$$\vartheta(\Omega + S\,; a, b) = e^{\frac{\pi i}{4}\,{}^t aSa}\ \vartheta(\Omega\,; a, b + Sa + S_0). \tag{2.5.9}$$

In fact, (2.5.9) follows from the following simple fact that ${}^t \xi S \xi \equiv {}^t S_0\, \xi \bmod 2$ for any $\xi \in \mathbb{Z}^n$ and $x^2 \equiv x \bmod 2$ for any $x \in \mathbb{Z}$. We know that $\vartheta(\Omega\,; 0, 0) \neq 0$ because $\vartheta(i\,Y\,; 0, 0) > 0$. □

Theorem 2.5.9 *We set*

$$k_n = \begin{cases} 8 & \text{if} \quad n = 1 \\ 2 & \text{if} \quad n = 2 \\ 1 & \text{if} \quad n \geq 3. \end{cases}$$

We define the function $\Delta^{(n)}(\Omega)$ *on* $\mathbb{H}_n$ *by*

$$\Delta^{(n)}(\Omega) := \prod_{\{a,b\}} \vartheta(\Omega\,; a, b)^{k_n},$$

where $\{a, b\}$ *runs through even theta characteristics. Then* $\Delta^{(n)}(\Omega)$ *is a nonzero Siegel modular form on* $\mathbb{H}_n$ *of weight 12, 10 and* $(2^n + 1)\, 2^{n-2}$ *respectively if* $n = 1, 2$ *and* $n \geq 3$ *respectively.*

Proof The proof can be found in [10]. □

Theorem 2.5.10 *Let m be an even positive integer. Let S be a positive definite even integral symmetric matrix of degree m and of level q. Then for all* $\gamma = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \Gamma_{n,0}(q)$ *with* $\det D > 0$,

$$\vartheta_S(\gamma\langle\Omega\rangle) = \nu_S(\gamma)\, \det(C\Omega + D)^{\frac{m}{2}}\, \vartheta_S(\Omega), \qquad \Omega \in \mathbb{H}_n,$$

where

$$\begin{aligned} \nu_S(\gamma) &= (\det D)^{\frac{m}{2} - mn} \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{\pi\, i\, \sigma(BD^{-1}\, {}^t\xi\, S\, \xi)} \\ &= (\operatorname{sgn} \det D)^{\frac{m}{2}} \left(\frac{(-1)^{\frac{m}{2}} \det S}{|\det D|} \right). \end{aligned}$$

Here $\left(\frac{a}{b}\right)$ *denotes the generalized Legendre symbol.*

Proof The proof can be found in [10, pp. 302–303]. □

Theorem 2.5.11 *Let m be an even positive integer. Let S be a positive definite even integral symmetric matrix of degree m and of level q. Then* $\vartheta_S(\Omega)$ *is a modular form with respect to the principal congruence subgroup* $\Gamma_n(q)$ *of* Γ_n.

Proof The proof follows from Theorem 2.5.10. □

Theorem 2.5.12 *Let S be a positive definite even integral symmetric matrix of degree m. Let A and B be $m \times n$ rational matrices. Then the theta series*

$$\vartheta_{S;A,B}(\Omega) = \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{\pi i \sigma\{S(\xi+\frac{1}{2}A)\,\Omega\,{}^t(\xi+\frac{1}{2}A)+{}^t B\,\xi\}}$$

is a modular form of weight $\frac{m}{2}$ with respect to a certain congruence subgroup $\Gamma_n(\ell)$ of Γ_n.

Proof The proof can be found in [1]. □

2.6 Relation Between Theta Series and the Weil Representation

Let (π, V_π) be a unitary projective representation of $Sp(2n, \mathbb{R})$ on the representation space V_π. We assume that (π, V_π) satisfies the following conditions (A) and (B):

(A) There exists a vector-valued map

$$\mathscr{F} : \mathbb{H}_n \longrightarrow V_\pi, \qquad \Omega \mapsto \mathscr{F}_\Omega := \mathscr{F}(\Omega)$$

satisfying the following covariance relation,

$$\pi(M)\mathscr{F}_\Omega = \psi(M)\, J(M, \Omega)^{-1}\, \mathscr{F}_{M\cdot\Omega}, \tag{2.6.1}$$

for all $M \in Sp(2n, \mathbb{R})$ and $\Omega \in \mathbb{H}_n$. Here ψ is a character of $Sp(2n, \mathbb{R})$ and $J : Sp(2n, \mathbb{R}) \times \mathbb{H}_n \longrightarrow GL(1, \mathbb{C})$ is a certain automorphic factor for $Sp(2n, \mathbb{R})$ on $\mathbb{H}_n$.

(B) Let Γ be an arithmetic subgroup of the Siegel modular group Γ_n. There exists a linear functional $\theta : V_\pi \longrightarrow \mathbb{C}$ which is semi-invariant under the action of Γ, in other words, for all $\gamma \in \Gamma$ and $\Omega \in \mathbb{H}_n$,

$$\langle \pi^*(\gamma)\theta,\ \mathscr{F}_\Omega \rangle = \langle \theta, \pi(\gamma)^{-1}\mathscr{F}_\Omega \rangle = \chi(\gamma)\, \langle \theta,\ \mathscr{F}_\Omega \rangle, \tag{2.6.2}$$

where π^* is the contragredient of π and $\chi : \Gamma \longrightarrow \mathbb{C}_1^\times$ is a unitary character of Γ.

Under the assumptions (A) and (B) on a unitary projective representation (π, V_π), we define the function Θ on $\mathbb{H}_n$ by

$$\Theta(\Omega) := \langle \theta, \mathscr{F}_\Omega \rangle = \theta\big(\mathscr{F}_\Omega\big), \qquad \Omega \in \mathbb{H}_n. \tag{2.6.3}$$

We shall see now that Θ is an automorphic form on $\mathbb{H}_n$ with respect to Γ for the automorphic factor J.

Lemma 2.6.1 *Let (π, V_π) be a unitary projective representation of $Sp(2n, \mathbb{R})$ satisfying the above assumptions* (A) *and* (B). *Then the function Θ on $\mathbb{H}_n$ defined by Formula* (2.6.3) *satisfies the following modular transformation behavior,*

$$\Theta(\gamma\cdot\Omega) = \psi(\gamma)^{-1}\,\chi(\gamma)^{-1}\,J(\gamma,\Omega)\,\Theta(\Omega), \tag{2.6.4}$$

for all $\gamma \in \Gamma$ and $\Omega \in \mathbb{H}_n$.

Proof For any $\gamma \in \Gamma$ and $\Omega \in \mathbb{H}_n$, according to the assumptions (2.6.1) and (2.6.2), we obtain

$$\begin{aligned}
&\Theta(\gamma\cdot\Omega) = \left\langle \theta, \mathscr{F}_{\gamma\cdot\Omega}\right\rangle \\
&= \left\langle \theta, \psi(\gamma)^{-1}\,J(\gamma,\Omega)\,\pi(\gamma)\mathscr{F}_\Omega\right\rangle \\
&= \psi(\gamma)^{-1}J(\gamma,\Omega)\left\langle \theta, \pi(\gamma)\mathscr{F}_\Omega\right\rangle \\
&= \psi(\gamma)^{-1}\,\chi(\gamma)^{-1}\,J(\gamma,\Omega)\left\langle \theta, \mathscr{F}_\Omega\right\rangle \\
&= \psi(\gamma)^{-1}\,\chi(\gamma)^{-1}\,J(\gamma,\Omega)\,\Theta(\Omega).
\end{aligned}$$

□

Now for a positive definite real symmetric matrix $\mathscr{M}$ of degree m, we define the holomorphic function $\Theta_{\mathscr{M}} : \mathbb{H}_n \longrightarrow \mathbb{C}$ by

$$\Theta_{\mathscr{M}}(\Omega) := \sum_{\xi\in\mathbb{Z}^{(m,n)}} e^{2\pi i\,\sigma(\mathscr{M}\xi\,\Omega\,{}^t\xi)}, \qquad \Omega \in \mathbb{H}_n. \tag{2.6.5}$$

Theorem 2.6.2 *Let $2\,\mathscr{M}$ be a symmetric positive definite, unimodular even integral matrix of degree m. Then for any $\gamma \in \Gamma_n$, the function $\Theta_{\mathscr{M}}$ satisfies the functional equation*

$$\Theta_{\mathscr{M}}(\gamma\langle\Omega\rangle) = \rho_{\mathscr{M}}(\gamma)\,J_m(\gamma,\Omega)\Theta_{\mathscr{M}}(\Omega), \qquad \Omega \in \mathbb{H}_n, \tag{2.6.6}$$

where $\rho_{\mathscr{M}}$ is a character of Γ with $|\rho_{\mathscr{M}}(\gamma)|^8 = 1$ for all $\gamma \in \Gamma_n$ and $J_m : Sp(2n,\mathbb{R})\times\mathbb{H}_n \longrightarrow \mathbb{C}_1^\times$ is the automorphic factor for $Sp(2n,\mathbb{R})$ on $\mathbb{H}_n$ defined by Formula (2.4.3) in Sect. 2.4.

Proof For an element $\gamma = \begin{pmatrix} A & B \\ C & D\end{pmatrix} \in \Gamma_n$ and $\Omega \in \mathbb{H}_n$, we put

$$\Omega_* = \gamma\cdot\Omega = (A\Omega + B)(C\Omega + D)^{-1}.$$

We define the linear functional ϑ on $L^2\big(\mathbb{R}^{(m,n)}\big)$ by

$$\vartheta(f) = \langle\vartheta, f\rangle := \sum_{\xi\in\mathbb{Z}^{(m,n)}} f(\xi), \qquad f \in L^2\big(\mathbb{R}^{(m,n)}\big).$$

We note that $\Theta_{\mathscr{M}}(\Omega) = \vartheta\left(\mathscr{F}_{\Omega}^{(\mathscr{M})}\right)$. Since $\mathscr{F}^{(\mathscr{M})}$ is a covariant map for the Weil representation $\omega_{\mathscr{M}}$ with respect to the automorphic factor J_m by Theorem 2.4.1, according to Lemma 2.6.1, it suffices to prove that ϑ is semi-invariant for $\omega_{\mathscr{M}}$ under the action of Γ_n, in other words, ϑ satisfies the following semi-invariance relation,

$$\left\langle \vartheta,\, R_{\mathscr{M}}(\gamma)\mathscr{F}_{\Omega}^{(\mathscr{M})} \right\rangle = \rho_{\mathscr{M}}(\gamma)^{-1} \left\langle \vartheta,\, \mathscr{F}_{\Omega}^{(\mathscr{M})} \right\rangle, \tag{2.6.7}$$

for all $\gamma \in \Gamma_n$ and $\Omega \in \mathbb{H}_n$.

We see that the following elements,

$$t_\beta = \begin{pmatrix} I_n & \beta \\ 0 & I_n \end{pmatrix} \text{ with any } \beta = {}^t\beta \in \mathbb{Z}^{(n,n)},$$

$$d_\alpha = \begin{pmatrix} {}^t\alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix} \text{ with any } \alpha \in GL(n, \mathbb{Z}),$$

$$\sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix},$$

generate the Siegel modular group Γ_n. Therefore it suffices to prove the semi-invariance relation (2.6.7) for the above generators of Γ_n.

Case I. $\gamma = t_\beta$ with $\beta = {}^t\beta \in \mathbb{Z}^{(n,n)}$.

In this case, we have

$$\Omega_* = \Omega + \beta \qquad \text{and} \qquad J_m(\gamma, \Omega) = 1.$$

According to the covariance relation (2.4.4) in Sect. 2.4, we obtain

$$\begin{aligned}
&\left\langle \vartheta,\, R_{\mathscr{M}}(\gamma)\mathscr{F}_{\Omega}^{(\mathscr{M})} \right\rangle \\
&= \left\langle \vartheta,\, J_m(\gamma, \Omega)^{-1}\mathscr{F}_{\gamma\cdot\Omega}^{(\mathscr{M})} \right\rangle \\
&= \left\langle \vartheta,\, \mathscr{F}_{\Omega+\beta}^{(\mathscr{M})} \right\rangle \\
&= \sum_{\xi \in \mathbb{Z}^{(m,n)}} \mathscr{F}_{\Omega+\beta}^{(\mathscr{M})}(\xi) \\
&= \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{2\pi i\, \sigma(\mathscr{M}\,\xi\,(\Omega+\beta)\,{}^t\xi))} \\
&= \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{2\pi i\, \sigma(\mathscr{M}\,\xi\,\Omega\,{}^t\xi)} \cdot e^{2\pi i\, \sigma(\mathscr{M}\,\xi\,\beta\,{}^t\xi)}
\end{aligned}$$

$$= \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{2\pi i \sigma(\mathscr{M} \xi \Omega {}^t\xi)}$$

$$= \left\langle \vartheta, \mathscr{F}_{\Omega}^{(\mathscr{M})} \right\rangle.$$

Here we used the fact that $2\sigma(\mathscr{M} \xi \beta {}^t\xi)$ is an even integer because $2\mathscr{M}$ is even integral. We put $\rho_{\mathscr{M}}(\gamma) = \rho_{\mathscr{M}}\left({}^t\beta\right) = 1$ for all $\beta = {}^t\beta \in \mathbb{Z}^{(n,n)}$. Therefore ϑ satisfies the semi-invariance relation (2.6.7) in the case $\gamma = t_\beta$ with $\beta = {}^t\beta \in \mathbb{Z}^{(n,n)}$.

Case II. $\gamma = d_\alpha$ with $\alpha \in GL(n, \mathbb{Z})$.

In this case, we have

$$\Omega_* = {}^t\alpha \Omega \alpha \qquad \text{and} \qquad J_m(d_\alpha, \Omega) = (\det \alpha)^{-\frac{m}{2}}.$$

According to the covariance relation (2.4.4) in Sect. 2.4, we obtain

$$\begin{aligned}
&\left\langle \vartheta, R_{\mathscr{M}}(\gamma) \mathscr{F}_{\Omega}^{(\mathscr{M})} \right\rangle \text{ with } \gamma = d_\alpha \\
&= \left\langle \vartheta, J_m(\gamma, \Omega)^{-1} \mathscr{F}_{\gamma \cdot \Omega}^{(\mathscr{M})} \right\rangle \\
&= (\det \alpha)^{\frac{m}{2}} \left\langle \vartheta, \mathscr{F}_{{}^t\alpha \Omega \alpha}^{(\mathscr{M})} \right\rangle \\
&= (\det \alpha)^{\frac{m}{2}} \sum_{\xi \in \mathbb{Z}^{(m,n)}} \mathscr{F}_{{}^t\alpha \Omega \alpha}^{(\mathscr{M})}(\xi) \\
&= (\det \alpha)^{\frac{m}{2}} \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{2\pi i \sigma(\mathscr{M} \xi {}^t\alpha \Omega \alpha {}^t\xi)} \\
&= (\det \alpha)^{\frac{m}{2}} \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{2\pi i \sigma\{\mathscr{M}((\xi {}^t\alpha) \Omega {}^t(\xi {}^t\alpha))\}} \\
&= (\det \alpha)^{\frac{m}{2}} \left\langle \vartheta, \mathscr{F}_{\Omega}^{(\mathscr{M})} \right\rangle.
\end{aligned}$$

Here we put $\rho_{\mathscr{M}}(d_\alpha) = (\det \alpha)^{-\frac{m}{2}}$. Therefore ϑ satisfies the semi-invariance relation (2.6.7) in the case $\gamma = d_\alpha$ with $\alpha \in GL(n, \mathbb{Z})$.

Case III. $\gamma = \sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$.

In this case, we have

$$\Omega_* = -\Omega^{-1} \qquad \text{and} \qquad J_m(\sigma_n, \Omega) = \left(\det \Omega\right)^{\frac{m}{2}}.$$

In the process of the proof of Theorem 2.4.1, using Lemma 2.4.2, we already showed that

$$\int_{\mathbb{R}^{(m,n)}} e^{2\pi i \sigma(\mathscr{M}(y\,\Omega^{t}y+2\,y^{t}x))}dy \tag{2.6.8}$$
$$= \left(\det \mathscr{M}\right)^{-\frac{n}{2}} \left(\det \frac{2\,\Omega}{i}\right)^{-\frac{m}{2}} e^{-2\pi i \sigma(\mathscr{M}\,x\,\Omega^{-1}\,{}^{t}x)}.$$

By Formula (2.6.8), we obtain

$$\begin{aligned}
&\widehat{\mathscr{F}_{\Omega}^{(\mathscr{M})}}(2\,\mathscr{M}x)\\
&= \int_{\mathbb{R}^{(m,n)}} \mathscr{F}_{\Omega}^{(\mathscr{M})}(y)\, e^{-2\pi i \sigma(y^{t}(2\,\mathscr{M}x))}\, dy\\
&= \int_{\mathbb{R}^{(m,n)}} e^{2\pi i \sigma(\mathscr{M}\,y\,\Omega^{t}y)} \cdot e^{-4\pi i \sigma(\mathscr{M}\,y^{t}x)}\, dy\\
&= \int_{\mathbb{R}^{(m,n)}} e^{2\pi i \sigma\{\mathscr{M}(y\,\Omega^{t}y+2\,y^{t}(-x))\}}\, dy\\
&= (\det \mathscr{M})^{-\frac{n}{2}} \left(\det \frac{2\,\Omega}{i}\right)^{-\frac{m}{2}} e^{-2\pi i \sigma(\mathscr{M}\,(-x)\,\Omega^{-1}\,{}^{t}(-x))}\\
&= (\det \mathscr{M})^{-\frac{n}{2}} \left(\det \frac{2\,\Omega}{i}\right)^{-\frac{m}{2}} e^{-2\pi i \sigma(\mathscr{M}\,x\,\Omega^{-1}\,{}^{t}x)}.
\end{aligned}$$

Thus we obtain

$$\widehat{\mathscr{F}_{\Omega}^{(\mathscr{M})}}(2\,\mathscr{M}x) = \left(\det \mathscr{M}\right)^{-\frac{n}{2}} \left(\det \frac{2\,\Omega}{i}\right)^{-\frac{m}{2}} e^{-2\pi i \sigma(\mathscr{M}\,x\,\Omega^{-1}\,{}^{t}x)}, \tag{2.6.9}$$

where $\widehat{f}$ is the Fourier transform of f defined by

$$\widehat{f}(x) = \int_{\mathbb{R}^{(m,n)}} f(y)\, e^{-2\pi i \sigma(y^{t}x)}\, dy, \quad x \in \mathbb{R}^{(m,n)}.$$

We prove the Poisson summation formula in our setting.

Lemma 2.6.3 *Let f be an element in $L^2\left(\mathbb{R}^{(m,n)}\right)$. Then*

$$\sum_{\xi \in \mathbb{Z}^{(m,n)}} \widehat{f}(\xi) = \sum_{\xi \in \mathbb{Z}^{(m,n)}} f(\xi). \tag{2.6.10}$$

Proof We define

$$h(x) = \sum_{\xi \in \mathbb{Z}^{(m,n)}} f(x+\xi), \qquad x \in \mathbb{R}^{(m,n)}. \tag{2.6.11}$$

We see that $h(x)$ is periodic in x_{ij} with period 1, where $x = (x_{ij})$ is a coordinate in $\mathbb{R}^{(m,n)}$. Thus $h(x)$ has the following Fourier series:

$$h(x) = \sum_{\xi \in \mathbb{Z}^{(m,n)}} c_\xi \, e^{2\pi i \sigma(x\,{}^t\xi)}, \tag{2.6.12}$$

where

$$\begin{aligned} c_\xi &= \int_0^1 \cdots \int_0^1 h(x)\, e^{-2\pi i \sigma(x\,{}^t\xi)}\, dx \\ &= \int_0^1 \cdots \int_0^1 \sum_{\xi \in \mathbb{Z}^{(m,n)}} f(x+\xi)\, e^{-2\pi i \sigma(x\,{}^t\xi)}\, dx \\ &= \int_{\mathbb{R}^{(m,n)}} f(x)\, e^{-2\pi i \sigma(x\,{}^t\xi)}\, dx \;=\; \widehat{f}(\xi). \end{aligned}$$

Here we interchanged summation and integration, and made a change of variables replacing $x + \xi$ by x to obtain the above equality.

By the definition (2.6.11), we have

$$h(0) = \sum_{\xi \in \mathbb{Z}^{(m,n)}} f(\xi).$$

On the other hand, from Formula (2.6.12), we get

$$h(0) = \sum_{\xi \in \mathbb{Z}^{(m,n)}} c_\xi \;=\; \sum_{\xi \in \mathbb{Z}^{(m,n)}} \widehat{f}(\xi).$$

Therefore we obtain the Poisson summation formula (2.6.10). □

According to the covariance relation (2.4.4) in Sect. 2.4, Formula (2.6.9) and Poisson summation formula, we obtain

$$\begin{aligned} &\left\langle \vartheta,\, R_{\mathscr{M}}(\gamma)\mathscr{F}_\Omega^{(\mathscr{M})} \right\rangle \quad \text{with } \gamma = \sigma_n \\ &= \left\langle \vartheta,\, J_m(\gamma,\Omega)^{-1}\mathscr{F}_{\gamma\cdot\Omega}^{(\mathscr{M})} \right\rangle \\ &= J_m(\gamma,\Omega)^{-1}\left\langle \vartheta,\, \mathscr{F}_{-\Omega^{-1}}^{(\mathscr{M})} \right\rangle \end{aligned}$$

$$= (\det \Omega)^{-\frac{m}{2}} \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{-2\pi i \sigma(\mathscr{M}\xi \Omega^{-1} {}^t\xi)}$$

$$= (\det \Omega)^{-\frac{m}{2}} \left(\det \mathscr{M}\right)^{\frac{n}{2}} \left(\det \frac{2\Omega}{i}\right)^{\frac{m}{2}} \sum_{\xi \in \mathbb{Z}^{(m,n)}} \widehat{\mathscr{F}_\Omega^{(\mathscr{M})}}(2\mathscr{M}\xi)$$

(by Formula (2.6.9))

$$= \left(\det 2\mathscr{M}\right)^{\frac{n}{2}} \left(\det \frac{I_n}{i}\right)^{\frac{m}{2}} \sum_{\xi \in \mathbb{Z}^{(m,n)}} \widehat{\mathscr{F}_\Omega^{(\mathscr{M})}}(\xi)$$

(because $2\mathscr{M}$ is unimodular)

$$= \left(\det \frac{I_n}{i}\right)^{\frac{m}{2}} \sum_{\xi \in \mathbb{Z}^{(m,n)}} \mathscr{F}_\Omega^{(\mathscr{M})}(\xi) \quad \text{(by Poisson summation formula)}$$

$$= (-i)^{\frac{mn}{2}} \left\langle \vartheta, \mathscr{F}_\Omega^{(\mathscr{M})} \right\rangle$$

$$= (-i)^{\frac{mn}{2}} \left\langle \vartheta, \mathscr{F}_\Omega^{(\mathscr{M})} \right\rangle.$$

We put $\rho_{\mathscr{M}}(\sigma_n) = (-i)^{-\frac{mn}{2}}$. Therefore ϑ satisfies the semi-invariance relation (2.6.7) in the case $\gamma = \sigma_n$. The proof of Case III is completed. Since J_m is an automorphic factor for $Sp(2n, \mathbb{R})$ on $\mathbb{H}_n$, we see that if Formula (2.6.6) holds for two elements γ_1, γ_2 in Γ, then it holds for $\gamma_1\gamma_2$. Finally, we complete the proof of Theorem 2.6.2. □

2.7 The Jacobi Group

In this section we describe an Iwasawa decomposition of the Jacobi group and the Lie algebra of the Jacobi group.

We let

$$G_{n,m}^J := Sp(2n, \mathbb{R}) \ltimes H_{\mathbb{R}}^{(n,m)}$$

be the semidirect product of the symplectic group $Sp(2n, \mathbb{R})$ and the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ endowed with the following multiplication law:

$$\begin{aligned} &(M, (\lambda, \mu, \kappa)) \cdot (M', (\lambda', \mu', \kappa')) \\ &= (MM', (\tilde{\lambda} + \lambda', \tilde{\mu} + \mu', \kappa + \kappa' + \tilde{\lambda}\,{}^t\mu' - \tilde{\mu}\,{}^t\lambda')) \end{aligned} \tag{2.7.1}$$

with $M, M' \in Sp(2n, \mathbb{R})$, $(\lambda, \mu, \kappa),\ (\lambda', \mu', \kappa') \in H_{\mathbb{R}}^{(n,m)}$ and $(\tilde{\lambda}, \tilde{\mu}) := (\lambda, \mu)M'$. We call $G_{n,m}^J$ the Jacobi group of degree n and index m. If there is no confusion about the degree n and the index m, we simply write G^J instead of $G_{n,m}^J$. We also put $G := Sp(2n, \mathbb{R})$ for brevity. It is easy to see that G^J acts on the Siegel–Jacobi space $\mathbb{H}_{n,m} := \mathbb{H}_n \times \mathbb{C}^{(m,n)}$ transitively by

$$(M, (\lambda, \mu, \kappa))\cdot(\Omega, Z) = (M\cdot\Omega, (Z + \lambda\Omega + \mu)(C\Omega + D)^{-1}), \tag{2.7.2}$$

where $M=\begin{pmatrix} A & B \\ C & D \end{pmatrix} \in G,\ (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$ and $(\Omega, Z) \in \mathbb{H}_{n,m}$.

Now we define the linear mapping

$$Q : \left(\mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}\right) \times \left(\mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}\right) \longrightarrow \mathbb{R}^{(m,m)}$$

by

$$Q((\lambda, \mu), (\lambda', \mu')) = \lambda\, {}^t\mu' - \mu\, {}^t\lambda',\quad \lambda, \mu, \lambda', \mu' \in \mathbb{R}^{(m,n)}.$$

Clearly, we have

$${}^t Q(\xi, \eta) = -Q(\eta, \xi) \text{ for all } \xi, \eta \in \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)},$$

$$Q(\xi M, \eta M) = Q(\xi, \eta) \text{ for all } M \in Sp(2n, \mathbb{R}).$$

For convenience, we write an element of G^J as

$$g = [M, (\lambda, \mu, \kappa)] := (I_{2n}, (\lambda, \mu, \kappa)) \cdot (M, (0, 0, 0)).$$

Then the multiplication becomes

$$[M, (\xi, \kappa)] \circ [M', (\xi', \kappa')] = [MM', (\xi + \xi' M^{-1}, \kappa + \kappa' + Q(\xi, \xi' M^{-1}))],$$

where $M,\ M' \in G$ and $(\xi, \kappa),\ (\xi', \kappa') \in H_{\mathbb{R}}^{(n,m)}$. We note that the stabilizer K of G at $i\, I_n$ under the symplectic action is given by

$$K = \left\{ \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \in GL(2n, \mathbb{R}) \,\middle|\, {}^t AB = {}^t BA,\ {}^t AA + {}^t BB = I_n \right\}$$

and is a maximal compact subgroup of G. We also recall that the Jacobi group G^J acts on $\mathbb{H}_{n,m}$ transitively via (2.7.2). Then it is easy to see that the stabilizer K^J of G^J at $(i\, I_n, 0)$ under this action is given by

$$K^J = \left\{ [k, (0, 0, \kappa)] \,\middle|\, k \in K,\ \kappa \in \mathrm{Sym}(m, \mathbb{R}) \right\}$$

$$\cong K \times \{(0,0,\kappa)|\ \kappa \in \mathrm{Sym}(m,\mathbb{R})\} \cong K \times \mathrm{Sym}(m,\mathbb{R}).$$

Thus on $G^J/K^J \cong \mathbb{H}_{n,m}$, we have the coordinate

$$g \cdot (i\, I_n, 0) := (\Omega, Z) := (X + i\, Y, \lambda\, \Omega + \mu), \quad g \in G^J.$$

In fact, if $g = [M, (\lambda, \mu, \kappa)] \in G^J$ with $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in G$,

$$\begin{aligned} \Omega &= M \cdot (i\, I_n) = (iA + B)(i\, C + D)^{-1} = X + i\, Y, \\ Z &= \{i\, (\lambda A + \mu\, C) + \lambda B + \lambda D\}(i\, C + D)^{-1} \\ &= \{\lambda\, (iA + B) + \mu(i\, C + D)\}(i\, C + D)^{-1} \\ &= \lambda\Omega + \mu. \end{aligned}$$

We set

$$dX = \begin{pmatrix} dX_{11} & \dots & dX_{1n} \\ \vdots & \ddots & \vdots \\ dX_{n1} & \dots & dX_{nn} \end{pmatrix}, \quad dZ = \begin{pmatrix} dZ_{11} & \dots & dZ_{1n} \\ \vdots & \ddots & \vdots \\ dZ_{m1} & \dots & dZ_{mn} \end{pmatrix}$$

and

$$\frac{\partial}{\partial X} = \begin{pmatrix} \frac{\partial}{\partial X_{11}} & \cdots & \frac{\partial}{\partial X_{1n}} \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial X_{n1}} & \cdots & \frac{\partial}{\partial X_{nn}} \end{pmatrix}, \quad \frac{\partial}{\partial Z} = \begin{pmatrix} \frac{\partial}{\partial Z_{11}} & \cdots & \frac{\partial}{\partial Z_{m1}} \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial Z_{1n}} & \cdots & \frac{\partial}{\partial Z_{mn}} \end{pmatrix}.$$

Similarly, we set $dY = (dY_{ij})$, $d\lambda = (d\lambda_{pq})$, $d\mu = (d\mu_{pq}), \cdots$ etc. By an easy calculation, we have

$$\begin{aligned} \frac{\partial}{\partial Z} &= \frac{1}{2i}\, Y^{-1} \left(\frac{\partial}{\partial \lambda} - \overline{\Omega} \frac{\partial}{\partial \mu} \right), \\ \frac{\partial}{\partial \overline{Z}} &= \frac{i}{2}\, Y^{-1} \left(\frac{\partial}{\partial \lambda} - \Omega \frac{\partial}{\partial \mu} \right), \\ \frac{\partial}{\partial X} &= \frac{\partial}{\partial \Omega} + \frac{\partial}{\partial \overline{\Omega}} + \frac{\partial}{\partial Z}\lambda + \frac{\partial}{\partial \overline{Z}}\lambda, \\ \frac{\partial}{\partial Y} &= i \frac{\partial}{\partial \Omega} - i \frac{\partial}{\partial \overline{\Omega}} + i \frac{\partial}{\partial Z}\lambda - i \frac{\partial}{\partial \overline{Z}}\lambda. \end{aligned}$$

We set

$$D_+ = \frac{1}{2}\left(\frac{\partial}{\partial X} - i\frac{\partial}{\partial Y}\right) = \frac{\partial}{\partial \Omega} + \frac{\partial}{\partial Z}\,(\mathrm{Im}\,Z)\,Y^{-1}$$

and

$$D_- = \frac{1}{2}\left(\frac{\partial}{\partial X} + i\frac{\partial}{\partial Y}\right) = \frac{\partial}{\partial \overline{\Omega}} + \frac{\partial}{\partial \overline{Z}}\,(\mathrm{Im}\,Z)\,Y^{-1}.$$

Let $\mathfrak{g}$ be the Lie algebra of G and $\mathfrak{g}_{\mathbb{C}}$ its complexification. Then

$$\mathfrak{g}_{\mathbb{C}} = \left\{ \begin{pmatrix} A & B \\ C & -{}^tA \end{pmatrix} \in \mathbb{C}^{(2n,2n)} \;\middle|\; B = {}^tB,\ \ C = {}^tC \right\}.$$

We let $\widehat{J} := i\,J_n$ with $J_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$. We define an involution τ of G by

$$\tau(g) := \widehat{J}\,g\,\widehat{J}^{-1}, \quad g \in G. \tag{2.7.3}$$

The differential map $d\tau = \mathrm{Ad}\,(\widehat{J})$ of τ extends complex linearly to the complexification $\mathfrak{g}_{\mathbb{C}}$ of $\mathfrak{g}$. $\mathrm{Ad}\,(\widehat{J})$ has 1 and -1 as eigenvalues. The $(+1)$-eigenspace of $\mathrm{Ad}\,(\widehat{J})$ is given by

$$\mathfrak{k}_{\mathbb{C}} = \left\{ \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \in \mathbb{C}^{(2n,2n)} \;\middle|\; {}^tA + A = 0,\ \ B = {}^tB \right\}.$$

We note that $\mathfrak{k}_{\mathbb{C}}$ is the complexification of the Lie algebra $\mathfrak{k}$ of a maximal compact subgroup $K = G \cap SO(2n, \mathbb{R}) \cong U(n)$ of G. The (-1)-eigenspace of $\mathrm{Ad}\,(\widehat{J})$ is given by

$$\mathfrak{p}_{\mathbb{C}} = \left\{ \begin{pmatrix} A & B \\ B & -A \end{pmatrix} \in \mathbb{C}^{(2n,2n)} \;\middle|\; A = {}^tA,\ \ B = {}^tB \right\}.$$

We observe that $\mathfrak{p}_{\mathbb{C}}$ is not a Lie algebra. But $\mathfrak{p}_{\mathbb{C}}$ has the following decomposition:

$$\mathfrak{p}_{\mathbb{C}} = \mathfrak{p}_+ \oplus \mathfrak{p}_-,$$

where

$$\mathfrak{p}_+ = \left\{ \begin{pmatrix} X & iX \\ iX & -X \end{pmatrix} \in \mathbb{C}^{(2n,2n)} \;\middle|\; X = {}^tX \right\}$$

and

$$\mathfrak{p}_- = \left\{ \begin{pmatrix} Y & -iY \\ -iY & -Y \end{pmatrix} \in \mathbb{C}^{(2n,2n)} \;\middle|\; Y = {}^tY \right\}.$$

We observe that $\mathfrak{p}_+$ and $\mathfrak{p}_-$ are abelian subalgebras of $\mathfrak{g}_{\mathbb{C}}$. Since $\mathrm{Ad}\,(\widehat{J})[X, Y] = [\mathrm{Ad}\,(\widehat{J})X,\ \mathrm{Ad}\,(\widehat{J})Y]$ for all $X, Y \in \mathfrak{g}_{\mathbb{C}}$, we have

$$[\mathfrak{k}_{\mathbb{C}}, \mathfrak{k}_{\mathbb{C}}] \subset \mathfrak{k}_{\mathbb{C}}, \quad [\mathfrak{k}_{\mathbb{C}}, \mathfrak{p}_{\mathbb{C}}] \subset \mathfrak{p}_{\mathbb{C}}, \quad [\mathfrak{p}_{\mathbb{C}}, \mathfrak{p}_{\mathbb{C}}] \subset \mathfrak{k}_{\mathbb{C}}.$$

Since $\mathrm{Ad}(k)X = kXk^{-1}$ ($k \in K,\ X \in \mathfrak{g}_{\mathbb{C}}$), we obtain

$$\mathrm{Ad}(k)\mathfrak{p}_+ \subset \mathfrak{p}_+, \quad \mathrm{Ad}(k)\mathfrak{p}_- \subset \mathfrak{p}_-.$$

For instance, if $k = \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \in K$, then

$$\mathrm{Ad}(k) \begin{pmatrix} X & \pm iX \\ \pm iX & -X \end{pmatrix} = \begin{pmatrix} X' & \pm iX' \\ \pm iX' & -X' \end{pmatrix}, \quad X = {}^tX,$$

where

$$X' = (A + iB)X\,{}^t(A + iB).$$

If we identify $\mathfrak{p}_-$ with $\mathrm{Sym}(n, \mathbb{C})$ and K with $U(n)$ as a subgroup of $GL(n, \mathbb{C})$ via the mapping $K \ni \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \longrightarrow A + iB \in U(n)$, then the action of K on $\mathfrak{p}_-$ is compatible with the natural representation $\rho^{[1]}$ of $GL(n, \mathbb{C})$ on $\mathrm{Sym}(n, \mathbb{C})$ given by

$$\rho^{[1]}(g)X = gX\,{}^tg, \quad g \in GL(n, \mathbb{C}), \quad X \in \mathrm{Sym}(n, \mathbb{C}).$$

The Lie algebra $\mathfrak{g}$ of G has a Cartan decomposition

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p},$$

where

$$\mathfrak{k} = \left\{ \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \in \mathbb{R}^{(2n,2n)} \;\middle|\; A + {}^tA = 0, \quad B = {}^tB \right\},$$

$$\mathfrak{p} = \left\{ \begin{pmatrix} A & B \\ B & -A \end{pmatrix} \in \mathbb{R}^{(2n,2n)} \;\middle|\; A = {}^tA, \quad B = {}^tB \right\}.$$

Then $\theta := \mathrm{Ad}\,(\widehat{J})|_{\mathfrak{g}}$ is a Cartan involution because

$$-\mathbb{B}(W, \theta(W)) = -\mathbb{B}(X, X) + \mathbb{B}(Y, Y) > 0$$

for all $W = X + Y \in \mathfrak{g}$ and $X \in \mathfrak{k},\ Y \in \mathfrak{p}$. Here $\mathbb{B}$ denotes the Killing form for $\mathfrak{g}$. Indeed,

$$\mathbb{B}(X, Y) = 2\,(n+1)\,\sigma(XY), \quad X, Y \in \mathfrak{g}.$$

The vector space $\mathfrak{p}$ is identified with the tangent space of $\mathbb{H}_n$ at $i\,I_n$. The correspondence

$$\frac{1}{2}\begin{pmatrix} B & A \\ A & -B \end{pmatrix} \longmapsto A + i\,B$$

yields an isomorphism of $\mathfrak{p}$ onto $\mathrm{Sym}(n, \mathbb{C})$. The Lie algebra $\mathfrak{g}^J$ of the Jacobi group G^J has a decomposition

$$\mathfrak{g}^J = \mathfrak{k}^J + \mathfrak{p}^J, \tag{2.7.4}$$

where

$$\mathfrak{k}^J = \{(X, (0, 0, \kappa) \mid\ X \in \mathfrak{k},\ \ \kappa \in \mathrm{Sym}(m, \mathbb{R})\},$$
$$\mathfrak{p}^J = \left\{(Y, (P, Q, 0) \mid\ Y \in \mathfrak{p},\ \ P, Q \in \mathbb{R}^{(m,n)}\right\}.$$

Thus the tangent space of the Siegel-Jacobi space $\mathbb{H}_{n,m} \cong G^J/K^J$ at $(i\,I_n, 0)$ is given by

$$\mathfrak{p}^J \cong \mathfrak{p} \oplus \left(\mathbb{R}^{(n,m)} \times \mathbb{R}^{(n,m)}\right) \cong \mathfrak{p} \oplus \mathbb{C}^{(n,m)}.$$

We define a complex structure I^J on the tangent space $\mathfrak{p}^J$ of $\mathbb{H}_{n,m}$ at $i\,I_n$ by

$$I^J\left(\begin{pmatrix} Y & X \\ X & -Y \end{pmatrix}, (P, Q)\right) := \left(\begin{pmatrix} X & -Y \\ -Y & -X \end{pmatrix}, (Q, -P)\right). \tag{2.7.5}$$

Identifying $\mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}$ with $\mathbb{C}^{(m,n)}$ via

$$(P, Q) \longmapsto i\,P + Q, \quad P, Q \in \mathbb{R}^{(m,n)},$$

we may regard the complex structure I^J as a real linear map

$$I^J(X + i\,Y, Q + i\,P) = (-Y + i\,X, -P + i\,Q),$$

where $X + iY \in \mathrm{Sym}(n, \mathbb{C})$ and $Q + iP \in \mathbb{C}^{(m,n)}$. I^J extends complex linearly on the complexification $\mathfrak{p}^J_{\mathbb{C}} = \mathfrak{p}^J \otimes_{\mathbb{R}} \mathbb{C}$ of $\mathfrak{p}^J$. $\mathfrak{p}^J_{\mathbb{C}}$ has a decomposition

$$\mathfrak{p}^J_{\mathbb{C}} = \mathfrak{p}^J_+ \oplus \mathfrak{p}^J_-,$$

where $\mathfrak{p}^J_+$ (resp. $\mathfrak{p}^J_-$) denotes the $(+i)$-eigenspace (resp. $(-i)$-eigenspace) of I^J. Precisely, both $\mathfrak{p}^J_+$ and $\mathfrak{p}^J_-$ are given by

$$\mathfrak{p}^J_+ = \left\{ \left(\begin{pmatrix} X & iX \\ iX & -X \end{pmatrix}, (P, iP) \right) \;\middle|\; X \in \mathrm{Sym}(n, \mathbb{C}), \quad P \in \mathbb{C}^{(m,n)} \right\}$$

and

$$\mathfrak{p}^J_- = \left\{ \left(\begin{pmatrix} X & -iX \\ -iX & -X \end{pmatrix}, (P, -iP) \right) \;\middle|\; X \in \mathrm{Sym}(n, \mathbb{C}), \quad P \in \mathbb{C}^{(m,n)} \right\}.$$

With respect to this complex structure I^J, we may say that f is *holomorphic* if and only if $\xi f = 0$ for all $\xi \in \mathfrak{p}^J_-$.

We fix $g = [M, (\lambda, \mu, \kappa)] \in G^J$ with $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in G$. Let $T_g : \mathbb{H}_{n,m} \longrightarrow \mathbb{H}_{n,m}$ be the mapping defined by (2.7.2). We consider the behavior of the differential map dT_g of T_g at $(iI_n, 0)$

$$dT_g : T_{(iI_n,0)}(\mathbb{H}_{n,m}) \longrightarrow T_{(\Omega,Z)}(\mathbb{H}_{n,m}), \quad (\Omega, Z) := g \cdot (iI_n, 0).$$

Now we let $\alpha(t) = (\Omega(t), Z(t))$ be a smooth curve in $\mathbb{H}_{n,m}$ passing through $(iI_n, 0)$ with $\alpha'(0) = (V, iP + Q) \in T_{(iI_n,0)}(\mathbb{H}_{n,m})$. Then

$$\begin{aligned} \gamma(t) &:= g \cdot \alpha(t) = (\Omega(g; t), Z(g; t)) \\ &= (M \cdot \Omega(t), (Z(t) + \tilde{\lambda}\,\Omega(t) + \tilde{\mu})(C\Omega(t) + D)^{-1}) \end{aligned}$$

is a curve in $\mathbb{H}_{n,m}$ passing through $\gamma(0) = (\Omega, Z)$ with $(\tilde{\lambda}, \tilde{\mu}) = (\lambda, \mu)M$. Using the relation

$$\left.\frac{\partial}{\partial t}\right|_{t=0} (C\,\Omega(t) + D)^{-1} = -(i\,C + D)^{-1} C\,\Omega'(0)(i\,C + D)^{-1},$$

we have

$$\begin{aligned} &\left.\frac{\partial}{\partial t}\right|_{t=0} \Omega(g; t) \\ &= A\,\Omega'(0)(i\,C + D)^{-1} + (i\,A + B) \left.\frac{\partial}{\partial t}\right|_{t=0} (C\,\Omega(t) + D)^{-1} \end{aligned}$$

$$
\begin{aligned}
&= A\,\Omega'(0)\,(i\,C + D)^{-1} - (i\,A + B)(i\,C + D)^{-1} C\,\Omega'(0)(i\,C + D)^{-1} \\
&= \{A\,{}^t(i\,C + D) - (i\,A + B)\,{}^t C\}\,{}^t(i\,C + D)^{-1}\,\Omega'(0)(i\,C + D)^{-1} \\
&= {}^t(i\,C + D)^{-1}\,\Omega'(0)(i\,C + D)^{-1}.
\end{aligned}
$$

and

$$
\begin{aligned}
&\frac{\partial}{\partial t}\bigg|_{t=0} Z(g;t) \\
&= (Z'(0) + \tilde{\lambda}\,\Omega'(0))(i\,C + D)^{-1} \\
&\quad + (Z(0) + i\,\tilde{\lambda} + \tilde{\mu})\,\frac{\partial}{\partial t}\bigg|_{t=0} (C\Omega(t) + D)^{-1} \\
&= (i\,P + Q + \tilde{\lambda}\,\Omega'(0))(i\,C + D)^{-1} \\
&\quad - (i\,\tilde{\lambda} + \tilde{\mu})(i\,C + D)^{-1} C\Omega'(0)(i\,C + D)^{-1} \\
&= (i\,P + Q)(i\,C + D)^{-1} \\
&\quad + \left\{\tilde{\lambda}\,{}^t(i\,C + D) - (i\,\tilde{\lambda} + \tilde{\mu})\,{}^t C\right\}\,{}^t(i\,C + D)^{-1}\,\Omega'(0)(i\,C + D)^{-1} \\
&= (i\,P + Q)(i\,C + D)^{-1} + \lambda\,{}^t(i\,C + D)^{-1}\,\Omega'(0)(i\,C + D)^{-1}.
\end{aligned}
$$

Here we used the fact that $(i\,C + D)^{-1}C$ is symmetric and the relation

$$\tilde{\lambda} = \lambda A + \mu\,C, \quad \tilde{\mu} = \lambda B + \mu D.$$

Therefore we obtain

$$
\begin{aligned}
\Omega'(g;0) &= {}^t(i\,C + D)^{-1}\Omega'(0)(i\,C + D)^{-1}, \\
Z'(g;0) &= Z'(0)(i\,C + D)^{-1} + \lambda\,{}^t(i\,C + D)^{-1}\Omega'(0)(i\,C + D)^{-1}.
\end{aligned}
$$

In summary, we have:

Proposition 2.7.1 *Let* $g = [M, (\lambda, \mu, \kappa)] \in G^J$ *with* $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in G$ *and let* $(\Omega, Z) = g \cdot (i\,I_n, 0)$. *Then the differential map*

$$dT_g : T_{(i\,I_n,0)}(\mathbb{H}_{n,m}) \longrightarrow T_{(\Omega,Z)}(\mathbb{H}_{n,m})$$

is given by

$$(v, w) \longmapsto (v(g), w(g)), \quad v \in \mathrm{Sym}(n, \mathbb{C}), \quad w \in \mathbb{C}^{(m,n)} \tag{2.7.6}$$

with

$$v(g) = {}^t(i\,C + D)^{-1} v\,(i\,C + D)^{-1},$$
$$w(g) = w(i\,C + D)^{-1} + \lambda\,{}^t(i\,C + D)^{-1} v\,(i\,C + D)^{-1}.$$

Now we are ready to give the Iwasawa decomposition of the Jacobi group G^J. For a positive diagonal matrix H of degree n, we put

$$t(H) = \begin{pmatrix} H & 0 \\ 0 & H^{-1} \end{pmatrix}$$

and for an upper triangular $n \times n$ matrix α with 1 in every diagonal entry and $\beta \in \mathbb{R}^{(n,n)}$, we write

$$n(\alpha, \beta) = \begin{pmatrix} \alpha & \beta \\ 0 & {}^t\alpha^{-1} \end{pmatrix}.$$

We let A be the set of such all $t(H)$ and let N be the set of such all $n(\alpha, \beta)$ such that $n(\alpha, \beta) \in G$, namely, $\alpha\,{}^t\beta = \beta\,{}^t\alpha$. It is clear that A is an abelian subgroup of G and N is a nilpotent subgroup of G. Then we have the so-called Iwasawa decomposition

$$G = NAK = KAN. \tag{2.7.7}$$

Now we define the subgroups A^J, N^J and $\tilde{N}^J$ of G^J by

$$A^J = \left\{ t(H, \lambda) := [t(H), (\lambda, 0, 0)] \in G^J \,\middle|\, t(H) \in A,\ \lambda \in \mathbb{R}^{(m,n)} \right\},$$
$$N^J = \left\{ n(\alpha, \beta; \mu) := [n(\alpha, \beta), (0, \mu, 0)] \in G^J \,\middle|\, n(\alpha, \beta) \in N,\ \mu \in \mathbb{R}^{(m,n)} \right\}$$

and

$$\widetilde{N}^J = \left\{ [n(\alpha, \beta), (0, \mu, \kappa)] \in G^J \,\middle|\, n(\alpha, \beta) \in N,\ \mu \in \mathbb{R}^{(m,n)},\ \kappa \in \mathbb{R}^{(m,m)} \right\}.$$

From now on, for brevity, we put

$$\widetilde{n}(\alpha, \beta; \mu, \kappa) := [n(\alpha, \beta), (0, \mu, \kappa)] \in G^J.$$

For $t(H, \lambda),\ t(H', \lambda') \in A^J$, we have

$$t(H, \lambda) \circ t(H', \lambda') = t(HH', \lambda + \lambda' H^{-1}).$$

Thus A^J is the semidirect product of $\mathbb{R}^{(m,n)}$ and $\mathbb{D}^+$, where $\mathbb{D}^+$ denotes the subgroup of $GL(n,\mathbb{R})$ consisting of positive diagonal matrices of degree n. Furthermore, we have for $t(H,\lambda) \in A^J$ and $\widetilde{n}(\alpha,\beta;\mu,\kappa) \in \widetilde{N}^J$

$$\widetilde{n}(\alpha,\beta;\mu,\kappa) \circ t(H,\lambda) = [n(\alpha,\beta)\, t(H), (\lambda\alpha^{-1},\ \mu - \lambda\alpha^{-1}\beta\, {}^t\alpha,\ -\mu\, {}^t\alpha^{-1}\, {}^t\lambda)]$$

and

$$\begin{aligned} t(H,\lambda) \circ \widetilde{n}(\alpha,\beta;\mu,\kappa) &= [t(H)\, n(\alpha,\beta), (\lambda,\ \mu H,\ \kappa + \lambda H\, {}^t\mu)] \\ &= \left[\begin{pmatrix} H\alpha & H\beta \\ 0 & H^{-1}\, {}^t\alpha^{-1} \end{pmatrix}, (\lambda,\ \mu H, \kappa + \lambda H\, {}^t\mu)\right]. \end{aligned}$$

Therefore we have

$$\begin{aligned} & t(H,\lambda) \circ \widetilde{n}(\alpha,\beta;\mu,\kappa) \circ t(H,\lambda)^{-1} \\ &= [t(H)\, n(X)\, t(H^{-1}), (\lambda - \lambda H\alpha^{-1}H^{-1},\ \mu H + \lambda H\alpha^{-1}\beta\, {}^t\alpha H, \\ &\quad \kappa + \lambda H\alpha\, {}^t\beta\, {}^t\alpha^{-1}H\, {}^t\lambda - \mu\, {}^t\alpha^{-1}H\, {}^t\lambda)] \\ &= [n(H\alpha H^{-1}, H\beta H),\ (\lambda - \lambda H\alpha^{-1}H^{-1}, \mu H + \lambda H\alpha^{-1}\beta\, {}^t\alpha H, \\ &\quad \kappa + \lambda H\alpha\, {}^t\beta\, {}^t\alpha^{-1}H\, {}^t\lambda - \mu\, {}^t\alpha^{-1}H\, {}^t\lambda)]. \end{aligned}$$

Thus there is a decomposition

$$G^J = \widetilde{N}^J A^J K. \tag{2.7.8}$$

For $g \in G^J$, one has

$$\begin{aligned} g &= [n(\alpha,\beta)\, t(H)\, k, (\lambda,\mu,\kappa)], \quad k \in K \\ &= \widetilde{n}(\alpha,\beta;\mu^*,\kappa^*) \circ t(H,\lambda^*) \circ k \end{aligned}$$

with

$$\lambda^* = \lambda H, \quad \mu^* = \mu + \lambda\beta\, {}^t\alpha \ \text{ and } \ \kappa^* = \kappa + \mu\, {}^t\lambda + \lambda\, \beta\, {}^t(\lambda\alpha).$$

Recalling the subgroup K^J of G^J defined by

$$K^J = \{\, [k, (0,0,\kappa)] \mid k \in K,\ \kappa \in \mathrm{Sym}(m,\mathbb{R}) \,\},$$

we also have a decomposition

$$G^J = N^J A^J K^J. \tag{2.7.9}$$

For $g \in G^J$, one has

$$\begin{aligned} g &= [n(\alpha, \beta)\, t(H)\, k, (\lambda, \mu, \kappa)] \\ &= n(\alpha, \beta; \tilde{\mu}) \circ t(H, \tilde{\lambda}) \circ [k, (0, 0, \tilde{\kappa})] \end{aligned}$$

with

$$\tilde{\lambda} = \lambda\, \alpha, \quad \tilde{\mu} = \mu + \lambda\, \alpha^{-1} \beta\, {}^t\alpha, \quad \tilde{\kappa} = \kappa + (\mu + \lambda\, \alpha^{-1} \beta\, {}^t\alpha)\, {}^t\lambda.$$

We call the decomposition (2.7.8) or (2.7.9) an Iwasawa decomposition of G^J. Finally, we note that the decomposition (2.7.8) or (2.7.9) may be understood as the product of the usual Iwasawa decomposition (2.7.7) of G with a decomposition

$$H_{\mathbb{R}}^{(n,m)} = \widetilde{N}_0 A_0 \tag{2.7.10}$$

of the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ into the group

$$A_0 = \left\{ (\lambda, 0, 0) \,\middle|\, \lambda \in \mathbb{R}^{(m,n)} \right\}$$

which normalizes the maximal abelian subgroup

$$\widetilde{N}_0 = \left\{ (0, \mu, \kappa) \,\middle|\, \mu \in \mathbb{R}^{(m,n)},\ \kappa = {}^t\kappa \in \mathbb{R}^{(m,m)} \right\}.$$

Now we describe the Lie algebra $\mathfrak{g}^J$ of the Jacobi group G^J briefly. First of all, we observe that $\mathfrak{g}$ of G may be regarded as a subalgebra of $\mathfrak{g}^J$ by identifying $\mathfrak{g}$ with $\mathfrak{g} \times \{0\}$ and the Lie algebra $\mathfrak{h}$ of the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ may be regarded as an ideal of $\mathfrak{g}^J$ by identifying $\mathfrak{h}$ with $\{0\} \times \mathfrak{h}$. We denote by E_{ij} the $n \times n$ matrix with entry 1 where the i-th row and the j-th column meet, all other entries 0.

For $1 \le a, b, p \le m,\ 1 \le i, j, q \le n$, we set

$$A_{ij} := \begin{pmatrix} E_{ij} + E_{ji} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -(E_{ij} + E_{ji}) & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$B_{ij} := \begin{pmatrix} 0 & 0 & E_{ij} + E_{ji} & 0 \\ 0 & 0 & 0 & 0 \\ E_{ij} + E_{ji} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$S_{ij} := \begin{pmatrix} E_{ij} - E_{ji} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & E_{ij} - E_{ji} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$T_{ij} := \begin{pmatrix} 0 & 0 & E_{ij} + E_{ji} & 0 \\ 0 & 0 & 0 & 0 \\ -(E_{ij} + E_{ji}) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$D^0_{ab} := \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2}(E_{ab} + E_{ba}) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$D_{pq} := \begin{pmatrix} 0 & 0 & 0 & 0 \\ E_{pq} & 0 & 0 & 0 \\ 0 & 0 & 0 & -E_{qp} \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\widehat{D}_{pq} := \begin{pmatrix} 0 & 0 & 0 & E_{qp} \\ 0 & 0 & E_{pq} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

We observe that the set

$$\left\{ S_{ij},\ T_{kl},\ D^0_{ab} \,\middle|\, 1 \le i < j \le n,\ 1 \le k \le l \le n,\ 1 \le a \le b \le m \right\}$$

forms a basis of $\mathfrak{k}^J$ and the set

$$\left\{ A_{ij},\ B_{ij},\ D_{pq},\ \widehat{D}_{rs} \,\middle|\, 1 \le i \le j \le n,\ 1 \le p, r \le m,\ 1 \le q, s \le n \right\}$$

forms a basis of $\mathfrak{p}^J$ (cf. (2.7.4)). We note that

$$A_{ij} = A_{ji},\quad B_{ij} = B_{ji},\quad S_{ij} = -S_{ji},\quad T_{ij} = T_{ji},$$
$$D^0_{ab} = D^0_{ba},\quad D^2_{pq} = \widehat{D}^2_{pq} = 0.$$

Lemma 2.7.2 *We have the following relation:*

$$[\mathfrak{k}^J, \mathfrak{k}^J] \subset \mathfrak{k}^J, \quad [\mathfrak{k}^J, \mathfrak{p}^J] \subset \mathfrak{p}^J,$$
$$[\mathfrak{p}, \mathfrak{h}] \subset \mathfrak{h}, \quad [\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}.$$

Here $\mathfrak{h}$ *is the Lie algebra of the Heisenberg group* $H_{\mathbb{R}}^{(n,m)}$.

Proof It follows from a straightforward calculation. We leave the proof to the reader. □

Remark 2.7.3 We remark that the relation

$$[\mathfrak{p}^J, \mathfrak{p}^J] \subset \mathfrak{k}^J$$

does not hold. The reason is that the Jacobi group G^J is not a reductive Lie group and hence its Lie algebra $\mathfrak{g}^J$ is not reductive. Unfortunately, we cannot define the notions of roots and root systems in the case of non-reductive Lie algebras as the case of reductive or semisimple Lie algebras.

Now we set

$$Z_{ab}^0 := -\sqrt{-1}\, D_{ab}^0,$$
$$Y_{pq}^{\pm} := \frac{1}{2}(D_{pq} \pm \sqrt{-1}\, \widehat{D}_{pq}),$$
$$Z_{ij}^{+} := -S_{ij},$$
$$Z_{ij}^{-} := -\sqrt{-1}\, T_{ij},$$
$$X_{ij}^{\pm} := \frac{1}{2}(A_{ij} \pm \sqrt{-1}\, B_{ij}).$$

Lemma 2.7.4 *The set*

$$\left\{ Z_{ab}^0,\ Z_{ij}^{+},\ Z_{kl}^{-} \,\middle|\, 1 \leq a \leq b \leq m,\ 1 \leq i < j \leq n,\ 1 \leq k < l \leq n \right\}$$

forms a basis of the complexification $\mathfrak{k}_{\mathbb{C}}^J$ *of* $\mathfrak{k}^J$ *and the set*

$$\left\{ X_{ij}^{\pm},\ Y_{pq}^{\pm} \,\middle|\, 1 \leq i \leq j \leq n,\ 1 \leq p \leq m,\ 1 \leq q \leq n \right\}$$

forms a basis of $\mathfrak{p}_{\mathbb{C}}^J$. *And the set*

$$\left\{ X_{ij}^{+},\ Y_{pq}^{+} \,\middle|\, 1 \leq i \leq j \leq n,\ 1 \leq p \leq m,\ 1 \leq q \leq n \right\}$$

forms a basis of $\mathfrak{p}_+^J$ *and the set*

$$\left\{ X_{ij}^-,\ Y_{pq}^- \;\middle|\; 1 \le i \le j \le n,\ 1 \le p \le m,\ 1 \le q \le n \right\}$$

forms a basis of $\mathfrak{p}_-^J$. *Both* $\mathfrak{p}_+^J$ *and* $\mathfrak{p}_-^J$ *are abelian subalgebras of* $\mathfrak{g}_{\mathbb{C}}^J$. *We have the relation*

$$[\mathfrak{k}_{\mathbb{C}}^J,\ \mathfrak{p}_+^J] \subset \mathfrak{p}_+^J, \qquad [\mathfrak{k}_{\mathbb{C}}^J,\ \mathfrak{p}_-^J] \subset \mathfrak{p}_-^J.$$

$\mathfrak{g}_{\mathbb{C}}$ *is a subalgebra of* $\mathfrak{g}_{\mathbb{C}}^J$ *and* $\mathfrak{h}_{\mathbb{C}}$, *the complexification of* $\mathfrak{h}$, *is an ideal of* $\mathfrak{g}_{\mathbb{C}}^J$.

Proof The proof follows from a straightforward calculation. We leave the proof to the reader. □

Finally, we would like to mention the fact that the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ and the Jacobi group G^J are of Harish-Chandra type. First we recall the notion of a group of Harish-Chandra type following Ichiro Satake [40, pp. 118–119].

Definition 2.7.5 Let G be a Zariski connected $\mathbb{R}$-group with Lie algebra $\mathfrak{g}$. The complexification $G_{\mathbb{C}}$ of G is a connected complex Lie group. Suppose there are given a Zariski connected $\mathbb{R}$-subgroup K with Lie algebra $\mathfrak{k}$ and two connected unipotent $\mathbb{C}$-subgroups $P_\pm$ of $G_{\mathbb{C}}$ with Lie algebra $\mathfrak{p}_\pm$.

We call G of Harish-Chandra type if the following two conditions are satisfied:

(HC1) (Infinitesimal condition) One has

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{p}_+ + \mathfrak{k}_{\mathbb{C}} + \mathfrak{p}_- \quad \text{(direct sum of vector spaces)}$$
$$[\mathfrak{k}_{\mathbb{C}}, \mathfrak{p}_\pm] \subset \mathfrak{p}_\pm, \quad \overline{\mathfrak{p}}_+ = \mathfrak{p}_-.$$

(HC2) (Global condition) One has a holomorphic injection

$$P_+ \times K_{\mathbb{C}} \times P_- \longrightarrow P_+ K_{\mathbb{C}} P_- \subset G_{\mathbb{C}}.$$

Namely, the natural mapping

$$(p_+, k, p_-) \longmapsto p_+ k p_-$$

gives a biholomorphic homeomorphism of $P_+ \times K_{\mathbb{C}} \times P_-$ onto its image $P_+ K_{\mathbb{C}} P_-$ in $G_{\mathbb{C}}$. It follows that $P_+ K_{\mathbb{C}} P_-$ is open in $G_{\mathbb{C}}$ and the $K_{\mathbb{C}} P_\pm$ are semidirect products. Moreover, one has

$$G^0 \subset P_+ K_{\mathbb{C}} P_- \quad \text{and} \quad G^0 \cap K_{\mathbb{C}} P_- = K^0.$$

Here G^0 (resp. K^0) denotes the identity connected component of G (resp. K) in the usual topology.

Thus we are in a situation quite similar to that for groups of Hermitian type except that here we do not assume K to be compact, nor impose any condition on $[\mathfrak{p}_+, \mathfrak{p}_-]$.

From the condition (HC2) we have the following natural inclusion mappings:

$$\mathscr{D} = G^0/K^0 \longrightarrow P_+K_{\mathbb{C}}P_-/K_{\mathbb{C}}P_- \hookrightarrow G_{\mathbb{C}}/K_{\mathbb{C}}P_-.$$

We note that $P_+K_{\mathbb{C}}P_-/K_{\mathbb{C}}P_-$ is diffeomorphic to P_+. Since $G_{\mathbb{C}}/K_{\mathbb{C}}P_-$ has an invariant complex structure, we may endow the homogeneous space $\mathscr{D}$ with a G^0-invariant complex structure. As the exponential map gives an analytic isomorphism $\exp : \mathfrak{p}_+ \longrightarrow P_+$, we may consider $\mathscr{D}$ as embedded in $\mathfrak{p}_+$, where a coset gK^0 ($g \in G^0$) is identified with $z \in \mathfrak{p}_+$, if and only if

$$\exp z = (g)_+ \quad (\text{the } P_+ - \text{part of } g).$$

This is a generalization of the Harish-Chandra embedding.

In this setting, we can also define the holomorphic action of G^0 on $\mathscr{D}$, the canonical automorphic factor $J : G^0 \times \mathscr{D} \longrightarrow K_{\mathbb{C}}$, and the canonical kernel function $\mathfrak{K}_c : \mathscr{D} \times \mathscr{D} \longrightarrow K_{\mathbb{C}}$ by the formulas:

$$\exp g(z) = (g \cdot \exp(z))_+, \quad J(g, z) = (g \cdot \exp(z))_0, \quad z \in \mathscr{D} \subset \mathfrak{p}_+$$

and

$$\mathfrak{K}_c(z, z') = \left\{ \left((\exp \overline{z}')^{-1} \exp z \right)_0 \right\}^{-1}, \quad z, z' \in \mathscr{D} \subset \mathfrak{p}_+.$$

Here the subscript 0 denotes the $K_{\mathbb{C}}$-component in the decomposition $P_+K_{\mathbb{C}}P_-$.

Satake showed that the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ and the Jacobi group G^J are of Harish-Chandra type. For detailed proofs we refer to [40, Example 1 and Example 2, pp. 119–126].

2.8 The Canonical Factor of Automorphy for the Jacobi Group

In this section, we construct the canonical automorphic factor for the Jacobi group in a geometrical way.

Definition 2.8.1 Let $\mathscr{D} = G/K$ be a bounded symmetric domain (due to Harish-Chandra), where G is a real semisimple Lie group with a maximal compact subgroup K. Let $K_{\mathbb{C}}$ be the complexification of K. Let $\pi : G \longrightarrow \mathscr{D}$ be the canonical projection and $z_0 = \pi(e)$, where e is the unit element of G. Then there is a $K_{\mathbb{C}}$-valued function $J : G \times \mathscr{D} \longrightarrow K_{\mathbb{C}}$ satisfying the following conditions (J1)–(J3):

(J1) $J(g,z)$ is C^∞ in $g \in G$ and holomorphic in $z \in \mathscr{D}$;
(J2) $J(g_1 g_2, z) = J(g_1, g_2(z))\, J(g_2, z)$ for $g_1, g_2 \in G,\ z \in \mathscr{D}$;
(J3) $J(k, z_0) = k$ for $k \in K$.

The function $J(g,z)$ is called a canonical automorphic factor for a pair $(G, \mathscr{D})$. We refer to [40, pp. 64–67] for more details.

Even though the Jacobi group $G^J = Sp(2n,\mathbb{R}) \ltimes H_{\mathbb{R}}^{(n,m)}$ is *not* a semisimple Lie group and $\mathbb{D}_{n,m} \cong \mathbb{H}_{n,m} = G^J/K^J$ is *not* a bounded symmetric domain, one can construct a $K_{\mathbb{C}}^J$-valued function on $G^J \times \mathbb{H}_{n,m}$ satisfying the above three conditions (J1)–(J3), where $K^J \cong U(n) \times \mathrm{Sym}(m,\mathbb{R})$ is not a compact subgroup of G^J and $K_{\mathbb{C}}^J$ is the complexification of K^J.

Let m and n be two positive integers. Then the automorphism group $\mathrm{Aut}(\mathbb{H}_{n+m})$ of the Siegel space $\mathbb{H}_{n+m}$ is given by

$$\mathrm{Aut}(\mathbb{H}_{n+m}) = Sp(2(m+n),\mathbb{R})/\{\pm I_{2(m+n)}\}.$$

We note that $\mathbb{H}_n$ is a rational boundary of $\mathbb{H}_{n+m}$. The normalizer $N(\mathbb{H}_n) := \{\widetilde{g} \in \mathrm{Aut}(\mathbb{H}_{n+m}) \mid \widetilde{g}\langle \mathbb{H}_n\rangle \subset \mathbb{H}_n\}$ of $\mathbb{H}_n$ is given by

$$N(\mathbb{H}_n) = P(\mathbb{H}_n)/\{\pm I_{2(m+n)}\},$$

where

$$\begin{aligned} P(\mathbb{H}_n) &:= \{\widetilde{g} \in Sp(2(m+n),\mathbb{R}) \mid \widetilde{g}(\mathbb{H}_n) \subset \mathbb{H}_n\} \\ &= \{[g,u,(\lambda,\mu,\kappa)] \in Sp(2(m+n),\mathbb{R})\}. \end{aligned}$$

Here we put

$$[g,u,(\lambda,\mu;\kappa)] := \begin{pmatrix} A & 0 & B & A\,{}^t\mu - B\,{}^t\lambda \\ u\lambda & u & u\mu & u\kappa \\ C & 0 & D & C\,{}^t\mu - D\,{}^t\lambda \\ 0 & 0 & 0 & {}^t u^{-1} \end{pmatrix},$$

where $g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$, $u \in GL(m,\mathbb{R})$, $\lambda, \mu \in \mathbb{R}^{(m,n)}$ and $\kappa \in \mathbb{R}^{(m,m)}$.

If $\begin{pmatrix} \Omega & {}^t Z \\ Z & T \end{pmatrix} \in \mathbb{H}_{m+n}$ with $\Omega \in \mathbb{H}_n$, $Z \in \mathbb{C}^{(m,n)}$ and $T \in \mathbb{H}_m$, we simply write

$$(\Omega, Z, T) := \begin{pmatrix} \Omega & {}^t Z \\ Z & T \end{pmatrix}.$$

We denote the symplectic action of $N(\mathbb{H}_n)$ on (Ω, Z, T) by

$$\widetilde{g} \cdot (\Omega, Z, T) := (\widetilde{\Omega}, \widetilde{Z}, \widetilde{T}), \quad \widetilde{g} \in N(\mathbb{H}_n).$$

It is easily seen that $(\widetilde{\Omega}, \widetilde{Z}, \widetilde{T})$ is of the form

$$\begin{aligned}
\widetilde{\Omega} &= \sigma_{\widetilde{g}}(\Omega), \\
\widetilde{Z} &= a(\widetilde{g}; \Omega)(Z) + b(\widetilde{g}; \Omega), \\
\widetilde{T} &= m_{\widetilde{g}}(T) + c(\widetilde{g}; \Omega, Z),
\end{aligned}$$

where $\widetilde{g} = [g, u, (\lambda, \mu, \kappa)] \in N(\mathbb{H}_n)$, $\sigma_{\widetilde{g}} \in \mathrm{Aut}(\mathbb{H}_n)$, $m_{\widetilde{g}} \in \mathrm{Aut}(\mathfrak{P}_m)$,

$$\begin{aligned}
a(\widetilde{g}; \cdot) &: \mathbb{H}_n \longrightarrow GL(\mathbb{C}^{(m,n)}) \quad \text{holomorphic}, \\
b(\widetilde{g}; \cdot) &: \mathbb{H}_n \longrightarrow \mathbb{C}^{(m,n)} \quad \text{holomorphic}, \\
c(\widetilde{g}; \cdot, \cdot) &: \mathbb{H}_n \times \mathbb{C}^{(m,n)} \longrightarrow \mathbb{C}^{(m,m)} \quad \text{holomorphic}.
\end{aligned}$$

Here $\mathfrak{P}_m := \{Y \in \mathbb{R}^{(m,m)} \mid Y = {}^tY > 0\}$ is an open convex cone in $\mathbb{R}^s$ with $s = \frac{m(m+1)}{2}$. We put

$$\mathrm{Aut}(\mathfrak{P}_m) := \left\{ \xi \in GL(\mathbb{C}^{(m,m)}) \mid \xi(\mathfrak{P}_m) \subset \mathfrak{P}_m \right\}.$$

Precisely, if $[g, u, (\lambda, \mu, \kappa)](\mathrm{mod}\,\{\pm I_{2(m+n)}\})$ with $\begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$ in $N(\mathbb{H}_n) \subset \mathrm{Aut}(\mathbb{H}_{n+m})$, then we have

$$\begin{aligned}
\sigma_{\widetilde{g}}(\Omega) &= (A\Omega + B)(C\Omega + D)^{-1}, \\
m_{\widetilde{g}}(T) &= u\, T\, {}^tu, \\
a(\widetilde{g}; \Omega)(Z) &= uZ(C\Omega + D)^{-1}, \\
b(\widetilde{g}; \Omega) &= u(\lambda\, \Omega + \mu)(C\Omega + D)^{-1}, \\
c(\widetilde{g}; \Omega, Z) &= u(\lambda\, {}^tZ + \kappa)\, {}^tu \\
&\quad - u \left\{ (Z + \lambda\Omega + \mu)(C\Omega + D)^{-1}(C\, {}^tW + C\, {}^t\mu - D\, {}^t\lambda) \right\} {}^tu.
\end{aligned}$$

Remark 2.8.2 In [36], Piateski-Sharpiro called the mapping $(\Omega, Z, T) \longmapsto (\widetilde{\Omega}, \widetilde{Z}, \widetilde{T})$ a quasi-linear transformation.

We observe that $\widetilde{g} = [g, u, (\lambda, \mu, \kappa)](\mathrm{mod}\,\{\pm I_{2(m+n)}\}) \in N(\mathbb{H}_n)$ acts on $\mathbb{H}_{n,m}$ by

$$(\Omega, Z) \longmapsto \left(\sigma_{\widetilde{g}}(\Omega), a(\widetilde{g}; \Omega)(Z) + b(\widetilde{g}; \Omega)\right), \quad \Omega \in \mathbb{H}_n,\ Z \in \mathbb{C}^{(m,n)}.$$

We note that the subgroup of $N(\mathbb{H}_n)$ consisting of elements $\widetilde{g} = [g, u, (\lambda, \mu, \kappa)] \pmod{\{\pm I_{2(m+n)}\}}$ with the property

$$m_{\widetilde{g}} = \text{Identity} \quad \text{on } \mathbb{H}_m$$

is just a Jacobi group, denoted by G^J. Namely,

$$G^J = \{[g, I_m, (\lambda, \mu, \kappa)] \in P(\mathbb{H}_n)\}.$$

It is easy to see that G^J is the semidirect product of $Sp(2n, \mathbb{R})$ and $H_{\mathbb{R}}^{(n,m)}$, where

$$H_{\mathbb{R}}^{(n,m)} := \{[I_{2n}, I_m, (\lambda, \mu, \kappa)] \in P(\mathbb{H}_n)\}.$$

We recall that

$$[I_{2n}, I_m, (\lambda, \mu, \kappa)] := \begin{pmatrix} I_n & 0 & 0 & {}^t\mu \\ \lambda & I_m & \mu & \kappa \\ 0 & 0 & I_n & -{}^t\lambda \\ 0 & 0 & 0 & I_m \end{pmatrix} \in P(\mathbb{H}_n) \subset Sp(2(m+n), \mathbb{R}).$$

The Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ is a 2-step nilpotent subgroup of $P(\mathbb{H}_n)$ and the unipotent radical of the parabolic subgroup $P(\mathbb{H}_n)$. This Heisenberg group plays an important role in the study of compactifications of the Siegel modular variety (cf. [2, 35]) and the Schrödinger–Weil representation (cf. [64, 65]).

From now on, we simply write, if there is no confusion,

$$(\lambda, \mu, \kappa) := [I_{2n}, I_m, (\lambda, \mu, \kappa)].$$

We observe that if $(\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$, then $\kappa + \mu\,{}^t\lambda$ is symmetric. Now it is easy to see that the multiplication law on G^J is given as follows:

$$\begin{aligned} &[g, I_m, (\lambda, \mu, \kappa)] \circ [g', I_m, (\lambda', \mu', \kappa')] \\ &= [gg', I_m, (\lambda + \widetilde{\lambda}, \mu + \widetilde{\mu}, \kappa + \kappa' + \widetilde{\lambda}\,{}^t\mu' - \widetilde{\mu}\,{}^t\lambda')], \end{aligned}$$

where $g, g' \in Sp(2n, \mathbb{R})$ and $(\widetilde{\lambda}, \widetilde{\mu}) = (\lambda, \mu)g'$.

By a simple calculation, we see that the action of $[g, I_m, (\lambda, \mu, \kappa)] \in G^J$ on (Ω, Z, T) is given by

$$\begin{aligned} \widetilde{\Omega} &= (A\Omega + B)(C\Omega + D)^{-1}, \\ \widetilde{Z} &= Z(C\Omega + D)^{-1} + (\lambda\,\Omega + \mu)(C\Omega + D)^{-1}, \\ \widetilde{T} &= T + \lambda\,{}^tZ + \kappa - (Z + \lambda\,\Omega + \mu)(C\Omega + D)^{-1}(C\,{}^tW + C\,{}^t\mu - D\,{}^t\lambda), \end{aligned}$$

where $g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$.

Now we consider another subgroup $\widehat{G}$ of G^J consisting of elements of G^J whose action is of the following form:

$$(\Omega, Z, T) \longmapsto \left(\sigma_{\tilde{g}}(\Omega), a(\tilde{g}; \Omega)(Z), T + c(\tilde{g}; \Omega, Z)\right), \quad c(\tilde{g}; \Omega, 0) = 0.$$

Then it is easily seen that

$$\widehat{G} = \left\{ [g, I_m, (0,0,0)] := \begin{pmatrix} A & 0 & B & 0 \\ 0 & I_m & 0 & 0 \\ C & 0 & D & 0 \\ 0 & 0 & 0 & I_m \end{pmatrix} \in G^J \right\}.$$

Here $g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$.

Lemma 2.8.3 *The map*

$$\widehat{J} : \widehat{G} \times \mathbb{H}_n \longrightarrow GL(\mathbb{C}^{(m,n)})$$

defined by

$$\widehat{J}(\widehat{g}, \Omega) := a(\widehat{g}; \Omega), \quad \widehat{g} \in \widehat{G}, \ \Omega \in \mathbb{H}_n$$

is a factor of automorphy for $\widehat{G}$.

Proof If $\widehat{g}_i = [g_i, I_m, (0,0,0)]$ with $g_i = \begin{pmatrix} A_i & B_i \\ C_i & D_i \end{pmatrix} \in Sp(2n, \mathbb{R})$ $(i = 1, 2)$ are two elements of $\widehat{G}$, then

$$\widehat{g}_1\widehat{g}_2 = [g_1 g_2, I_m, (0,0,0)] \text{ and}$$
$$g_1 g_2 = \begin{pmatrix} * & * \\ C_1A_2 + D_1C_2 & C_1B_2 + D_1D_2 \end{pmatrix}.$$

If $\Omega \in \mathbb{H}_n$ and $Z \in \mathbb{C}^{(m,n)}$, then one has

$$\widehat{J}(\widehat{g}_1\widehat{g}_2, \Omega)Z = Z\{(C_1A_2 + D_1C_2)\Omega + C_1B_2 + D_1D_2\}^{-1}.$$

On the other hand, if $\Omega \in \mathbb{H}_n$ and $Z \in \mathbb{C}^{(m,n)}$, one has the following:

$$\widehat{J}(\widehat{g}_1, \widehat{g}_2\langle\Omega\rangle)\widehat{J}(\widehat{g}_2, \Omega)Z = \widehat{J}(\widehat{g}_1, g_2\langle\Omega\rangle)\widehat{J}(\widehat{g}_2, \Omega)Z$$
$$= \widehat{J}(\widehat{g}_1, g_2\langle\Omega\rangle)Z(C_2\Omega + D_2)^{-1}$$
$$= Z(C_2\Omega + D_2)^{-1}\left\{C_1(A_2\Omega + B_2)(C_2\Omega + D_2)^{-1} + D_1\right\}^{-1}$$

$$= Z(C_2\Omega + D_2)^{-1}\left\{\{C_1(A_2\Omega + B_2) + D_1(C_2\Omega + D_2)\}(C_2\Omega + D_2)^{-1}\right\}$$
$$= Z(C_2\Omega + D_2)^{-1}(C_2\Omega + D_2)\{(C_1A_2 + D_1C_2)\Omega + C_1B_2 + D_1D_2\}^{-1}$$
$$= Z\{(C_1A_2 + D_1C_2)\Omega + C_1B_2 + D_1D_2\}^{-1}.$$

Thus for any $\Omega \in \mathbb{H}_n$ and $Z \in \mathbb{C}^{(m,n)}$, one has

$$\widehat{J}(\widehat{g}_1\widehat{g}_2, \Omega)Z = \widehat{J}(\widehat{g}_1, \widehat{g}_2\langle\Omega\rangle)\widehat{J}(\widehat{g}_2, \Omega)Z.$$

This completes the proof. □

We note that by the definition of $c(\tilde{g}; \Omega, Z)$, the following mapping,

$$A(\tilde{g}, (\Omega, Z)) := c(\tilde{g}; \Omega, Z), \quad \tilde{g} \in G^J, \ (\Omega, Z) \in \mathbb{H}_{n,m}, \tag{2.8.1}$$

is a summand of automorphy, that is,

$$A(\tilde{g}_1\tilde{g}_2, (\Omega, Z)) := A(\tilde{g}_1, \tilde{g}_2 \cdot (\Omega, Z)) + A(\tilde{g}_2, (\Omega, Z)), \tag{2.8.2}$$

where $\tilde{g}_1, \ \tilde{g}_2 \in G^J$ and $(\Omega, Z) \in \mathbb{H}_{n,m}$.

We let

$$K_{\mathbb{C}} \subset GL(\mathbb{C}^{(m,n)})$$

be the complex Lie group generated by the linear mappings

$$\{a(\tilde{g}; \Omega) \mid \tilde{g} \in G^J\}.$$

Then $K_{\mathbb{C}}$ is isomorphic to $GL(n, \mathbb{C})$.

Lemma 2.8.4 *Let*

$$\rho : GL(n, \mathbb{C}) \longrightarrow GL(V_\rho)$$

be a finite-dimensional representation of $GL(n, \mathbb{C})$ on a finite-dimensional complex vector space V_ρ and let $\chi : \mathbb{C}^{(m,m)} \longrightarrow \mathbb{C}^\times$ be a character on the additive group $\mathbb{C}^{(m,m)}$. Then the mapping $J_\rho : \widehat{G} \times \mathbb{H}_n \longrightarrow GL(V_\rho)$ defined by

$$J_\rho(\widehat{g}, \Omega) := \rho(J(\widehat{g}, \Omega), \quad \widehat{g} \in \widehat{G}, \ \Omega \in \mathbb{H}_n$$

is a factor of automorphy for $\widehat{G}$. Furthermore, the mapping $J_{\chi,\rho} : G^J \times \mathbb{H}_{n,m} \longrightarrow GL(V_\rho)$ defined by

$$J_{\chi,\rho}(\tilde{g}, (\Omega, Z) := \chi(c(\tilde{g}; \Omega, Z))\,\rho(a(\tilde{g}, \Omega)), \quad \tilde{g} \in G^J, \ (\Omega, Z) \in \mathbb{H}_{n,m}$$

is a factor of automorphy for G^J with respect to χ and ρ.

Proof The proof of the first statement is obvious. The proof of the second statement follows immediately from the fact that $A(g,(\Omega,Z)) := c(g;\Omega,Z)$ is a summand of automorphy (cf. (2.8.1) and (2.8.2)) and that J_ρ is a factor of automorphy for $\widehat{G}$. □

Definition 2.8.5 J_ρ and $J_{\chi,\rho}$ are called the canonical automorphic factor for $\widehat{G}$ with respect to ρ and the canonical automorphic factor for G^J with respect to a pair (χ,ρ) respectively.

Remark 2.8.6 We take a unitary character $\chi : \mathbb{C}^{(m,m)} \longrightarrow \mathbb{C}_1^\times$ defined by

$$\chi(\alpha) := \exp(2\pi i\,\sigma(\alpha)), \quad \alpha \in \mathbb{C}^{(m,m)},$$

where $\mathbb{C}_1^\times := \{z \in \mathbb{C} \mid |z| = 1\}$ and a rational representation ρ of $GL(n,\mathbb{C})$ on a finite-dimensional complex vector space V_ρ. In this case, using the canonical automorphic factor $J_{\chi,\rho}$, we define the notion of Jacobi forms of type (χ,ρ) in Sect. 2.9.

Ichiro Satake (1927–2014) constructed the canonical automorphic factor for G^J in the Lie group theoretical way (cf. [39, pp. 87–94] or [40, pp. 64–71, pp. 121–125]). We will briefly describe his method of constructing the canonical automorphic factor for G^J without proof.

Let (V,A) be a symplectic vector space over $\mathbb{R}$ of dimension $2n$ with a symplectic form A. We observe that $A : V \times V \longrightarrow \mathbb{R}$ is a non-degenerate bilinear alternating form on $V \times V$. Let

$$Sp(V,A) := \{g \in GL(V) \mid A(gx,gy) = A(x,y) \quad \text{for all } x,y \in V\}$$

be the symplectic group for (V,A). For brevity, we set $G = Sp(V,A)$ and let K be a maximal compact subgroup of G. Then there corresponds uniquely a complex structure I in V such that $A(x,Iy)$ $(x,y \in V)$ is symmetric and positive definite, and K is the centralizer of I in G. Let $G_{\mathbb{C}}(= Sp(V_{\mathbb{C}},A))$ (resp. $K_{\mathbb{C}}$) be the complexification of G (resp. K) with Lie algebra $\mathfrak{g}_{\mathbb{C}}$ (resp. $\mathfrak{k}_{\mathbb{C}}$). Let $\mathfrak{m}_+$ be the i-eigenspace of $\frac{1}{2}\,\mathrm{Ad}(I)$ in $\mathfrak{g}_{\mathbb{C}}$. Then we have

$$[\mathfrak{k}_{\mathbb{C}},\mathfrak{m}_+] \subset \mathfrak{m}_+, \qquad \mathfrak{g}_{\mathbb{C}} = \mathfrak{m}_+ \oplus \mathfrak{k}_{\mathbb{C}} \oplus \mathfrak{m}_-, \qquad \mathfrak{m}_- := \overline{\mathfrak{m}_+}.$$

We denote by $W_\pm$ the $(\pm i)$-eigenspace of I in $V_{\mathbb{C}}$. Then it is easy to see that $W_\pm$ is totally isotropic, that is, $A|_{W_\pm \times W_\pm}$ is identically zero and one has

$$\mathfrak{m}_+ W_- \subset W_+, \qquad \mathfrak{m}_+ W_+ = \{0\}.$$

Therefore, an element $X \in \mathfrak{m}_+$ may be viewed as a $\mathbb{C}$-linear map from W_- into W_+. More precisely, if W_- is identified with the dual space of W_+ with respect to the non-degenerate inner product

$$\langle \overline{w}, w' \rangle = A(\overline{w}, w') \qquad (w, w' \in W_+),$$

then $\mathfrak{m}_+$ is identified with the space of all symmetric $\mathbb{C}$-linear maps from W_- into W_+. We note that $\mathfrak{m}_+$ is an abelian Lie subalgebra of $\mathfrak{g}_{\mathbb{C}}$. Thus the exponential map $\exp : \mathfrak{m}_+ \longrightarrow M_+ (:= \exp(\mathfrak{m}_+))$ is bijective, and so one has the Harish-Chandra embedding:

$$\mathscr{D} = G/K \hookrightarrow M_+ K_{\mathbb{C}} M_- / K_{\mathbb{C}} M_- \cong M_+ \xrightarrow{\log} \mathfrak{m}_+ \subset \mathfrak{g}_{\mathbb{C}}. \tag{2.8.3}$$

Therefore we may identify $z \in \mathscr{D}$ with $\log(m(z)) \in \mathfrak{m}_+$, where $m : \mathscr{D} \longrightarrow M_+$ is an injection given by (2.8.3). Then $\mathscr{D}$ is realized as a bounded symmetric domain in $\mathrm{Hom}_{\mathbb{C}}(W_-, W_+)$ defined by the conditions

$$z = {}^t z, \qquad 1 - z\overline{z} > 0.$$

If we write $g \in G$ in the matrix form $g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ with $\alpha : W_+ \longrightarrow W_+$, $\beta : W_- \longrightarrow W_+$, $\gamma : W_+ \longrightarrow W_-$ and $\delta : W_- \longrightarrow W_-$, then it is well-known that one has

$$g(z) = (\alpha z + \beta)(\gamma z + \delta)^{-1}$$

and the canonical automorphic factor $J : G \times \mathscr{D} \longrightarrow K_{\mathbb{C}}$ is given by

$$J(g, z) = \begin{pmatrix} {}^t(\gamma z + \delta)^{-1} & 0 \\ 0 & (\gamma z + \delta) \end{pmatrix} \in K_{\mathbb{C}} \tag{2.8.4}$$

for all $g \in G$ and $z \in \mathscr{D}$. He also computed the canonical kernel function $\mathfrak{K} : \mathscr{D} \times \mathscr{D} \longrightarrow K_{\mathbb{C}}$ given by

$$\mathfrak{K}(z', z) = \begin{pmatrix} 1 - z'\overline{z} & 0 \\ 0 & {}^t(1 - z'\overline{z})^{-1} \end{pmatrix} \in K_{\mathbb{C}} \tag{2.8.5}$$

for all $z, z' \in \mathscr{D}$.

In our situation, $V = \mathbb{R}^{2n}$ and $A : V \times V \longrightarrow \mathbb{R}$ is given by

$$A(\alpha, \beta) = {}^t\beta J_n \alpha = {}^t x_2\, y_1 - {}^t y_2\, x_1,$$

where

$$J_n := \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} \quad \text{and} \quad \alpha = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix},\ \beta = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \in \mathbb{R}^{2n}.$$

A complex structure $I : V \longrightarrow V$ is given by $I(\alpha) = J_n \alpha$ ($\alpha \in \mathbb{R}^{2n}$). Thus $AI : V \times V \longrightarrow \mathbb{R}$ is given by

$$AI(\alpha, \beta) := A(\alpha, I\beta) = {}^t\beta\alpha, \qquad \alpha, \beta \in V.$$

And $G = Sp(V, A) = Sp(2n, \mathbb{R})/\{\pm I_{2n}\}$ and $K \cong U(n)/\{\pm I_n\}$. Here $\mathfrak{m}_+$ is given by

$$\mathfrak{m}_+ = \left\{ \begin{pmatrix} x & i\,x \\ i\,x & -x \end{pmatrix} \in \mathbb{C}^{(2n,2n)} \;\middle|\; {}^t x = x \in \mathbb{C}^{(n,n)} \right\}.$$

$W_\pm$ are given by

$$W_\pm = \left\{ \begin{pmatrix} x \\ \pm i\,x \end{pmatrix} \in \mathbb{C}^{2n} \;\middle|\; x \in \mathbb{C}^n \right\}.$$

Returning to the Satake case, we denote by $\widetilde{V}$ the central extension of the additive group V by the one-dimensional torus $\mathbf{T} = \mathbb{R}/\mathbb{Z}$ defined by the factor set $e^{\pi i\, A(u,v)}$ $(u, v \in V)$. Then $\widetilde{V}$ is a 2-step nilpotent Lie group with center $\mathbf{T}$ such that $\widetilde{V}/\mathbf{T} \cong V$. We see that the group $G = Sp(V, A)$ acts on $\widetilde{V}$ naturally as a group of automorphisms, so that one can construct the semidirect product $\widetilde{G} = \widetilde{V} \rtimes G$ with the multiplication law given by

$$(\xi, u, g)(\eta, v, h) := (\xi\eta\, e^{\pi i\, A(u, g(v))}, u + g(v), gh),$$

where $\xi, \eta \in \mathbf{T}$, $u, v \in V$ and $g, h \in G$.

Obviously, $\widetilde{K} = \mathbf{T} \cdot K (\cong \mathbf{T} \times K)$ is a maximal compact subgroup of $\widetilde{G}$. One also has a natural complexification $\widetilde{G}_{\mathbb{C}}$ of $\widetilde{G}$ consisting of all triples $\widetilde{g} = (\xi, u, g)$ with $\xi \in \mathbb{C}^\times$, $u \in V_{\mathbb{C}}$, $g \in G_{\mathbb{C}} (= Sp(V_{\mathbb{C}}, A))$ and inheriting the same law of multiplication as above. It is easily seen that the Lie algebra $\widetilde{g}_{\mathbb{C}}$ of $\widetilde{G}_{\mathbb{C}}$ consists of all triples $\widetilde{X} = (\xi, u, X)$ with $\xi \in \mathbb{C}, u \in V_{\mathbb{C}}$, $X \in \mathfrak{g}_{\mathbb{C}}$, the bracket product being given by

$$[(\xi, u, X), (\eta, v, Y)] = (2\pi\, i\, A(u, v), Xv - Yu, [X, Y])$$

for $\xi, \eta \in \mathbb{C}$, $u, v \in V_{\mathbb{C}}$ and $X, Y \in \mathfrak{g}_{\mathbb{C}}$. If one denotes by $\widetilde{\mathfrak{m}}_+$ the abelian subalgebra of $\widetilde{\mathfrak{g}}_{\mathbb{C}}$ consisting of all triples of the form $(0, w, X)$ with $w \in W_+$ and $X \in \mathfrak{m}_+$. The exponential map $\widetilde{\mathfrak{m}}_+ \longrightarrow \widetilde{M}_+$ is bijective and one has

$$\exp(0, w, X) = (1, w, \exp(X)) \quad \text{for all } w \in W_+,\ X \in \mathfrak{m}_+.$$

If one denotes by $\widetilde{\pi}$ and $\widetilde{m}$ the canonical projection $\widetilde{G} \longrightarrow \widetilde{\mathscr{D}} = \widetilde{G}/\widetilde{K}$ and the natural injection

$$\widetilde{\mathscr{D}} \hookrightarrow \widetilde{M}_+ \cong \widetilde{\mathfrak{m}}_+$$

respectively. then one has for $\widetilde{g} = (\xi, u, g) \in \widetilde{G}$, $\widetilde{z} = \widetilde{\pi}(\widetilde{g})$, $z = \pi(g)$

$$\widetilde{m}(\widetilde{z}) = (1, u_z, m(z)),$$

where $u_z = u_+ - zu_-,\ u = u_+ + u_-$ and $u_\pm \in W_\pm$. Thus, when one regards $\widetilde{\mathscr{D}}$ as a vector bundle over $\mathscr{D}$ with typical fibre V, the injection $\widetilde{m} : \mathscr{D} \longrightarrow M_+ \cong \mathfrak{m}_+$, while on each fibre over $z \in \mathscr{D}$ it induces a $\mathbb{C}$-linear isomorphism $(V, I_z) \cong W_+$ given by the correspondence $u \mapsto u_z$, where $I_z = g \circ I \circ g^{-1}$ is a complex structure on V corresponding to $z \in \mathscr{D}$. From now on, we identify $\widetilde{\mathscr{D}}$ with the subset $W_+ \times \mathscr{D}$ of $\widetilde{\mathfrak{m}}_+ \cong W_+ \times \mathfrak{m}_+$ by the above correspondence and write a generic element of $\widetilde{\mathscr{D}}$ in the form $\widetilde{z} = (w, z)$ with $w \in W_+$ and $z \in \mathscr{D}$. We also set $\widetilde{z}_0 = \widetilde{\pi}(e) = (0, z_0)$. Satake proved that for $\widetilde{g} = (\xi, u, g) \in \widetilde{G}$ and $\widetilde{z} = (w, z) \in \mathscr{D}$, one has

$$\widetilde{g}(\widetilde{z}) = (u_{g(z)} + J(g, z)w, g(z)),$$

where

$$u_{g(z)} = u_+ - g(z)u_- \qquad \text{for } u = u_+ + u_- \text{ and } u_\pm \in W_\pm.$$

He also showed that a canonical automorphy factor $\widetilde{J} : \widetilde{G} \times \mathscr{D} \longrightarrow \widetilde{K}_{\mathbb{C}}$ is given by

$$\widetilde{J}(\widetilde{g}, \widetilde{z}) = (\eta(\widetilde{g}, \widetilde{z}), J(g, z))$$

with

$$\eta(\widetilde{g}, \widetilde{z}) = \xi \cdot e^{\{\pi i\,(A(u,u_{g(z)})+2A(u,J_1(g,z)w)+A(gw,J_1(g,z)w))\}},$$

where $J_1(g, z) = (\gamma z + \delta)^{-1}$ with $g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in G$. By the definition one has

$$\widetilde{g} \cdot \widetilde{m}(\widetilde{z}) = \widetilde{m}(\widetilde{g}(\widetilde{z})) \cdot \widetilde{J}(\widetilde{g}, \widetilde{z}) \cdot \widetilde{m}_-$$

with $\widetilde{m}_- \in \widetilde{M}_-$. We put

$$\widetilde{g}(\widetilde{z}) = (w', z'), \quad \widetilde{J}(\widetilde{g}, \widetilde{z}) = (\eta, 0, J) \quad \text{and} \quad \widetilde{m}_- = (1, w_-, m_-).$$

Then one has

$$(\xi, u, g)(1, w, \exp(z)) = (1, w', \exp(z'))\,(\eta, 0, J)\,(1, w_-, m_-).$$

It is easy to see that $\eta : \widetilde{G} \times \widetilde{\mathscr{D}} \longrightarrow \mathbb{C}^\times$ is the automorphy factor such that $\eta(\widetilde{k}, \widetilde{z}_0) = \xi$ for $\widetilde{k} = (\xi, 0, k) \in \widetilde{K}$.

In our situation, we chose $\widetilde{V} = H_{\mathbb{R}}^{(n,m)}$ with $V = \mathbb{R}^{(m,n)} \times \mathbb{R}^{(m,n)}$. We observe that the unit circle $\mathbf{T} = \{t \in \mathbb{C} \mid |t| = 1\}$ was replaced by the additive group

$$S(m) := \{(0, 0, \kappa) \mid \kappa = {}^t\kappa \in \mathbb{R}^{(m,m)}\}$$

of dimension $\frac{m(m+1)}{2}$. Moreover, in our case, we have

$$\widetilde{K} = K \times S(m).$$

Thus $\widetilde{K}$ is not a compact subgroup of G^J. Since $G = Sp(2n, \mathbb{R})$ acts on the Heisenberg group $\widetilde{V} = H_{\mathbb{R}}^{(n,m)}$, we considered the semidirect product $G^J := G \ltimes \widetilde{V}$, called the Jacobi group.

Remark 2.8.7 Satake also constructed the canonical kernel function $\widetilde{\mathfrak{K}} : \widetilde{\mathscr{D}} \times \widetilde{\mathscr{D}} \longrightarrow \widetilde{K}_{\mathbb{C}}$ associated with $\widetilde{J}$ explicitly. For the reader, we write explicitly $\widetilde{\mathfrak{K}}(\widetilde{z}', \widetilde{z})$ obtained by Satake.

$$\widetilde{\mathfrak{K}}(\widetilde{z}', \widetilde{z}) = (\kappa(\widetilde{z}', \widetilde{z}), \mathfrak{K}(z', z))$$

with

$$\kappa(\widetilde{z}', \widetilde{z})) = e^{\{\pi i\,(A(\overline{z}(1-z'\overline{z})^{-1}w', w') + 2\,A(\overline{w}, (1-z'\overline{z})^{-1}w') + A(\overline{w}, (1-z'\overline{z})^{-1}z'\overline{w}))\}},$$

with $\widetilde{z}' = (w', z') \in \widetilde{\mathscr{D}}$ and $\mathfrak{K}(z', z)$ is the canonical kernel function associated with the automorphic factor $J(g, z)$.

Remark 2.8.8 Let (V_1, V_2) be a pair of totally isotropic real subspaces of V such that $V = V_1 \oplus V_2$. Then there exists a complex symplectic transformation $\varrho \in G_{\mathbb{C}}$ such that $\varrho(V_{1\mathbb{C}}) = W_+$ and $\varrho(V_{2\mathbb{C}}) = W_-$. We already know that we have the decomposition $V_{\mathbb{C}} = W_+ \oplus W_-$. We can construct the partial Cayley transform $\Phi : \widetilde{\mathscr{D}} \longrightarrow \widetilde{\mathscr{D}}^*$, where $\widetilde{\mathscr{D}} = W_+ \times \mathscr{D}$ and $\widetilde{\mathscr{D}}^* = V_{1\mathbb{C}} \times \mathscr{D}^*$. Then one can show that Φ is a biholomorphic mapping of $\widetilde{\mathscr{D}}$ onto $\widetilde{\mathscr{D}}^*$. Here $\mathscr{D}^*$ is the Siegel upper half plane, i.e., a domain in $\mathrm{Hom}_{\mathbb{C}}(V_{2\mathbb{C}}, V_{1\mathbb{C}})$ defined by the conditions

$$z^* = {}^t z^*, \qquad \mathrm{Im}\, z^* > 0,$$

where the transpose is taken with respect to the inner product

$$\langle u_1, u_2 \rangle = -A(u_1, u_2) \qquad (u_i \in V_{i\mathbb{C}},\ i = 1, 2).$$

Using the partial Cayley transform Φ, he constructed the canonical automorphic factor $J^* : \widetilde{G} \times \widetilde{\mathscr{D}}^* \longrightarrow \widetilde{K}_{\mathbb{C}}$ for $(\widetilde{G}, \widetilde{\mathscr{D}}^*)$ and the canonical kernel function $\widetilde{\mathfrak{K}}^* : \widetilde{\mathscr{D}}^* \times \widetilde{\mathscr{D}}^* \longrightarrow \widetilde{K}_{\mathbb{C}}$ associated with J^* explicitly.

2.9 Factor of Automorphy of Weight $\frac{1}{2}$

Let ρ be a rational representation of $GL(n, \mathbb{C})$ on a finite dimensional complex vector space V_ρ. Let $\mathscr{M} \in \mathbb{R}^{(m,m)}$ be a symmetric half-integral semi-positive definite matrix of degree m. Let $C^\infty(\mathbb{H}_{n,m}, V_\rho)$ be the algebra of all C^∞ functions on $\mathbb{H}_{n,m}$ with values in V_ρ. For $f \in C^\infty(\mathbb{H}_{n,m}, V_\rho)$, we define

$$\begin{aligned}
&(f|_{\rho,\mathscr{M}}[(g,(\lambda,\mu,\kappa))])(\Omega, Z)\\
&\stackrel{\text{def}}{=} e^{-2\pi i\,\sigma\left(\mathscr{M}(Z+\lambda\,\Omega+\mu)(C\Omega+D)^{-1}C\,{}^t(Z+\lambda\,\Omega+\mu)\right)}\\
&\quad\times e^{2\pi i\,\sigma\left(\mathscr{M}(\lambda\,\Omega\,{}^t\lambda+2\,\lambda\,{}^tZ+\kappa+\mu\,{}^t\lambda)\right)}\\
&\quad\times \rho(C\,\Omega+D)^{-1} f((A\Omega+B)(C\Omega+D)^{-1}, (Z+\lambda\,\Omega+\mu)(C\,\Omega+D)^{-1}),
\end{aligned} \tag{2.9.1}$$

where $g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n, \mathbb{R})$, $(\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$ and $(\Omega, Z) \in \mathbb{H}_{n,m}$.

Definition 2.9.1 Let ρ and $\mathscr{M}$ be as above. Let

$$H_{\mathbb{Z}}^{(n,m)} := \left\{(\lambda,\mu,\kappa) \in H_{\mathbb{R}}^{(n,m)} \mid \lambda, \mu \in \mathbb{Z}^{(m,n)},\ \kappa \in \mathbb{Z}^{(m,m)}\right\}$$

be the discrete subgroup of $H_{\mathbb{R}}^{(n,m)}$. A Jacobi form of index $\mathscr{M}$ with respect to ρ on a subgroup Γ of Γ_n of finite index is a holomorphic function $f \in C^\infty(\mathbb{H}_{n,m}, V_\rho)$ satisfying the following conditions (A) and (B):

(A) $f|_{\rho,\mathscr{M}}[\tilde{\gamma}] = f$ for all $\tilde{\gamma} \in \widetilde{\Gamma} := \Gamma \ltimes H_{\mathbb{Z}}^{(n,m)}$.

(B) For each $M \in \Gamma_n$, $f|_{\rho,\mathscr{M}}[M]$ has a Fourier expansion of the following form:

$$(f|_{\rho,\mathscr{M}}[M])(\Omega, Z) = \sum_{\substack{T = {}^tT \geq 0\\ \text{half-integral}}} \sum_{R \in \mathbb{Z}^{(n,m)}} c(T,R) \cdot e^{\frac{2\pi i}{\lambda_\Gamma}\sigma(T\Omega)} \cdot e^{2\pi i\,\sigma(RZ)}$$

with a suitable $\lambda_\Gamma (\neq 0) \in \mathbb{Z}$ and $c(T, R) \neq 0$ only if $\begin{pmatrix} \frac{1}{\lambda_\Gamma}T & \frac{1}{2}R \\ \frac{1}{2}\,{}^tR & \mathscr{M} \end{pmatrix} \geq 0$.

If $n \geq 2$, the condition (B) is superfluous by the Köcher principle (cf. [66, Lemma 1.6]). We denote by $J_{\rho,\mathscr{M}}(\Gamma)$ the vector space of all Jacobi forms of index $\mathscr{M}$ with respect to ρ on Γ. Ziegler (cf. [66, Theorem 1.8] or [8, Theorem 1.1]) proves that the vector space $J_{\rho,\mathscr{M}}(\Gamma)$ is finite-dimensional. In the special case $\rho(A) = (\det(A))^k$ with $A \in GL(n, \mathbb{C})$ and a fixed $k \in \mathbb{Z}$, we write $J_{k,\mathscr{M}}(\Gamma)$ instead of $J_{\rho,\mathscr{M}}(\Gamma)$ and call k the weight of the corresponding Jacobi forms. For more results about Jacobi forms with $n > 1$ and $m > 1$, we refer to [53–55, 57–59, 61] and [66]. Jacobi forms play an important role in lifting elliptic cusp forms to Siegel cusp forms of degree $2n$ (cf. [19]).

Definition 2.9.2 A Jacobi form $f \in J_{\rho,\mathscr{M}}(\Gamma)$ is said to be a cusp (or cuspidal) form if $\begin{pmatrix} \frac{1}{\lambda_\Gamma} T & \frac{1}{2} R \\ \frac{1}{2}\,{}^t R & \mathscr{M} \end{pmatrix} > 0$ for any T, R with $c(T, R) \neq 0$. A Jacobi form $f \in J_{\rho,\mathscr{M}}(\Gamma)$ is said to be singular if it admits a Fourier expansion such that a Fourier coefficient $c(T, R)$ vanishes unless $\det \begin{pmatrix} \frac{1}{\lambda_\Gamma} T & \frac{1}{2} R \\ \frac{1}{2}\,{}^t R & \mathscr{M} \end{pmatrix} = 0$.

Without loss of generality we may assume that ρ is irreducible. Then we choose a hermitian inner product $\langle\ ,\ \rangle$ on V_ρ that is preserved under the unitary group $U(n) \subset GL(n, \mathbb{C})$. For two Jacobi forms f_1 and f_2 in $J_{\rho,\mathscr{M}}(\Gamma)$, putting $\tilde{\Gamma} = \Gamma \ltimes H_{\mathbb{Z}}^{(n,m)}$, we define the Petersson inner product formally by

$$(f_1, f_2) := \int_{\tilde{\Gamma}\backslash \mathbb{H}_{n,m}} \langle \rho(Y^{\frac{1}{2}}) f_1(\Omega, Z), \rho(Y^{\frac{1}{2}}) f_2(\Omega, Z) \rangle \kappa_{\mathscr{M}}(\Omega, Z)\, dv. \tag{2.9.2}$$

Here

$$dv = (\det Y)^{-(n+m+1)} [dX] \wedge [dY] \wedge [dU] \wedge [dV] \tag{2.9.3}$$

is a G^J-invariant volume element on $\mathbb{H}_{n,m}$ and

$$\kappa_{\mathscr{M}}(\Omega, Z) := e^{-4\pi i\, \sigma({}^t(\operatorname{Im} Z)\, \mathscr{M}\, \operatorname{Im} Z\, (\operatorname{Im} \Omega)^{-1})} = e^{-4\pi i\, \sigma({}^t V \mathscr{M} V Y^{-1})}, \tag{2.9.4}$$

where $\Omega = X + i\,Y \in \mathbb{H}_n$, $Z = U + i\,V \in \mathbb{C}^{(m,n)}$, $X = (x_{ij})$, $Y = (y_{ij})$, $U = (u_{kl})$, $V = (v_{kl})$ real and

$$[dX] = \bigwedge_{i \leq j} dx_{ij}, \quad [dY] = \bigwedge_{i \leq j} dy_{ij}, \quad [dU] = \bigwedge_{k,l} du_{kl} \quad \text{and} \quad [dV] = \bigwedge_{k,l} dv_{kl}.$$

A Jacobi form f in $J_{\rho,\mathscr{M}}(\Gamma)$ is said to be square integrable if $(f, f) < \infty$. We note that cusp Jacobi forms are square integrable and that (f_1, f_2) is finite if one of f_1 and f_2 is a cusp Jacobi form (cf. [66, p. 203]).

For brevity, we put $G = Sp(2n, \mathbb{R})$. For $g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in G$, we set

$$J(g, \Omega) = C\Omega + D, \quad \Omega \in \mathbb{H}_n. \tag{2.9.5}$$

We define the map $J_{\mathscr{M}} : G^J \times \mathbb{H}_{n,m} \longrightarrow \mathbb{C}^\times$ by

$$\begin{aligned} J_{\mathscr{M}}\big(\tilde{g}, (\Omega, Z)\big) := {} & e^{2\pi i\, \sigma(\mathscr{M}[Z+\lambda\Omega+\mu](C\Omega+D)^{-1}C)} \\ & \times e^{-2\pi i\, \sigma(\mathscr{M}(\lambda\, \Omega\, {}^t\lambda + 2\lambda\, {}^t Z + \kappa + \mu\, {}^t\lambda))}, \end{aligned} \tag{2.9.6}$$

where $\widetilde{g} = (g, (\lambda, \mu, \kappa)) \in G^J$ with $g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in G$ and $(\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$. Here we use the Siegel's notation $S[X] := {}^tXSX$ for two matrices S and X.

We define the map $J_{\rho,\mathscr{M}} : G^J \times \mathbb{H}_{n,m} \longrightarrow GL(V_\rho)$ by

$$J_{\rho,\mathscr{M}}(\widetilde{g}, (\Omega, Z)) = J_{\mathscr{M}}(\widetilde{g}, (\Omega, Z))\, \rho(J(g, \Omega)), \tag{2.9.7}$$

where $\widetilde{g} = (g, h) \in G^J$ with $g \in G$ and $h \in H_{\mathbb{R}}^{(n,m)}$. For a function f on $\mathbb{H}_n$ with values in V_ρ, we can lift f to a function Φ_f on G^J:

$$\begin{aligned} \Phi_f(\sigma) :&= (f|_{\rho,\mathscr{M}}[\sigma])(iI_n, 0) \\ &= J_{\rho,\mathscr{M}}(\sigma, (iI_n, 0))^{-1} f(\sigma \cdot (iI_n, 0)), \quad \sigma \in G^J. \end{aligned}$$

We allow a weight k to be half-integral. Let

$$\mathfrak{S} = \left\{ S \in \mathbb{C}^{(n,n)} \mid S = {}^tS,\ \mathrm{Re}\,(S) > 0 \right\}$$

be a connected simply connected complex manifold. Then there is a uniquely determined holomorphic function $\det^{1/2}$ on $\mathfrak{S}$ such that

$$\left(\det{}^{1/2} S\right)^2 = \det S \qquad \text{for all } S \in \mathfrak{S}, \tag{2.9.8}$$

$$\det{}^{1/2} S = (\det S)^{1/2} \qquad \text{for all } S \in \mathfrak{S} \cap \mathbb{R}^{(n,n)}. \tag{2.9.9}$$

For each integer $k \in \mathbb{Z}$ and $S \in \mathfrak{S}$, we put

$$\det{}^{k/2} S = \left(\det{}^{1/2} S\right)^k.$$

We recall that for $g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in G$ and $\Omega \in \mathbb{H}_n$,

$$g \cdot \Omega = (A\Omega + B)(C\Omega + D)^{-1}.$$

For any $g \in G$ and $\Omega, \Omega' \in \mathbb{H}_n$, we put

$$\begin{aligned} \varepsilon(g; \Omega', \Omega) = \det{}^{-1/2} \left(\frac{g \cdot \Omega' - \overline{g \cdot \Omega}}{2i} \right) \det{}^{1/2} \left(\frac{\Omega' - \overline{\Omega}}{2i} \right) \\ \times |\det J(g, \Omega')|^{-1/2} |\det J(g, \Omega)|^{-1/2}. \end{aligned} \tag{2.9.10}$$

For each $\Omega \in \mathbb{H}_n$, we define the function $\beta_\Omega : G \times G \longrightarrow T$ by

$$\beta_\Omega(g_1, g_2) = \varepsilon(g_1; \Omega, g_2(\Omega)), \quad g_1, g_2 \in G. \tag{2.9.11}$$

Then β_Ω satisfies the cocycle condition (cf. [46, p. 129]) and the cohomology class of β_Ω of order two:

$$\beta_\Omega(g_1, g_2)^2 = \alpha_\Omega(g_2)\, \alpha_\Omega(g_1 g_2)^{-1}\, \alpha_\Omega(g_1), \tag{2.9.12}$$

where

$$\alpha_\Omega(g) = \frac{\det J(g, \Omega)}{|\det J(g, \Omega)|}, \quad g \in G,\ \Omega \in \mathbb{H}_n. \tag{2.9.13}$$

For any $\Omega \in \mathbb{H}_n$, we let

$$G_\Omega = \left\{ (g, \epsilon) \in G \times T \mid \epsilon^2 = \alpha_\Omega(g)^{-1} \right\}$$

be the two-fold covering group with the multiplication law

$$(g_1, \epsilon_1)(g_2, \epsilon_2) = \big(g_1 g_2,\ \epsilon_1 \epsilon_2\, \beta_\Omega(g_1, g_2)\big).$$

The covering group G_Ω depends on the choice of $\Omega \in \mathbb{H}_n$, i.e., the choice of a maximal compact subgroup of G. However, for any two elements $\Omega_1, \Omega_2 \in \mathbb{H}_n$, G_{Ω_1} is isomorphic to G_{Ω_2} (cf. [46]). We put

$$G_* := G_{iI_n}. \tag{2.9.14}$$

Takase [46, p. 131] defined the automorphic factor $J_{1/2} : G_* \times \mathbb{H}_n \longrightarrow \mathbb{C}^\times$ by

$$J_{1/2}(g_\epsilon, \Omega) := \epsilon^{-1} \varepsilon(g; \Omega, iI_n)\, |\det J(g, \Omega)|^{1/2}, \tag{2.9.15}$$

where $g_\epsilon = (g, \epsilon) \in G_*$ with $g \in G$ and $\Omega \in \mathbb{H}_n$. Takase proved the following proposition (cf. [46, Proposition 5.1, p. 131]). We give his proof below.

Proposition 2.9.3 *$J_{1/2}$ is real analytic on G_* and holomorphic on $\mathbb{H}_n$. We have also*

(a) $J_{1/2}(g_*, \Omega)^2 = J(g, \Omega)$ *for all* $g_* = (g, \epsilon) \in G_*$ and $\Omega \in \mathbb{H}_n$;
(b) *For all* $g_* = (g, \epsilon)$, $h_* = (h, \eta) \in G_*$ *and* $\Omega \in \mathbb{H}_n$,

$$J_{1/2}(g_* h_*, \Omega) = J_{1/2}(g_*, h \cdot \Omega) J_{1/2}(h_*, \Omega). \tag{2.9.16}$$

Proof For the definition of $J(g, \Omega)$, we refer to (2.9.5). It is clear that $J_{1/2}$ is real analytic on $G_* \times \mathbb{H}_n$. For each $g_* = (g, \epsilon) \in G_*$, we have

$$\varepsilon(g; \Omega, iI_n)^2 = \alpha_\Omega(g)\,\alpha_{iI_n}(g)^{-1} = \epsilon^2 \det J(g, \Omega)/|J(g, \Omega)|.$$

So we have $J_{1/2}(g_*, \Omega)^2 = \det J(g, \Omega)$. Then $J_{1/2}$ is holomorphic on $\mathbb{H}_n$. For any $g_* = (g, \epsilon)$ and $h_* = (h, \eta)$ in G_*, we have $g_* h_* = (gh, \epsilon\,\eta\,\beta_{iI_n}(g, h))$ and

$$\begin{aligned}
&J_{1/2}(g_* h_*, \Omega)\\
=&\epsilon^{-1}\eta^{-1}\beta_{iI_n}(g, h)^{-1}\varepsilon(gh; \Omega, iI_n)\,|\det J(gh, \Omega)|^{1/2}\\
=&\epsilon^{-1}\eta^{-1}\varepsilon(g; iI_n, h\cdot\Omega)^{-1}\,\varepsilon(g; h\cdot\Omega, h\cdot(iI_n))\\
&\times \varepsilon(h; \Omega, iI_n)\,|\det J(g, h\cdot\Omega)\,J(h, \Omega)|^{1/2}\\
=&\epsilon^{-1}\varepsilon(g; h\cdot\Omega, iI_n)\,|\det J(g, h\cdot\Omega)|^{1/2}\,\eta^{-1}\varepsilon(h; \Omega, iI_n)\,|\det J(h, \Omega)|^{1/2}\\
=&J_{1/2}(g_*, h\cdot\Omega)J_{1/2}(h_*, \Omega).
\end{aligned}$$

Therefore $J_{1/2}$ is a factor of automorphy for G_* on $\mathbb{H}_n$. □

Let $\pi_* : G_* \longrightarrow G$ be the projection defined by $\pi_*(g, \epsilon) = g$. Let Γ be a subgroup of the Siegel modular group Γ_n of finite index. Let $\Gamma_* = \pi_*^{-1}(\Gamma) \subset G_*$. Let χ be a finite order unitary character of Γ_*. Let $k \in \mathbb{Z}^+$ be a positive integer. We say that a holomorphic function $\phi : \mathbb{H}_n \longrightarrow \mathbb{C}$ is a Siegel modular form of a half-integral weight $k/2$ with level Γ if it satisfies the condition

$$\phi(\gamma_*\langle\Omega\rangle) = \chi(\gamma_*)J_{1/2}(\gamma_*, \Omega)^k\phi(\Omega) \tag{2.9.17}$$

for all $\gamma_* \in \Gamma_*$ and $\Omega \in \mathbb{H}_n$. We denote by $M_{k/2}(\Gamma, \chi)$ the vector space of all Siegel modular forms of weight $k/2$ with level Γ. Let $S_{k/2}(\Gamma, \chi)$ be the subspace of $M_{k/2}(\Gamma, \chi)$ consisting of $\phi \in M_{k/2}(\Gamma, \chi)$ such that

$$|\phi(\Omega)|\det(\mathrm{Im}\,\Omega)^{k/4} \text{ is bounded on } \mathbb{H}_n.$$

An element of $S_{k/2}(\Gamma, \chi)$ is called a Siegel cusp form of weight $k/2$. The Petersson norm on $S_{k/2}(\Gamma, \chi)$ is defined by

$$||\phi||^2 = \int_{\Gamma\backslash\mathbb{H}_n} |\phi(\Omega)|^2\,\det(\mathrm{Im}\,\Omega)^{k/2}\,dv_\Omega,$$

where

$$dv_\Omega = (\det Y)^{-(n+1)}[dX][dY]$$

is a G-invariant volume element on $\mathbb{H}_n$.

2.10 The Schrödinger–Weil Representation

Throughout this section we assume that $\mathscr{M}$ is a positive definite symmetric real $m \times m$ matrix. We let

$$L = \left\{ (0, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)} \;\middle|\; \mu \in \mathbb{R}^{(m,n)},\ \kappa = {}^t\kappa \in \mathbb{R}^{(m,m)} \right\}.$$

be a commutative normal subgroup of $H_{\mathbb{R}}^{(n,m)}$ and $\chi_{\mathscr{M}} : L \longrightarrow \mathbb{C}^\times$ be the unitary character of L defined by

$$\chi_{\mathscr{M}}((0, \mu, \kappa)) := e^{\pi\, i\, \sigma(\mathscr{M}\kappa)}, \quad (0, \mu, \kappa) \in L.$$

The representation $\mathscr{W}_{\mathscr{M}} = \mathrm{Ind}_{L}^{H_{\mathbb{R}}^{(n,m)}} \chi_{\mathscr{M}}$ induced by $\chi_{\mathscr{M}}$ from L is realized on the Hilbert space $H(\chi_{\mathscr{M}}) \cong L^2\left(\mathbb{R}^{(m,n)}, d\xi\right)$. $\mathscr{W}_{\mathscr{M}}$ is irreducible (cf. Formula (1.2.11) and Theorem 1.2.1) and is called the Schrödinger representation $\mathscr{W}_{\mathscr{M}}$ of the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ with the central character $\chi_{\mathscr{M}}$. We refer to [52, 56, 60, 62, 63] for more representations of the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ and their related topics. Then $\mathscr{W}_{\mathscr{M}}$ is expressed explicitly as

$$[\mathscr{W}_{\mathscr{M}}(h_0) f](\lambda) = e^{\pi\, i\, \sigma\{\mathscr{M}(\kappa_0 + \mu_0\, {}^t\lambda_0 + 2\lambda\, {}^t\mu_0)\}} f(\lambda + \lambda_0), \tag{2.10.1}$$

where $h_0 = (\lambda_0, \mu_0, \kappa_0) \in H$ and $\lambda \in \mathbb{R}^{(m,n)}$. We note that the symplectic group $Sp(2n, \mathbb{R})$ acts on $H_{\mathbb{R}}^{(n,m)}$ by conjugation inside G^J (see Formula (2.3.1)). For a fixed element $g \in Sp(2n, \mathbb{R})$, the irreducible unitary representation $\mathscr{W}_{\mathscr{M}}^g$ of $H_{\mathbb{R}}^{(n,m)}$ defined by

$$\mathscr{W}_{\mathscr{M}}^g(h) = \mathscr{W}_{\mathscr{M}}(ghg^{-1}), \quad h \in H_{\mathbb{R}}^{(n,m)}$$

has the property that

$$\mathscr{W}_{\mathscr{M}}^g((0,0,\kappa)) = \mathscr{W}_{\mathscr{M}}((0,0,\kappa)) = e^{\pi i\, \sigma(\mathscr{M}\kappa)}\, \mathrm{Id}_{H(\chi_{\mathscr{M}})}, \quad \kappa \in \mathrm{Sym}(m, \mathbb{R}).$$

Here $\mathrm{Id}_{H(\chi_{\mathscr{M}})}$ denotes the identity operator on the Hilbert space $H(\chi_{\mathscr{M}})$. According to the Stone–von Neumann theorem, there exists a unitary operator $R_{\mathscr{M}}(g)$ on $H(\chi_{\mathscr{M}})$ with $R_{\mathscr{M}}(I_{2n}) = \mathrm{Id}_{H(\chi_{\mathscr{M}})}$ such that

$$R_{\mathscr{M}}(g)\mathscr{W}_{\mathscr{M}}(h) = \mathscr{W}_{\mathscr{M}}^g(h) R_{\mathscr{M}}(g) \qquad \text{for all } h \in H_{\mathbb{R}}^{(n,m)}. \tag{2.10.2}$$

We observe that $R_{\mathscr{M}}(g)$ is determined uniquely up to a scalar of modulus one.

From now on, for brevity, we put $G = Sp(2n, \mathbb{R})$. According to Schur's lemma, we have a map $c_{\mathscr{M}} : G \times G \longrightarrow T$ satisfying the relation

$$R_{\mathscr{M}}(g_1 g_2) = c_{\mathscr{M}}(g_1, g_2) R_{\mathscr{M}}(g_1) R_{\mathscr{M}}(g_2) \quad \text{for all } g_1, g_2 \in G. \tag{2.10.3}$$

For brevity, we put $T := \mathbb{C}_1^{\times}$. We recall that T denotes the multiplicative group of complex numbers of modulus one. Therefore $R_{\mathscr{M}}$ is a projective representation of G on $H(\chi_{\mathscr{M}})$ and $c_{\mathscr{M}}$ defines the cocycle class in $H^2(G, T)$. The cocycle $c_{\mathscr{M}}$ yields the central extension $G_{\mathscr{M}}$ of G by T. The group $G_{\mathscr{M}}$ is a set $G \times T$ equipped with the following multiplication:

$$(g_1, t_1) \cdot (g_2, t_2) = \left(g_1 g_2, t_1 t_2 \, c_{\mathscr{M}}(g_1, g_2)^{-1}\right), \quad g_1, g_2 \in G, \ t_1, t_2 \in T. \tag{2.10.4}$$

We see immediately that the map $\widetilde{R}_{\mathscr{M}} : G_{\mathscr{M}} \longrightarrow GL(H(\chi_{\mathscr{M}}))$ defined by

$$\widetilde{R}_{\mathscr{M}}(g, t) = t \, R_{\mathscr{M}}(g) \quad \text{for all } (g, t) \in G_{\mathscr{M}} \tag{2.10.5}$$

is a *true* representation of $G_{\mathscr{M}}$. As in [26, pp. 70–71, Section 1.7], we can define the map $s_{\mathscr{M}} : G \longrightarrow T$ satisfying the relation

$$c_{\mathscr{M}}(g_1, g_2)^2 = s_{\mathscr{M}}(g_1)^{-1} s_{\mathscr{M}}(g_2)^{-1} s_{\mathscr{M}}(g_1 g_2) \quad \text{for all } g_1, g_2 \in G.$$

Thus we see that

$$G_{2,\mathscr{M}} = \left\{ (g, t) \in G_{\mathscr{M}} \mid t^2 = s_{\mathscr{M}}(g)^{-1} \right\} \tag{2.10.6}$$

is the metaplectic group associated with $\mathscr{M}$ which is a two-fold covering group of G. The restriction $R_{2,\mathscr{M}}$ of $\widetilde{R}_{\mathscr{M}}$ to $G_{2,\mathscr{M}}$ is the Weil representation of G associated with $\mathscr{M}$.

If we identify $h = (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$ (resp. $g \in G$) with $(I_{2n}, (\lambda, \mu, \kappa)) \in G^J$ (resp. $(g, (0, 0, 0)) \in G^J$), every element $\tilde{g}$ of G^J can be written as $\tilde{g} = hg$ with $h \in H_{\mathbb{R}}^{(n,m)}$ and $g \in G$. In fact,

$$(g, (\lambda, \mu, \kappa)) = (I_{2n}, ((\lambda, \mu) g^{-1}, \kappa)) \, (g, (0, 0, 0)) = ((\lambda, \mu) g^{-1}, \kappa) \cdot g.$$

Therefore we define the projective representation $\pi_{\mathscr{M}}$ of the Jacobi group G^J with cocycle $c_{\mathscr{M}}(g_1, g_2)$ by

$$\pi_{\mathscr{M}}(hg) = \mathscr{W}_{\mathscr{M}}(h) \, R_{\mathscr{M}}(g), \quad h \in H_{\mathbb{R}}^{(n,m)}, \ g \in G. \tag{2.10.7}$$

Indeed, since $H_{\mathbb{R}}^{(n,m)}$ is a normal subgroup of G^J, for any $h_1, h_2 \in H_{\mathbb{R}}^{(n,m)}$ and $g_1, g_2 \in G$,

$$\begin{aligned}
&\pi_{\mathscr{M}}(h_1 g_1 h_2 g_2)\\
&= \pi_{\mathscr{M}}(h_1 g_1 h_2 g_1^{-1} g_1 g_2)\\
&= \mathscr{W}_{\mathscr{M}}\big(h_1(g_1 h_2 g_1^{-1})\big) R_{\mathscr{M}}(g_1 g_2) \qquad \text{(by Formula (2.10.7))}\\
&= c_{\mathscr{M}}(g_1, g_2)\mathscr{W}_{\mathscr{M}}(h_1)\mathscr{W}_{\mathscr{M}}^{g_1}(h_2) R_{\mathscr{M}}(g_1) R_{\mathscr{M}}(g_2) \qquad \text{(by Formula (2.10.3))}\\
&= c_{\mathscr{M}}(g_1, g_2)\mathscr{W}_{\mathscr{M}}(h_1) R_{\mathscr{M}}(g_1)\mathscr{W}_{\mathscr{M}}(h_2) R_{\mathscr{M}}(g_2) \qquad \text{(by Formula (2.10.2))}\\
&= c_{\mathscr{M}}(g_1, g_2)\pi_{\mathscr{M}}(h_1 g_1)\pi_{\mathscr{M}}(h_2 g_2) \qquad \text{(by Formula (2.10.7))}.
\end{aligned}$$

We let

$$G_{\mathscr{M}}^J = G_{\mathscr{M}} \ltimes H_{\mathbb{R}}^{(n,m)}$$

be the semidirect product of $G_{\mathscr{M}}$ and $H_{\mathbb{R}}^{(n,m)}$ with the multiplication law

$$\begin{aligned}
&\big((g_1, t_1), (\lambda_1, \mu_1, \kappa_1)\big) \cdot \big((g_2, t_2), (\lambda_2, \mu_2, \kappa_2)\big)\\
&= \big((g_1, t_1)(g_2, t_2), (\tilde{\lambda} + \lambda_2, \tilde{\mu} + \mu_2, \kappa_1 + \kappa_2 + \tilde{\lambda}\,{}^t\mu_2 - \tilde{\mu}\,{}^t\lambda_2)\big),
\end{aligned}$$

where $(g_1, t_1), (g_2, t_2) \in G_{\mathscr{M}}$, $(\lambda_1, \mu_1, \kappa_1), (\lambda_2, \mu_2, \kappa_2) \in H_{\mathbb{R}}^{(n,m)}$ and $(\tilde{\lambda}, \tilde{\mu}) = (\lambda, \mu)g_2$. If we identify $h = (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$ (resp. $(g, t) \in G_{\mathscr{M}}$) with $((I_{2n}, 1), (\lambda, \mu, \kappa)) \in G_{\mathscr{M}}^J$ (resp. $((g, t), (0, 0, 0)) \in G_{\mathscr{M}}^J$), we see easily that every element $\big((g, t), (\lambda, \mu, \kappa)\big)$ of $G_{\mathscr{M}}^J$ can be expressed as

$$\begin{aligned}
\big((g, t), (\lambda, \mu, \kappa)\big) &= \big((I_{2n}, 1), ((\lambda, \mu)g^{-1}, \kappa)\big)\big((g, t), (0, 0, 0)\big)\\
&= ((\lambda, \mu)g^{-1}, \kappa)(g, t).
\end{aligned}$$

Now we can define the true representation $\widetilde{\omega}_{\mathscr{M}}$ of $G_{\mathscr{M}}^J$ by

$$\widetilde{\omega}_{\mathscr{M}}(h \cdot (g, t)) = t\,\pi_{\mathscr{M}}(hg) = t\,\mathscr{W}_{\mathscr{M}}(h)\,R_{\mathscr{M}}(g), \tag{2.10.8}$$

where $h \in H_{\mathbb{R}}^{(n,m)}$ and $(g, t) \in G_{\mathscr{M}}$. Indeed, since $H_{\mathbb{R}}^{(n,m)}$ is a normal subgroup of $G_{\mathscr{M}}^J$,

$$\begin{aligned}
&\widetilde{\omega}_{\mathscr{M}}\big(h_1(g_1, t_1)h_2(g_2, t_2)\big)\\
&= \widetilde{\omega}_{\mathscr{M}}\big(h_1(g_1, t_1)h_2(g_1, t_1)^{-1}(g_1, t_1)(g_2, t_2)\big)\\
&= \widetilde{\omega}_{\mathscr{M}}\big(h_1(g_1, t_1)h_2(g_1, t_1)^{-1}\big(g_1 g_2, t_1 t_2\, c_{\mathscr{M}}(g_1, g_2)^{-1}\big)\big)
\end{aligned}$$

$$
\begin{aligned}
&= t_1 t_2\, c_{\mathscr{M}}(g_1, g_2)^{-1}\, \mathscr{W}_{\mathscr{M}}\big(h_1(g_1, t_1) h_2(g_1, t_1)^{-1}\big) R_{\mathscr{M}}(g_1 g_2) \\
&= t_1 t_2\, \mathscr{W}_{\mathscr{M}}(h_1) \mathscr{W}_{\mathscr{M}}\big((g_1, t_1) h_2(g_1, t_1)^{-1}\big)\, R_{\mathscr{M}}(g_1) R_{\mathscr{M}}(g_2) \\
&= t_1 t_2\, \mathscr{W}_{\mathscr{M}}(h_1) \mathscr{W}_{\mathscr{M}}\big(g_1 h_2 g_1^{-1}\big)\, R_{\mathscr{M}}(g_1) R_{\mathscr{M}}(g_2) \\
&= t_1 t_2\, \mathscr{W}_{\mathscr{M}}(h_1) R_{\mathscr{M}}(g_1)\, \mathscr{W}_{\mathscr{M}}(h_2) R_{\mathscr{M}}(g_2) \qquad \text{(by Formula (2.10.2))} \\
&= \{t_1\, \pi_{\mathscr{M}}(h_1 g_1)\}\, \{t_2\, \pi_{\mathscr{M}}(h_2 g_2)\} \\
&= \widetilde{\omega}_{\mathscr{M}}\big(h_1(g_1, t_1)\big)\, \tilde{\omega}_{\mathscr{M}}\big(h_2(g_2, t_2)\big).
\end{aligned}
$$

Here we used the fact that $(g_1, t_1) h_2(g_1, t_1)^{-1} = g_1 h_2 g_1^{-1}$.

We recall that the following matrices,

$$
t(b) = \begin{pmatrix} I_n & b \\ 0 & I_n \end{pmatrix} \text{ with any } b = {}^t b \in \mathbb{R}^{(n,n)},
$$

$$
g(\alpha) = \begin{pmatrix} {}^t\alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix} \text{ with any } \alpha \in GL(n, \mathbb{R}),
$$

$$
\sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix},
$$

generate the symplectic group $G = Sp(2n, \mathbb{R})$ (cf. [10, p. 326], [32, p. 210]). Therefore the following elements $h_t(\lambda, \mu\,;\kappa)$, $t(b\,;t)$, $g(\alpha\,;t)$ and $\sigma_{n\,;t}$ of $G_{\mathscr{M}} \ltimes H_{\mathbb{R}}^{(n,m)}$ defined by

$$
\begin{aligned}
&h_t(\lambda, \mu\,;\kappa) = \big((I_{2n}, t), (\lambda, \mu, \kappa)\big) \text{ with } t \in T,\ \lambda, \mu \in \mathbb{R}^{(m,n)} \text{ and } \kappa \in \mathbb{R}^{(m,m)}, \\
&t(b\,;t) = \big((t(b), t), (0, 0, 0)\big) \text{ with any } b = {}^t b \in \mathbb{R}^{(n,n)},\ t \in T, \\
&g(\alpha\,;t) = \big(\big(g(\alpha), t\big), (0, 0, 0)\big) \text{ with any } \alpha \in GL(n, \mathbb{R}) \text{ and } t \in T, \\
&\sigma_{n\,;t} = ((\sigma_n, t), (0, 0, 0)) \text{ with } t \in T
\end{aligned}
$$

generate the group $G_{\mathscr{M}} \ltimes H_{\mathbb{R}}^{(n,m)}$. We can show that the representation $\widetilde{\omega}_{\mathscr{M}}$ is realized on the representation $H(\chi_{\mathscr{M}})$ as follows: for each $f \in L^2\big(\mathbb{R}^{(m,n)}\big)$ and $x \in \mathbb{R}^{(m,n)}$, the actions of $\widetilde{\omega}_{\mathscr{M}}$ on the generators are given by

$$
\big[\widetilde{\omega}_{\mathscr{M}}\big(h_t(\lambda, \mu\,;\kappa)\big) f\big](x) = t\, e^{\pi i\, \sigma\{\mathscr{M}(\kappa + \mu\, {}^t\lambda + 2x\, {}^t\mu)\}} f(x + \lambda), \tag{2.10.9}
$$

$$
\big[\widetilde{\omega}_{\mathscr{M}}\big(t(b\,;t)\big) f\big](x) = t\, e^{\pi i\, \sigma(\mathscr{M}\, x\, b\, {}^t x)} f(x), \tag{2.10.10}
$$

$$
\big[\widetilde{\omega}_{\mathscr{M}}\big(g(\alpha\,;t)\big) f\big](x) = t\, |\det \alpha|^{\frac{m}{2}}\, f(x\, {}^t\alpha), \tag{2.10.11}
$$

$$
\big[\widetilde{\omega}_{\mathscr{M}}\big(\sigma_{n\,;t}\big) f\big](x) = t\, \big(\det \mathscr{M}\big)^{\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} f(y)\, e^{-2\pi i\, \sigma(\mathscr{M}\, y\, {}^t x)}\, dy. \tag{2.10.12}
$$

Let

$$G^J_{2,\mathscr{M}} = G_{2,\mathscr{M}} \ltimes H^{(n,m)}_{\mathbb{R}}$$

be the semidirect product of $G_{2,\mathscr{M}}$ and $H^{(n,m)}_{\mathbb{R}}$. Then $G^J_{2,\mathscr{M}}$ is a subgroup of $G^J_{\mathscr{M}}$ which is a two-fold covering group of the Jacobi group G^J. The restriction $\omega_{\mathscr{M}}$ of $\widetilde{\omega}_{\mathscr{M}}$ to $G^J_{2,\mathscr{M}}$ is called the Schrödinger–Weil representation of G^J associated with $\mathscr{M}$.

We denote by $L^2_+\big(\mathbb{R}^{(m,n)}\big)$ $\big($resp. $L^2_-\big(\mathbb{R}^{(m,n)}\big)\big)$ the subspace of $L^2\big(\mathbb{R}^{(m,n)}\big)$ consisting of even (resp. odd) functions in $L^2\big(\mathbb{R}^{(m,n)}\big)$. According to Formulas (2.10.10)–(2.10.12), $R_{2,\mathscr{M}}$ is decomposed into representations of $R^{\pm}_{2,\mathscr{M}}$:

$$R_{2,\mathscr{M}} = R^+_{2,\mathscr{M}} \oplus R^-_{2,\mathscr{M}},$$

where $R^+_{2,\mathscr{M}}$ and $R^-_{2,\mathscr{M}}$ are the even Weil representation and the odd Weil representation of G that are realized on $L^2_+\big(\mathbb{R}^{(m,n)}\big)$ and $L^2_-\big(\mathbb{R}^{(m,n)}\big)$ respectively. Obviously, the center $\mathscr{Z}^J_{2,\mathscr{M}}$ of $G^J_{2,\mathscr{M}}$ is given by

$$\mathscr{Z}^J_{2,\mathscr{M}} = \big\{\big((I_{2n}, 1), (0, 0, \kappa)\big) \in G^J_{2,\mathscr{M}}\big\} \cong \mathrm{Sym}(m, \mathbb{R}).$$

We note that the restriction of $\omega_{\mathscr{M}}$ to $G_{2,\mathscr{M}}$ coincides with $R_{2,\mathscr{M}}$ and $\omega_{\mathscr{M}}(h) = \mathscr{W}_{\mathscr{M}}(h)$ for all $h \in H^{(n,m)}_{\mathbb{R}}$.

Remark 2.10.1 In the case $n = m = 1$, $\omega_{\mathscr{M}}$ is dealt in [4] and [30]. We refer to [11] and [20] for more details about the Weil representation $R_{2,\mathscr{M}}$.

Remark 2.10.2 The Schrödinger–Weil representation is applied usefully to the theory of Maaß–Jacobi forms [37].

As before we let $\mathscr{M}$ be a positive definite symmetric $m \times m$ real matrix. We define the mapping $\mathscr{F}^{(\mathscr{M})} : \mathbb{H}_{n,m} \longrightarrow L^2\big(\mathbb{R}^{(m,n)}\big)$ by

$$\mathscr{F}^{(\mathscr{M})}(\Omega, Z)(x) = e^{\pi i\, \sigma\{\mathscr{M}(x\,\Omega\,{}^t x + 2\,x\,{}^t Z)\}}, \tag{2.10.13}$$

where $(\Omega, Z) \in \mathbb{H}_{n,m}$ and $x \in \mathbb{R}^{(m,n)}$. For brevity we put $\mathscr{F}^{(\mathscr{M})}_{\Omega,Z} := \mathscr{F}^{(\mathscr{M})}(\Omega, Z)$ for $(\Omega, Z) \in \mathbb{H}_{n,m}$.

We set for brevity

$$G^J_* := G_{2,\mathscr{M}} \ltimes H^{(n,m)}_{\mathbb{R}}. \tag{2.10.14}$$

We note that G^J_* acts on $\mathbb{H}_{n,m}$ via the canonical projection of G^J_* onto G^J.

We define the automorphic factor $J^*_{\mathscr{M}} : G^J_* \times \mathbb{H}_{n,m} \longrightarrow \mathbb{C}^\times$ for G^J_* on $\mathbb{H}_{n,m}$ by

$$J^*_{\mathscr{M}}(\widetilde{g}_*, (\Omega, Z)) = e^{\pi i\, \sigma\left(\mathscr{M}(Z+\lambda\,\Omega+\mu)(C\Omega+D)^{-1}C\,{}^t(Z+\lambda\,\Omega+\mu)\right)} \times e^{-\pi i\, \sigma\left(\mathscr{M}(\lambda\,\Omega\,{}^t\lambda+2\,\lambda\,{}^tZ+\kappa+\mu\,{}^t\lambda)\right)} \left\{J_{1/2}((g,\epsilon),\Omega)\right\}^m, \tag{2.10.15}$$

where $\widetilde{g}_* = ((g,\epsilon),(\lambda,\mu,\kappa)) \in G^J_*$ with $g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in G$, $(\lambda,\mu,\kappa) \in H^{(n,m)}_{\mathbb{R}}$ and $(\Omega, Z) \in \mathbb{H}_{n,m}$.

Theorem 2.10.3 *Let $\mathscr{M}$ be a positive definite symmetric $m \times m$ real matrix. Then the map $\mathscr{F}^{(\mathscr{M})} : \mathbb{H}_{n,m} \longrightarrow L^2\big(\mathbb{R}^{(m,n)}\big)$ defined by (2.10.13) is a* covariant map *for the Schrödinger–Weil representation $\omega_{\mathscr{M}}$ of G^J and the automorphic factor $J^*_{\mathscr{M}}$ for G^J_* on $\mathbb{H}_{n,m}$ defined by Formula (2.10.15). In other words, $\mathscr{F}^{(\mathscr{M})}$ satisfies the following covariance relation,*

$$\omega_{\mathscr{M}}(\widetilde{g}_*)\mathscr{F}^{(\mathscr{M})}_{\Omega} = c(\widetilde{g}_*)\, J^*_{\mathscr{M}}\big(\widetilde{g}_*, (\Omega, Z)\big)^{-1} \mathscr{F}^{(\mathscr{M})}_{\widetilde{g}_*\cdot(\Omega,Z)}, \tag{2.10.16}$$

for all $\widetilde{g}_* \in G^J_*$ and $(\Omega, Z) \in \mathbb{H}_{n,m}$. Here $c(\widetilde{g}_*)$ is the constant determined uniquely by $\widetilde{g}_*$.

Proof For an element $\widetilde{g}_* = ((g,\epsilon),(\lambda,\mu,\kappa)) \in G^J_*$ with $g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in G$, we put $(\Omega_*, Z_*) = \widetilde{g}_* \cdot (\Omega, Z)$ for $(\Omega, Z) \in \mathbb{H}_{n,m}$. Then we have

$$\Omega_* = g\langle\Omega\rangle = (A\Omega + B)(C\Omega + D)^{-1},$$
$$Z_* = (Z + \lambda\,\Omega + \mu)(C\Omega + D)^{-1}.$$

Since the following elements $h(\lambda,\mu\,;\kappa)_\epsilon$, $t(b\,;\epsilon)$ and $\sigma_{n,\epsilon}$ of G^J_* defined by

$$h(\lambda,\mu\,;\kappa)_\epsilon = \big((I_{2n},\epsilon),(\lambda,\mu,\kappa)\big) \quad \text{with } \epsilon = \pm 1,\ \lambda,\mu \in \mathbb{R}^{(m,n)},\ \kappa \in \mathbb{R}^{(m,m)},$$
$$t(b\,;\epsilon) = \big((t(b),\epsilon),(0,0,0)\big) \quad \text{with } \epsilon = \pm 1,\ b \in \mathrm{Sym}(n,\mathbb{R}),$$
$$\sigma_{n,\epsilon} = \big((\sigma_n,\epsilon),(0,0,0)\big) \text{ with } \epsilon^2 = (-i)^n.$$

generate the group G^J_*, it suffices to prove the covariance relation (2.10.16) for the above generators.

Case I. $\widetilde{g}_* = h(\lambda,\mu,\kappa)_\epsilon$ with $\epsilon = \pm 1$ and $\lambda,\mu \in \mathbb{R}^{(m,n)}$, $\kappa \in \mathbb{R}^{(m,m)}$.

In this case, we have

$$\Omega_* = \Omega, \quad Z_* = Z + \lambda\,\Omega + \mu.$$

We put

$$h(\lambda, \mu, \kappa)_+ := \big((I_{2n}, 1), (\lambda, \mu, \kappa)\big)$$

and

$$h(\lambda, \mu, \kappa)_- := \big((I_{2n}, -1), (\lambda, \mu, \kappa)\big).$$

It is easily seen that according to Formula (2.9.15), we obtain

$$J_{1/2}((I_{2n}, 1), \Omega) = 1 \quad \text{and} \quad J_{1/2}((I_{2n}, -1), \Omega) = -1.$$

Therefore we get

$$J^*_{\mathscr{M}}\big(h(\lambda, \mu, \kappa)_+, (\Omega, Z)\big) = e^{-\pi i\, \sigma\{\mathscr{M}(\lambda\, \Omega\, {}^t\lambda + 2\, \lambda\, {}^t Z + \kappa + \mu\, {}^t\lambda)\}} \tag{2.10.17}$$

and

$$J^*_{\mathscr{M}}\big(h(\lambda, \mu, \kappa)_-, (\Omega, Z)\big) = (-1)^m\, e^{-\pi i\, \sigma\{\mathscr{M}(\lambda\, \Omega\, {}^t\lambda + 2\, \lambda\, {}^t Z + \kappa + \mu\, {}^t\lambda)\}}. \tag{2.10.18}$$

According to Formulas (2.10.1), (2.10.8) and (2.10.9), for $x \in \mathbb{R}^{(m,n)}$,

$$\begin{aligned}
&\Big[\omega_{\mathscr{M}}\big(h(\lambda, \mu, \kappa)_+\big)\mathscr{F}_\Omega^{(\mathscr{M})}\Big](x) \\
&= e^{\pi i\, \sigma\{\mathscr{M}(\kappa + \mu\, {}^t\lambda + 2\, x\, {}^t\mu)\}}\, \mathscr{F}_\Omega^{(\mathscr{M})}(x + \lambda) \\
&= e^{\pi i\, \sigma\{\mathscr{M}(\kappa + \mu\, {}^t\lambda + 2\, x\, {}^t\mu)\}}\, e^{\pi i\, \sigma\{\mathscr{M}((x+\lambda)\Omega\, {}^t(x+\lambda) + 2\, (x+\lambda)\, {}^t Z)\}}.
\end{aligned}$$

On the other hand, according to Formula (2.10.17), for $x \in \mathbb{R}^{(m,n)}$,

$$\begin{aligned}
&J^*_{\mathscr{M}}\big(h(\lambda, \mu, \kappa)_+, (\Omega, Z)\big)^{-1}\, \mathscr{F}^{(\mathscr{M})}_{\widetilde{g}_* \cdot (\Omega, Z)}(x) \\
&= J^*_{\mathscr{M}}\big(h(\lambda, \mu, \kappa)_+, (\Omega, Z)\big)^{-1}\, \mathscr{F}^{(\mathscr{M})}_{\Omega, Z + \lambda\, \Omega + \mu}(x) \\
&= e^{\pi i\, \sigma\{\mathscr{M}(\lambda \Omega\, {}^t\lambda + 2\, \lambda\, {}^t Z + \kappa + \mu\, {}^t\lambda)\}} \cdot e^{\pi i\, \sigma\{\mathscr{M}(x\, \Omega\, {}^t x + 2\, x\, {}^t(Z + \lambda\Omega + \mu))\}} \\
&= e^{\pi i\, \sigma\{\mathscr{M}(\kappa + \mu\, {}^t\lambda + 2\, x\, {}^t\mu)\}}\, e^{\pi i\, \sigma\{\mathscr{M}((x+\lambda)\Omega\, {}^t(x+\lambda) + 2\, (x+\lambda)\, {}^t Z)\}}.
\end{aligned}$$

Therefore we prove the covariance relation (2.10.16) in the case $\widetilde{g}_* = h(\lambda, \mu, \kappa)_+$.

Similarly, we can prove the covariance relation (2.10.16) in the $\widetilde{g}_* = h(\lambda, \mu, \kappa)_-$. In fact, according to Formulas (2.10.1), (2.10.8), (2.10.9) and (2.10.10), we obtain the following covariance relation:

$$\begin{aligned}
&\left[\omega_{\mathscr{M}}\big(h(\lambda, \mu, \kappa)_-\big)\mathscr{F}_{\Omega}^{(\mathscr{M})}\right](x) \\
&= -e^{\pi i\,\sigma\{\mathscr{M}(\kappa+\mu\,{}^t\lambda+2x\,{}^t\mu)\}}\, e^{\pi i\,\sigma\{\mathscr{M}((x+\lambda)\Omega\,{}^t(x+\lambda)+2(x+\lambda)\,{}^tZ)\}} \\
&= (-1)^{m+1}\, J_{\mathscr{M}}^*\big(h(\lambda, \mu, \kappa)_-, (\Omega, Z)\big)^{-1}\mathscr{F}^{(\mathscr{M})}_{\widetilde{g}_*\cdot(\Omega,Z)}(x).
\end{aligned}$$

Case II. $\widetilde{g}_* = t(b; \epsilon)$ with $\epsilon = \pm 1$ and $b \in \mathrm{Sym}(n, \mathbb{R})$.

In this case, we have

$$\Omega_* = \Omega + b \quad \text{and} \quad Z_* = Z.$$

We put

$$t(b)_+ = \big((t(b), 1), (0, 0, 0)\big)$$

and

$$t(b)_- = \big((t(b), -1), (0, 0, 0)\big).$$

It is easily seen that

$$J_{1/2}((t(b), 1), \Omega) = 1 \quad \text{and} \quad J_{1/2}((t(b), -1), \Omega) = -1.$$

Therefore we get

$$J_{\mathscr{M}}^*\big(t(b)_+, (\Omega, Z)\big) = 1 \quad \text{and} \quad J_{\mathscr{M}}^*\big(t(b)_-, (\Omega, Z)\big) = -1. \tag{2.10.19}$$

According to Formulas (2.10.8) and (2.10.10), we obtain

$$\left[\omega_{\mathscr{M}}\big(t(b)_+\big)\mathscr{F}_{\Omega}^{(\mathscr{M})}\right](x) = e^{\pi\, i\,\sigma(\mathscr{M}\, xb\,{}^tx)}\mathscr{F}_{\Omega}^{(\mathscr{M})}(x), \quad x \in \mathbb{R}^{(m,n)}.$$

On the other hand, according to Formula (2.10.19), for $x \in \mathbb{R}^{(m,n)}$, we obtain

$$\begin{aligned}
&J_{\mathscr{M}}^*\big(t(b)_+, (\Omega, Z)\big)^{-1}\mathscr{F}^{(\mathscr{M})}_{t(b)_+\cdot(\Omega,Z)}(x) \\
&= \mathscr{F}^{(\mathscr{M})}_{\Omega+b,Z}(x) \\
&= e^{\pi i\,\sigma\left(\mathscr{M}\left(x(\Omega+b)\,{}^tx+2x\,{}^tZ\right)\right)} \\
&= e^{\pi\, i\,\sigma(\mathscr{M}\, xb\,{}^tx)}\mathscr{F}_{\Omega}^{(\mathscr{M})}(x).
\end{aligned}$$

Therefore we prove the covariance relation (2.10.16) in the case $\widetilde{g}_* = t(b)_+$ with $b \in \mathrm{Sym}(n, \mathbb{R})$.

Similarly, we can prove the covariance relation (2.10.16) in the $\widetilde{g}_* = t(b)_-$. In fact, according to Formula (2.10.19), we obtain the following covariance relation:

$$\begin{aligned}
&\left[\omega_{\mathscr{M}}\left(t(b)_-\right)\mathscr{F}_{\Omega}^{(\mathscr{M})}\right](x) \\
&= -e^{\pi\, i\, \sigma(\mathscr{M}\, x b\, {}^t x)}\, \mathscr{F}_{\Omega}^{(\mathscr{M})}(x) \\
&= (-1)^{m+1}\, J_{\mathscr{M}}^*\left(t(b)_-, (\Omega, Z)\right)^{-1}\, \mathscr{F}_{t(b)_-\cdot(\Omega,Z)}^{(\mathscr{M})}(x).
\end{aligned}$$

Case III. $\widetilde{g}_* = \left((\sigma_n, \epsilon), (0,0,0)\right)$ with $\sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$ and $\epsilon^2 = (-i)^n$.

In this case, we have

$$\Omega_* = -\Omega^{-1} \quad \text{and} \quad Z_* = Z\,\Omega^{-1}.$$

In order to prove the covariance relation (2.10.16), we need the following useful lemma.

Lemma 2.10.4 *For a fixed element $\Omega \in \mathbb{H}_n$ and a fixed element $Z \in \mathbb{C}^{(m,n)}$, we obtain the following property:*

$$\begin{aligned}
&\int_{\mathbb{R}^{(m,n)}} e^{\pi\, i\, \sigma(x\,\Omega\, {}^t x + 2\, x\, {}^t Z)} dx_{11} \cdots dx_{mn} \qquad (2.10.20) \\
&= \left(\det \frac{\Omega}{i}\right)^{-\frac{m}{2}}\, e^{-\pi\, i\, \sigma(Z\,\Omega^{-1}\, {}^t Z)},
\end{aligned}$$

where $x = (x_{ij}) \in \mathbb{R}^{(m,n)}$.

Proof of Lemma 2.10.4. By a simple computation, we see that

$$e^{\pi i\, \sigma(x\Omega\, {}^t x + 2x\, {}^t Z)} = e^{-\pi i\, \sigma(Z\Omega^{-1}\, {}^t Z)} \cdot e^{\pi i\, \sigma\{(x + Z\Omega^{-1})\Omega\, {}^t (x + Z\Omega^{-1})\}}.$$

Since the real Jacobi group $G^J = G \ltimes H_{\mathbb{R}}^{(n,m)}$ acts on $\mathbb{H}_{n,m}$ transitively and holomorphically, we may put

$$\Omega = i\, A\, {}^t A, \quad Z = i\, V, \quad A \in \mathbb{R}^{(n,n)}, \quad V = (v_{ij}) \in \mathbb{R}^{(m,n)}.$$

Then we obtain

$$
\begin{aligned}
&\int_{\mathbb{R}^{(m,n)}} e^{\pi i\,\sigma(x\Omega\,{}^t x+2x\,{}^t Z)}dx_{11}\cdots dx_{mn}\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\int_{\mathbb{R}^{(m,n)}} e^{\pi i\,\sigma[\{x+iV(iA\,{}^tA)^{-1}\}(iA\,{}^tA)\,{}^t\{x+iV(iA\,{}^tA)^{-1}\}]}\,dx_{11}\cdots dx_{mn}\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\int_{\mathbb{R}^{(m,n)}} e^{\pi i\,\sigma[\{x+V(A\,{}^tA)^{-1}\}A\,{}^tA\,{}^t\{x+V(A\,{}^tA)^{-1}\}]}\,dx_{11}\cdots dx_{mn}\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\int_{\mathbb{R}^{(m,n)}} e^{-\pi\,\sigma\{(uA)\,{}^t(uA)\}}\,du_{11}\cdots du_{mn}\\
&\qquad\qquad \big(\text{put } u = x + V(A\,{}^tA)^{-1} = (u_{ij})\big)\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\int_{\mathbb{R}^{(m,n)}} e^{-\pi\,\sigma(w\,{}^tw)}(\det A)^{-m}\,dw_{11}\cdots dw_{mn}\\
&\qquad\qquad \big(\text{put } w = uA = (w_{ij})\big)\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\,(\det A)^{-m}\cdot\left(\prod_{i=1}^{m}\prod_{j=1}^{n}\int_{\mathbb{R}} e^{-\pi\,w_{ij}^2}\,dw_{ij}\right)\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\,(\det A)^{-m}\quad \Big(\text{because } \int_{\mathbb{R}} e^{-\pi\,w_{ij}^2}\,dw_{ij}=1\quad \text{for all } i,\,j\Big)\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\,\big(\det\big(A\,{}^tA\big)\big)^{-\frac{m}{2}}\\
&= e^{-\pi i\,\sigma(Z\Omega^{-1}\,{}^t Z)}\,\left(\det\left(\frac{\Omega}{i}\right)\right)^{-\frac{m}{2}}.
\end{aligned}
$$

This completes the proof of Lemma 2.10.4. □

According to Formulas (2.10.8) and (2.10.12), for $x\in\mathbb{R}^{(m,n)}$, we obtain

$$
\begin{aligned}
&\left[\omega_{\mathscr{M}}(\widetilde{g}_*)\mathscr{F}_{\Omega}^{(\mathscr{M})}\right](x)\\
&= \epsilon\,\big(\det\mathscr{M}\big)^{\frac{n}{2}}\int_{\mathbb{R}^{(m,n)}}\mathscr{F}_{\Omega}^{(\mathscr{M})}(y)\,e^{-2\pi\,i\,\sigma\,(\mathscr{M}\,y\,{}^t x)}dy\\
&= \epsilon\,\big(\det\mathscr{M}\big)^{\frac{n}{2}}\int_{\mathbb{R}^{(m,n)}} e^{\pi\,i\,\sigma\{\mathscr{M}(y\,\Omega\,{}^t y+2\,y\,{}^t Z)\}}\,e^{-2\pi\,i\,\sigma(\mathscr{M}\,y\,{}^t x)}dy\\
&= \epsilon\,\big(\det\mathscr{M}\big)^{\frac{n}{2}}\int_{\mathbb{R}^{(m,n)}} e^{\pi\,i\,\sigma\{\mathscr{M}(y\,\Omega\,{}^t y+2\,y\,{}^t(Z-x))\}}dy.
\end{aligned}
$$

If we substitute $u = \mathscr{M}^{\frac{1}{2}} y$, then $du = (\det \mathscr{M})^{\frac{n}{2}}\, dy$. Therefore according to Lemma 2.10.4, we obtain

$$\begin{aligned}
&\left[\omega_{\mathscr{M}}\big(\widetilde{g}_*\big)\mathscr{F}_{\Omega}^{(\mathscr{M})}\right](x)\\
&= \epsilon\,\big(\det \mathscr{M}\big)^{\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} e^{\pi\, i\, \sigma\left(u\,\Omega\,{}^t u + 2\,\mathscr{M}^{1/2}\,u\,{}^t(Z-x)\right)}\,(\det \mathscr{M})^{-\frac{n}{2}}\, du\\
&= \epsilon \int_{\mathbb{R}^{(m,n)}} e^{\pi\, i\, \sigma\left(u\,\Omega\,{}^t u + 2\,u\,{}^t(\mathscr{M}^{1/2}\,(Z-x))\right)}\, du\\
&= \epsilon \left(\det \frac{\Omega}{i}\right)^{-\frac{m}{2}} e^{-\pi\, i\, \sigma\left\{\mathscr{M}^{1/2}(Z-x)\,\Omega^{-1}\,{}^t(Z-x)\,\mathscr{M}^{1/2}\right\}} \quad \text{(by Lemma 2.10.4)}\\
&= \epsilon\; i^{\frac{mn}{2}}\; (\det \Omega)^{-\frac{m}{2}}\; e^{-\pi\, i\, \sigma\left(\mathscr{M}\,(Z-x)\,\Omega^{-1}\,{}^t(Z-x)\right)}\\
&= \epsilon\; i^{\frac{mn}{2}}\; (\det \Omega)^{-\frac{m}{2}}\; e^{-\pi\, i\, \sigma\left(\mathscr{M}(Z\,\Omega^{-1}\,{}^t Z + x\,\Omega^{-1}\,{}^t x - 2\,Z\,\Omega^{-1}\,{}^t x)\right)}.
\end{aligned}$$

On the other hand,

$$\begin{aligned}
&J_{1/2}\big((\sigma_n, \epsilon), \Omega\big)\\
&= \epsilon^{-1} \det{}^{-1/2}\left(\frac{\sigma_n\cdot\Omega - \overline{\sigma_n\cdot(i\,I_n)}}{2\,i}\right) \det{}^{1/2}\left(\frac{\Omega - \overline{(i\,I_n)}}{2\,i}\right) |J(\sigma_n, i\,I_n)|^{-1/2}\\
&= \epsilon^{-1} \det{}^{-1/2}\left(\frac{i\,\Omega^{-1}\big(\Omega - \overline{i\,I_n}\big)}{2\,i}\right) \det{}^{1/2}\left(\frac{\Omega - \overline{(i\,I_n)}}{2\,i}\right)\\
&= \epsilon^{-1} \det{}^{-1/2}\left(i\,\Omega^{-1}\right)\\
&= \epsilon^{-1}\left(\frac{1}{i}\right)^{n/2} (\det\,\Omega)^{1/2}.
\end{aligned}$$

Therefore, according to Formula (2.10.15), for $x \in \mathbb{R}^{(m,n)}$, we obtain

$$\begin{aligned}
&J^*_{\mathscr{M}}\big(\widetilde{g}_*, (\Omega, Z)\big)^{-1}\, \mathscr{F}^{(\mathscr{M})}_{\widetilde{g}_*\cdot(\Omega,Z)}(x)\\
&= e^{-\pi\, i\, \sigma(\mathscr{M}\,Z\,\Omega^{-1}\,{}^t Z)}\; J_{1/2}\big((\sigma_n, \epsilon), \Omega\big)^{-m}\; \mathscr{F}^{(\mathscr{M})}_{-\Omega^{-1},\, Z\,\Omega^{-1}}(x)\\
&= \epsilon^m\; i^{\frac{mn}{2}}\; (\det \Omega)^{-\frac{m}{2}}\; e^{-\pi\, i\, \sigma(\mathscr{M}\,Z\,\Omega^{-1}\,{}^t Z)}\, e^{\pi\, i\, \sigma\left\{\mathscr{M}\left(x\,(-\Omega^{-1})\,{}^t x + 2\,x\,{}^t(Z\,\Omega^{-1})\right)\right\}}\\
&= \epsilon^m\; i^{\frac{mn}{2}}\; (\det \Omega)^{-\frac{m}{2}}\; e^{-\pi\, i\, \sigma\left(\mathscr{M}(Z\,\Omega^{-1}\,{}^t Z + x\,\Omega^{-1}\,{}^t x - 2\,Z\,\Omega^{-1}\,{}^t x)\right)}.
\end{aligned}$$

Hence we obtain the covariance relation

$$\omega_{\mathscr{M}}\big(\widetilde{g}_*\big)\mathscr{F}_{\Omega}^{(\mathscr{M})} = \epsilon^{1-m}\, J^*_{\mathscr{M}}\big(\widetilde{g}_*, (\Omega, Z)\big)^{-1}\, \mathscr{F}^{(\mathscr{M})}_{\widetilde{g}_*\cdot(\Omega,Z)}$$

in the case $\widetilde{g}_* = \big((\sigma_n, \epsilon), (0, 0; 0)\big)$ with $\epsilon^2 = (-i)^n$. Since $J^*_{\mathscr{M}}$ is an automorphic factor for G^J_* on $\mathbb{H}_{n,m}$, we see that if the covariance relation (2.10.16) holds for two elements $\widetilde{g}_*$, $\widetilde{h}_*$ in G^J_*, then it holds for $\widetilde{g}_*\widetilde{h}_*$. Finally, we complete the proof. □

Remark 2.10.5 Using the Schrödinger–Weil representation, Takase [47] established a bijective correspondence between the space of cuspidal Jacobi forms and the space of Siegel cusp forms of half integral weight which is compatible with the action of Hecke operators. For example, if m is a positive integer, the classical result (cf. [8] and [17])

$$J^{cusp}_{m,1}(\Gamma_n) \cong S_{m-1/2}(\Gamma_0(4))$$

can be obtained by the method of the representation theory. Here $\Gamma_0(4)$ is the Hecke subgroup of the Siegel modular group Γ_n and $J^{\mathrm{cusp}}_{m,1}(\Gamma_n)$ denotes the vector space of cuspidal Jacobi forms of weight m and index 1.

Let (π, V_π) be a unitary representation of G^J_* on the representation space V_π. Let Γ be an arithmetic subgroup of the Siegel modular group Γ_n. We set $\Gamma_* = \pi_*^{-1}(\Gamma)$ and

$$\Gamma^J_* = \Gamma_* \ltimes H^{(n,m)}_{\mathbb{Z}}.$$

We assume that (π, V_π) satisfies the following conditions (A) and (B):

(A) There exists a vector-valued map

$$\mathscr{F} : \mathbb{H}_{n,m} \longrightarrow V_\pi, \qquad (\Omega, Z) \mapsto \mathscr{F}_{\Omega,Z} := \mathscr{F}(\Omega, Z)$$

satisfying the following covariance relation,

$$\pi\big(\widetilde{g}_*\big)\mathscr{F}_{\Omega,Z} = \psi\big(\widetilde{g}_*\big)\, J_*\big(\widetilde{g}_*, (\Omega, Z)\big)^{-1}\, \mathscr{F}_{\widetilde{g}_*\cdot(\Omega,Z)}, \tag{2.10.21}$$

for all $\widetilde{g}_* \in G^J_*$ and $(\Omega, Z) \in \mathbb{H}_{n,m}$, where ψ is a character of G^J_* and $J_* : G^J_* \times \mathbb{H}_{n,m} \longrightarrow \mathbb{C}^\times$ is a certain automorphic factor for G^J_* on $\mathbb{H}_{n,m}$.

(B) There exists a linear functional $\theta : V_\pi \longrightarrow \mathbb{C}$ which is semi-invariant under the action of Γ^J_*, in other words, for all $\widetilde{\gamma}_* \in \Gamma^J_*$ and $(\Omega, Z) \in \mathbb{H}_{n,m}$,

$$\langle\, \pi^*\big(\widetilde{\gamma}_*\big)\theta,\ \mathscr{F}_{\Omega,Z}\,\rangle = \langle\, \theta, \pi\big(\widetilde{\gamma}_*\big)^{-1}\mathscr{F}_{\Omega,Z}\,\rangle = \chi\big(\widetilde{\gamma}_*\big)\,\langle\, \theta,\ \mathscr{F}_{\Omega,Z}\,\rangle, \tag{2.10.22}$$

where π^* is the contragredient of π and $\chi : \Gamma^J_* \longrightarrow T$ is a unitary character of Γ^J_*.

Under the assumptions (A) and (B) on a unitary representation (π, V_π), we define the function Θ on $\mathbb{H}_{n,m}$ by

$$\Theta(\Omega, Z) := \langle \theta, \mathscr{F}_{\Omega,Z} \rangle = \theta\big(\mathscr{F}_{\Omega,Z}\big), \qquad (\Omega, Z) \in \mathbb{H}_{n,m}. \tag{2.10.23}$$

We now shall see that Θ is an automorphic form on $\mathbb{H}_{n,m}$ with respect to Γ_*^J for the automorphic factor J_*.

Lemma 2.10.6 *Let (π, V_π) be a unitary representation of G_*^J satisfying the above assumptions (A) and (B). Then the function Θ on $\mathbb{H}_{n,m}$ defined by Formula (2.10.23) satisfies the following modular transformation behavior,*

$$\Theta\big(\widetilde{\gamma}_* \cdot (\Omega, Z)\big) = \psi\big(\widetilde{\gamma}_*\big)^{-1} \chi\big(\widetilde{\gamma}_*\big)^{-1} J_*\big(\widetilde{\gamma}_*, (\Omega, Z)\big)\, \Theta(\Omega, Z), \tag{2.10.24}$$

for all $\widetilde{\gamma}_* \in \Gamma_*^J$ and $(\Omega, Z) \in \mathbb{H}_{n,m}$.

Proof For any $\widetilde{\gamma}_* \in \Gamma_*^J$ and $(\Omega, Z) \in \mathbb{H}_{n,m}$, according to the assumptions (2.10.21) and (2.10.22), we obtain

$$\begin{aligned}
&\Theta\big(\widetilde{\gamma}_* \cdot (\Omega, Z)\big) = \big\langle \theta, \mathscr{F}_{\widetilde{\gamma}_* \cdot (\Omega,Z)} \big\rangle \\
&= \big\langle \theta, \psi\big(\widetilde{\gamma}_*\big)^{-1} J_*\big(\widetilde{\gamma}_*, (\Omega, Z)\big)\, \pi\big(\widetilde{\gamma}_*\big) \mathscr{F}_{\Omega,Z} \big\rangle \\
&= \psi\big(\widetilde{\gamma}_*\big)^{-1} J_*\big(\widetilde{\gamma}_*, (\Omega, Z)\big) \big\langle \theta, \pi\big(\widetilde{\gamma}_*\big) \mathscr{F}_{\Omega,Z} \big\rangle \\
&= \psi\big(\widetilde{\gamma}_*\big)^{-1} \chi\big(\widetilde{\gamma}_*\big)^{-1} J_*\big(\widetilde{\gamma}_*, (\Omega, Z)\big) \big\langle \theta, \mathscr{F}_{\Omega,Z} \big\rangle \\
&= \psi\big(\widetilde{\gamma}_*\big)^{-1} \chi\big(\widetilde{\gamma}_*\big)^{-1} J_*\big(\widetilde{\gamma}_*, (\Omega, Z)\big)\, \Theta(\Omega, Z).
\end{aligned}$$

□

Now for a positive definite integral symmetric matrix $\mathscr{M}$ of degree m, we define the holomorphic function $\Theta_{\mathscr{M}} : \mathbb{H}_{n,m} \longrightarrow \mathbb{C}$ by

$$\Theta_{\mathscr{M}}(\Omega, Z) := \sum_{\xi \in \mathbb{Z}^{(m,n)}} e^{\pi i\, \sigma(\mathscr{M}(\xi\, \Omega\, {}^t\xi + 2\xi\, {}^t Z))}, \qquad (\Omega, Z) \in \mathbb{H}_{n,m}. \tag{2.10.25}$$

Theorem 2.10.7 *Let m be an odd positive integer. Let $\mathscr{M}$ be a symmetric positive definite integral matrix of degree m such that* $\det \mathscr{M} = 1$. *Let Γ be an arithmetic subgroup of Γ_n generated by all the following elements:*

$$t(b) = \begin{pmatrix} I_n & b \\ 0 & I_n \end{pmatrix}, \quad g(\alpha) = \begin{pmatrix} {}^t\alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix}, \quad \sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix},$$

where $b = {}^t b \in \mathrm{Sym}(n, \mathbb{Z})$ with even diagonal and $\alpha \in GL(n, \mathbb{Z})$. Then for any $\widetilde{\gamma}_* \in \Gamma_*^J$, the function $\Theta_{\mathscr{M}}$ defined by Formula (2.10.25) satisfies the functional

equation

$$\Theta_{\mathscr{M}}\big(\widetilde{\gamma}_*\cdot(\Omega,Z)\big)=\rho_{\mathscr{M}}(\widetilde{\gamma}_*)\,J^*_{\mathscr{M}}\big(\widetilde{\gamma}_*,(\Omega,Z)\big)\Theta_{\mathscr{M}}(\Omega,Z) \tag{2.10.26}$$

for all $(\Omega,Z)\in\mathbb{H}_{n,m}$, *where* $J^*_{\mathscr{M}}:G^J_*\times\mathbb{H}_{n,m}\longrightarrow\mathbb{C}^\times$ *is the automorphic factor for* G^J_* *on* $\mathbb{H}_{n,m}$ *defined by Formula* (2.10.15) *and* $\rho_{\mathscr{M}}$ *is a character of* Γ^J_* *determined uniquely by the formulas*

$$\rho_{\mathscr{M}}\big(((I_{2n},1),(\lambda,\mu,\kappa))\big)=e^{-\pi i\,\sigma(\mathscr{M}(\kappa+\mu\,{}^t\lambda))}\ \text{with}\ \lambda,\mu,\kappa\ \text{integral},$$

$$\rho_{\mathscr{M}}\big(((I_{2n},-1),(\lambda,\mu,\kappa))\big)=-e^{-\pi i\,\sigma(\mathscr{M}(\kappa+\mu\,{}^t\lambda))}\ \text{with}\ \lambda,\mu,\kappa\ \text{integral},$$

$$\rho_{\mathscr{M}}\big(((t(b),1),(0,0,0))\big)=1\ \text{with}\ b\in\mathrm{Sym}(n,\mathbb{Z})\ \textit{even diagonal},$$

$$\rho_{\mathscr{M}}\big(((t(b),-1),(0,0,0))\big)=1\ \textit{with}\ b\in\mathrm{Sym}(n,\mathbb{Z})\ \textit{even diagonal},$$

$$\rho_{\mathscr{M}}\big(((g(\alpha),\epsilon),(0,0,0))\big)=\epsilon^m(\det\alpha)^{\frac{m}{2}}\ \text{with}\ \alpha\in GL(n,\mathbb{Z}),\ \epsilon=\pm1,\ \pm i,$$

$$\rho_{\mathscr{M}}\big(((\sigma_n,\epsilon),(0,0,0))\big)=\epsilon^m(-i)^{\frac{mn}{2}}\ \text{with}\ \epsilon^2=(-i)^n.$$

Proof The proof follows from Theorem 2.10.3 and Lemma 2.10.6. We leave the details to the reader. □

2.11 Transformation Formula of Theta Series with Spherical Harmonics

Let S be a positive definite symmetric $m\times m$ rational matrix, and let α and β be two $m\times n$ rational matrices. We define the theta series

$$\vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}:\mathbb{H}_n\times\mathbb{C}^{(m,n)}\longrightarrow\mathbb{C}$$

by

$$\vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega,Z):=\sum_{N\in\mathbb{Q}^{(m,n)}}\chi\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(N)\,e^{\pi i\,\sigma({}^tNSN\Omega+2\,{}^tNZ)}, \tag{2.11.1}$$

where $\Omega\in\mathbb{H}_n$, $Z\in\mathbb{C}^{(m,n)}$ and

$$\chi\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(N)=\begin{cases}0 & \text{if } N-\alpha\notin\mathbb{Z}^{(m,n)}\\ e^{2\pi i\,\sigma({}^tN\beta)} & \text{otherwise.}\end{cases}$$

Let $\mathfrak{P}_{m,n}$ be the algebra of complex-valued polynomial functions on $\mathbb{C}^{(m,n)}$. We take a coordinate $Z = (z_{kj})$ in $\mathbb{C}^{(m,n)}$.

Definition 2.11.1 Let S, α and β be as above. For a homogeneous polynomial $P \in \mathfrak{P}_{m,n}$, we define

$$\vartheta_{S,P}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega, Z) = \sum_{N\in\mathbb{Q}^{(m,n)}} \chi\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(N)\, P(N)\, e^{\pi i\, \sigma({}^tNSN\Omega + 2\,{}^tNZ)}, \tag{2.11.2}$$

$$\vartheta_{S,P}(\Omega, Z) = \vartheta_{S,P}\begin{bmatrix}0\\ 0\end{bmatrix}(\Omega, Z), \tag{2.11.3}$$

$$\vartheta_{S,P}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega) = \vartheta_{S,P}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega, 0). \tag{2.11.4}$$

For any homogeneous polynomial P in $\mathfrak{P}_{m,n}$, we put

$$P(\partial) \;=\; P\left(\frac{\partial}{\partial z_{kj}}\right), \quad 1 \le k \le m,\ 1 \le j \le n.$$

Then we get

$$P(\partial)\vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega, Z) \tag{2.11.5}$$

$$= \sum_{N\in\mathbb{Q}^{(m,n)}} \chi\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(N)\, P(2\pi i N)\, e^{\pi i\, \sigma({}^tNSN\Omega + 2\,{}^tNZ)}.$$

Definition 2.11.2 Let $T = (t_{kj})$ be the inverse matrix of S. Then a polynomial P in $\mathfrak{P}_{m,n}$ is said to be pluriharmonic with respect to S if it satisfies the equations

$$\sum_{k,l=1}^{m} \frac{\partial^2 P}{\partial z_{ki}\,\partial z_{lj}}\, t_{kl} \;=\; 0 \qquad \text{for all } i,\ j = 1, 2, \ldots, n.$$

Theorem 2.11.3 *Let S, α and β be as above. Then for all $\begin{pmatrix}A & B\\ C & D\end{pmatrix}$ in a suitable congruence subgroup Γ of Γ_n, we have*

$$\vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\big((A\Omega + B)(C\Omega + D)^{-1}, Z(C\Omega + D)^{-1}\big) \tag{2.11.6}$$

$$= \det(C\Omega + D)^{\frac{m}{2}}\, e^{\pi i\, \sigma(Z(C\Omega+D)^{-1}C\,{}^tZS^{-1})}\, \vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega, Z).$$

See Sect. 2.4 for the definition of the function $\det(C\Omega + D)^{\frac{m}{2}}$.

Proof $Sp(2n,\mathbb{R})$ acts on the homogeneous space $\mathbb{H}_n \times \mathbb{C}^{(m,n)}$ by

$$M\cdot(\Omega, Z) = \big((A\Omega + B)(C\Omega + D)^{-1},\ Z(C\Omega + D)^{-1}\big),$$

where $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$, $\Omega \in \mathbb{H}_n$ and $Z \in \mathbb{C}^{(m,n)}$. It is known that $Sp(2n,\mathbb{R})$ is generated by the translations t_b with $b = {}^t b$ and the inversion σ_n. Thus it suffices to prove the functional equation (2.11.6) for the generators t_b and σ_n in a suitable congruence subgroup Γ of Γ_n.

For $t_b = \begin{pmatrix} I_n & b \\ 0 & I_n \end{pmatrix} \in Sp(2n,\mathbb{R})$,

$$\vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega + b, Z) = \sum_{N\in\mathbb{Q}^{(m,n)}} \chi\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(N)\, e^{\pi i\,\sigma({}^tNSN(\Omega+b)+2\,{}^tNZ)}$$

$$= \vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega, Z)$$

if we choose suitable b's so that $e^{\pi i\,\sigma({}^tNSNb)} = 1$. This is possible because S, α and β are *rational* matrices.

For the inversion $\sigma_n = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$, we can prove the functional equation (2.11.6) following the argument in the proof of Theorem 2.5.4. We leave the details to the reader. Another representation theoretic proof can be found in [34]. □

Lemma 2.11.4 *Let $\mathfrak{P}_N := \mathbb{C}[X_1, \ldots, X_N]$. For $P \in \mathfrak{P}_N$, we let $P(\partial)$ denote the differential operator $P\left(\frac{\partial}{\partial X_1}, \ldots, \frac{\partial}{\partial X_N}\right)$. For $P,\ Q \in \mathfrak{P}_N$, we define*

$$\langle P, Q\rangle = (P(\partial)Q)(0).$$

Then $\langle\ ,\ \rangle$ is a symmetric nondegenerate bilinear form on $\mathfrak{P}_N$ which satisfies the property $\langle P, QR\rangle = \langle Q(\partial)P, R\rangle = \langle R(\partial)P, Q\rangle$ for all $P, Q, R \in \mathfrak{P}_N$.

Proof We first observe that

$$\langle X_1^{a_1}\cdots X_N^{a_N}, X_1^{b_1}\cdots X_N^{b_N}\rangle = \begin{cases} a_1!\cdots a_N! & \text{if } (a_1,\ldots,a_N) = (b_1,\ldots,b_N),\\ 0 & \text{otherwise.}\end{cases}$$

Thus $\langle P, Q\rangle$ is a symmetric nondegenerate bilinear form on $\mathfrak{P}_N$. Similarly, $\langle P, QR\rangle = \langle Q(\partial)P, R\rangle = \langle R(\partial)P, Q\rangle$ is easily shown for monomials P, Q, R. Hence we complete the proof. □

Lemma 2.11.5 *Let $\mathfrak{H}(S) \subset \mathfrak{P}_{m,n}$ be the space of pluriharmonic polynomials with respect to S, and $I \subset \mathfrak{P}_{m,n}$ be the ideal generated by the $h_{ij} = \sum_{k,l=1}^{m} t_{kl} z_{ki} z_{lj}$ for*

all $i,\ j = 1, \ldots, n$, where $T = (t_{kl}) = S^{-1}$ as before in Definition 2.11.2. Then $\mathfrak{H}(S) = I^{\perp}$ with respect to the pairing $\langle\ ,\ \rangle$ introduced in Lemma 2.11.4, and

$$\mathfrak{P}_{m,n} = \mathfrak{H}(S) \oplus I \qquad (\textit{orthogonal sum}).$$

Proof Let $P \in \mathfrak{P}_{m,n}$. Then $\langle fh_{ij}, P\rangle = \big(f(\partial)h_{ij}(\partial)P\big)(0) = 0$ for all $f \in \mathfrak{P}_{m,n}$ if and only if $h_{ij}(\partial)P = 0$ for all $i,\ j$ if and only if P is pluriharmonic with respect to S. Thus $\mathfrak{H}(S) = I^{\perp}$. Let $\mathfrak{P}_{m,n}(\mathbb{R}) = \mathbb{R}[z_{11}, z_{12}, \ldots, z_{mn}]$. By the same argument, we have $\mathfrak{H}(S)_{\mathbb{R}} = I_{\mathbb{R}}^{\perp}$, where $\mathfrak{H}(S)_{\mathbb{R}} = \mathfrak{H}(S) \cap \mathfrak{P}_{m,n}(\mathbb{R})$ and $I_{\mathbb{R}} = I \cap \mathfrak{P}_{m,n}(\mathbb{R})$. It is easy to see that $\langle\ ,\ \rangle$ is positive definite on $\mathfrak{P}_{m,n}(\mathbb{R})$. So $\mathfrak{P}_{m,n}(\mathbb{R}) = \mathfrak{H}(S)_{\mathbb{R}} \oplus I_{\mathbb{R}}$. Therefore we have $\mathfrak{P}_{m,n} = \mathfrak{H}(S) \oplus I$. □

Lemma 2.11.6 *If P is a pluriharmonic polynomial in $\mathfrak{H}(S) \subset \mathfrak{P}_{m,n}$, then*

$$\Big(P(\partial)\Big[g(Z)\, e^{\sigma(ZC\,{}^tZS^{-1})}\Big]\Big)(0) = \big(P(\partial)g(Z)\big)(0) \tag{2.11.7}$$

for any $C \in \mathbb{C}^{(n,n)}$ and any analytic function g defined in a neighborhood of 0.

Proof We put $h(Z) = \sigma(ZC\,{}^tZS^{-1})$ and $T = (t_{kl}) = S^{-1}$. It suffices to prove Formula (2.11.7) for any polynomial $g(Z)$.

$$\begin{aligned}
\Big(P(\partial)\Big[g(Z)\, e^{h(Z)}\Big]\Big)(0) &= \sum_{n=0}^{\infty} \frac{1}{n!}\ \big(P(\partial)\big[g(Z)h(Z)^n\big]\big)(0) \\
&= \sum_{n=0}^{\infty} \frac{1}{n!}\ \langle P, gh^n\rangle \\
&= \sum_{n=0}^{\infty} \frac{1}{n!}\ \langle h(\partial)^n P, g\rangle \qquad \text{(by Lemma 2.11.4).}
\end{aligned}$$

By the way, $h(\partial)P = 0$ because P is pluriharmonic. Indeed, if we put $C = (c_{ij})$ and $Z = (z_{ki})$, then we have

$$\begin{aligned}
h(Z) &= \sigma(ZC\,{}^tZS^{-1}) \\
&= \sum_{i,j=1}^{n} c_{ij} \left(\sum_{k,l=1}^{m} t_{kl}\, z_{ki}\, z_{lj}\right).
\end{aligned}$$

We put $f_{ij}(Z) = \sum_{k,l=1}^{m} t_{kl}\, z_{ki}\, z_{lj}$. Then $h(\partial)P = \sum_{i,j=1}^{n} c_{ij}\big(f_{ij}(\partial)P\big) = 0$ because P is pluriharmonic. Therefore we get

$$\Big(P(\partial)\Big[g(Z)\, e^{h(Z)}\Big]\Big)(0) = \langle P, g\,\rangle = \big(P(\partial)g(Z)\big)(0).$$

□

Corollary 2.11.7 *If P is a pluriharmonic polynomial in $\mathfrak{H}(S) \subset \mathfrak{P}_{m,n}$ and C is an $n \times n$ symmetric complex matrix, then*

$$P(\partial)e^{\sigma(ZC\,{}^tZS^{-1})} = P(2\,C\,{}^tZS^{-1})\,e^{\sigma(ZC\,{}^tZS^{-1})}.$$

Proof We put $h(Z) = e^{\sigma(ZC\,{}^tZS^{-1})}$. For any $A \in \mathbb{C}^{(m,n)}$, we let

$$f(Z) = h(Z+A) = h(Z)\,h(A)\,g(Z),$$

where $g(Z) = e^{2\sigma(AC\,{}^tZS^{-1})}$. Then

$$\begin{aligned}\big(P(\partial)h(Z)\big)(A) &= \big(P(\partial)f(Z)\big)(0)\\ &= h(A)\,\big(P(\partial)[h(Z)\,g(Z)]\big)(0)\\ &= h(A)\,\big(P(\partial)g(Z)\big)(0) \qquad \text{(by Lemma 2.11.6).}\end{aligned}$$

But

$$\frac{\partial g}{\partial z_{ki}} = (2\,C\,{}^tA\,S^{-1})_{ki}\,g(Z).$$

By a repeated application of this, we have

$$P(\partial)g(Z) = P(2\,C\,{}^tA\,S^{-1})\,g(Z).$$

Therefore

$$\begin{aligned}\big(P(\partial)h(Z)\big)(A) &= h(A)\,\big(P(\partial)g(Z)\big)(0)\\ &= h(A)\,P(2\,C\,{}^tA\,S^{-1})\,g(0)\\ &= h(A)\,P(2\,C\,{}^tA\,S^{-1}).\end{aligned}$$

Hence $P(\partial)h(Z) = P(2\,C\,{}^tA\,S^{-1})\,h(Z)$. □

Lemma 2.11.8 *Let f be an analytic function on $\mathbb{C}^{(m,n)}$ and let $P \in \mathfrak{P}_{m,n}$. For $A \in \mathbb{C}^{(n,n)}$ and $B \in \mathbb{C}^{(m,m)}$, we let*

$$f_{A,B}(Z) = f(BZA) \qquad \text{and} \qquad P_{A,B}(Z) = P({}^tBZA^{-1}).$$

Then

$$P(\partial)f_{A,B}(Z) = \big(P_{A,B}(\partial)f\big)(BZA).$$

In particular, $\langle P, f_{A,B}\rangle = \langle P_{A,B}, f\rangle$.

Proof We let $A = (a_{ij}) \in \mathbb{C}^{(n,n)}$, $b = (b_{kl}) \in \mathbb{C}^{(m,m)}$ and $Z = (z_{lp})$. By an easy computation, we get

$$\frac{\partial f_{A,B}}{\partial z_{lp}}(Z) = \sum_{k=1}^{m}\sum_{i=1}^{n} b_{kl}\, a_{pi}\, \frac{\partial f}{\partial z_{ki}}(BZA), \quad 1 \le l \le m,\ 1 \le p \le n.$$

We put

$$\widetilde{Z} = {}^t BZA^{-1} \quad \text{with } \widetilde{Z} = \left(\widetilde{z}_{lp}\right).$$

Since

$$\frac{\partial}{\partial \widetilde{z}_{lp}} = \sum_{k=1}^{m}\sum_{i=1}^{n} b_{kl}\, a_{pi}\, \frac{\partial}{\partial z_{ki}}, \quad 1 \le l \le m,\ 1 \le p \le n,$$

we have

$$\frac{\partial f_{A,B}}{\partial z_{lp}}(Z) = \frac{\partial f}{\partial \widetilde{z}_{lp}}(BZA) \qquad \text{for all } l, p.$$

Therefore we have $P(\partial) f_{A,B}(Z) = \left(P_{A,B}(\partial) f\right)(BZA)$. □

Lemma 2.11.9 *$GL(n, \mathbb{C}) \times O(S)$ acts on $\mathfrak{P}_{m,n}$ by*

$$(A, B)P(Z) = P(B^{-1}ZA), \tag{2.11.8}$$

where $A \in GL(n, \mathbb{C})$, $B \in O(S)$ and $P \in \mathfrak{P}_{m,n}$. The space $\mathfrak{H}(S)$ of pluriharmonic polynomials in $\mathfrak{P}_{m,n}$ is invariant under the action (2.11.8).

Proof According to Lemma 2.11.5, $I = \mathfrak{H}(S)^{\perp}$ is the ideal of $\mathfrak{P}_{m,n}$ generated by $h_{ij}(Z) = \sum_{k,l=1}^{m} t_{kl}\, z_{ki}\, z_{lj}$ for all i, j. So by Lemma 2.11.8, it suffices to show that $h_{ij}(ZA)$ and $h_{ij}(BZ)$ belong to I for all $A \in GL(n, \mathbb{C})$ and $B \in O(S)$. If $A = (a_{ij}) \in GL(n, \mathbb{C})$, $Z = (z_{ki}) \in \mathbb{C}^{(m,n)}$ and $T = (t_{kl}) = S^{-1}$, then

$$\begin{aligned} h_{ij}(ZA) &= \sum_{p,q=1}^{n} a_{pi}\, a_{qj} \left(\sum_{k,l=1}^{m} t_{kl} z_{kp} z_{lq} \right) \\ &= \sum_{p,q=1}^{n} a_{pi}\, a_{qj}\, h_{pq}(Z) \in I. \end{aligned}$$

If $B = (b_{kl}) \in O(S)$, then

$$h_{ij}(BZ) = \sum_{p,q=1}^{n} z_{pi}\, z_{qj} \left(\sum_{k,l=1}^{m} t_{kl} b_{kp} b_{lq} \right)$$
$$= \sum_{p,q=1}^{n} z_{pi}\, z_{qj} \left({}^t BTB \right)_{pq}.$$

Since $B \in O(S)$, we have $T = {}^t BTB$. Indeed $BS\,{}^t B = S$ and hence ${}^t B\, S^{-1}\, B^{-1} = S^{-1}$. Thus ${}^t BTB = T$. Hence we have

$$h_{ij}(BZ) = \sum_{p,q=1}^{n} t_{pq}\, z_{pi}\, z_{qj} = h_{ij}(Z) \in I.$$

Therefore we complete the proof. □

Theorem 2.11.10 *Let S, α and β be as above. Let P be a pluriharmonic polynomial in $\mathfrak{P}_{m,n}$ with respect to S. Then*

$$\vartheta_{S,P}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega) = \det(C\Omega+D)^{-\frac{m}{2}}\, \vartheta_{S,\widetilde{P}}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\big((A\Omega+B)(C\Omega+D)^{-1}\big), \qquad (2.11.9)$$

where $\widetilde{P}(Z) = P(Z(C\Omega+D))$, for all $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ in a suitable congruence subgroup Γ of Γ_n.

Proof Let P be a homogeneous pluriharmonic polynomial of degree k. Then according to Formula (2.11.5), we get

$$(2\pi i)^{-k} P(\partial)\vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega, Z)$$
$$= (2\pi i)^{-k} \sum_{N \in \mathbb{Q}^{(m,n)}} \chi\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(N)\, P(2\pi i N)\, e^{\pi i\, \sigma({}^t NSN\Omega + 2\,{}^t NZ)}$$
$$= \sum_{N \in \mathbb{Q}^{(m,n)}} \chi\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(N)\, P(N)\, e^{\pi i\, \sigma({}^t NSN\Omega + 2\,{}^t NZ)}$$
$$= \vartheta_{S,P}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega, Z).$$

Here the fact that P is homogeneous of degree k was used in the second equality. Putting $Z = 0$, we get

$$(2\pi i)^{-k}\left(P(\partial)\vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\right)(\Omega, 0) = \vartheta_{S,P}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega). \tag{2.11.10}$$

By Theorem 2.11.3,

$$\begin{aligned}&\vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega, Z)\\ &= \det(C\Omega + D)^{-\frac{m}{2}} e^{-\pi i\,\sigma(Z(C\Omega+D)^{-1}C\,{}^tZS^{-1})}\\ &= \vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\big((A\Omega + B)(C\Omega + D)^{-1}, Z(C\Omega + D)^{-1}\big).\end{aligned} \tag{2.11.11}$$

If we apply the differential operator $(2\pi i)^{-k}P(\partial)$ to both sides of Formula (2.11.11) and put $Z = 0$, according to Formula (2.11.10), Lemma 2.11.6 and Lemma 2.11.8, we obtain

$$\begin{aligned}&\vartheta_{S,P}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega)\\ &= (2\pi i)^{-k}\det(C\Omega + D)^{-\frac{m}{2}}\\ &\quad\times\left[P(\partial)\vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\big((A\Omega + B)(C\Omega + D)^{-1}, Z(C\Omega + D)^{-1}\big)\right]_{Z=0}\\ &= (2\pi i)^{-k}\det(C\Omega + D)^{-\frac{m}{2}}\left[\widetilde{P}(\partial)\vartheta_S\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\big((A\Omega + B)(C\Omega + D)^{-1}, Z\big)\right]_{Z=0}\\ &= \det(C\Omega + D)^{-\frac{m}{2}}\vartheta_{S,\widetilde{P}}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\big((A\Omega + B)(C\Omega + D)^{-1}, 0\big)\\ &= \det(C\Omega + D)^{-\frac{m}{2}}\vartheta_{S,\widetilde{P}}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\big((A\Omega + B)(C\Omega + D)^{-1}\big),\end{aligned}$$

where $\widetilde{P}(Z) = P(Z(C\Omega + D))$. We note that we used Formula (2.11.6) and Lemma 2.11.6 in the first equality, and Lemma 2.11.8 in the second equality. In the third equality we used the fact that $\widetilde{P}$ is homogeneous of degree k. Consequently, we complete the proof. □

Definition 2.11.11 Let (ρ, V_ρ) be a finite-dimensional rational representation of $GL(n,\mathbb{C})$. A vector-valued function $f : \mathbb{H}_n \longrightarrow V_\rho$ is called a modular form with respect to ρ if it is a holomorphic function on $\mathbb{H}_n$ such that

$$f((A\Omega + B)(C\Omega + D)^{-1}) = \rho(C\Omega + D) f(\Omega), \quad \Omega \in \mathbb{H}_n$$

for all $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ in a suitable congruence subgroup of Γ_n.

We recall that $\mathfrak{H}(S)$ denotes the space of all pluriharmonic polynomials in $\mathfrak{P}_{m,n}$ with respect to S. Let W be some $GL(n,\mathbb{C})$-stable subspace of $\mathfrak{H}(S)$. We define the W^*-valued function $\vartheta_W \begin{bmatrix} \alpha \\ \beta \end{bmatrix} : \mathbb{H}_n \longrightarrow W^*$ by

$$\left(\vartheta_W \begin{bmatrix} \alpha \\ \beta \end{bmatrix}(\Omega)\right)(P) := \vartheta_{S,P} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}(\Omega) \tag{2.11.12}$$

for all $\Omega \in \mathbb{H}_n$ and $P \in W \subset \mathfrak{H}(S)$. Here W^* denotes the dual space of W.

Now we introduce the homogeneous line bundle $\mathfrak{M}^{\frac{1}{2}}$ over $\mathbb{H}_n$. First of all we consider the double covering $\widetilde{GL(n,\mathbb{C})}$ of $GL(n,\mathbb{C})$ defined by

$$\widetilde{GL(n,\mathbb{C})} = \left\{(g,\alpha) \mid \alpha^2 = \det(g),\ g \in GL(n,\mathbb{C}),\ \alpha \in \mathbb{C}^\times\right\}$$

equipped with the multiplication

$$(g_1,\alpha_1)(g_2,\alpha_2) = (g_1 g_2, \alpha_1\alpha_2), \quad g_1, g_2 \in GL(n,\mathbb{C}),\ \alpha_1,\alpha_2 \in \mathbb{C}^\times.$$

Let ρ be a one-dimensional representation of $GL(n,\mathbb{C})$ defined by

$$\rho(g,\alpha) = \alpha = (\det(g))^{\frac{1}{2}}, \quad g \in GL(n,\mathbb{C}),\ \alpha \in \mathbb{C}^\times.$$

Then ρ yields the homogeneous line bundle on $\mathbb{H}_n$, denoted by $\mathfrak{M}^{\frac{1}{2}}$. The complex manifold

$$\mathfrak{M}^{\frac{1}{2}} = \mathbb{H}_n \times \mathbb{C}$$

is a holomorphic line bundle over $\mathbb{H}_n$ with the action of the metaplectic group $Mp(2n,\mathbb{R})$ given by

$$\widetilde{M} \cdot (\Omega, z) = \left((A\Omega + B)(C\Omega + D)^{-1},\ \det(C\Omega + D)^{1/2}\, z\right),$$

where $\widetilde{M} \in Mp(2n,\mathbb{R})$ and $\begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp(2n,\mathbb{R})$ is the image of $\widetilde{M}$ under the surjective homomorphism of $Mp(2n,\mathbb{R})$ onto $Sp(2n,\mathbb{R})$. See Sect. 2.4 for the

definition of the function $\det(C\Omega + D)^{\frac{m}{2}}$. For a positive integer k, we define

$$\mathfrak{M}^{\frac{k}{2}} = \left(\mathfrak{M}^{\frac{1}{2}}\right)^{\otimes k} = \mathfrak{M}^{\frac{1}{2}} \otimes \cdots \otimes \mathfrak{M}^{\frac{1}{2}} \quad (k-\text{times}).$$

Let τ be the representation of $GL(n, \mathbb{C})$ on W defined by

$$\big(\tau(g)P\big)(Z) := P(Zg), \quad g \in GL(n, \mathbb{C}),\ P \in W,\ Z \in \mathbb{C}^{(m,n)}.$$

We observe that if $\widetilde{P}$ is a homogeneous pluriharmonic polynomial given by Theorem 2.11.10, then $\widetilde{P} = \tau(C\Omega + D)P$. Let τ^* be the contragredient of τ. That is,

$$\big(\tau^*(g)\ell\big)(P) = \ell\big(\tau(g)^{-1}P\big), \quad g \in GL(n, \mathbb{C}),\ \ell \in W^*,\ P \in W.$$

Theorem 2.11.12 *Let α and β be as above. Then the function $\vartheta_W\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega)$ defined in (2.11.12) is a modular form with values in $W^* \otimes \mathfrak{M}^{\frac{m}{2}}$ with respect to the representation $\tau^* \otimes \det^{\frac{m}{2}}$ for a suitable congruence subgroup Γ. For any W and Ω, it is non-zero for suitable α and β. Here the function $\chi : GL(n, \mathbb{R}) \longrightarrow \mathbb{C}^\times$ given by*

$$\chi(a) = \det{}^{\frac{1}{2}}(a), \qquad a \in GL(n, \mathbb{R})$$

is a one-dimensional representation of $GL(n, \mathbb{R})$ satisfying the following properties (i) and (ii):

(i) $\chi(a)^2 = \det a$ *for all* $a \in GL(n, \mathbb{R})$;
(ii) $\chi(a) = (\sqrt{\det a})$ *for all* $a \in GL(n, \mathbb{R})$ *with* $\det a > 0$.

For a positive integer m, we define

$$\det{}^{\frac{m}{2}} \overset{\text{def}}{=} \left(\det{}^{\frac{1}{2}}\right)^m.$$

Proof By Theorem 2.11.10, for all $P \in W \subset \mathfrak{H}(S)$ and for all $\begin{pmatrix}A & B\\ C & D\end{pmatrix}$ in a suitable congruence subgroup Γ of Γ_n, we get

$$\left(\vartheta_W\begin{bmatrix}\alpha\\ \beta\end{bmatrix}(\Omega)\right)(P)$$

$$= \det(C\Omega + D)^{-\frac{m}{2}} \cdot \vartheta_{S,\widetilde{P}}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}\big((A\Omega + B)(C\Omega + D)^{-1}\big)$$

$$= \det(C\Omega + D)^{-\frac{m}{2}} \cdot \left(\vartheta_W \begin{bmatrix} \alpha \\ \beta \end{bmatrix}\left((A\Omega + B)(C\Omega + D)^{-1}\right)\right)(\widetilde{P})$$

$$= \det(C\Omega + D)^{-\frac{m}{2}} \cdot \left(\vartheta_W \begin{bmatrix} \alpha \\ \beta \end{bmatrix}\left((A\Omega + B)(C\Omega + D)^{-1}\right)\right)(\tau(C\Omega + D)P)$$

$$= \det(C\Omega + D)^{-\frac{m}{2}} \cdot \left(\tau^*(C\Omega + D)^{-1}\vartheta_W \begin{bmatrix} \alpha \\ \beta \end{bmatrix}\left((A\Omega + B)(C\Omega + D)^{-1}\right)\right)(P),$$

where $\widetilde{P}$ is a homogeneous pluriharmonic polynomial defined by $\widetilde{P}(Z) = P(Z(C\Omega + D))$. Therefore

$$\vartheta_W \begin{bmatrix} \alpha \\ \beta \end{bmatrix}\left((A\Omega + B)(C\Omega + D)^{-1}\right) = \det(C\Omega + D)^{\frac{m}{2}} \cdot \tau^*(C\Omega + D)\vartheta_W \begin{bmatrix} \alpha \\ \beta \end{bmatrix}(\Omega).$$

Hence $\vartheta_W \begin{bmatrix} \alpha \\ \beta \end{bmatrix}(\Omega)$ is a modular form on $\mathbb{H}_n$ with values in $W^* \otimes \mathfrak{M}^{\frac{m}{2}}$ with respect to a suitable congruence subgroup Γ of Γ_n. □

Remark 2.11.13 Using Theorem 2.11.12, we can prove that for all $n \geq 2$ and $1 \leq r \leq n-1$, there are congruence subgroups $\Gamma \subset \Gamma_n$ and Γ-invariant non-vanishing holomorphic k-forms on $\mathbb{H}_n$, where $k = \frac{n(n+1)}{2} - \frac{r(r+1)}{2}$. The proof can be found in [34]. This fact was proved by Freitag and Stillman.

Definition 2.11.14 Let (ρ, V_ρ) be a finite-dimensional rational representation of $GL(n, \mathbb{C})$. A pluriharmonic form with respect to ρ is a polynomial P from $\mathbb{C}^{(m,n)}$ to V_ρ if it satisfies the following conditions:

$$\sum_{k=1}^{m} \frac{\partial^2 P}{\partial z_{ki}\partial z_{kj}} = 0 \qquad \text{for all } i, j = 1, 2, \ldots, n$$

and

$$P(ZA) = \rho({}^tA)P(Z) \qquad \text{for all } A \in GL(n, \mathbb{C}).$$

We denote by $\mathfrak{H}_{m,n}(\rho)$ the space of all pluriharmonic forms with respect to ρ.

Freitag proved the following.

Theorem 2.11.15 *Let S be a positive definite even unimodular matrix of degree m and let (ρ, V_ρ) be a finite-dimensional rational representation of $GL(n, \mathbb{C})$. Let $P \in \mathfrak{H}_{m,n}(\rho)$ be a pluriharmonic form with respect to ρ. Then the theta series*

$$\Theta_{S,P}(\Omega) := \sum_{N \in \mathbb{Z}^{(m,n)}} P(S^{1/2}N)\, e^{\pi i \sigma({}^tNSN\Omega)}$$

is a modular form with respect to the representation ρ_* of $GL(n,\mathbb{C})$ defined by

$$\rho_*(A) \;=\; \rho(A)\,(\det A)^{\frac{m}{2}}, \qquad A \in GL(n,\mathbb{C})$$

for the Siegel modular group Γ_n.

Proof We will omit the proof. The proof can be found in [10]. □

2.12 Theta Sums of Higher Index

Let $\mathscr{M}$ be a positive definite symmetric real matrix of degree m. We recall the Schrödinger representation $\mathscr{W}_{\mathscr{M}}$ of the Heisenberg group $H_{\mathbb{R}}^{(n,m)}$ associated with $\mathscr{M}$ which is given by Formula (2.10.1) in Sect. 2.10. We note that for an element (λ,μ,κ) of $H_{\mathbb{R}}^{(n,m)}$, we have the decomposition

$$(\lambda,\mu,\kappa) = (\lambda,0,0)\circ(0,\mu,0)\circ(0,0,\kappa-\lambda\,{}^t\mu).$$

We consider the embedding $\Phi_n : SL(2,\mathbb{R}) \longrightarrow Sp(2n,\mathbb{R})$ defined by

$$\Phi_n\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) := \begin{pmatrix} aI_n & bI_n \\ cI_n & dI_n \end{pmatrix}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{R}). \tag{2.12.1}$$

For $x, y \in \mathbb{R}^{(m,n)}$, we put

$$(x,y)_{\mathscr{M}} := \sigma({}^t x\mathscr{M} y) \qquad \text{and} \qquad \|x\|_{\mathscr{M}} := \sqrt{(x,x)_{\mathscr{M}}}.$$

According to Formulas (2.10.10), (2.10.11) and (2.10.12), for any $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{R}) \hookrightarrow Sp(2n,\mathbb{R})$ and $f \in L^2\left(\mathbb{R}^{(m,n)}\right)$, we have the following explicit representation:

$$[R_{\mathscr{M}}(M)f](x) \tag{2.12.2}$$
$$= \begin{cases} |a|^{\frac{mn}{2}} e^{ab\|x\|^2_{\mathscr{M}}\pi i} f(ax) & \text{if } c = 0, \\ (\det\mathscr{M})^{\frac{n}{2}}\,|c|^{-\frac{mn}{2}} \int_{\mathbb{R}^{(m,n)}} e^{\frac{\alpha(M,x,y,\mathscr{M})}{c}\pi i} f(y)dy & \text{if } c \neq 0, \end{cases}$$

where

$$\alpha(M,x,y,\mathscr{M}) = a\,\|x\|^2_{\mathscr{M}} + d\,\|y\|^2_{\mathscr{M}} - 2(x,y)_{\mathscr{M}}.$$

Indeed, if $a = 0$ and $c \neq 0$, using the decomposition

$$M = \begin{pmatrix} 0 & -c^{-1} \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} c & d \\ 0 & c^{-1} \end{pmatrix}$$

and if $a \neq 0$ and $c \neq 0$, using the decomposition

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & c^{-1} \\ 0 & a^{-1} \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} ac & ad \\ 0 & (ac)^{-1} \end{pmatrix},$$

we obtain Formula (2.12.2).

If

$$M_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}, \quad M_2 = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \text{ and } M_3 = \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix} \in SL(2, \mathbb{R})$$

with $M_3 = M_1 M_2$, the corresponding cocycle is given by

$$c_{\mathscr{M}}(M_1, M_2) = e^{-i\,\pi\, mn\, \mathrm{sign}(c_1 c_2 c_3)/4}, \tag{2.12.3}$$

where

$$\mathrm{sign}(x) = \begin{cases} -1 & (x < 0) \\ 0 & (x = 0) \\ 1 & (x > 0). \end{cases}$$

In the special case when

$$M_1 = \begin{pmatrix} \cos\phi_1 & -\sin\phi_1 \\ \sin\phi_1 & \cos\phi_1 \end{pmatrix} \quad \text{and} \quad M_2 = \begin{pmatrix} \cos\phi_2 & -\sin\phi_2 \\ \sin\phi_2 & \cos\phi_2 \end{pmatrix},$$

we find

$$c_{\mathscr{M}}(M_1, M_2) = e^{-i\,\pi\, mn\, (\sigma_{\phi_1} + \sigma_{\phi_2} - \sigma_{\phi_1 + \phi_2})/4},$$

where

$$\sigma_\phi = \begin{cases} 2\nu & \text{if } \phi = \nu\pi \\ 2\nu + 1 & \text{if } \nu\pi < \phi < (\nu + 1)\pi. \end{cases}$$

It is well known that every $M \in SL(2,\mathbb{R})$ admits the unique Iwasawa decomposition

$$M = \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v^{1/2} & 0 \\ 0 & v^{-1/2} \end{pmatrix} \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}, \tag{2.12.4}$$

where $\tau = u + iv \in \mathbb{H}_1$ and $\phi \in [0, 2\pi)$. This parametrization $M = (\tau, \phi)$ in $SL(2,\mathbb{R})$ leads to the natural action of $SL(2,\mathbb{R})$ on $\mathbb{H}_1 \times [0, 2\pi)$ defined by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} (\tau, \phi) := \left(\frac{a\tau + b}{c\tau + d},\ \phi + \arg(c\tau + d) \bmod 2\pi \right). \tag{2.12.5}$$

Lemma 2.12.1 *For two elements g_1 and g_2 in $SL(2,\mathbb{R})$, we let*

$$g_1 = \begin{pmatrix} 1 & u_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v_1^{1/2} & 0 \\ 0 & v_1^{-1/2} \end{pmatrix} \begin{pmatrix} \cos\phi_1 & -\sin\phi_1 \\ \sin\phi_1 & \cos\phi_1 \end{pmatrix}$$

and

$$g_2 = \begin{pmatrix} 1 & u_2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v_2^{1/2} & 0 \\ 0 & v_2^{-1/2} \end{pmatrix} \begin{pmatrix} \cos\phi_2 & -\sin\phi_2 \\ \sin\phi_2 & \cos\phi_2 \end{pmatrix}$$

be the Iwasawa decompositions of g_1 and g_2 respectively, where $u_1, u_2 \in \mathbb{R},\ v_1 > 0,\ v_2 > 0$ and $0 \le \phi_1, \phi_2 < 2\pi$. Let

$$g_3 = g_1 g_2 = \begin{pmatrix} 1 & u_3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v_3^{1/2} & 0 \\ 0 & v_3^{-1/2} \end{pmatrix} \begin{pmatrix} \cos\phi_3 & -\sin\phi_3 \\ \sin\phi_3 & \cos\phi_3 \end{pmatrix}$$

be the Iwasawa decomposition of $g_3 = g_1 g_2$. Then we have

$$u_3 = \frac{A}{(u_2 \sin\phi_1 + \cos\phi_1)^2 + (v_2 \sin\phi_1)^2},$$
$$v_3 = \frac{v_1 v_2}{(u_2 \sin\phi_1 + \cos\phi_1)^2 + (v_2 \sin\phi_1)^2}$$

and

$$\phi_3 = \tan^{-1}\left[\frac{(v_2 \cos\phi_2 + u_2 \sin\phi_2)\tan\phi_1 + \sin\phi_2}{(-v_2 \sin\phi_2 + u_2 \cos\phi_2)\tan\phi_1 + \cos\phi_2} \right],$$

where

$$A = u_1(u_2 \sin\phi_1 + \cos\phi_1)^2 + (u_1 v_2 - v_1 u_2)\sin^2\phi_1$$

$$+ v_1 u_2 \cos^2\phi_1 + v_1(u_2^2 + v_2^2 - 1)\sin\phi_1 \cos\phi_1.$$

Proof If $g \in SL(2,\mathbb{R})$ has the unique Iwasawa decomposition (2.12.4), then we get the following:

$$\begin{aligned}
a &= v^{1/2}\cos\phi + uv^{-1/2}\sin\phi,\\
b &= -v^{1/2}\sin\phi + uv^{-1/2}\cos\phi,\\
c &= v^{-1/2}\sin\phi, \quad d = v^{-1/2}\cos\phi,\\
u &= (ac+bd)\left(c^2+d^2\right)^{-1}, \quad v = \left(c^2+d^2\right)^{-1}, \quad \tan\phi = \frac{c}{d}.
\end{aligned}$$

We set

$$g_3 = g_1 g_2 = \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix}.$$

Since

$$u_3 = (a_3c_3 + b_3d_3)\left(c_3^2 + d_3^2\right)^{-1}, \quad v_3 = \left(c_3^2 + d_3^2\right)^{-1}, \quad \tan\phi_3 = \frac{c_3}{d_3},$$

by an easy computation, we obtain the desired results. □

Now we use the new coordinates ($\tau = u + iv, \phi$) with $\tau \in \mathbb{H}_1$ and $\phi \in [0, 2\pi)$ in $SL(2,\mathbb{R})$. According to Formulas (2.10.10), (2.10.11) and (2.10.12), the projective representation $R_{\mathscr{M}}$ of $SL(2,\mathbb{R}) \hookrightarrow Sp(2n,\mathbb{R})$ reads in these coordinates ($\tau = u + iv, \phi$) as follows:

$$[R_{\mathscr{M}}(\tau,\phi)f](x) = v^{\frac{mn}{4}} e^{u\|x\|^2_{\mathscr{M}}\pi i}\,[R_{\mathscr{M}}(i,\phi)f]\left(v^{1/2}x\right), \tag{2.12.6}$$

where $f \in L^2\left(\mathbb{R}^{(m,n)}\right)$, $x \in \mathbb{R}^{(m,n)}$ and

$$[R_{\mathscr{M}}(i,\phi)f](x) \tag{2.12.7}$$

$$= \begin{cases} f(x) & \text{if } \phi \equiv 0 \bmod 2\pi,\\ f(-x) & \text{if } \phi \equiv \pi \bmod 2\pi,\\ (\det\mathscr{M})^{\frac{n}{2}}\,|\sin\phi|^{-\frac{mn}{2}} \int_{\mathbb{R}^{(m,n)}} e^{B(x,y,\phi,\mathscr{M})\pi i} f(y)dy & \text{if } \phi \not\equiv 0 \bmod \pi. \end{cases}$$

Here

$$B(x,y,\phi,\mathscr{M}) = \frac{\left(\|x\|^2_{\mathscr{M}} + \|y\|^2_{\mathscr{M}}\right)\cos\phi - 2(x,y)_{\mathscr{M}}}{\sin\phi}.$$

Now we set

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

We note that

$$\left[R_{\mathscr{M}}\left(i, \frac{\pi}{2}\right) f\right](x) = [R_{\mathscr{M}}(S) f](x) \tag{2.12.8}$$

$$= (\det \mathscr{M})^{\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} f(y)\, e^{-2\,(x,\,y)_{\mathscr{M}}\,\pi\, i}\, dy$$

for $f \in L^2\left(\mathbb{R}^{(m,n)}\right)$.

Remark 2.12.2 For Schwartz functions $f \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$, we have

$$\lim_{\phi \longrightarrow 0\pm} |\sin\phi|^{-\frac{mn}{2}} \int_{\mathbb{R}^{(m,n)}} e^{B(x,y,\phi,\mathscr{M})\,\pi\, i}\, f(y) dy = e^{\pm i\,\pi\, mn/4} f(x) \neq f(x).$$

Therefore the projective representation $R_{\mathscr{M}}$ is not continuous at $\phi = \nu\pi$ ($\nu \in \mathbb{Z}$) in general. If we set

$$\tilde{R}_{\mathscr{M}}(\tau, \phi) = e^{-i\,\pi\, mn\sigma_\phi/4} R_{\mathscr{M}}(\tau, \phi),$$

$\tilde{R}_{\mathscr{M}}$ corresponds to a unitary representation of the double cover of $SL(2, \mathbb{R})$ (cf. Formula (2.10.5) and [26]). We note that

$$\tilde{R}_{\mathscr{M}}(i, \phi)\tilde{R}_{\mathscr{M}}(i, \phi') = \tilde{R}_{\mathscr{M}}(i, \phi + \phi'), \qquad \phi, \phi' \in [0, 4\pi).$$

We observe that for any element $(g, (\lambda, \mu, \kappa)) \in G^J$ with $g \in Sp(2n, \mathbb{R})$ and $(\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$, we have the following decomposition:

$$(g, (\lambda, \mu, \kappa)) = (I_{2n}, ((\lambda, \mu)g^{-1}, \kappa))\,(g, (0, 0, 0)) = ((\lambda, \mu)g^{-1}, \kappa) \cdot g.$$

Thus $Sp(2n, \mathbb{R})$ acts on $H_{\mathbb{R}}^{(n,m)}$ naturally by

$$g \cdot (\lambda, \mu, \kappa) = \left((\lambda, \mu)g^{-1}, \kappa\right), \quad g \in Sp(2n, \mathbb{R}),\ (\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}.$$

Definition 2.12.3 For any Schwartz function $f \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$, we define the function $\Theta_f^{[\mathscr{M}]}$ on the Jacobi group $SL(2, \mathbb{R}) \ltimes H_{\mathbb{R}}^{(n,m)} \hookrightarrow G^J$ by

$$\Theta_f^{[\mathscr{M}]}(\tau, \phi\,; \lambda, \mu, \kappa) := \sum_{\omega \in \mathbb{Z}^{(m,n)}} [\pi_{\mathscr{M}}((\lambda, \mu, \kappa)(\tau, \phi))\, f](\omega), \tag{2.12.9}$$

where $(\tau, \phi) \in SL(2, \mathbb{R})$ and $(\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$. The function $\Theta_f^{[\mathscr{M}]}$ is called the theta sum of index $\mathscr{M}$ associated with a Schwartz function f. The projective representation $\pi_{\mathscr{M}}$ of the Jacobi group G^J was already defined by Formula (2.10.7). More precisely, for $\tau = u + iv \in \mathbb{H}_1$ and $(\lambda, \mu, \kappa) \in H_{\mathbb{R}}^{(n,m)}$, we have

$$\Theta_f^{[\mathscr{M}]}(\tau, \phi\,; \lambda, \mu, \kappa) = v^{\frac{mn}{4}}\ e^{2\pi i\, \sigma(\mathscr{M}(\kappa + \mu^t \lambda))}$$
$$\times \sum_{\omega \in \mathbb{Z}^{(m,n)}} e^{\pi i \left\{u[\omega+\lambda]_{\mathscr{M}}^2 + 2(\omega, \mu)_{\mathscr{M}}\right\}}\ [R_{\mathscr{M}}(i, \phi) f]\left(v^{1/2}(\omega + \lambda)\right).$$

Lemma 2.12.4 *We set $f_\phi := \tilde{R}_{\mathscr{M}}(i, \phi) f$ for $f \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$. Then for any $R > 1$, there exists a constant C_R such that for all $x \in \mathbb{R}^{(m,n)}$ and $\phi \in \mathbb{R}$,*

$$|f_\phi(x)| \le C_R\ (1 + [x]_{\mathscr{M}})^{-R}.$$

Proof Following the arguments in the proof of Lemma 4.3 in [30, pp. 428–429], we get the desired result. □

Theorem 2.12.5 (Jacobi 1) *Let $\mathscr{M}$ be a positive definite symmetric integral matrix of degree m such that $\mathscr{M}\mathbb{Z}^{(m,n)} = \mathbb{Z}^{(m,n)}$. Then for any Schwartz function $f \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$, we have*

$$\Theta_f^{[\mathscr{M}]}\left(-\frac{1}{\tau},\ \phi + \arg\tau\,; -\mu, \lambda, \kappa\right)$$
$$= \left(\det \mathscr{M}\right)^{-\frac{n}{2}} c_{\mathscr{M}}(S, (\tau, \phi))\ \Theta_f^{[\mathscr{M}]}(\tau, \phi\,; \lambda, \mu, \kappa),$$

where

$$c_{\mathscr{M}}(S, (\tau, \phi)) := e^{i\,\pi mn\, \mathrm{sign}(\sin\phi\ \sin(\phi + \arg\tau))}.$$

Proof First we recall that for any Schwartz function $\varphi \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$, the Fourier transform $\mathscr{F}\varphi$ of φ is given by

$$\left(\mathscr{F}\varphi\right)(x) = \int_{\mathbb{R}^{(m,n)}} \varphi(y)\, e^{-2\pi i\, \sigma(y\,{}^t x)} dy.$$

Now we put

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in SL(2, \mathbb{Z}) \hookrightarrow Sp(2n, \mathbb{R})$$

and for any $F \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$, we put

$$F_{\mathscr{M}}(x) := F(\mathscr{M}^{-1}x), \quad x \in \mathbb{R}^{(m,n)}.$$

According to Formula (2.10.12), for any $F \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$,

$$\begin{aligned}
[R_{\mathscr{M}}(S)F](x) &= \left(\det \mathscr{M}\right)^{\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} F(y)\, e^{-2\pi i\, \sigma(\mathscr{M} y\, {}^t x)} dy \\
&= \left(\det \mathscr{M}\right)^{-\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} F(\mathscr{M}^{-1} y)\, e^{-2\pi i\, \sigma(y\, {}^t x)} dy \\
&= \left(\det \mathscr{M}\right)^{-\frac{n}{2}} \int_{\mathbb{R}^{(m,n)}} F_{\mathscr{M}}(y)\, e^{-2\pi i\, \sigma(y\, {}^t x)} dy \\
&= \left(\det \mathscr{M}\right)^{-\frac{n}{2}} [\mathscr{F} F_{\mathscr{M}}](x).
\end{aligned}$$

Thus we have

$$\mathscr{F} F_{\mathscr{M}} = \left(\det \mathscr{M}\right)^{\frac{n}{2}} R_{\mathscr{M}}(S)F \quad \text{for } F \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right). \tag{2.12.10}$$

By Lemma 2.12.1, we get easily

$$S\cdot(\tau,\phi) = \left(-\frac{1}{\tau}, \phi + \arg\tau\right). \tag{2.12.11}$$

If we take $F = \pi_{\mathscr{M}}((\lambda,\mu,\kappa)(\tau,\phi))f$ for $f \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$, a fixed element $(\lambda,\mu,\kappa) \in H_{\mathbb{R}}^{(n,m)}$ and an fixed element $(\tau,\phi) \in SL(2,\mathbb{R})$, then it is easily seen that $F \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$.

According to Formula (2.12.11), if we take $F = \pi_{\mathscr{M}}((\lambda,\mu,\kappa)(\tau,\phi))f$ for $f \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$,

$$\begin{aligned}
\left[R_{\mathscr{M}}(S)F\right](x) &= \left[R_{\mathscr{M}}(S)\pi_{\mathscr{M}}\left((\lambda,\mu,\kappa)(\tau,\phi)\right)f\right](x), \quad x \in \mathbb{R}^{(m,n)} \\
&= [R_{\mathscr{M}}(S)\mathscr{W}_{\mathscr{M}}(\lambda,\mu,\kappa)R_{\mathscr{M}}(\tau,\phi)f](x) \\
&= \left[\mathscr{W}_{\mathscr{M}}\left((\lambda,\mu)S^{-1},\kappa\right)R_{\mathscr{M}}(S)R_{\mathscr{M}}(\tau,\phi)f\right](x) \\
&= c_{\mathscr{M}}(S,(\tau,\phi))^{-1}\left[\mathscr{W}_{\mathscr{M}}(-\mu,\lambda,\kappa)R_{\mathscr{M}}\left(S\cdot(\tau,\phi)\right)f\right](x) \\
&= c_{\mathscr{M}}(S,(\tau,\phi))^{-1}\left[\mathscr{W}_{\mathscr{M}}(-\mu,\lambda,\kappa)R_{\mathscr{M}}\left(-\frac{1}{\tau},\phi+\arg\tau\right)f\right](x) \\
&= c_{\mathscr{M}}(S,(\tau,\phi))^{-1}\left[\pi_{\mathscr{M}}\left((-\mu,\lambda,\kappa)\left(-\frac{1}{\tau},\phi+\arg\tau\right)\right)f\right](x).
\end{aligned}$$

Thus we obtain

$$\begin{aligned}&\left[R_{\mathscr{M}}(S)F\right](x) \qquad (2.12.12)\\ &= c_{\mathscr{M}}(S,(\tau,\phi))^{-1}\left[\pi_{\mathscr{M}}\left((-\mu,\lambda,\kappa)\left(-\frac{1}{\tau},\phi+\arg\tau\right)\right)f\right](x).\end{aligned}$$

According to Poisson summation formula, we have

$$\sum_{\omega\in\mathbb{Z}^{(m,n)}}\left[\mathscr{F}F_{\mathscr{M}}\right](\omega)=\sum_{\omega\in\mathbb{Z}^{(m,n)}}F_{\mathscr{M}}(\omega). \qquad (2.12.13)$$

It follows from (2.12.10) and (2.12.12) that

$$\begin{aligned}\sum_{\omega\in\mathbb{Z}^{(m,n)}}\left[\mathscr{F}F_{\mathscr{M}}\right](\omega)&=\left(\det\mathscr{M}\right)^{\frac{n}{2}}\sum_{\omega\in\mathbb{Z}^{(m,n)}}\left[R_{\mathscr{M}}(S)F\right](\omega)\\&=\left(\det\mathscr{M}\right)^{\frac{n}{2}}c_{\mathscr{M}}(S,(\tau,\phi))^{-1}\\&\quad\times\sum_{\omega\in\mathbb{Z}^{(m,n)}}\left[\pi_{\mathscr{M}}\left((-\mu,\lambda,\kappa)\left(-\frac{1}{\tau},\phi+\arg\tau\right)\right)f\right](x)\\&=\left(\det\mathscr{M}\right)^{\frac{n}{2}}c_{\mathscr{M}}(S,(\tau,\phi))^{-1}\,\Theta_f^{[\mathscr{M}]}\\&\quad\left(-\frac{1}{\tau},\ \phi+\arg\tau\,;-\mu,\lambda,\kappa\right).\end{aligned}$$

On the other hand,

$$\begin{aligned}\sum_{\omega\in\mathbb{Z}^{(m,n)}}F_{\mathscr{M}}(\omega)&=\sum_{\omega\in\mathbb{Z}^{(m,n)}}F\left(\mathscr{M}^{-1}\omega\right)\\&=\sum_{\omega\in\mathbb{Z}^{(m,n)}}\left[\pi_{\mathscr{M}}((\lambda,\mu,\kappa)(\tau,\phi))f\right]\left(\mathscr{M}^{-1}\omega\right)\\&=\sum_{\omega\in\mathbb{Z}^{(m,n)}}\left[\pi_{\mathscr{M}}((\lambda,\mu,\kappa)(\tau,\phi))f\right](\omega)\quad\left(\because\ \mathscr{M}^{-1}\mathbb{Z}^{(m,n)}=\mathbb{Z}^{(m,n)}\right)\\&=\Theta_f^{[\mathscr{M}]}(\tau,\phi\,;\lambda,\mu,\kappa).\end{aligned}$$

Hence from (2.12.13) we obtain the desired formula

$$\begin{aligned}&\Theta_f^{[\mathscr{M}]}\left(-\frac{1}{\tau},\ \phi+\arg\tau\,;-\mu,\lambda,\kappa\right)\\&\quad=\left(\det\mathscr{M}\right)^{-\frac{n}{2}}c_{\mathscr{M}}(S,(\tau,\phi))\,\Theta_f^{[\mathscr{M}]}(\tau,\phi\,;\lambda,\mu,\kappa).\end{aligned}$$

If

$$S=\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix},\ \ (\tau,\phi)=\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}\ \text{ and }\ S\cdot(\tau,\phi)=\begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix}\in SL(2,\mathbb{R}),$$

according to Lemma 2.12.1, we get easily

$$c_1c_2c_3=\left(u^2+v^2\right)^{1/2}\sin\phi\ \sin(\phi+\arg\tau),$$

where

$$(\tau,\phi)=\begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}\begin{pmatrix} v^{1/2} & 0 \\ 0 & v^{-1/2} \end{pmatrix}\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$$

is the Iwasawa decomposition of $(\tau,\phi)\in SL(2,\mathbb{R})$. Thus we obtain

$$c_{\mathscr{M}}(S,(\tau,\phi))=e^{i\,\pi mn\,\mathrm{sign}(c_1c_2c_3)}=e^{i\,\pi mn\,\mathrm{sign}(\sin\phi\ \sin(\phi+\arg\tau))}.$$

This completes the proof. □

Theorem 2.12.6 (Jacobi 2) *Let $\mathscr{M}=(\mathscr{M}_{kl})$ be a positive definite symmetric integral $m\times m$ matrix and let $s=(s_{kj})\in\mathbb{Z}^{(m,n)}$ be integral. Then we have*

$$\Theta_f^{[\mathscr{M}]}(\tau+2,\phi\,;\lambda,s-2\,\lambda+\mu,\kappa-s\,{}^t\lambda)=\Theta_f^{[\mathscr{M}]}(\tau,\phi\,;\lambda,\mu,\kappa)$$

for all $(\tau,\phi)\in SL(2,\mathbb{R})$ and $(\lambda,\mu,\kappa)\in H_{\mathbb{R}}^{(n,m)}$.

Proof For brevity, we put $T_*=\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. According to Lemma 2.12.1, for any $(\tau,\phi)\in SL(2,\mathbb{R})$, the multiplication of T_* and (τ,ϕ) is given by

$$T_*(\tau,\phi)=(\tau+2,\phi).\tag{2.12.14}$$

For $s\in\mathbb{R}^{(m,n)}$, $(\lambda,\mu,\kappa)\in H_{\mathbb{R}}^{(n,m)}$ and $(\tau,\phi)\in SL(2,\mathbb{R})$, according to (2.12.14),

$$\begin{aligned}
&\pi_{\mathscr{M}}((0,s,0)T_*)\,\pi_{\mathscr{M}}((\lambda,\mu,\kappa)(\tau,\phi))\\
&=\mathscr{W}_{\mathscr{M}}(0,s,0)R_{\mathscr{M}}(T_*)\mathscr{W}_{\mathscr{M}}(\lambda,\mu,\kappa)R_{\mathscr{M}}(\tau,\phi)\\
&=\mathscr{W}_{\mathscr{M}}(0,s,0)\mathscr{W}_{\mathscr{M}}\left((\lambda,\mu)T_*^{-1},\kappa\right)R_{\mathscr{M}}(T_*)R_{\mathscr{M}}(\tau,\phi)\\
&=c_{\mathscr{M}}(T_*,(\tau,\phi))^{-1}\mathscr{W}_{\mathscr{M}}(\lambda,s-2\,\lambda+\mu,\kappa-s\,{}^t\lambda)R_{\mathscr{M}}\left(T_*(\tau,\phi)\right)\\
&=\mathscr{W}_{\mathscr{M}}(\lambda,s-2\,\lambda+\mu,\kappa-s\,{}^t\lambda)R_{\mathscr{M}}(\tau+2,\phi)\\
&=\pi_{\mathscr{M}}\left((\lambda,s-2\,\lambda+\mu,\kappa-s\,{}^t\lambda)(\tau+2,\phi)\right).
\end{aligned}$$

Here we used the fact that $c_{\mathscr{M}}(T_*,(\tau,\phi))=1$ because T_* is upper triangular.

On the other hand, according to the assumptions on $\mathscr{M}$ and s, for $f \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$ and $\omega \in \mathbb{Z}^{(m,n)}$, using Formulas (2.10.1), (2.10.10) or (2.12.6), we have

$$\begin{aligned}
&\left[\pi_{\mathscr{M}}\left((0,s,0)T_*\right)\pi_{\mathscr{M}}\left((\lambda,\mu,\kappa)(\tau,\phi)\right)f\right](\omega)\\
&=\left[\mathscr{W}_{\mathscr{M}}(0,s,0)R_{\mathscr{M}}(T_*)\,\pi_{\mathscr{M}}\left((\lambda,\mu,\kappa)(\tau,\phi)\right)f\right](\omega)\\
&=e^{2\pi i\,\sigma(\mathscr{M}\omega\,{}^t s)}\cdot e^{2\,\|\omega\|^2_{\mathscr{M}}\pi\,i}\,\left[R_{\mathscr{M}}(i,0)\,\pi_{\mathscr{M}}\left((\lambda,\mu,\kappa)(\tau,\phi)\right)f\right](\omega)\\
&=\left[\pi_{\mathscr{M}}\left((\lambda,\mu,\kappa)(\tau,\phi)\right)f\right](\omega).
\end{aligned}$$

Here we used the facts that

$$e^{2\pi i\,\sigma(\mathscr{M}\omega\,{}^t s)}=1,\quad e^{2\,\|\omega\|^2_{\mathscr{M}}\pi\,i}=1\quad\text{and}\quad R_{\mathscr{M}}(i,0)f=f\ \text{(cf. (2.12.7))}.$$

Therefore for $f\in\mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$,

$$\begin{aligned}
&\Theta_f^{[\mathscr{M}]}(\tau+2,\phi\,;\lambda,s-2\,\lambda+\mu,\kappa-s\,{}^t\lambda)\\
&=\sum_{\omega\in\mathbb{Z}^{(m,n)}}\left[\pi_{\mathscr{M}}\left((\lambda,s-2\,\lambda+\mu,\kappa-s\,{}^t\lambda)(\tau+2,\phi)\right)f\right](\omega)\\
&=\sum_{\omega\in\mathbb{Z}^{(m,n)}}\left[\pi_{\mathscr{M}}\left((0,s,0)T_*\right)\pi_{\mathscr{M}}\left((\lambda,\mu,\kappa)(\tau,\phi)\right)f\right](\omega)\\
&=\sum_{\omega\in\mathbb{Z}^{(m,n)}}\left[\pi_{\mathscr{M}}\left((\lambda,\mu,\kappa)(\tau,\phi)\right)f\right](\omega)\\
&=\Theta_f^{[\mathscr{M}]}(\tau,\phi\,;\lambda,\mu,\kappa).
\end{aligned}$$

This completes the proof. □

Theorem 2.12.7 (Jacobi 3) *Let $\mathscr{M}=(\mathscr{M}_{kl})$ be a positive definite symmetric integral $m\times m$ matrix and let $(\lambda_0,\mu_0,\kappa_0)\in H_{\mathbb{Z}}^{(n,m)}$ be an integral element of $H_{\mathbb{R}}^{(n,m)}$. Then we have*

$$\begin{aligned}
&\Theta_f^{[\mathscr{M}]}(\tau,\phi\,;\lambda+\lambda_0,\mu+\mu_0,\kappa+\kappa_0+\lambda_0\,{}^t\mu-\mu_0\,{}^t\lambda)\\
&=e^{\pi\,i\,\sigma(\mathscr{M}(\kappa_0+\mu_0\,{}^t\lambda_0))}\Theta_f^{[\mathscr{M}]}(\tau,\phi\,;\lambda,\mu,\kappa)
\end{aligned}$$

for all $(\tau,\phi)\in SL(2,\mathbb{R})$ and $(\lambda,\mu,\kappa)\in H_{\mathbb{R}}^{(n,m)}$.

Proof For any $f\in\mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$, we have

$$\sum_{\omega\in\mathbb{Z}^{(m,n)}}\left[\mathscr{W}_{\mathscr{M}}(\lambda_0,\mu_0,\kappa_0)\pi_{\mathscr{M}}\left((\lambda,\mu,\kappa)(\tau,\phi)\right)f\right](\omega)$$

$$= \sum_{\omega\in\mathbb{Z}^{(m,n)}} [\mathscr{W}_{\mathscr{M}}(\lambda_0,\mu_0,\kappa_0)\mathscr{W}_{\mathscr{M}}(\lambda,\mu,\kappa)R_{\mathscr{M}}(\tau,\phi)f](\omega)$$

$$= \sum_{\omega\in\mathbb{Z}^{(m,n)}} \left[\mathscr{W}_{\mathscr{M}}(\lambda_0+\lambda,\mu_0+\mu,\kappa_0+\kappa+\lambda_0{}^t\mu-\mu_0{}^t\lambda))R_{\mathscr{M}}(\tau,\phi)f\right](\omega)$$

$$= \sum_{\omega\in\mathbb{Z}^{(m,n)}} \left[\pi_{\mathscr{M}}\left((\lambda_0+\lambda,\mu_0+\mu,\kappa_0+\kappa+\lambda_0{}^t\mu-\mu_0{}^t\lambda)(\tau,\phi)\right)f\right](\omega)$$

$$= \Theta_f^{[\mathscr{M}]}(\tau,\phi\,;\lambda+\lambda_0,\mu+\mu_0,\kappa+\kappa_0+\lambda_0{}^t\mu-\mu_0{}^t\lambda).$$

On the other hand, for any $f\in\mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$, we have

$$\sum_{\omega\in\mathbb{Z}^{(m,n)}} \left[\mathscr{W}_{\mathscr{M}}(\lambda_0,\mu_0,\kappa_0)\pi_{\mathscr{M}}\left((\lambda,\mu,\kappa)(\tau,\phi)\right)f\right](\omega)$$

$$= \sum_{\omega\in\mathbb{Z}^{(m,n)}} e^{\pi i\sigma\{\mathscr{M}(\kappa_0+\mu_0{}^t\lambda_0+2\,\omega\,{}^t\mu_0)\}}\,[\pi_{\mathscr{M}}(\tau,\phi\,;\lambda,\mu,\kappa)f](\omega+\lambda_0)$$

$$= e^{\pi i\sigma\{\mathscr{M}(\kappa_0+\mu_0{}^t\lambda_0)\}} \sum_{\omega\in\mathbb{Z}^{(m,n)}} [\pi_{\mathscr{M}}(\tau,\phi\,;\lambda,\mu,\kappa)f](\omega+\lambda_0) \quad (\because\ \mu_0 \text{ is integral})$$

$$= e^{\pi i\sigma\{\mathscr{M}(\kappa_0+\mu_0{}^t\lambda_0)\}} \sum_{\omega\in\mathbb{Z}^{(m,n)}} [\pi_{\mathscr{M}}(\tau,\phi\,;\lambda,\mu,\kappa)f](\omega) \quad (\because\ \lambda_0 \text{ is integral})$$

$$= e^{\pi i\sigma\{\mathscr{M}(\kappa_0+\mu_0{}^t\lambda_0)\}}\,\Theta_f^{[\mathscr{M}]}(\tau,\phi\,;\lambda,\mu,\kappa).$$

Finally, we obtain the desired result. □

We put $V(m,n)=\mathbb{R}^{(m,n)}\times\mathbb{R}^{(m,n)}$. Let

$$G^{(m,n)} := SL(2,\mathbb{R})\ltimes V(m,n)$$

be the group with the following multiplication law:

$$(g_1,(\lambda_1,\mu_1))\cdot(g_2,(\lambda_2,\mu_2)) = (g_1g_2,(\lambda_1,\mu_1)g_2+(\lambda_2,\mu_2)), \tag{2.12.15}$$

where $g_1,g_2\in SL(2,\mathbb{R})$ and $\lambda_1,\lambda_2,\mu_1,\mu_2\in\mathbb{R}^{(m,n)}$.

We define

$$\Gamma^{(m,n)} := SL(2,\mathbb{Z})\ltimes H_{\mathbb{Z}}^{(n,m)}.$$

Then $\Gamma^{(m,n)}$ acts on $G^{(m,n)}$ naturally through the multiplication law (2.12.15).

Lemma 2.12.8 *$\Gamma^{(m,n)}$ is generated by the elements*

$$(S,(0,0)),\quad (T_{\mathfrak{b}},(0,s)) \quad\text{and}\quad (I_2,(\lambda_0,\mu_0)),$$

where

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T_\flat = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and } s, \lambda_0, \mu_0 \in \mathbb{Z}^{(m,n)}.$$

Proof Since $SL(2,\mathbb{Z})$ is generated by S and $T_\flat$, we get the desired result. □

We define

$$\begin{aligned} &\Theta_f^{[\mathscr{M}]}(\tau, \phi; \lambda, \mu) \\ &= v^{\frac{mn}{4}} \sum_{\omega \in \mathbb{Z}^{(m,n)}} e^{\pi i \left\{ u \|\omega + \lambda\|^2_{\mathscr{M}} + 2(\omega, \mu)_{\mathscr{M}} \right\}} \left[R_{\mathscr{M}}(i, \phi) f \right] \left(v^{1/2}(\omega + \lambda) \right). \end{aligned}$$

Theorem 2.12.9 *Let $\Gamma^{(m,n)}_{[2]}$ be the subgroup of $\Gamma^{(m,n)}$ generated by the elements*

$$(S, (0,0)), \quad (T_*, (0,s)) \quad \text{and} \quad (I_2, (\lambda_0, \mu_0)),$$

where

$$T_* = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \quad \text{and } s, \lambda_0, \mu_0 \in \mathbb{Z}^{(m,n)}.$$

Let $\mathscr{M} = (\mathscr{M}_{kl})$ be a positive definite symmetric unimodular integral $m \times m$ matrix such that $\mathscr{M}\mathbb{Z}^{(m,n)} = \mathbb{Z}^{(m,n)}$. Then for $f, g \in \mathscr{S}\left(\mathbb{R}^{(m,n)}\right)$, the function

$$\Theta_f^{[\mathscr{M}]}(\tau, \phi; \lambda, \mu)\, \overline{\Theta_g^{[\mathscr{M}]}(\tau, \phi; \lambda, \mu)}$$

is invariant under the action of $\Gamma^{(m,n)}_{[2]}$ on $G^{(m,n)}$.

Proof The proof follows directly from Theorem 2.12.5 (Jacobi 1), Theorem 2.12.6 (Jacobi 2) and Theorem 2.12.7 (Jacobi 3) because the left actions of the generators of $\Gamma^{(m,n)}_{[2]}$ are given by

$$\begin{aligned} ((\tau, \phi), (\lambda, \mu)) &\longmapsto \left(\left(-\frac{1}{\tau}, \phi + \arg \tau \right), (-\mu, \lambda) \right), \\ ((\tau, \phi), (\lambda, \mu)) &\longmapsto ((\tau + 2, \phi), (\lambda, s - 2\lambda + \mu)) \end{aligned}$$

and

$$((\tau, \phi), (\lambda, \mu)) \longmapsto ((\tau, \phi), (\lambda + \lambda_0, \mu + \mu_0)).$$

□

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